

Mathematics for Economists

A Graduate Reference Volume

Prepared for the Swiss Program for Beginning Doctoral Students in Economics

Study Center Gerzensee

Abstract

These notes develop the mathematics required for the first year of an economics doctoral program, and somewhat beyond it. They follow the topics of the Mathematics Review Course taught at the Study Center Gerzensee within the Swiss Program for Beginning Doctoral Students in Economics — probability and conditional expectation, single- and multi-variable calculus, static optimisation, real analysis, and discrete-time dynamic programming — and extend each of them to the level of rigor at which the results are actually used in microeconomic theory, macroeconomics, and econometrics.

Every definition carries its standing hypotheses, every theorem states the structure required of the underlying space, and proofs are given in full except where a result is quoted from a source cited at the point of use. Economic applications are worked rather than gestured at: the environment, the correspondence between mathematical and economic objects, the derivation, and the consequences of relaxing the hypotheses are all stated.

Preface

A first-year economics doctoral student meets three kinds of mathematical difficulty. The first is technical: an argument uses the implicit function theorem, or the Banach fixed point theorem, or the fact that a continuous function on a compact set attains its maximum, and the student has either never seen the result or has seen it without its hypotheses. The second is translational: the mathematics is familiar but the economic object it stands for is not, so that a Lagrange multiplier is manipulated correctly without being recognised as a shadow price and a fixed point is computed without being recognised as an equilibrium. The third is architectural: the results are known individually but not as a system, so that the student does not see that the existence of a Walrasian equilibrium, the existence of a value function solving a Bellman equation, and the existence of a least-squares projection are three instances of the same handful of theorems about compactness, completeness, and convexity.

These notes address all three. They follow the topics of the Mathematics Review Course of the Swiss Program for Beginning Doctoral Students in Economics, taught at the Study Center Gerzensee: probability and conditional expectation, calculus in one and several variables, static optimisation, real analysis, and discrete-time dynamic programming (Fukuda 2026e). That course compresses these subjects into four days, and its slides are deliberately economical — they state what is needed, illustrate it, and move on. A student who wants to know why a hypothesis is there, what fails without it, or how a result generalises must look elsewhere. The purpose of this volume is to be that elsewhere, while keeping the course’s sequence, its notation, and above all its economic motivation.

The mathematical standard aimed at is that of a graduate text rather than that of a review course. Every theorem states the space it lives on and the regularity it requires. Necessary conditions are distinguished from sufficient ones, existence from uniqueness, and local statements from global ones. Where a teaching-level simplification appears in the course slides, the simplification is retained for its intuitive value and the exact statement is given alongside it, with the relation between the two made explicit; the Kuhn–Tucker theorem in [Chapter 12](#) and the envelope formula in [Chapter 15](#) are the clearest instances. Proofs are given in full, except where a result is quoted, in which case the source is named at the point of use and the reason for omission is stated.

The economic applications are worked rather than named. Each one specifies the environment, the preferences or technology, the constraint set, the correspondence between mathematical and economic objects, the derivation, and — this is the part usually left out — what goes wrong when a hypothesis fails. The applications are drawn from the course itself: profit maximisation by price-taking and price-setting firms, the risk premium and the Arrow–Pratt coefficient, ordinary least squares as an orthogonal projection, the consumer-choice problem, intertemporal consumption and the Euler equation, Nash bargaining, Bayesian updating in a market-microstructure setting, normal-normal learning, and the cake-eating, Ramsey growth, and Lucas asset-pricing problems. Each is developed to the point where the mathematics is doing visible work.

Sources

The volume rests on three layers of source material, and the reader should know which is which.

The first layer is the set of six slide decks of the Gerzensee course (Fukuda 2026a,b,c,e,f,g). These fix the scope: no topic is included that the course does not need, and no result stated in the slides is omitted here.

The second layer is the accompanying reference volume (Fukuda 2026d), a 768-page set of lecture notes of which the slides are a distillation. It supplies definitions the slides state informally, proofs the slides sketch, exercises the slides mark optional, and the technical conditions the slides suppress. Where a result appears in both, the numbering of the reference volume is given so that the two can be read together.

The third layer consists of the mathematical texts that carry each topic beyond what a review course can reach: Rudin (1976) and Amann and Escher (2005, 2008, 2009) for real analysis; Axler (2024) for linear algebra; Boyd and Vandenberghe (2004) and Rockafellar (1970) for convexity and optimisation; Le Gall (2022) and Le Gall (2016) for measure-theoretic probability; Corbae et al. (2009) and Ok (2007) for analysis written with economic applications in view; and Ljungqvist and Sargent (2018) and Stokey et al. (1989) for recursive methods. Each is cited, with the relevant chapter or section, at the points where its content is used. Le Gall (2016) is listed because it is the natural continuation for readers who go on to continuous-time finance; nothing in this volume requires stochastic calculus, and it is therefore not cited again.

How to Use These Notes

Order of reading

The chapters are ordered by logical dependence, not by the order in which the four-day course presents them. The course opens with probability because Bayes' rule is immediately useful; but the sharpest statement about conditional expectation — that it is the orthogonal projection of a square-integrable random variable onto the space of functions of the conditioning variable — needs completeness, convexity, and a projection theorem. Those are developed in Chapters 6 and 9, so probability appears here as Chapter 14. The dependence structure is as follows.

- **Foundations.** Chapter 1 (sets, relations, logic, the real number system) and Chapter 2 (vector spaces, linear maps, matrices, spectra) presuppose nothing.
- **Analysis.** Chapter 3 (inner products, norms, metrics) requires Chapter 2. Chapter 4 (open and closed sets, sequences, compactness) requires Chapter 3. Chapter 5 (continuity, semicontinuity, the Weierstrass theorem) requires Chapter 4. Chapter 6 (completeness, the contraction mapping theorem, Blackwell's conditions) requires Chapter 5.
- **Calculus.** Chapter 7 (one variable) requires only Chapter 1 and the limit notions of Chapter 4. Chapter 8 (several variables, the implicit function theorem) requires Chapter 7, Chapter 2 and, for the proof of the implicit function theorem, Chapter 6. Chapter 10 (integration) requires Chapter 7.
- **Convexity and optimisation.** Chapter 9 (convex sets and functions, separation) requires Chapter 4 and Chapter 8. Chapter 11 (projection, orthogonality, least squares) requires Chapter 6 and Chapter 9. Chapter 12 (Lagrange and Kuhn–Tucker) requires Chapter 9 and Chapter 8. Chapter 13 (correspondences, the maximum theorem, fixed points) requires Chapter 5 and Chapter 9.
- **Probability and dynamics.** Chapter 14 requires Chapter 10 and Chapter 11. Chapter 15 (discrete-time dynamic programming) requires Chapter 6, Chapter 12, Chapter 13, and, for the stochastic case, Chapter 14. Chapter 16 (continuous-time optimisation) requires Chapter 8 and Chapter 15.

A reader following the four-day course can read Chapter 14, then Chapters 7, 8 and 10, then Chapter 12, then Chapters 3–6, then Chapter 15, consulting earlier chapters as the cross-references demand.

Conventions used throughout

Definitions are boxed in green, formal statements in blue, and remarks and examples are set plainly. A statement labelled **Assumption** is a standing hypothesis that remains in force until it is explicitly withdrawn; a statement labelled **Theorem**, **Proposition**, **Lemma**, or **Corollary** carries all of its hypotheses within it and depends on no standing assumption except those stated in its own text or in an assumption it cites by number.

Every result is numbered within its section, so that Theorem 7.3.2 is the second numbered statement of Section 7.3. Cross-references are hyperlinked. The two indices at the end of the volume list, respectively, defined terms and named results, and the symbols with the page on which each is introduced.

Mathematical Prerequisites

The volume is self-contained in the sense that every object it uses is defined before it is used. It is not self-contained in the sense of assuming no prior exposure: a reader meeting the derivative or the determinant for the first time here will find the treatment too fast. What is assumed is the content of a solid undergraduate program in mathematics for economics, namely

- (i) elementary set notation and the manipulation of quantifiers;
- (ii) single-variable differential and integral calculus at the computational level — differentiating and integrating elementary functions, and the statement of the fundamental theorem of calculus;
- (iii) matrix algebra: multiplication, inversion, determinants, and the solution of linear systems;
- (iv) elementary probability: random variables, expectation, variance, and the common families of distributions.

Each of these is revisited from a higher standpoint rather than assumed silently. [Chapter 1](#) restates set theory in the language of relations and mappings and constructs the real numbers axiomatically; [Chapter 2](#) replaces matrix algebra by the theory of linear maps between vector spaces, recovering matrices as coordinate representations; [Chapter 7](#) rebuilds the derivative as a local linear approximation; and [Chapter 14](#) builds probability on a measure space. A reader who is comfortable with (i)–(iv) can read the corresponding chapters quickly, but should not skip them: the higher standpoint is what the later chapters use.

No prior exposure to metric-space topology, convex analysis, measure theory, or dynamic programming is assumed.

Notation and Conventions

The notation follows the Gerzensee course materials wherever they fix a convention, and standard mathematical usage elsewhere. Conflicts between sources have been resolved in favour of the version that makes hypotheses most visible; where a competing convention is common in economics, it is noted at the point of first use.

Sets and numbers

Symbol	Meaning
$\mathbb{N}, \mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}$	natural numbers (excluding 0), integers, rationals, reals, complex numbers
$\mathbb{N}_0$	$\mathbb{N} \cup \{0\}$
$\mathbb{R}_+, \mathbb{R}_{++}$	$[0, \infty)$ and $(0, \infty)$
$\overline{\mathbb{R}}$	extended real line $\mathbb{R} \cup \{-\infty, +\infty\}$
$\mathbb{R}^n$	n -fold Cartesian product of $\mathbb{R}$; elements are column vectors
$\mathbb{R}_+^n, \mathbb{R}_{++}^n$	vectors with all coordinates ≥ 0 , respectively > 0
$A^c, A \setminus B$	complement of A , set difference
$\mathcal{P}(X)$	power set of X
$\text{Int } A, \overline{A}, \partial A$	interior, closure, boundary of A
$\text{ri } A, \text{co } A, \text{aff } A$	relative interior, convex hull, affine hull of A
$\mathbb{1}_A$	indicator function of the set A
$:=$	equality that defines the left-hand side

Vectors, matrices and maps

Vectors in $\mathbb{R}^n$ are columns; $x^\top$ denotes the transpose, so that $x^\top y = \langle x, y \rangle = \sum_{i=1}^n x_i y_i$. Matrices are written A, B, X , with $A^\top$ the transpose, A^{-1} the inverse when it exists, $\text{rank } A$ the rank, $\text{tr } A$ the trace, $\det A$ the determinant, $\text{Ker } A$ the kernel and $\text{Im } A$ the image. The i -th standard basis vector of $\mathbb{R}^n$ is e^i . For a matrix A , the notation $A \succeq 0$ ($A \succ 0$) means that A is symmetric positive semidefinite (positive definite).

For $f: A \rightarrow B$ the notation $f(S)$ denotes the image of $S \subseteq A$ and $f^{-1}(T)$ the preimage of $T \subseteq B$; f^{-1} is used for the inverse map only where f is stated to be a bijection. A correspondence, that is a set-valued map, is written $\Gamma: A \rightrightarrows B$ and its graph is $\text{gr } \Gamma$.

Derivatives use $f'(x)$ in one variable, $\nabla f(x) \in \mathbb{R}^n$ for the gradient of a real-valued f on $\mathbb{R}^n$, $Df(x)$ for the $m \times n$ Jacobian matrix of $f: \mathbb{R}^n \rightarrow \mathbb{R}^m$, and $D^2 f(x)$ for the $n \times n$ Hessian. The gradient is a column vector and $Df(x) = \nabla f(x)^\top$ when $m = 1$; this is the one place where a transpose is easy to lose, and it is stated explicitly whenever it matters.

Analysis

A metric space is a pair (X, d) and a normed space a pair $(X, \|\cdot\|)$; the pair is abbreviated to X when the metric or norm is clear. Open balls are $B_\varepsilon(x) = \{y \in X : d(x, y) < \varepsilon\}$. Sequences are $(x_n)_{n \in \mathbb{N}}$ and subsequences $(x_{n_k})_{k \in \mathbb{N}}$. The spaces of continuous and of bounded functions from X to Y are $\mathcal{C}(X, Y)$ and $\mathcal{B}(X, Y)$; both carry the supremum norm $\|f\|_\infty = \sup_{x \in X} |f(x)|$ unless stated otherwise.

Optimisation

A constrained problem is written

$$\max_{x \in \mathbb{R}^m} f(x) \quad \text{subject to} \quad g_j(x) \geq 0, \quad j = 1, \dots, n,$$

with the constraints in the ≥ 0 orientation used throughout the Gerzensee course (Fukuda 2026c, slide 3). The feasible set is D , the Lagrangian is $\mathcal{L}$, and multipliers are $y = (y_1, \dots, y_n) \in \mathbb{R}_+^n$. Writing constraints as $g_j \geq 0$ rather than $g_j \leq 0$ fixes the sign of the multiplier at a maximum to be non-negative, which is what makes the shadow-price reading correct without a case distinction; the opposite convention, common in the optimisation literature, is noted in Chapter 12.

Probability

A probability space is $(\Omega, \mathcal{F}, \mathbb{P})$, random variables are X, Y, Z , and realisations are x, y, z . Expectation, variance and covariance are $\mathbb{E}$, Var and Cov ; $X \perp\!\!\!\perp Y$ denotes independence. Conditional expectation given an event H is $\mathbb{E}[X \mid H]$ and given a random variable Y is $\mathbb{E}[X \mid Y]$, the latter being a random variable, that is, a function of Y . Precision is $\tau = 1/\sigma^2$, following the treatment of normal-normal updating in (Fukuda 2026f, slides 27–28).

Dynamics

Time is $t \in \mathbb{N}_0$ in discrete time and $t \in [0, \infty)$ in continuous time. State variables are x and controls u ; the discount factor is $\beta \in (0, 1)$; the transition is $x' = g(x, u)$, with a prime denoting the next period, as in (Fukuda 2026b, slide 8). The value function is V , the policy function is h , and the Bellman operator is T .

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Chapter 1

Sets, Relations, Logic, and the Real Number System

1.1 Motivation: Why an Economist Starts with Sets

An economic model is a statement about a set of alternatives, a rule for comparing them, and a subset of them that is feasible. A consumer facing prices $p \in \mathbb{R}_{++}^n$ and wealth $m > 0$ chooses from the budget set $\mathcal{B}(p, m) = \{x \in \mathbb{R}_+^n : \langle p, x \rangle \leq m\}$; a firm chooses from a production set; a player in a game chooses from a strategy set. The comparison rule is a binary relation on the set of alternatives, and the object economists call a utility function is a numerical representation of that relation. Every subsequent chapter presupposes that these three objects — a set, a relation, a distinguished subset — have been given a precise meaning.

This chapter therefore does three things. It fixes set-theoretic language (Section 1.2–Section 1.3); it records the forms of argument used throughout the volume (Section 1.4); and it constructs the real number system axiomatically (Section 1.5), because the single property that distinguishes $\mathbb{R}$ from $\mathbb{Q}$ — the least upper bound property — is what makes every existence theorem in later chapters true. The chapter closes with the first substantive economic application, the representation of a preference relation by a utility function (Section 1.6).

1.2 Sets, Operations, and Mappings

1.2.1 Sets and their specification

We take the notion of a set as primitive: a *set* is a collection of objects, called its *elements* or members. We write $a \in A$ when a is an element of A and $a \notin A$ otherwise. The set with no elements is the *empty set*, written $\emptyset$.

A set may be specified by enumeration, as in $\{a_1, \dots, a_n\}$, or by a defining property relative to an ambient set, as in $\{x \in X : P(x)\}$. The ambient set matters: the collection of all x with $P(x)$, taken without an ambient set, is not in general a set, and the restriction to subsets of a given set is what avoids the classical paradoxes. All constructions in this volume take place inside an ambient set that is either named or clear from context.

Definition 1.2.1 (Inclusion and equality). Let A and B be sets. We say A is a *subset* of B , written $A \subseteq B$, if $a \in A$ implies $a \in B$. We say $A = B$ if $A \subseteq B$ and $B \subseteq A$. If $A \subseteq B$ and $A \neq B$, then A is a *proper subset* of B , written $A \subsetneq B$.

The definition of equality dictates the standard proof technique: to establish $A = B$ one establishes two inclusions. This is used so often below that it is worth naming once.

Definition 1.2.2 (Operations on sets). Let A, B be subsets of an ambient set X , and let $(A_i)_{i \in I}$ be a family of subsets of X indexed by a set I .

- (i) The *complement* of A in X is $A^c := \{x \in X : x \notin A\}$.
- (ii) The *union* and *intersection* of the family are

$$\bigcup_{i \in I} A_i := \{x \in X : x \in A_i \text{ for some } i \in I\}, \quad \bigcap_{i \in I} A_i := \{x \in X : x \in A_i \text{ for all } i \in I\}.$$
- (iii) The *set difference* is $A \setminus B := A \cap B^c$.
- (iv) The *Cartesian product* of sets $A_1, \dots, A_n$ is $\prod_{i=1}^n A_i := \{(a_1, \dots, a_n) : a_i \in A_i \text{ for each } i\}$, the set of ordered n -tuples.
- (v) The *power set* of X is $\mathcal{P}(X) := \{A : A \subseteq X\}$.

The index set I in (ii) is arbitrary: it may be finite, countably infinite, or uncountable. This generality is not idle. In [Chapter 4](#) the collection of open sets is required to be closed under arbitrary unions but only under finite intersections, and the whole of point-set topology turns on that asymmetry.

Proposition 1.2.1 (De Morgan's laws). Let $(A_i)_{i \in I}$ be a family of subsets of X with $I \neq \emptyset$. Then

$$\left(\bigcup_{i \in I} A_i \right)^c = \bigcap_{i \in I} A_i^c, \quad \left(\bigcap_{i \in I} A_i \right)^c = \bigcup_{i \in I} A_i^c.$$

Proof. For the first identity, $x \in \left(\bigcup_{i \in I} A_i \right)^c$ holds if and only if $x \in X$ and it is not the case that $x \in A_i$ for some $i \in I$; that is, if and only if $x \in X$ and $x \notin A_i$ for every $i \in I$; that is, if and only if $x \in A_i^c$ for every i , which is $x \in \bigcap_{i \in I} A_i^c$.

The second identity follows from the first applied to the family $(A_i^c)_{i \in I}$ together with $(A^c)^c = A$: indeed $\left(\bigcup_{i \in I} A_i^c \right)^c = \bigcap_{i \in I} A_i$, and taking complements of both sides gives the claim. ■

Remark 1.2.1. The hypothesis $I \neq \emptyset$ is needed. With $I = \emptyset$ the convention $\bigcup_{i \in \emptyset} A_i = \emptyset$ and $\bigcap_{i \in \emptyset} A_i = X$ makes both identities hold, but only because the intersection over the empty family is defined to be the ambient set. Outside an ambient set that convention has no meaning, which is a second reason for insisting on one.

1.2.2 Mappings

Definition 1.2.3 (Mapping). Let A and B be sets. A *mapping* (or *function*) $f: A \rightarrow B$ assigns to each $a \in A$ exactly one element $f(a) \in B$. The set A is the *domain* and B the *codomain*. For $S \subseteq A$ and $T \subseteq B$,

$$f(S) := \{f(a) : a \in S\} \subseteq B, \quad f^{-1}(T) := \{a \in A : f(a) \in T\} \subseteq A$$

are the *image* of S and the *preimage* of T . The set $f(A)$ is the *range* of a mapping.

The preimage is better behaved than the image, and the difference is the reason that continuity is defined in [Chapter 5](#) through preimages rather than images.

Proposition 1.2.2 (Images and preimages under set operations). Let $f: A \rightarrow B$, let $(S_i)_{i \in I}$ be subsets of A , and let $(T_i)_{i \in I}$ be subsets of B with $I \neq \emptyset$. Then

- (i) $f^{-1}(\bigcup_i T_i) = \bigcup_i f^{-1}(T_i)$ and $f^{-1}(\bigcap_i T_i) = \bigcap_i f^{-1}(T_i)$;
- (ii) $f^{-1}(T^c) = (f^{-1}(T))^c$ for every $T \subseteq B$;
- (iii) $f(\bigcup_i S_i) = \bigcup_i f(S_i)$, whereas in general only $f(\bigcap_i S_i) \subseteq \bigcap_i f(S_i)$.

Proof. (i) We have $a \in f^{-1}(\bigcap_i T_i)$ if and only if $f(a) \in T_i$ for every i , if and only if $a \in f^{-1}(T_i)$ for every i . The union case is identical with “some” in place of “every”.

(ii) $a \in f^{-1}(T^c)$ if and only if $f(a) \notin T$, if and only if $a \notin f^{-1}(T)$.

(iii) If $b \in f(\bigcup_i S_i)$ then $b = f(a)$ for some $a \in S_{i_0}$, so $b \in f(S_{i_0})$; conversely each $f(S_i) \subseteq f(\bigcup_j S_j)$. For intersections, if $a \in \bigcap_i S_i$ then $f(a) \in f(S_i)$ for every i , giving the stated inclusion. ■

Example 1.2.1 (Failure of equality for images of intersections). Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be $f(x) = x^2$, $S_1 = (-\infty, 0)$ and $S_2 = (0, \infty)$. Then $S_1 \cap S_2 = \emptyset$, so $f(S_1 \cap S_2) = \emptyset$, while $f(S_1) = f(S_2) = (0, \infty)$, so $\bigcap_i f(S_i) = (0, \infty)$. The inclusion in Proposition 1.2.2(iii) is therefore strict in general, and it is an equality for all families if and only if f is injective.

Definition 1.2.4 (Injection, surjection, bijection). A mapping $f: A \rightarrow B$ is *injective mapping* if $f(a) = f(a')$ implies $a = a'$; *surjective mapping* if $f(A) = B$; and *bijective mapping* if it is both. A bijection f has a unique inverse $f^{-1}: B \rightarrow A$ satisfying $f^{-1} \circ f = \text{id}_A$ and $f \circ f^{-1} = \text{id}_B$.

Definition 1.2.5 (Finite, countable, uncountable). A set A is *finite* if $A = \emptyset$ or there is a bijection $\{1, \dots, n\} \rightarrow A$ for some $n \in \mathbb{N}$; *countably infinite set* if there is a bijection $\mathbb{N} \rightarrow A$; *countable* if it is finite or countably infinite; and *uncountable* otherwise.

Proposition 1.2.3. A countable union of countable sets is countable. The set $\mathbb{Q}$ is countably infinite; the set $\mathbb{R}$ is uncountable.

Proof. The first two assertions are standard diagonal enumerations; see Rudin (1976, Thm. 2.12 and its corollary). For the third, Cantor's diagonal argument shows that no map $\mathbb{N} \rightarrow (0, 1)$ is surjective: given $f: \mathbb{N} \rightarrow (0, 1)$ with decimal expansions $f(n) = 0.d_{n1}d_{n2}\dots$ (choosing the expansion that does not terminate in all nines), the number $x = 0.c_1c_2\dots$ with $c_n = 5$ if $d_{nn} \neq 5$ and $c_n = 4$ otherwise lies in $(0, 1)$ and differs from every $f(n)$ in its n -th digit. Since $(0, 1) \subseteq \mathbb{R}$, the set $\mathbb{R}$ is uncountable. ■

Remark 1.2.2 (Where countability matters in economics). The distinction is not decorative. In Chapter 14 a probability measure can be assigned to every subset of a countable sample space, but on an uncountable sample space such as $[0, 1]$ no countably additive probability measure exists on the full power set that extends length; one must restrict attention to a σ -algebra of measurable sets (Le Gall 2022, Ch. 1). The Gerzensee treatment makes exactly this point when it moves from “an at most countable set Ω ” (Fukuda 2026f, slide 3) to the remark that a continuum sample space requires a σ -algebra (Fukuda 2026g, slide 37).

1.3 Relations, Orders, and Preferences

Definition 1.3.1 (Binary relation). A *binary relation* R on a set X is a subset $R \subseteq X \times X$; one writes xRy for $(x, y) \in R$. The relation is

- (i) *reflexive relation* if xRx for all $x \in X$;
- (ii) *complete relation* if for all $x, y \in X$, xRy or yRx (or both);
- (iii) *transitive relation* if xRy and yRz imply xRz ;
- (iv) *symmetric relation* if xRy implies yRx ;
- (v) *antisymmetric relation* if xRy and yRx imply $x = y$.

Completeness implies reflexivity (take $y = x$), so the two are not independent hypotheses. Economics texts frequently list both; nothing is gained by it, and we list completeness alone where it is available.

Definition 1.3.2 (Equivalence relation and partition). An *equivalence relation* on X is a reflexive, symmetric and transitive relation, usually written $\sim$. For $x \in X$ the *equivalence class* of x is $[x] := \{y \in X : y \sim x\}$. A *partition* of X is a family of non-empty, pairwise disjoint subsets whose union is X .

Proposition 1.3.1. The equivalence classes of an equivalence relation on X form a partition of X ; conversely, every partition of X arises in this way from exactly one equivalence relation.

Proof. Reflexivity gives $x \in [x]$, so the classes are non-empty and cover X . Suppose $[x] \cap [y] \neq \emptyset$ and take z in the intersection. Then $z \sim x$ and $z \sim y$, so by symmetry and transitivity $x \sim y$; hence for any $w \in [x]$ we get $w \sim x \sim y$ and $w \in [y]$, and symmetrically, so $[x] = [y]$. Thus distinct classes are disjoint.

Conversely, given a partition $\mathcal{P}$, define $x \sim y$ if x and y lie in the same cell. This is reflexive, symmetric and transitive because the cells are non-empty, pairwise disjoint, and cover X ; and its classes are exactly the cells. Uniqueness holds because the relation is determined by its classes. ■

This proposition is used twice in later chapters: in [Chapter 14](#), a finite partition $(H_j)_{j=1}^m$ of the sample space is precisely the information structure underlying the law of iterated expectations ([Fukuda 2026f](#), slide 18); and in the theory of choice, the indifference relation derived from a preference partitions the consumption set into indifference curves.

1.3.1 Preference relations

Definition 1.3.3 (Preference relation). Let X be a non-empty set of alternatives. A *preference relation* $\succsim$ on X is a complete and transitive binary relation. Its *strict part* $\succ$ and *indifference part* $\sim$ are defined by

$$x \succ y \iff (x \succsim y \text{ and not } y \succsim x), \quad x \sim y \iff (x \succsim y \text{ and } y \succsim x).$$

Proposition 1.3.2. Let $\succsim$ be a preference relation on X . Then $\sim$ is an equivalence relation, $\succ$ is transitive and irreflexive, and for all $x, y \in X$ exactly one of $x \succ y$, $y \succ x$, $x \sim y$ holds.

Proof. Reflexivity of $\sim$ follows from completeness. Symmetry is immediate from the definition. For transitivity of $\sim$, if $x \sim y$ and $y \sim z$ then $x \succsim y \succsim z$ gives $x \succsim z$, and $z \succsim y \succsim x$ gives $z \succsim x$.

For $\succ$: irreflexivity is immediate since $x \not\succ x$. For transitivity, suppose $x \succ y$ and $y \succ z$. Transitivity of $\succsim$ gives $x \succsim z$. If also $z \succsim x$, then $z \succsim x \succsim y$ gives $z \succsim y$, contradicting $y \succ z$. Hence $x \succ z$.

For the trichotomy, completeness gives $x \succsim y$ or $y \succsim x$. If both, then $x \sim y$; if only the first, $x \succ y$; if only the second, $y \succ x$. These three cases are mutually exclusive by construction. ■

Definition 1.3.4 (Utility representation). A function $u: X \rightarrow \mathbb{R}$ *utility representation* the preference relation $\succsim$ if

$$x \succsim y \iff u(x) \geq u(y) \quad \text{for all } x, y \in X.$$

Theorem 1.3.1 (Representation on a countable set). Let X be a countable non-empty set and let $\succsim$ be a preference relation on X . Then there exists $u: X \rightarrow \mathbb{R}$ representing $\succsim$, and u may be chosen with $u(X) \subseteq (0, 1)$.

Proof. Enumerate $X = \{x_1, x_2, \dots\}$ (finite or infinite) and choose weights $w_n > 0$ with $W := \sum_n w_n < \infty$, for instance $w_n = 2^{-n}$. Define

$$u(x) := \frac{1}{W} \sum_{n: x \succ x_n} w_n.$$

Suppose $x \succsim y$. If $y \succ x_n$ then $x \succ x_n$: indeed $x \succsim y \succ x_n$ gives $x \succ x_n$, and were $x_n \succ x$ we would get $x_n \succ x \succ y$, contradicting $y \succ x_n$. Hence $\{n: y \succ x_n\} \subseteq \{n: x \succ x_n\}$ and $u(x) \geq u(y)$.

Conversely, suppose $u(x) \geq u(y)$ and, for a contradiction, that $y \succ x$. Then as above $\{n: x \succ x_n\} \subseteq \{n: y \succ x_n\}$. The inclusion is strict: since $y \succ x$ and $x = x_{n_0}$ for some n_0 , we have $n_0 \in \{n: y \succ x_n\}$ while $n_0 \notin \{n: x \succ x_n\}$ by irreflexivity of $\succ$. As $w_{n_0} > 0$, this gives $u(y) > u(x)$, a contradiction. Hence $x \succsim y$.

Finally $0 \leq u \leq 1$; replacing u by $(1 + u)/3$ places the range in $(0, 1)$ without disturbing the representation, since the transformation is strictly increasing. ■

Remark 1.3.1 (Why countability is not innocuous). [Theorem 1.3.1](#) fails on uncountable X without

further structure. Let $X = \mathbb{R}^2$ with the *lexicographic preference*

$$(x_1, x_2) \succsim (y_1, y_2) \iff x_1 > y_1, \text{ or } (x_1 = y_1 \text{ and } x_2 \geq y_2).$$

This relation is complete and transitive but admits no utility representation. Indeed, suppose u represented it. For each $t \in \mathbb{R}$ we have $(t, 1) \succ (t, 0)$, so $u(t, 1) > u(t, 0)$, and we may pick a rational $q(t) \in (u(t, 0), u(t, 1))$ by [Proposition 1.5.2\(ii\)](#). If $t < t'$ then $(t', 0) \succ (t, 1)$, so $q(t) < u(t, 1) < u(t', 0) < q(t')$; hence $t \mapsto q(t)$ is a strictly increasing injection $\mathbb{R} \rightarrow \mathbb{Q}$, contradicting [Proposition 1.2.3](#).

The lesson carries into [Chapter 5](#): a continuous utility representation requires the preference to be continuous, in the sense that the sets $\{y : y \succsim x\}$ and $\{y : x \succsim y\}$ are closed, and lexicographic preference violates exactly this. The general statement is Debreu's theorem; see [Ok \(2007, Ch. 2\)](#) and [Corbae et al. \(2009, Ch. 4\)](#).

1.4 Logic and Methods of Proof

1.4.1 Connectives and quantifiers

For propositions P, Q , the implication $P \Rightarrow Q$ is by definition the proposition $(\neg P) \vee Q$. This is the source of the “vacuous truth” convention: $P \Rightarrow Q$ is true whenever P is false, regardless of Q . The truth table is

P	Q	$\neg P$	$(\neg P) \vee Q$	$P \Rightarrow Q$
T	T	F	T	T
T	F	F	F	F
F	T	T	T	T
F	F	T	T	T

which reproduces the table given in ([Fukuda 2026g](#), slide 40). The convention is used there to show that $\emptyset$ is open: the statement “if $x \in \emptyset$ then $B_\varepsilon(x) \subseteq \emptyset$ ” is vacuously true, so $\emptyset$ satisfies the definition of an open set. That argument recurs in [Chapter 4](#).

Quantifier order is not commutative and is a frequent source of error. The statement

$$\forall \varepsilon > 0 \exists \delta > 0 \forall x \in A : d(x, x_0) < \delta \Rightarrow d(f(x), f(x_0)) < \varepsilon$$

defines continuity of f at x_0 , where δ may depend on both ε and x_0 . Moving the quantifier $\forall x_0 \in A$ inside, to

$$\forall \varepsilon > 0 \exists \delta > 0 \forall x_0 \in A \forall x \in A : d(x, x_0) < \delta \Rightarrow d(f(x), f(x_0)) < \varepsilon,$$

gives uniform continuity, a strictly stronger property ([Chapter 5](#)). The negation of a quantified statement exchanges $\forall$ and $\exists$ and negates the matrix: $\neg(\forall x P(x))$ is $\exists x \neg P(x)$.

1.4.2 Three methods of proof

Proposition 1.4.1 (Principle of mathematical induction). Let $S \subseteq \mathbb{N}$ satisfy $1 \in S$ and $n \in S \Rightarrow n + 1 \in S$. Then $S = \mathbb{N}$.

The principle is equivalent to the well-ordering of $\mathbb{N}$ — every non-empty subset of $\mathbb{N}$ has a least element — and either may be taken as an axiom. Strong induction, from “ $\{1, \dots, n\} \subseteq S$ implies $n + 1 \in S$ ” conclude $S = \mathbb{N}$, follows by applying [Proposition 1.4.1](#) to $S' = \{n : \{1, \dots, n\} \subseteq S\}$.

Example 1.4.1 (Induction in a dynamic argument). The error bound in the contraction mapping theorem ([Theorem 6.4.1](#)) rests on the estimate $d(x_n, x_{n+1}) \leq \beta^n d(x_0, x_1)$ for all n , which is an induction: the case $n = 0$ is trivial, and $d(x_{n+1}, x_{n+2}) = d(Tx_n, Tx_{n+1}) \leq \beta d(x_n, x_{n+1}) \leq \beta^{n+1} d(x_0, x_1)$.

Proof by contradiction establishes P by assuming $\neg P$ and deriving a false statement; *proof by contraposition* establishes $P \Rightarrow Q$ by establishing $\neg Q \Rightarrow \neg P$, the two being equivalent by the truth table above. The uniqueness half of the contraction mapping theorem is a proof by contradiction; [Proposition 1.3.2](#) used contraposition twice.

Remark 1.4.1 (A common error). “Suppose not; then ...; hence P ” is not a proof by contradiction; it is a proof of P with an unused hypothesis. A genuine argument by contradiction must terminate in a statement known to be false. Confusing the two conceals which hypotheses are actually used, which matters when one later asks whether a hypothesis can be weakened — the recurring question of [Chapter 12](#).

1.5 The Real Number System

The real numbers are characterised, up to isomorphism, by three groups of axioms. We state them because the third is used in essentially every existence theorem in this volume.

Definition 1.5.1 (Ordered field). A *field* is a set F with operations $+$ and $\cdot$ satisfying the usual commutative, associative and distributive laws, with additive identity 0 , multiplicative identity $1 \neq 0$, additive inverses, and multiplicative inverses for every element other than 0 . An *ordered field* is a field with a complete transitive antisymmetric relation $\leq$ such that $x \leq y$ implies $x + z \leq y + z$ for all z , and $0 \leq x, 0 \leq y$ imply $0 \leq xy$.

Definition 1.5.2 (Bounds, supremum, infimum). Let $A \subseteq \mathbb{R}$ be non-empty. An *upper bound* for A is $b \in \mathbb{R}$ with $a \leq b$ for all $a \in A$; A is *bounded above* if such a b exists. The *supremum* $\sup A$ is the least upper bound: an upper bound s such that $s \leq b$ for every upper bound b . *Lower bound*, *bounded below* and *infimum* $\inf A$ are defined symmetrically. A set bounded above and below is *bounded*.

Assumption 1.5.1 (Completeness axiom). $\mathbb{R}$ is an ordered field in which every non-empty subset that is bounded above has a supremum in $\mathbb{R}$.

This is the *least upper bound property*. It fails in $\mathbb{Q}$: the set $\{q \in \mathbb{Q} : q^2 < 2\}$ is bounded above in $\mathbb{Q}$ but has no least upper bound there. The whole difference between $\mathbb{Q}$ and $\mathbb{R}$, and with it every convergence theorem below, rests on this one axiom.

Proposition 1.5.1 (Characterisation of the supremum). Let $A \subseteq \mathbb{R}$ be non-empty and bounded above, and let $s \in \mathbb{R}$. Then $s = \sup A$ if and only if

- (i) $a \leq s$ for all $a \in A$; and
- (ii) for every $\varepsilon > 0$ there exists $a \in A$ with $a > s - \varepsilon$.

Proof. Suppose $s = \sup A$. Then (i) holds since s is an upper bound. For (ii), if some $\varepsilon > 0$ had $a \leq s - \varepsilon$ for all $a \in A$, then $s - \varepsilon$ would be an upper bound smaller than s , contradicting leastness.

Conversely, assume (i) and (ii). By (i), s is an upper bound. Let b be any upper bound and suppose $b < s$. Applying (ii) with $\varepsilon = s - b > 0$ produces $a \in A$ with $a > s - (s - b) = b$, contradicting that b is an upper bound. Hence $s \leq b$ for every upper bound b , so s is least. ■

Part (ii) is the form in which the supremum is actually used: it converts the abstract existence of a least upper bound into the ability to produce elements of A arbitrarily close to it. Every “choose $a_n \in A$ with $a_n > \sup A - 1/n$ ” argument in later chapters is an application of it.

Proposition 1.5.2 (Archimedean property and density of $\mathbb{Q}$). (i) For all $x, y \in \mathbb{R}$ with $x > 0$ there exists $n \in \mathbb{N}$ with $nx > y$.
(ii) For all $x < y$ in $\mathbb{R}$ there exists $q \in \mathbb{Q}$ with $x < q < y$.

Proof. (i) Suppose not: then $A = \{nx : n \in \mathbb{N}\}$ is bounded above by y . By [Assumption 1.5.1](#) it has a supremum s . Since $x > 0$, the number $s - x$ is not an upper bound, so $mx > s - x$ for some $m \in \mathbb{N}$, whence $(m + 1)x > s$, contradicting that s is an upper bound.

(ii) Since $y - x > 0$, part (i) gives $n \in \mathbb{N}$ with $n(y - x) > 1$, that is $ny > nx + 1$. Again by (i) there are integers $m_1 > nx$ and $m_2 > -nx$, so $-m_2 < nx < m_1$; let m be the least integer in $\{k \in \mathbb{Z} : -m_2 \leq k \leq m_1\}$ with $k > nx$, which exists by well-ordering. Then $m - 1 \leq nx$, so $nx < m \leq nx + 1 < ny$, and $q = m/n$ satisfies $x < q < y$. ■

Remark 1.5.1 (The extended real line). It is convenient to write $\sup A = +\infty$ when A is unbounded above and $\inf A = -\infty$ when A is unbounded below, and to set $\sup \emptyset = -\infty$, $\inf \emptyset = +\infty$. The resulting object $\overline{\mathbb{R}} = \mathbb{R} \cup \{\pm\infty\}$ is not a field — $\infty - \infty$ is undefined — and no algebraic identity may be applied to it without checking that the terms are finite. In optimisation this is not pedantry: the value of an infeasible maximisation problem is $-\infty$ by the convention $\sup \emptyset = -\infty$, whereas the value of a feasible but unbounded problem is $+\infty$, and the two failures call for different remedies.

1.6 Economic Application: Choice, Feasibility, and the Existence Question

We can now state, in the language of this chapter, the problem that occupies the rest of the volume.

Example 1.6.1 (The consumer's problem, stated set-theoretically). Let $X = \mathbb{R}_+^n$ be the *consumption set*, let $\succsim$ be a preference relation on X , and let prices $p \in \mathbb{R}_{++}^n$ and wealth $m > 0$ be given. The *budget set* is

$$\mathcal{B}(p, m) := \{x \in \mathbb{R}_+^n : \langle p, x \rangle \leq m\}.$$

The consumer's problem is to find

$$x^* \in \mathcal{B}(p, m) \quad \text{such that} \quad x^* \succsim x \text{ for all } x \in \mathcal{B}(p, m),$$

and the *demand correspondence* $x^*(p, m)$ is the set of all such x^* .

Four questions arise immediately, and each is answered by a theorem in a later chapter.

- Does a maximal element exist?* If $\succsim$ is represented by a continuous u and $\mathcal{B}(p, m)$ is compact, the Weierstrass theorem ([Theorem 5.4.1](#)) gives existence. Compactness of $\mathcal{B}(p, m)$ requires $p \in \mathbb{R}_{++}^n$: if some $p_i = 0$ the set is unbounded and existence can fail. This is why the strict positivity of prices is a hypothesis and not a normalisation.
- Is the maximiser unique?* Uniqueness requires strict quasi-concavity of u , equivalently strict convexity of preferences ([Chapter 9](#)).
- Can it be characterised by first-order conditions?* This requires differentiability of u and a constraint qualification ([Chapter 12](#)); the Cobb–Douglas case worked in ([Fukuda 2026c](#), slides 21–32) is developed there in full.
- How does it move with (p, m) ?* Continuity of $x^*(\cdot, \cdot)$ as a correspondence is Berge's maximum theorem ([Theorem 13.5.1](#)); its differentiability, where it holds, follows from the implicit function theorem ([Theorem 8.6.1](#)).

The point of listing them together is that they are the same four questions — existence, uniqueness, characterisation, comparative statics — asked of every optimisation problem in economics, including the dynamic ones of [Chapter 15](#), where the object being chosen is an infinite sequence and the “budget set” is a set of feasible paths.

1.7 Chapter Summary

Sets are specified relative to an ambient set; equality of sets is proved by double inclusion; preimages commute with all set operations while images commute only with unions. A preference relation is a complete, transitive binary relation; its indifference part partitions the alternatives; and it admits a utility representation on any countable set, but not in general on an uncountable one, as lexicographic preference shows. Implication is $(\neg P) \vee Q$, which makes statements with false hypotheses vacuously true — the device by which $\emptyset$ is open. The real numbers form an ordered field with the least upper bound property; the working form of that property is [Proposition 1.5.1\(ii\)](#), and the Archimedean property and the density of the rationals are its first two consequences.

Exercise 1.7.1. Show that $f: A \rightarrow B$ is injective if and only if $f(\bigcap_{i \in I} S_i) = \bigcap_{i \in I} f(S_i)$ for every non-empty family $(S_i)_{i \in I}$ of subsets of A . (One direction is [Example 1.2.1](#); for the other, consider two-element families.)

Exercise 1.7.2. Let $\succsim$ be a preference relation on a finite set X . Show directly, by induction on the number of elements of X , that a $\succsim$ -maximal element exists. Where does the argument break down when X is infinite, and which hypothesis of the Weierstrass theorem repairs it?

Exercise 1.7.3. Let $A, B \subseteq \mathbb{R}$ be non-empty and bounded above. Prove that $\sup(A + B) = \sup A + \sup B$, where $A + B = \{a + b : a \in A, b \in B\}$. Show by example that $\sup(A \cdot B) = \sup A \cdot \sup B$ can fail, and identify a sign condition under which it holds.

Chapter 2

Vector Spaces, Linear Maps, and Matrices

2.1 Motivation: Linearity as the First Approximation

Linear algebra enters economics along two distinct routes, and it is worth separating them because they require different parts of the theory.

The first route is exact. Some economic models are linear by construction: a Leontief input–output system, a linear-quadratic control problem, a system of linear difference equations describing the dynamics of a linearised macroeconomic model, the normal equations of ordinary least squares. Here one needs solvability of $Ax = b$, invertibility, rank, and — for dynamics — the spectrum of A .

The second route is approximate, and it is the more pervasive. The central idea of differential calculus is that a smooth function is, near a point, well approximated by a linear one; the derivative *is* that linear map. The Gerzensee slides make this explicit when they present the total differential $dy = \langle \nabla f, dx \rangle$ as an inner product and the Jacobian as the matrix representing df (Fukuda 2026g, slides 10–15). Every comparative-statics computation in economics is the solution of a linear system obtained in this way, and the invertibility condition in the implicit function theorem (Theorem 8.6.1) is exactly the requirement that the relevant linear system be uniquely solvable.

This chapter develops the theory in the coordinate-free language of vector spaces and linear maps, with matrices appearing as coordinate representations. The payoff is that the same theorems apply verbatim in the infinite-dimensional function spaces of Chapter 6 and Chapter 15, where no matrix exists.

2.2 Vector Spaces

Definition 2.2.1 (Vector space). A *vector space* over $\mathbb{R}$ is a set V with an addition $+: V \times V \rightarrow V$ and a scalar multiplication $\cdot: \mathbb{R} \times V \rightarrow V$ such that $(V, +)$ is an abelian group with neutral element 0, and for all $\lambda, \mu \in \mathbb{R}$ and $v, w \in V$,

$$\lambda(v + w) = \lambda v + \lambda w, \quad (\lambda + \mu)v = \lambda v + \mu v, \quad (\lambda\mu)v = \lambda(\mu v), \quad 1 \cdot v = v.$$

Elements of V are *vectors*.

Example 2.2.1 (Vector spaces used in this volume). (i) $\mathbb{R}^n$ with coordinatewise operations. This is the space of commodity bundles, price vectors, and parameter vectors.

(ii) The set $M_{m \times n}(\mathbb{R})$ of $m \times n$ real matrices.

(iii) The set $\mathbb{R}^S$ of all functions $f: S \rightarrow \mathbb{R}$ on a set S , with pointwise operations. When $S = \{1, \dots, n\}$ this is $\mathbb{R}^n$; when $S = \mathbb{N}$ it is the space of real sequences, the natural home of an infinite consumption path $(c_t)_{t \in \mathbb{N}_0}$; when $S = \Omega$ is a sample space it is the space of random variables.

(iv) The set $\mathcal{C}([a, b], \mathbb{R})$ of continuous real functions on $[a, b]$, and the set $\mathcal{B}(X, \mathbb{R})$ of bounded real functions on a set X . These are the spaces on which the Bellman operator of Chapter 15 acts.

Items (iii) and (iv) are infinite-dimensional. That is precisely why the coordinate-free treatment is worth the effort.

Definition 2.2.2 (Subspace, span, linear independence, basis). Let V be a vector space over $\mathbb{R}$.

- (i) A *subspace* of V is a non-empty $W \subseteq V$ closed under addition and scalar multiplication. Every subspace contains 0 .
- (ii) The *span* of $S \subseteq V$ is $\text{span } S := \{\sum_{i=1}^k \lambda_i v_i : k \in \mathbb{N}, \lambda_i \in \mathbb{R}, v_i \in S\}$, with $\text{span } \emptyset := \{0\}$. It is the smallest subspace containing S .
- (iii) A finite list $v_1, \dots, v_k$ is *linear independence* if $\sum_i \lambda_i v_i = 0$ implies $\lambda_1 = \dots = \lambda_k = 0$, and *linear dependence* otherwise.
- (iv) A *basis* of V is a linearly independent list spanning V . If V has a finite basis it is *finite-dimensional space* and its *dimension* $\dim V$ is the common length of all its bases.

Theorem 2.2.1 (Well-definedness of dimension). In a finite-dimensional vector space, every two bases have the same length, and every linearly independent list can be extended to a basis.

Proof. The engine is the exchange lemma: in a vector space spanned by a list of length m , every linearly independent list has length at most m . Given that, if B_1 and B_2 are bases of lengths m_1 and m_2 , then B_1 is independent and B_2 spans, giving $m_1 \leq m_2$; exchanging roles gives $m_2 \leq m_1$.

For the exchange lemma itself, let $w_1, \dots, w_m$ span V and let $v_1, \dots, v_k$ be independent with $k > m$. Adjoin v_1 to the spanning list: the list $v_1, w_1, \dots, w_m$ is dependent (since $v_1 \in \text{span}\{w_j\}$), so some w_j lies in the span of its predecessors and can be deleted, leaving a spanning list of length m containing v_1 . Repeating with $v_2, \dots, v_k$ — at each step the deleted vector must be some w_j rather than a previously inserted v_i , because the v_i are independent — exhausts the w_j after m steps while v_{m+1} still remains, a contradiction. The extension statement follows by adjoining vectors outside the current span until the span is all of V , which terminates by the same length bound. A complete treatment is Axler (2024, Ch. 2). ■

2.3 Linear Maps and their Matrix Representation

Definition 2.3.1 (Linear map). Let V, W be vector spaces over $\mathbb{R}$. A map $T: V \rightarrow W$ is *linear* if $T(\lambda v + \mu v') = \lambda T(v) + \mu T(v')$ for all $\lambda, \mu \in \mathbb{R}$ and $v, v' \in V$. Its *kernel* and *image* are

$$\text{Ker } T := \{v \in V : T(v) = 0\}, \quad \text{Im } T := \{T(v) : v \in V\},$$

subspaces of V and W respectively. The *rank* of T is $\text{rank } T := \dim \text{Im } T$.

Theorem 2.3.1 (Matrix representation theorem). A map $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ is linear if and only if there exists a unique $m \times n$ matrix A with $T(x) = Ax$ for every $x \in \mathbb{R}^n$. Moreover

$$A = [T(e^1) \ T(e^2) \ \dots \ T(e^n)],$$

where e^i is the i -th standard basis vector, so that the i -th column of A is $Ae^i = T(e^i)$.

Proof. If $T(x) = Ax$ then linearity is the distributive law for matrix multiplication.

Conversely, let T be linear and define A by the displayed formula. Writing $x = \sum_{i=1}^n x_i e^i$ and using linearity $n - 1$ times,

$$T(x) = T\left(\sum_{i=1}^n x_i e^i\right) = \sum_{i=1}^n x_i T(e^i) = [T(e^1) \ \dots \ T(e^n)] \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = Ax.$$

For uniqueness, suppose $Ax = Bx$ for all x . Taking $x = e^i$ gives $Ae^i = Be^i$, that is, the i -th columns of A and B agree; as this holds for every i , $A = B$. ■

This is Theorem 5.1 of Fukuda (2026d) and is presented on slides 12–14 of (Fukuda 2026g). Two readings of Ax are useful and both appear later. Row by row, $(Ax)_i = \langle a^i, x \rangle$ where a^i is the i -th row of A ; this is the reading under which each row is a linear functional, used in Chapter 3 for the Riesz representation theorem. Column by column, $Ax = \sum_i x_i A e^i$ is a linear combination of the columns of A with weights x_i ; this is the reading under which $\text{Im } A$ is the column space, used in Chapter 11 for least squares.

Theorem 2.3.2 (Rank–nullity). Let V be finite-dimensional and $T: V \rightarrow W$ linear. Then

$$\dim V = \dim \text{Ker } T + \text{rank } T.$$

Proof. Let $u_1, \dots, u_k$ be a basis of $\text{Ker } T$ and extend it, by Theorem 2.2.1, to a basis $u_1, \dots, u_k, v_1, \dots, v_r$ of V , so $\dim V = k + r$. We claim $T(v_1), \dots, T(v_r)$ is a basis of $\text{Im } T$.

They span: any $T(v)$ with $v = \sum_i \alpha_i u_i + \sum_j \beta_j v_j$ equals $\sum_j \beta_j T(v_j)$ because $T(u_i) = 0$. They are independent: if $\sum_j \beta_j T(v_j) = 0$ then $T(\sum_j \beta_j v_j) = 0$, so $\sum_j \beta_j v_j \in \text{Ker } T$ and hence $\sum_j \beta_j v_j = \sum_i \alpha_i u_i$ for some α_i ; independence of the full basis forces all β_j (and α_i) to vanish. Therefore $\text{rank } T = r$ and $\dim V = k + r$. ■

Corollary 2.3.1 (Solvability of $Ax = b$). Let A be $m \times n$ and $b \in \mathbb{R}^m$.

- (i) $Ax = b$ has a solution if and only if $b \in \text{Im } A$, equivalently $\text{rank}[A \ b] = \text{rank } A$.
- (ii) If a solution exists, the solution set is $x_0 + \text{Ker } A$ for any particular solution x_0 ; it is a singleton if and only if $\text{rank } A = n$.
- (iii) For $m = n$: A is invertible $\iff \text{rank } A = n \iff \text{Ker } A = \{0\} \iff \det A \neq 0 \iff Ax = b$ has exactly one solution for every b .

Proof. (i) is the definition of the image. (ii) If $Ax_0 = b$ then $Ax = b$ iff $A(x - x_0) = 0$. The solution is unique iff $\text{Ker } A = \{0\}$, which by Theorem 2.3.2 is $\text{rank } A = n$. (iii) For $m = n$, Theorem 2.3.2 makes $\text{Ker } A = \{0\}$ and $\text{Im } A = \mathbb{R}^n$ equivalent; the equivalence with $\det A \neq 0$ is the standard determinant criterion, and invertibility is then existence and uniqueness for every b . ■

Remark 2.3.1 (Square matters). Part (iii) is a statement about square matrices only. For $m \neq n$ injectivity and surjectivity are genuinely different, and the distinction is economically substantive: in the least-squares problem of Chapter 11 the design matrix X is $n \times k$ with $n \gg k$, so $Xb = y$ typically has no solution — which is exactly why one projects instead of solving. In econometrics the condition $\text{rank } X = k$ is the no-perfect-multicollinearity assumption, and by Corollary 2.3.1(ii) it is precisely what makes the estimator unique.

2.4 Determinants and Volume

Definition 2.4.1 (Determinant). The *determinant* is the unique function $\det: M_{n \times n}(\mathbb{R}) \rightarrow \mathbb{R}$ that is linear in each column separately, vanishes when two columns coincide, and satisfies $\det I_n = 1$. Explicitly

$$\det A = \sum_{\sigma \in S_n} \text{sgn}(\sigma) \prod_{i=1}^n a_{\sigma(i)i},$$

the sum running over permutations of $\{1, \dots, n\}$.

Proposition 2.4.1 (Properties). For $A, B \in M_{n \times n}(\mathbb{R})$: $\det(AB) = \det A \det B$; $\det A^T = \det A$; A is invertible if and only if $\det A \neq 0$, in which case $\det(A^{-1}) = (\det A)^{-1}$; and for $\lambda \in \mathbb{R}$, $\det(\lambda A) = \lambda^n \det A$.

The geometric content is what the Gerzensee slides emphasise: $|\det A|$ is the factor by which the linear map $x \mapsto Ax$ magnifies n -dimensional volume, and the sign of $\det A$ records whether orientation is preserved (Fukuda 2026g, slide 15). Two consequences are used later.

Example 2.4.1 (Change of variables). In the change-of-variables formula for multiple integrals ([Theorem 10.4.2](#)), the factor $|\det J|$ appears exactly because J is the linear approximation of the substitution and $|\det J|$ is its volume magnification. The polar-coordinate computation of the Gaussian integral in ([Fukuda 2026a](#), slides 91–93), where $dx dy = r dr d\theta$, is this statement with

$$J = \begin{pmatrix} \partial x / \partial r & \partial x / \partial \theta \\ \partial y / \partial r & \partial y / \partial \theta \end{pmatrix} = \begin{pmatrix} \cos \theta & -r \sin \theta \\ \sin \theta & r \cos \theta \end{pmatrix}, \quad \det J = r(\cos^2 \theta + \sin^2 \theta) = r > 0.$$

The full derivation is given in [subsection 10.7.2](#).

Example 2.4.2 (Cramer's rule and comparative statics). If $\det A \neq 0$, the unique solution of $Ax = b$ has i -th coordinate $x_i = \det A_i / \det A$, where A_i is A with its i -th column replaced by b . In comparative statics the system $Ax = b$ arises by differentiating equilibrium conditions, x collects the responses of endogenous variables, and b the direct effect of the parameter; Cramer's rule then delivers the sign of an individual response from the signs of two determinants. This is how sign restrictions on a Hessian or a Jacobian translate into comparative-statics predictions in [Chapter 8](#).

2.5 Eigenvalues, Diagonalisation, and Symmetric Matrices

Definition 2.5.1 (Eigenvalue and eigenvector). Let $A \in M_{n \times n}(\mathbb{R})$. A scalar $\lambda \in \mathbb{C}$ is an *eigenvalue* of A if $Av = \lambda v$ for some $v \neq 0$; such a v is an *eigenvector*. The eigenvalues are the roots of the *characteristic polynomial* $p_A(\lambda) = \det(A - \lambda I)$. The *spectral radius* is $\rho(A) := \max\{|\lambda| : \lambda \text{ an eigenvalue of } A\}$.

Theorem 2.5.1 (Spectral theorem for symmetric matrices). Let $A \in M_{n \times n}(\mathbb{R})$ be symmetric. Then all eigenvalues of A are real, and there is an orthogonal matrix Q (that is, $Q^T Q = I$) and a diagonal matrix $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_n)$ with $A = Q\Lambda Q^T$. Equivalently, $\mathbb{R}^n$ has an orthonormal basis of eigenvectors of A .

Proof sketch. Reality of the eigenvalues: if $Av = \lambda v$ with $v \in \mathbb{C}^n \setminus \{0\}$, then $\bar{v}^T Av = \lambda \bar{v}^T v$; taking conjugate transposes and using $A = A^T = \bar{A}$ gives $\bar{v}^T Av = \bar{\lambda} \bar{v}^T v$; since $\bar{v}^T v = \|v\|^2 > 0$ we get $\lambda = \bar{\lambda}$.

Diagonalisation proceeds by induction on n . Let λ_1 be an eigenvalue with unit eigenvector q_1 ; the subspace $q_1^\perp$ is A -invariant, because $\langle q_1, Aw \rangle = \langle Aq_1, w \rangle = \lambda_1 \langle q_1, w \rangle = 0$ for $w \perp q_1$. Applying the induction hypothesis to the restriction of A to $q_1^\perp$, which is symmetric with respect to the induced inner product, and adjoining q_1 produces the orthonormal eigenbasis. The full argument, in the form of the real spectral theorem, is [Axler \(2024, Ch. 7\)](#). ■

Definition 2.5.2 (Quadratic form and definiteness). Let A be symmetric. The *quadratic form* associated with A is $q_A(x) = x^T Ax$. The matrix A is *positive definite matrix* ($A \succ 0$) if $q_A(x) > 0$ for all $x \neq 0$; *positive semidefinite matrix* ($A \succeq 0$) if $q_A(x) \geq 0$ for all x ; and negative (semi)definite if $-A$ is positive (semi)definite. Otherwise A is *indefinite*.

Proposition 2.5.1 (Definiteness via the spectrum). A symmetric A is positive definite if and only if all its eigenvalues are strictly positive, and positive semidefinite if and only if all are non-negative. Moreover

$$\lambda_{\min} \|x\|^2 \leq x^T Ax \leq \lambda_{\max} \|x\|^2 \quad \text{for all } x \in \mathbb{R}^n,$$

where $\lambda_{\min}, \lambda_{\max}$ are the smallest and largest eigenvalues.

Proof. Write $A = Q\Lambda Q^T$ as in [Theorem 2.5.1](#) and set $y = Q^T x$. Since Q is orthogonal, $\|y\| = \|x\|$ and $x \neq 0 \iff y \neq 0$, and

$$x^T Ax = y^T \Lambda y = \sum_{i=1}^n \lambda_i y_i^2.$$

If every $\lambda_i > 0$ this is strictly positive for $y \neq 0$. Conversely, if $\lambda_j \leq 0$ for some j , taking $y = e^j$, that is $x = Qe^j$, gives $x^T Ax = \lambda_j \leq 0$ with $x \neq 0$. The semidefinite case is identical with weak inequalities. The two-sided bound follows from $\lambda_{\min} \sum_i y_i^2 \leq \sum_i \lambda_i y_i^2 \leq \lambda_{\max} \sum_i y_i^2$. ■

Remark 2.5.1 (Only the symmetric part matters). For any square A , $x^\top Ax = x^\top (\frac{1}{2}(A + A^\top))x$, because $x^\top Ax$ is a scalar and hence equals its own transpose $x^\top A^\top x$. A quadratic form therefore determines and is determined by the symmetric part of A alone, and one may always assume A symmetric. This is the content of the gradient computation in (Fukuda 2026a, slide 49): for $f(x) = x^\top Ax$,

$$\nabla f(x) = (A + A^\top)x,$$

which reduces to $2Ax$ exactly when A is symmetric. Forgetting the factor $A + A^\top$ is one of the most common errors in matrix differentiation.

Proposition 2.5.2 (Leading principal minors). Let A be symmetric $n \times n$ and let D_k denote the determinant of its upper-left $k \times k$ submatrix. Then $A \succ 0$ if and only if $D_k > 0$ for $k = 1, \dots, n$; and $A \prec 0$ if and only if $(-1)^k D_k > 0$ for $k = 1, \dots, n$, that is, the leading principal minors alternate in sign starting negative.

Remark 2.5.2 (The semidefinite case is different). The analogous criterion for *semidefiniteness* requires all principal minors, not merely the leading ones, to be non-negative. The matrix $A = \begin{pmatrix} 0 & 0 \\ 0 & -1 \end{pmatrix}$ has $D_1 = D_2 = 0$, so all leading principal minors are non-negative, yet A is negative semidefinite and not positive semidefinite. This trap matters when checking second-order conditions at a boundary optimum in Chapter 12, where semidefiniteness rather than definiteness is what the theory delivers.

2.6 Economic Application: Linear Dynamics and Stability

Consider the linear difference system

$$x_{t+1} = Ax_t, \quad x_0 \in \mathbb{R}^n \text{ given}, \quad (2.1)$$

which arises as the linearisation of a nonlinear dynamic model around a steady state — for instance the Ramsey model of Section 15.6 linearised about (k^*, c^*) . Iterating, $x_t = A^t x_0$.

Proposition 2.6.1 (Stability). For system (2.1): $x_t \rightarrow 0$ for every $x_0 \in \mathbb{R}^n$ if and only if $\rho(A) < 1$. If $\rho(A) > 1$, then x_t diverges for every x_0 outside a proper subspace of $\mathbb{R}^n$.

Proof for the diagonalisable case. Suppose $A = P\Lambda P^{-1}$ with $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_n)$. Then $A^t = P\Lambda^t P^{-1}$ with $\Lambda^t = \text{diag}(\lambda_1^t, \dots, \lambda_n^t)$. Writing $z_t = P^{-1}x_t$ decouples the system into $z_{t,i} = \lambda_i^t z_{0,i}$. Each coordinate tends to 0 for every initial condition if and only if $|\lambda_i| < 1$, and grows without bound whenever $|\lambda_i| > 1$ and $z_{0,i} \neq 0$. The set of x_0 with $z_{0,i} = 0$ for all i with $|\lambda_i| > 1$ — the stable subspace — is a proper subspace when some $|\lambda_i| > 1$. The non-diagonalisable case is handled by the Jordan form, where a λ -block of size k contributes terms $t^j \lambda^t$ with $j < k$; these still tend to 0 when $|\lambda| < 1$, so the criterion is unchanged. ■

Remark 2.6.1 (Saddle-path stability). In macroeconomic models the interesting case is neither $\rho(A) < 1$ nor $\rho(A) > 1$ but a mixture: some eigenvalues inside the unit circle and some outside. The state vector then splits into predetermined variables, whose initial values are given, and jump variables, whose initial values are chosen. A unique bounded equilibrium path exists when the number of eigenvalues outside the unit circle equals the number of jump variables — the Blanchard–Kahn count. Economically, the jump variables must be placed on the stable subspace, and the counting condition says that the placement problem has exactly one solution, which by Corollary 2.3.1(iii) is a statement about a square invertible system. The recursive treatment is Ljungqvist and Sargent (2018, Ch. 2).

Example 2.6.1 (Leontief input–output). Let $A \in M_{n \times n}(\mathbb{R}_+)$ have a_{ij} equal to the units of good i required per unit of good j , and let $d \in \mathbb{R}_+^n$ be final demand. Gross output x must satisfy $x = Ax + d$, that is $(I - A)x = d$. A non-negative solution exists for every $d \in \mathbb{R}_+^n$ if and only if $(I - A)^{-1}$ exists and is

non-negative, which holds if and only if $\rho(A) < 1$; in that case

$$(I - A)^{-1} = \sum_{k=0}^{\infty} A^k,$$

the Neumann series, whose k -th term is the input requirement generated at the k -th round of production. The condition $\rho(A) < 1$ is thus the productivity condition that the economy generates a surplus. Convergence of the series is a fixed-point statement: $x \mapsto Ax + d$ is a contraction in a suitable norm, and [Theorem 6.4.1](#) applies. The same series reappears in [Chapter 15](#) as the present-value expansion of a value function.

2.7 Chapter Summary

A linear map between finite-dimensional spaces is completely described, once bases are fixed, by a matrix whose columns are the images of the basis vectors ([Theorem 2.3.1](#)). Rank-nullity converts questions about solvability into questions about dimension ([Theorem 2.3.2](#), [Corollary 2.3.1](#)); for square matrices, and only for square matrices, injectivity, surjectivity, invertibility, and non-vanishing determinant coincide. The determinant measures oriented volume magnification, which is why it appears in the change-of-variables formula. Symmetric matrices are orthogonally diagonalisable with real eigenvalues ([Theorem 2.5.1](#)), so definiteness of a quadratic form is a statement about the sign of the spectrum ([Proposition 2.5.1](#)) — the fact used for second-order conditions throughout [Chapter 12](#). Only the symmetric part of a matrix affects its quadratic form, and $\nabla(x^\top Ax) = (A + A^\top)x$. Stability of a linear dynamic system is governed by the spectral radius.

Exercise 2.7.1. Verify directly, without invoking [Remark 2.5.1](#), that for $f(x_1, x_2) = a_{11}x_1^2 + (a_{12} + a_{21})x_1x_2 + a_{22}x_2^2$ one has $\nabla f(x) = (A + A^\top)x$, and conclude that two distinct matrices can induce the same quadratic form. (This is Exercise 3.3 of Fukuda (2026d); see (Fukuda 2026a, slide 49).)

Exercise 2.7.2. Let X be $n \times k$ with $\text{rank } X = k \leq n$. Show that $X^\top X$ is symmetric positive definite, hence invertible. Where is the rank hypothesis used, and what fails without it? (The result is what makes the OLS formula $\hat{\beta} = (X^\top X)^{-1}X^\top y$ of (Fukuda 2026a, slide 60) well posed.)

Exercise 2.7.3. Let $A \in M_{n \times n}(\mathbb{R}_+)$ with all row sums strictly less than 1. Show $\rho(A) < 1$, and deduce from [Example 2.6.1](#) that the Leontief system is productive. Interpret the row-sum condition economically.

Chapter 3

Inner Products, Norms, and Metric Spaces

3.1 Motivation: Geometry Without Coordinates

The Gerzensee course opens its real-analysis block with the inner product and immediately lists what it buys: length, angle, and orthogonality in $\mathbb{R}^n$ and in function spaces; the projection theorem, which delivers both ordinary least squares and conditional expectation; and the separation theorems, which deliver the second welfare theorem and the existence of state prices in a no-arbitrage economy (Fukuda 2026g, slide 3). All of these are geometric statements, and all of them survive the passage from $\mathbb{R}^n$ to spaces of functions and of random variables — provided the geometry is set up abstractly.

The structures form a strict hierarchy:

$$\text{inner product spaces} \subsetneq \text{normed spaces} \subsetneq \text{metric spaces},$$

reproduced on slide 25 of (Fukuda 2026g). Each step forgets something. A metric gives distance but no notion of adding points; a norm gives length and a vector structure but no notion of angle; an inner product gives angle as well. Knowing which structure a theorem needs is what tells us how far it generalises, and this chapter states each result at the weakest level at which it is true.

3.2 Inner Product Spaces

Definition 3.2.1 (Inner product). Let V be a real vector space. An *inner product* on V is a map $\langle \cdot, \cdot \rangle : V \times V \rightarrow \mathbb{R}$ such that for all $x, y, z \in V$ and $\lambda \in \mathbb{R}$:

- (i) $\langle x, x \rangle \geq 0$, with $\langle x, x \rangle = 0$ if and only if $x = 0$ (positive definiteness);
- (ii) $\langle x, y \rangle = \langle y, x \rangle$ (symmetry);
- (iii) $\langle x, \lambda y \rangle = \lambda \langle x, y \rangle$ and $\langle x, y + z \rangle = \langle x, y \rangle + \langle x, z \rangle$ (linearity in the second argument; with (ii) this gives linearity in each argument separately).

The pair $(V, \langle \cdot, \cdot \rangle)$ is an *inner product space*.

These are exactly the five properties verified in (Fukuda 2026g, slides 5–6) for the Euclidean inner product $\langle a, x \rangle = \sum_{i=1}^n a_i x_i$ on $\mathbb{R}^n$; the abstraction consists in taking them as the definition.

Example 3.2.1 (Inner product spaces used in this volume). (i) $\mathbb{R}^n$ with $\langle x, y \rangle = x^\top y = \sum_i x_i y_i$.

(ii) $\mathbb{R}^n$ with $\langle x, y \rangle_W = x^\top W y$ for a symmetric $W \succ 0$. This is an inner product precisely because W is positive definite (Proposition 2.5.1), and it is the metric under which generalised least squares is an orthogonal projection.

(iii) $\mathcal{L}^2(\mathbb{R}) = \{f : \mathbb{R} \rightarrow \mathbb{R} \text{ measurable} : \int_{-\infty}^{\infty} |f(x)|^2 dx < \infty\}$ with $\langle f, g \rangle = \int_{-\infty}^{\infty} f(x)g(x) dx$.

- (iv) The space of square-integrable random variables on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$, with $\langle X, Y \rangle = \mathbb{E}[XY]$; or, on the subspace of centred variables, $\langle X, Y \rangle = \mathbb{E}[(X - \mathbb{E}X)(Y - \mathbb{E}Y)] = \text{Cov}(X, Y)$. Items (iii) and (iv) are stated in (Fukuda 2026g, slide 7).

Remark 3.2.1 (The qualification the slides leave out). On $\mathcal{L}^2$ the map $\langle f, g \rangle = \int fg$ satisfies (ii) and (iii) but only half of (i): $\langle f, f \rangle = 0$ forces $f = 0$ almost everywhere, not everywhere. The slide’s hedge that “roughly, the inner product satisfies the properties” (Fukuda 2026g, slide 7) refers to exactly this. The remedy is standard: work with the quotient of $\mathcal{L}^2$ by the subspace of functions vanishing almost everywhere, writing L^2 for the quotient, whose elements are equivalence classes under the relation of Proposition 1.3.1. On L^2 , positive definiteness holds. In the random-variable case (iv) the same convention identifies random variables that agree $\mathbb{P}$ -almost surely, and it is what makes conditional expectation in Chapter 14 a well-defined element rather than a function defined up to null sets.

Lemma 3.2.1 (Cauchy–Schwarz inequality). Let $(V, \langle \cdot, \cdot \rangle)$ be an inner product space and set $\|x\| = \sqrt{\langle x, x \rangle}$. Then

$$|\langle x, y \rangle| \leq \|x\| \|y\| \quad \text{for all } x, y \in V,$$

with equality if and only if x and y are linearly dependent.

Proof. If $y = 0$ both sides vanish and x, y are dependent. Assume $y \neq 0$ and set $\lambda := \langle x, y \rangle / \langle y, y \rangle$. Then

$$0 \leq \|x - \lambda y\|^2 = \langle x - \lambda y, x - \lambda y \rangle = \|x\|^2 - 2\lambda \langle x, y \rangle + \lambda^2 \|y\|^2 = \|x\|^2 - \frac{\langle x, y \rangle^2}{\|y\|^2},$$

which rearranges to $\langle x, y \rangle^2 \leq \|x\|^2 \|y\|^2$.

Equality holds if and only if $\|x - \lambda y\| = 0$, that is $x = \lambda y = \frac{\langle x, y \rangle}{\langle y, y \rangle} y$, which is linear dependence; and conversely if $x = \mu y$ then both sides equal $|\mu| \|y\|^2$. ■

The choice of λ is the one that makes $x - \lambda y$ orthogonal to y , so the proof is the projection argument of Chapter 11 in miniature. This is the content of the geometric derivation on (Fukuda 2026g, slide 17), where the condition $\langle \lambda x, y - \lambda x \rangle = 0$ is solved for λ .

Definition 3.2.2 (Orthogonality and angle). Vectors x, y in an inner product space are *orthogonal*, written $x \perp y$, if $\langle x, y \rangle = 0$. For non-zero x, y the *angle* between them is the unique $\theta \in [0, \pi]$ with $\cos \theta = \langle x, y \rangle / (\|x\| \|y\|)$, which is well defined precisely because Lemma 3.2.1 places the ratio in $[-1, 1]$.

Example 3.2.2 (Correlation is a cosine). Let $x, y \in \mathbb{R}^n$ with sample means μ_x, μ_y , and put $\tilde{x} = x - \mu_x \mathbf{1}$, $\tilde{y} = y - \mu_y \mathbf{1}$ where $\mathbf{1} = (1, \dots, 1)^\top$. With $\sigma_{x,y} = \frac{1}{n} \langle \tilde{x}, \tilde{y} \rangle$, $\sigma_x^2 = \frac{1}{n} \|\tilde{x}\|^2$ and $\sigma_y^2 = \frac{1}{n} \|\tilde{y}\|^2$, the Pearson correlation coefficient is

$$\rho = \frac{\sigma_{x,y}}{\sigma_x \sigma_y} = \frac{\langle \tilde{x}, \tilde{y} \rangle}{\|\tilde{x}\| \|\tilde{y}\|} = \cos \theta.$$

Lemma 3.2.1 therefore gives $|\rho| \leq 1$ immediately, and the equality case gives $|\rho| = 1$ if and only if $\tilde{y} = c\tilde{x}$ for some $c \in \mathbb{R}$, that is

$$y_i - \mu_y = \frac{\sigma_{x,y}}{\sigma_x^2} (x_i - \mu_x) \quad \text{for all } i,$$

since c must equal $\langle \tilde{x}, \tilde{y} \rangle / \|\tilde{x}\|^2$. This settles Exercise 7.6 of (Fukuda 2026g, slide 18) and identifies the constant as the population regression slope of Chapter 11. The same computation with $\langle X, Y \rangle = \text{Cov}(X, Y)$ shows that the correlation of two random variables is the cosine of the angle between them in L^2 .

3.2.1 Riesz representation in $\mathbb{R}^n$

Theorem 3.2.1 (Riesz representation theorem for $\mathbb{R}^n$). (i) For each $a \in \mathbb{R}^n$ the map $f_a: \mathbb{R}^n \rightarrow \mathbb{R}$, $f_a(x) = \langle a, x \rangle$, is linear.

- (ii) Conversely, for every linear $f: \mathbb{R}^n \rightarrow \mathbb{R}$ there is a unique $a \in \mathbb{R}^n$ with $f(x) = \langle a, x \rangle$ for all $x \in \mathbb{R}^n$; explicitly $a = (f(e^1), \dots, f(e^n))^T$.

Proof. (i) is linearity of the inner product in its second argument.

(ii) Existence: writing $x = \sum_i x_i e^i$ and using linearity, $f(x) = \sum_i x_i f(e^i) = \langle a, x \rangle$ with a as displayed. Uniqueness: if $\langle a, x \rangle = \langle a', x \rangle$ for all x , then $\langle a - a', x \rangle = 0$ for all x ; taking $x = a - a'$ gives $\|a - a'\|^2 = 0$, hence $a = a'$ by positive definiteness. ■

This is Theorem 7.1 of Fukuda (2026d); the two-dimensional picture is (Fukuda 2026g, slide 9). It is the special case $m = 1$ of Theorem 2.3.1, but stated this way it says something the matrix version does not: *a linear functional is the same thing as a vector*. Three economic readings follow.

Example 3.2.3 (Expected utility as an inner product). Let $\Omega = \{\omega_1, \dots, \omega_n\}$ and let $p \in \mathbb{R}^n$ be a probability vector. Expected utility $\mathbb{E}[u] = \sum_i p_i u_i = \langle u, p \rangle$ is linear in the probability vector p for fixed utility index u , and linear in u for fixed p (Fukuda 2026g, slide 8). Read in the first way, Theorem 3.2.1 says that *any* functional on lotteries that is linear in probabilities has an expected-utility representation, with the representing vector a being the utility index. The von Neumann–Morgenstern theorem is the statement that the behavioural axioms force linearity; the Riesz theorem then supplies the index.

Example 3.2.4 (State prices and the stochastic discount factor). Let there be n states and let an asset be a payoff vector $z \in \mathbb{R}^n$. Suppose the pricing functional $\pi: \mathbb{R}^n \rightarrow \mathbb{R}$ is linear — which is exactly the law of one price plus portfolio additivity. By Theorem 3.2.1 there is a unique $q \in \mathbb{R}^n$ with $\pi(z) = \langle q, z \rangle = \sum_i q_i z_i$; the q_i are *state price*. If moreover $\pi(z) > 0$ whenever $z \geq 0$, $z \neq 0$ — absence of arbitrage — then $q \in \mathbb{R}_{++}^n$, and writing $q_i = p_i m_i$ for a probability p gives $\pi(z) = \mathbb{E}[mz]$, exhibiting m as the *stochastic discount factor*. The step from “no arbitrage” to “a strictly positive linear functional exists” is not Riesz but a separation theorem, proved in Chapter 9; Riesz converts the functional into a vector. See LeRoy and Werner (2000, Ch. 2).

Example 3.2.5 (The total differential). For $f: \mathbb{R}^n \rightarrow \mathbb{R}$ differentiable at x , the differential

$$dy = \sum_{i=1}^n \frac{\partial f}{\partial x_i}(x) dx_i$$

is a linear functional of the displacement dx . By Theorem 3.2.1 it is represented by a unique vector, and that vector is by definition the gradient: $dy = \langle \nabla f(x), dx \rangle$ (Fukuda 2026g, slide 10). This is the cleanest statement of what a gradient is, and it explains why the gradient is a column vector while the derivative $Df(x)$ is a row: the derivative is the functional, the gradient is its Riesz representative.

3.3 Normed Spaces

Definition 3.3.1 (Norm). Let V be a real vector space. A *norm* is a map $\|\cdot\|: V \rightarrow \mathbb{R}$ such that for all $x, y \in V$, $\lambda \in \mathbb{R}$:

- (i) $\|x\| \geq 0$;
- (ii) $\|x\| = 0$ if and only if $x = 0$;
- (iii) $\|\lambda x\| = |\lambda| \|x\|$ (absolute homogeneity);
- (iv) $\|x + y\| \leq \|x\| + \|y\|$ (triangle inequality).

The pair $(V, \|\cdot\|)$ is a *normed space*.

Proposition 3.3.1 (An inner product induces a norm). If $\langle \cdot, \cdot \rangle$ is an inner product on V , then $\|x\| := \sqrt{\langle x, x \rangle}$ is a norm.

Proof. (i) and (ii) are positive definiteness; (iii) follows from bilinearity, since $\|\lambda x\|^2 = \lambda^2 \|x\|^2$. For (iv),

$$\|x + y\|^2 = \|x\|^2 + 2\langle x, y \rangle + \|y\|^2 \leq \|x\|^2 + 2\|x\|\|y\| + \|y\|^2 = (\|x\| + \|y\|)^2,$$

using Lemma 3.2.1; taking square roots gives the claim. ■

Proposition 3.3.2 (Reverse triangle inequality). In any normed space, $|\|x\| - \|x_0\|| \leq \|x - x_0\|$.

Proof. $\|x\| = \|(x - x_0) + x_0\| \leq \|x - x_0\| + \|x_0\|$ gives $\|x\| - \|x_0\| \leq \|x - x_0\|$. Exchanging x and x_0 and using $\|x_0 - x\| = \|x - x_0\|$ gives $\|x_0\| - \|x\| \leq \|x - x_0\|$. Together these bound the absolute value. ■

This is Exercise 7.4 of (Fukuda 2026g, slide 20). It is used at once in Chapter 5 to show that the norm is a continuous function — indeed Proposition 3.3.2 says it is Lipschitz with constant 1.

Example 3.3.1 (Norms used in this volume). On $\mathbb{R}^n$: the Euclidean norm $\|x\|_2 = (\sum_i x_i^2)^{1/2}$, induced by the Euclidean inner product; the ℓ^1 norm $\|x\|_1 = \sum_i |x_i|$; the supremum norm $\|x\|_\infty = \max_i |x_i|$. On $\mathcal{C}([a, b], \mathbb{R})$ and on $\mathcal{B}(X, \mathbb{R})$: the *supremum norm*

$$\|f\|_\infty := \sup_x |f(x)|,$$

which is the norm used for value functions in Chapter 15 (Fukuda 2026g, slides 21–22, 73).

Proposition 3.3.3 (The supremum norm is a norm). Let X be a non-empty set and $\mathcal{B}(X, \mathbb{R})$ the set of bounded functions $f: X \rightarrow \mathbb{R}$. Then $\mathcal{B}(X, \mathbb{R})$ is a vector space under pointwise operations and $\|f\|_\infty = \sup_{x \in X} |f(x)|$ is a norm on it.

Proof. Boundedness is preserved by sums and scalar multiples, so $\mathcal{B}(X, \mathbb{R})$ is a subspace of $\mathbb{R}^X$, and $\|f\|_\infty < \infty$ for every f in it. Property (i) is clear. For (ii), $\|f\|_\infty = 0$ forces $|f(x)| = 0$ for every x , that is $f = 0$ — note that here, unlike in Remark 3.2.1, no almost-everywhere qualification arises. For (iii), $\sup_x |\lambda f(x)| = |\lambda| \sup_x |f(x)|$. For (iv), for each x , $|f(x) + g(x)| \leq |f(x)| + |g(x)| \leq \|f\|_\infty + \|g\|_\infty$; as the right-hand side does not depend on x , it is an upper bound for $\{|(f + g)(x)|\}$, so by Proposition 1.5.1 it dominates the supremum. ■

Remark 3.3.1 (Not every norm comes from an inner product). A norm is induced by an inner product if and only if it satisfies the *parallelogram law*

$$\|x + y\|^2 + \|x - y\|^2 = 2\|x\|^2 + 2\|y\|^2.$$

Necessity is immediate by expanding both sides. The supremum norm fails it: in $\mathbb{R}^2$ with $x = (1, 0)$, $y = (0, 1)$ we get $\|x + y\|_\infty = \|x - y\|_\infty = 1$, so the left side is 2 while the right side is 4. Hence $\mathcal{B}(X, \mathbb{R})$ with the supremum norm carries no compatible inner product, and no notion of angle. This is why the value-function arguments of Chapter 15 use the contraction mapping theorem, which needs only a metric, rather than a projection argument, which needs an inner product.

3.4 Metric Spaces

Definition 3.4.1 (Metric). Let X be a non-empty set. A *metric* (or *distance*) on X is a map $d: X \times X \rightarrow \mathbb{R}$ such that for all $x, y, z \in X$:

- (i) $d(x, y) \geq 0$;
- (ii) $d(x, y) = 0$ if and only if $x = y$;
- (iii) $d(x, y) = d(y, x)$ (symmetry);
- (iv) $d(x, z) \leq d(x, y) + d(y, z)$ (triangle inequality).

The pair (X, d) is a *metric space*.

Proposition 3.4.1 (A norm induces a metric). If $(V, \|\cdot\|)$ is a normed space, then $d(x, y) := \|x - y\|$ is a metric on V .

Proof. (i)–(iii) are immediate from Definition 3.3.1(i)–(iii), using $\|y - x\| = \|-(x - y)\| = \|x - y\|$. For (iv), $d(x, z) = \|(x - y) + (y - z)\| \leq \|x - y\| + \|y - z\| = d(x, y) + d(y, z)$. ■

Combining [Proposition 3.3.1](#) and [Proposition 3.4.1](#) gives the hierarchy of (Fukuda 2026g, slide 25): every inner product space is a normed space and every normed space is a metric space. Both inclusions are strict — [Remark 3.3.1](#) shows the first, and the next example the second.

Example 3.4.1 (A metric that comes from no norm). On any non-empty set X the *discrete metric*

$$d(x, y) = \begin{cases} 0 & x = y \\ 1 & x \neq y \end{cases}$$

is a metric. If X is a vector space with more than one point, d is induced by no norm: a norm satisfies $\|2x\| = 2\|x\|$, so $d(2x, 0) = 2d(x, 0)$, but the discrete metric gives $d(2x, 0) = d(x, 0) = 1$ for any $x \neq 0$. More generally, a metric on a vector space comes from a norm if and only if it is translation invariant and absolutely homogeneous.

The discrete metric is not a curiosity: under it every subset is open, every sequence that converges is eventually constant, and the compact sets are exactly the finite ones. It is the standard source of counterexamples in [Chapter 4](#).

Example 3.4.2 (Metrics on economic objects). (i) Consumption bundles in $\mathbb{R}^n$: Euclidean metric.

- (ii) Consumption *plans* $c: [0, \infty) \rightarrow \mathbb{R}$ or $(c_t)_{t \in \mathbb{N}_0}$: supremum metric on $\mathcal{B}(\cdot, \mathbb{R})$. The slide’s remark that one chooses “a consumption function $c: [0, \infty) \rightarrow \mathbb{R}$ instead of $(c_1, c_2) \in \mathbb{R}^2$ ” (Fukuda 2026g, slide 4) is precisely the passage from (i) to (ii), and it is the passage that makes the infinite-dimensional theory necessary.
- (iii) Value functions $V: \mathbb{R}^n \rightarrow \mathbb{R}$: supremum metric, under which the Bellman operator is a contraction ([Theorem 6.5.1](#)).
- (iv) Probability distributions: many metrics are in use (total variation, Kolmogorov–Smirnov, Wasserstein), and they are not equivalent in infinite dimensions; which one is meant must always be stated.

Definition 3.4.2 (Open ball, bounded set). Let (X, d) be a metric space, $x_0 \in X$ and $\lambda > 0$. The *open ball* with centre x_0 and radius λ is

$$B_\lambda(x_0) := \{x \in X : d(x_0, x) < \lambda\}.$$

A set $A \subseteq X$ is *bounded* if $A = \emptyset$ or $A \subseteq B_r(x_0)$ for some $x_0 \in X$ and $r > 0$.

Proposition 3.4.2. A set $A \subseteq \mathbb{R}^n$ is bounded in the sense of [Definition 3.4.2](#) if and only if there exists $\alpha > 0$ with $\|x\| \leq \alpha$ for all $x \in A$.

Proof. If $\|x\| \leq \alpha$ on A , then $A \subseteq B_{\alpha+1}(0)$. Conversely if $A \subseteq B_r(x_0)$ then for $x \in A$, $\|x\| \leq \|x - x_0\| + \|x_0\| < r + \|x_0\|$, so $\alpha = r + \|x_0\|$ works. ■

This settles Exercise 7.20 of (Fukuda 2026g, slide 28). The economic instance given there is the budget set: for $p \in \mathbb{R}_{++}^n$ and $m > 0$, every $x \in \mathcal{B}(p, m)$ satisfies $x_i \leq m/p_i$, so $\|x\|_\infty \leq m/\min_i p_i$ and $\mathcal{B}(p, m)$ is bounded. If some $p_i = 0$, the bound fails and so does the conclusion — the substantive reason for requiring strictly positive prices, flagged already in [Example 1.6.1](#).

3.5 Complete Normed and Inner Product Spaces

The definitions below are stated here for completeness of the hierarchy; the notion of a Cauchy sequence on which they rest is developed in [Chapter 4](#) and their consequences in [Chapter 6](#).

Definition 3.5.1 (Banach and Hilbert spaces). A normed space that is complete as a metric space ([Definition 6.3.1](#)) is a *Banach space*. An inner product space that is complete in the induced norm is a *Hilbert space*.

These are Definitions 7.12 and 7.13 of Fukuda (2026d), given on slide 77 of (Fukuda 2026g). The two examples that matter here are L^2 , which is a Hilbert space and is where Chapter 11 operates, and $\mathcal{B}(X, \mathbb{R})$ with the supremum norm, which is a Banach space but not a Hilbert space by Remark 3.3.1, and is where Chapter 15 operates.

Remark 3.5.1 (Why completeness is stated at the level of the metric). Completeness is a metric notion: it makes sense in (X, d) without any vector structure. Banach and Hilbert spaces are therefore not new kinds of object but conjunctions — “normed and complete”, “inner product and complete”. This matters when one restricts attention to a subset: a closed subset of a complete metric space is complete (Proposition 6.3.2), and a closed subset of a Banach space need not be a subspace, but is still a complete metric space. The contraction mapping theorem applies to it regardless, which is what allows Chapter 15 to work on the set of bounded, continuous, *increasing and concave* functions — a closed subset, not a subspace.

3.6 Chapter Summary

An inner product supplies length, angle and orthogonality; the Cauchy–Schwarz inequality is the statement that the resulting cosine lies in $[-1, 1]$, and its equality case characterises linear dependence, which in the statistical reading is perfect correlation. Every linear functional on $\mathbb{R}^n$ is an inner product with a unique vector (Theorem 3.2.1); this identification is what turns linearity in probabilities into an expected-utility index, linearity of a pricing functional into a vector of state prices, and the differential into the gradient. A norm forgets angle, a metric forgets the vector structure as well; both inclusions in the hierarchy are strict, as the supremum norm and the discrete metric show. Boundedness of the budget set requires strictly positive prices. Completeness, defined in Chapter 4, upgrades normed and inner product spaces to Banach and Hilbert spaces, the two settings in which Chapters 11 and 15 respectively work.

Exercise 3.6.1. Prove the polarisation identity $\langle x, y \rangle = \frac{1}{4}(\|x + y\|^2 - \|x - y\|^2)$ in any inner product space, and deduce that an inner product is determined by its norm. Combine this with Remark 3.3.1 to give a complete criterion for when a norm comes from an inner product.

Exercise 3.6.2. Show that on $\mathbb{R}^n$ the norms $\|\cdot\|_1$, $\|\cdot\|_2$ and $\|\cdot\|_\infty$ satisfy $\|x\|_\infty \leq \|x\|_2 \leq \|x\|_1 \leq n\|x\|_\infty$, so that all three induce the same bounded sets and, by Chapter 4, the same open sets. Explain why the analogous statement fails on $\mathcal{C}([0, 1], \mathbb{R})$ for $\|f\|_1 = \int_0^1 |f|$ versus $\|f\|_\infty$, and give a sequence witnessing the failure.

Exercise 3.6.3. Let $W \in M_{n \times n}(\mathbb{R})$ be symmetric. Show that $\langle x, y \rangle_W = x^\top W y$ is an inner product on $\mathbb{R}^n$ if and only if $W \succ 0$, and that positive *semidefiniteness* gives only a seminorm. Identify the subspace on which the seminorm vanishes in terms of $\text{Ker } W$.

Chapter 4

Topology of Metric Spaces: Open Sets, Sequences, and Compactness

4.1 Motivation: What “Nearby” Has to Mean

Three of the most-used theorems in economic theory — that a continuous function on a compact set attains its maximum, that a bounded sequence has a convergent subsequence, and that a contraction on a complete space has a unique fixed point — are statements about which sets contain their limit points and which sequences converge. This chapter builds that vocabulary.

The order of business follows the slides (Fukuda 2026g, slides 26–48): open balls, then interior and closure, then the three axioms that isolate what a topology is, then sequences and their limits, then the two results that make compactness usable — Bolzano–Weierstrass and Heine–Borel. Throughout, (X, d) is a metric space; the specialisation to $\mathbb{R}^n$ is stated whenever it is genuinely used, because several of the results below are false in general metric spaces and knowing which is the point of stating them abstractly.

4.2 Open and Closed Sets

Definition 4.2.1 (Interior, closure, boundary). Let (X, d) be a metric space and $A \subseteq X$.

- (i) A point $a \in X$ is *interior point* A if $B_\varepsilon(a) \subseteq A$ for some $\varepsilon > 0$. The *interior* $\text{Int } A$ is the set of interior points of A ; by definition $\text{Int } A \subseteq A$. The set A is *open set* if $A = \text{Int } A$.
- (ii) A point $a \in X$ is a *point of closure* of A if $A \cap B_\varepsilon(a) \neq \emptyset$ for every $\varepsilon > 0$. The *closure* $\bar{A}$ is the set of points of closure; by definition $A \subseteq \bar{A}$. The set A is *closed set* if $A = \bar{A}$.
- (iii) The *boundary* is $\partial A := \bar{A} \setminus \text{Int } A$.

Example 4.2.1 (Closure can add points that are “reached but not attained”). In $\mathbb{R}$ with $d(x, y) = |x - y|$ let $A = \{1/n : n \in \mathbb{N}\}$. Then $0 \in \bar{A}$: for any $\varepsilon > 0$, Proposition 1.5.2 supplies n with $1/n < \varepsilon$, so $1/n \in A \cap B_\varepsilon(0)$. Since $0 \notin A$, A is not closed, and $\bar{A} = A \cup \{0\}$. Also $\text{Int } A = \emptyset$, since every ball around $1/n$ contains irrational numbers. This is Example 7.1 of Fukuda (2026d), illustrated on (Fukuda 2026g, slide 33).

Proposition 4.2.1 (Duality of interior and closure). For every $A \subseteq X$,

$$(\bar{A})^c = \text{Int}(A^c), \quad \text{equivalently} \quad (\text{Int } A)^c = \overline{(A^c)}.$$

Consequently A is open if and only if A^c is closed, and A is closed if and only if A^c is open.

Proof. A point a fails to be a point of closure of A if and only if some $\varepsilon > 0$ has $A \cap B_\varepsilon(a) = \emptyset$, that is $B_\varepsilon(a) \subseteq A^c$, that is $a \in \text{Int}(A^c)$. This is the first identity; the second follows by applying the first to A^c and taking complements.

For the consequence: A open means $A = \text{Int } A$, hence $A^c = (\text{Int } A)^c = \overline{A^c}$, which is A^c closed; and conversely. ■

This is Proposition 7.5 and Corollary 7.1 of Fukuda (2026d) (Fukuda 2026g, slide 35). It is worth stating explicitly because almost every proof about closed sets is obtained by complementing a proof about open sets.

Proposition 4.2.2 (Open balls are open). In any metric space, $B_\lambda(a)$ is open.

Proof. Let $x \in B_\lambda(a)$ and put $\rho := d(a, x) < \lambda$, so $\lambda - \rho > 0$. For $y \in B_{\lambda-\rho}(x)$ the triangle inequality gives

$$d(a, y) \leq d(a, x) + d(x, y) < \rho + (\lambda - \rho) = \lambda,$$

so $y \in B_\lambda(a)$. Hence $B_{\lambda-\rho}(x) \subseteq B_\lambda(a)$ and $x \in \text{Int } B_\lambda(a)$. As x was arbitrary, $B_\lambda(a) = \text{Int } B_\lambda(a)$. ■

Proposition 4.2.3 (Int A is open and $\overline{A}$ is closed). For every $A \subseteq X$: $\text{Int } A$ is open, $\overline{A}$ is closed, $\text{Int } A$ is the largest open set contained in A , and $\overline{A}$ is the smallest closed set containing A .

Proof. Let $a \in \text{Int } A$, so $B_\varepsilon(a) \subseteq A$ for some $\varepsilon > 0$. By Proposition 4.2.2, every $x \in B_\varepsilon(a)$ has a ball $B_\delta(x) \subseteq B_\varepsilon(a) \subseteq A$, so $x \in \text{Int } A$; hence $B_\varepsilon(a) \subseteq \text{Int } A$ and $\text{Int } A$ is open. Applying this to A^c and using Proposition 4.2.1 shows $\overline{A}$ is closed.

If $G \subseteq A$ is open then every $a \in G$ has a ball inside $G \subseteq A$, so $G \subseteq \text{Int } A$; this is maximality. The statement for $\overline{A}$ follows by complementation. ■

Theorem 4.2.1 (The three axioms of a topology). Let $\mathcal{O}$ denote the collection of open subsets of a metric space (X, d) . Then

- (i) the union of any subfamily of $\mathcal{O}$ belongs to $\mathcal{O}$;
- (ii) the intersection of any *finite* subfamily of $\mathcal{O}$ belongs to $\mathcal{O}$;
- (iii) $\emptyset \in \mathcal{O}$ and $X \in \mathcal{O}$.

Proof. (i) Let $(G_i)_{i \in I} \subseteq \mathcal{O}$ and $x \in G = \bigcup_i G_i$. Then $x \in G_{i_0}$ for some i_0 , and openness of G_{i_0} supplies $\varepsilon > 0$ with $B_\varepsilon(x) \subseteq G_{i_0} \subseteq G$.

(ii) Let $G_1, \dots, G_n \in \mathcal{O}$ and $x \in G = \bigcap_{i=1}^n G_i$. For each i pick $\varepsilon_i > 0$ with $B_{\varepsilon_i}(x) \subseteq G_i$ and set $\varepsilon = \min\{\varepsilon_1, \dots, \varepsilon_n\}$. Then $\varepsilon > 0$ — this is where finiteness is used — and $B_\varepsilon(x) \subseteq B_{\varepsilon_i}(x) \subseteq G_i$ for every i , so $B_\varepsilon(x) \subseteq G$.

(iii) For X : any ε works since $B_\varepsilon(x) \subseteq X$ always. For $\emptyset$: the statement “ $x \in \emptyset$ implies $B_\varepsilon(x) \subseteq \emptyset$ ” is vacuously true, by the truth table of Section 1.4. ■

Example 4.2.2 (Infinitely many open sets can intersect in a closed set). In $\mathbb{R}$, $\bigcap_{n \in \mathbb{N}} (-\frac{1}{n}, \frac{1}{n}) = \{0\}$, an intersection of open sets that is not open. The minimum in the proof of Theorem 4.2.1(ii) becomes an infimum equal to 0, and the argument collapses. Dually, $\bigcup_{n \in \mathbb{N}} [\frac{1}{n}, 1] = (0, 1]$ is a union of closed sets that is not closed.

Definition 4.2.2 (Topological space). Let X be a set. A collection $\mathcal{O} \subseteq \mathcal{P}(X)$ satisfying (i)–(iii) of Theorem 4.2.1 is a *topology* on X , the pair $(X, \mathcal{O})$ is a *topological space*, and the members of $\mathcal{O}$ are the open sets. A topology arising from a metric as above is *metrisable topology*.

Remark 4.2.1 (Two abstractions of the same list of axioms). Compare Definition 4.2.2 with the definition of a σ -algebra: a collection $\mathcal{F} \subseteq \mathcal{P}(\Omega)$ containing $\emptyset$ and Ω , closed under complementation, and closed under *countable* unions. The two are different closure requirements on a collection of sets, and the difference is exactly where the two theories diverge: topology tolerates arbitrary unions but only finite intersections; measure theory tolerates countable unions and countable intersections but requires complements. On $\mathbb{R}$ the Borel σ -algebra is generated by the open sets, which is how the two are linked (Fukuda 2026g, slide 37). The parallel between the two definitions of “measurable/continuous via preimages” drawn there — f is continuous if $f^{-1}(B)$ is open whenever B is open; X is a random variable if $X^{-1}(B)$ is measurable whenever B is measurable — is made

precise in [Theorem 5.2.1](#) and [Definition 14.2.1](#).

4.3 Sequences and Limits

Definition 4.3.1 (Sequence, subsequence, limit). A *sequence* in X is a map $\mathbb{N} \rightarrow X$, written $(x_n)_{n \in \mathbb{N}}$. If $\tau: \mathbb{N} \rightarrow \mathbb{N}$ is strictly increasing, then $(x_{\tau(n)})_{n \in \mathbb{N}}$ is a *subsequence*, usually written $(x_{n_k})_{k \in \mathbb{N}}$.

The sequence *convergent sequence* to $x^* \in X$, written $x_n \rightarrow x^*$ or $x^* = \lim_{n \rightarrow \infty} x_n$, if for every $\varepsilon > 0$ there is $n_0 \in \mathbb{N}$ such that $n \geq n_0$ implies $d(x^*, x_n) < \varepsilon$.

Proposition 4.3.1 (Basic facts). Let (x_n) be a sequence in a metric space X .

- (i) A limit, if it exists, is unique.
- (ii) If $x_n \rightarrow x^*$ then every subsequence converges to x^* .
- (iii) A convergent sequence is bounded.

Proof. (i) If $x_n \rightarrow x$ and $x_n \rightarrow y$, then for every $\varepsilon > 0$ we may choose n with $d(x, x_n) < \varepsilon/2$ and $d(y, x_n) < \varepsilon/2$, so $d(x, y) \leq d(x, x_n) + d(x_n, y) < \varepsilon$. As $\varepsilon > 0$ was arbitrary, $d(x, y) = 0$, hence $x = y$.

(ii) Since τ is strictly increasing, $\tau(k) \geq k$ by induction. Given $\varepsilon > 0$ take n_0 from the definition; for $k \geq n_0$ we have $\tau(k) \geq k \geq n_0$, so $d(x^*, x_{\tau(k)}) < \varepsilon$.

(iii) Take $\varepsilon = 1$ and the corresponding n_0 . Then $d(x^*, x_n) < 1$ for $n \geq n_0$, and setting $M := \max\{1, d(x^*, x_1), \dots, d(x^*, x_{n_0-1})\}$ gives $(x_n) \subseteq B_{M+1}(x^*)$; the maximum is over a finite set and hence finite. ■

Parts (ii) and (iii) are Exercises 7.30 and 7.31 of (Fukuda 2026g, slide 43).

Theorem 4.3.1 (Sequential characterisation of closedness). A set $A \subseteq X$ is closed if and only if the limit of every convergent sequence of elements of A lies in A .

Proof. ($\Rightarrow$) Let A be closed, $(x_n) \subseteq A$ and $x_n \rightarrow x^*$. For every $\varepsilon > 0$ there is n with $x_n \in B_\varepsilon(x^*)$, and $x_n \in A$, so $A \cap B_\varepsilon(x^*) \neq \emptyset$. Hence $x^* \in \bar{A} = A$.

($\Leftarrow$) Suppose the stated condition holds and let $x^* \in \bar{A}$. For each $n \in \mathbb{N}$ the set $A \cap B_{1/n}(x^*)$ is non-empty; pick x_n in it. Then $d(x^*, x_n) < 1/n \rightarrow 0$, so $x_n \rightarrow x^*$ with $(x_n) \subseteq A$, and the hypothesis gives $x^* \in A$. Hence $\bar{A} \subseteq A$, and the reverse inclusion always holds. ■

This is Proposition 7.11 of Fukuda (2026d) (Fukuda 2026g, slide 43). It converts a statement about balls into a statement about sequences, and it is the form in which closedness is checked in practice.

4.3.1 Monotone subsequences and Bolzano–Weierstrass

Proposition 4.3.2 (Every real sequence has a monotone subsequence). Every sequence $(x_n)_{n \in \mathbb{N}}$ in $\mathbb{R}$ has a subsequence that is increasing or a subsequence that is decreasing (possibly both).

Proof. Call $m \in \mathbb{N}$ a *peak* if $x_m > x_n$ for all $n > m$, and let $S := \{m \in \mathbb{N} : m \text{ is a peak}\}$.

If S is infinite, enumerate it increasingly as $n_1 < n_2 < \dots$. For each k , since n_k is a peak and $n_{k+1} > n_k$, we get $x_{n_k} > x_{n_{k+1}}$; hence (x_{n_k}) is strictly decreasing.

If S is finite, let N exceed every element of S and set $n_1 := N$. Since n_1 is not a peak, there is $n_2 > n_1$ with $x_{n_2} \geq x_{n_1}$; since $n_2 > N$ is not a peak either, there is $n_3 > n_2$ with $x_{n_3} \geq x_{n_2}$, and so on. This produces an increasing subsequence. ■

This is Proposition 7.12 of Fukuda (2026d); the argument is the one sketched on (Fukuda 2026g, slides 45–46), made explicit.

Proposition 4.3.3 (Monotone convergence). An increasing sequence in $\mathbb{R}$ that is bounded above converges to $\sup_n x_n$; a decreasing sequence bounded below converges to $\inf_n x_n$.

Proof. Let (x_n) be increasing and bounded above and put $s := \sup_n x_n$, which exists by [Assumption 1.5.1](#). Given $\varepsilon > 0$, [Proposition 1.5.1\(ii\)](#) supplies n_0 with $x_{n_0} > s - \varepsilon$. For $n \geq n_0$, monotonicity and the upper-bound property give $s - \varepsilon < x_{n_0} \leq x_n \leq s$, so $|s - x_n| < \varepsilon$. The decreasing case follows by applying this to $(-x_n)$. ■

Theorem 4.3.2 (Bolzano–Weierstrass). Every bounded sequence in $\mathbb{R}$ has a convergent subsequence. More generally, every bounded sequence in $\mathbb{R}^n$ has a convergent subsequence.

Proof. In $\mathbb{R}$: by [Proposition 4.3.2](#) extract a monotone subsequence, which inherits boundedness, and apply [Proposition 4.3.3](#).

In $\mathbb{R}^n$: let $(x_m)_{m \in \mathbb{N}}$ be bounded, with coordinates $x_m = (x_m^1, \dots, x_m^n)$. Each coordinate sequence is bounded in $\mathbb{R}$, since $|x_m^i| \leq \|x_m\|$. Apply the one-dimensional case to $(x_m^1)_m$ to get a subsequence along which the first coordinate converges; apply it again, along that subsequence, to the second coordinate; and so on. As n is finite, after n extractions we obtain a single subsequence along which every coordinate converges. By [Proposition 4.3.4](#) the subsequence converges in $\mathbb{R}^n$. ■

Proposition 4.3.4 (Coordinatewise convergence). A sequence (x_m) in $\mathbb{R}^n$ converges if and only if each coordinate sequence $(x_m^i)_m$ converges in $\mathbb{R}$, and then $\lim_m x_m = (\lim_m x_m^1, \dots, \lim_m x_m^n)$.

Proof. If $x_m \rightarrow x^*$, then $|x_m^i - x^{i*}| \leq \|x_m - x^*\| \rightarrow 0$ for each i . Conversely, if $x_m^i \rightarrow x^{i*}$ for each i , then given $\varepsilon > 0$ choose n_i with $|x_m^i - x^{i*}| < \varepsilon/\sqrt{n}$ for $m \geq n_i$; for $m \geq \max_i n_i$, $\|x_m - x^*\| = (\sum_i |x_m^i - x^{i*}|^2)^{1/2} < \varepsilon$. ■

This is Proposition 7.14 of Fukuda (2026d) (Fukuda 2026g, slide 48). The finiteness of n is essential: the argument in [Theorem 4.3.2](#) performs n successive extractions, and in an infinite-dimensional space it cannot be completed. Indeed Bolzano–Weierstrass *fails* there, as [Example 4.3.1](#) shows.

Example 4.3.1 (Bolzano–Weierstrass fails in infinite dimensions). In the space of square-summable sequences with $\|x\|^2 = \sum_i x_i^2$, consider the orthonormal vectors $e^1, e^2, \dots$, where e^k has a 1 in position k and zeros elsewhere. Then $\|e^k\| = 1$ for all k , so the sequence is bounded, while $\|e^j - e^k\| = \sqrt{2}$ for $j \neq k$, so no subsequence is Cauchy and hence none converges. Bounded and closed therefore does not imply compact outside finite dimensions, which is why [Theorem 4.4.2](#) is stated for $\mathbb{R}^n$ and why the fixed-point arguments of [Chapter 15](#) use completeness and contraction rather than compactness.

4.4 Compactness

Definition 4.4.1 (Compactness). Let (X, d) be a metric space and $K \subseteq X$.

- (i) An *open cover* of K is a family $(G_i)_{i \in I}$ of open sets with $K \subseteq \bigcup_{i \in I} G_i$. The set K is *compact* if every open cover of K admits a finite subcover, that is, a finite $J \subseteq I$ with $K \subseteq \bigcup_{i \in J} G_i$.
- (ii) K is *sequentially compact* if every sequence in K has a subsequence converging to a point of K .

Theorem 4.4.1 (Compactness and sequential compactness agree in metric spaces). In a metric space, a set is compact if and only if it is sequentially compact.

The proof is standard and somewhat long; see Rudin (1976, Thm. 3.28 and its converse) or Fukuda (2026d, Sec. 8.1). In general *topological* spaces the two notions differ, which is one reason for staying within metric spaces throughout this volume.

Proposition 4.4.1 (Compact implies closed and bounded). In any metric space, a compact set is closed and bounded.

Proof. Boundedness: fix $x_0 \in X$; the balls $B_n(x_0)$, $n \in \mathbb{N}$, cover K , and a finite subcover is contained in the largest of them.

Closedness: let $x^* \in \overline{K}$ and take a sequence $(x_n) \subseteq K$ with $x_n \rightarrow x^*$, as in the proof of [Theorem 4.3.1](#). By sequential compactness ([Theorem 4.4.1](#)) some subsequence converges to a point of K ; by [Proposition 4.3.1\(ii\)](#) that limit is x^* . Hence $x^* \in K$ and K is closed by [Theorem 4.3.1](#). ■

Theorem 4.4.2 (Heine–Borel). A subset of $\mathbb{R}^n$ is compact if and only if it is closed and bounded.

Proof. Necessity is [Proposition 4.4.1](#). For sufficiency, let $K \subseteq \mathbb{R}^n$ be closed and bounded and let $(x_m) \subseteq K$. Boundedness and [Theorem 4.3.2](#) give a convergent subsequence $x_{m_k} \rightarrow x^*$; closedness and [Theorem 4.3.1](#) give $x^* \in K$. So K is sequentially compact, hence compact by [Theorem 4.4.1](#). ■

Remark 4.4.1 (The equivalence is special to $\mathbb{R}^n$). [Example 4.3.1](#) exhibits a closed bounded set — the closed unit ball of an infinite-dimensional space — that is not compact. In fact the closed unit ball of a normed space is compact if and only if the space is finite-dimensional (Riesz’s lemma; see Aliprantis and Border (2006, Ch. 5)). Whenever a proof in economics invokes “closed and bounded, hence compact”, the underlying space must be checked to be finite-dimensional.

Proposition 4.4.2 (Stability properties). (i) A closed subset of a compact set is compact.

(ii) A finite union of compact sets is compact; an arbitrary intersection of compact sets is compact.

(iii) A finite product of compact sets is compact in the product metric.

Proof. (i) Let $F \subseteq K$ be closed with K compact, and let $(x_n) \subseteq F$. Compactness of K gives a subsequence converging to some $x^* \in K$; closedness of F and [Theorem 4.3.1](#) give $x^* \in F$.

(ii) For a finite union, a sequence in $\bigcup_{j=1}^m K_j$ has infinitely many terms in some K_{j_0} , and that subsequence has a further subsequence converging in K_{j_0} . An intersection of compact sets is a closed subset (by [Proposition 4.4.1](#) and [Proposition 4.2.1](#)) of any one of them, so (i) applies.

(iii) Extract successively in each coordinate, as in the proof of [Theorem 4.3.2](#); finiteness of the number of factors is what makes the extraction terminate. ■

Example 4.4.1 (The budget set is compact). For $p \in \mathbb{R}_{++}^n$ and $m > 0$, $\mathcal{B}(p, m) = \{x \in \mathbb{R}_+^n : \langle p, x \rangle \leq m\}$ is compact. It is bounded, as shown after [Proposition 3.4.2](#). It is closed: if $(x_k) \subseteq \mathcal{B}(p, m)$ and $x_k \rightarrow x^*$, then $x^* \in \mathbb{R}_+^n$ because each coordinate is a limit of non-negative numbers, and $\langle p, x^* \rangle = \lim_k \langle p, x_k \rangle \leq m$ by continuity of the inner product ([Proposition 5.2.1](#)). [Theorem 4.4.2](#) applies.

Every hypothesis is doing work. If $p_i = 0$ for some i the set is unbounded. If the constraint were $\langle p, x \rangle < m$ the set would not be closed and the supremum of a strictly increasing utility would not be attained. If consumption were not restricted to $\mathbb{R}_+^n$, the set would again be unbounded whenever some $p_i < 0$. The existence half of [Example 1.6.1](#) rests on exactly this proposition, via the Weierstrass theorem of [Chapter 5](#).

4.5 Connectedness

Definition 4.5.1 (Connected set). A metric space X is *connected* if the only subsets of X that are both open and closed are $\emptyset$ and X ; equivalently, X cannot be written as the union of two disjoint non-empty open sets. A subset $A \subseteq X$ is connected if it is connected as a metric space in the induced metric.

Proposition 4.5.1. A subset of $\mathbb{R}$ is connected if and only if it is an interval, that is, if $x, y \in A$ and $x < z < y$ imply $z \in A$. Every convex subset of $\mathbb{R}^n$ is connected.

Proof. If A is not an interval, take $x < z < y$ with $x, y \in A$, $z \notin A$; then $A \cap (-\infty, z)$ and $A \cap (z, \infty)$ are disjoint, non-empty, relatively open, and cover A , so A is not connected. The converse, and the convexity statement, follow from the observation that a continuous image of a connected set is connected ([Proposition 5.2.5](#)) applied to the segment map $t \mapsto (1-t)x + ty$, whose domain $[0, 1]$ is connected. ■

Connectedness is what makes the intermediate value theorem true, and hence what underlies both the existence of an interior solution between two boundary regimes and Debreu’s continuous utility representation theorem alluded to in [Remark 1.3.1](#).

4.6 Chapter Summary

Open sets are those equal to their interior, closed sets those equal to their closure, and the two notions are exchanged by complementation. The open sets are closed under arbitrary unions and finite intersections and contain $\emptyset$ and X ; abstracting these three properties gives a topology, and the countable-union-plus-complement variant gives a σ -algebra. A set is closed if and only if it contains the limits of its convergent sequences. Every bounded real sequence has a monotone, hence convergent, subsequence ([Theorem 4.3.2](#)); by coordinatewise extraction the same holds in $\mathbb{R}^n$, and the argument requires n to be finite. Compactness is equivalent to sequential compactness in metric spaces, implies closed and bounded always, and is equivalent to closed and bounded exactly in $\mathbb{R}^n$ ([Theorem 4.4.2](#)). The budget set with strictly positive prices is compact, which is the hypothesis under which [Chapter 5](#) produces an optimal consumption bundle.

Exercise 4.6.1. Show that $\partial A = \overline{A} \cap \overline{A^c}$, that ∂A is always closed, and that A is closed if and only if $\partial A \subseteq A$. Compute $\text{Int } A$, $\overline{A}$ and ∂A for $A = \mathbb{Q} \cap [0, 1]$ in $\mathbb{R}$, and comment on which of the three is empty.

Exercise 4.6.2. Prove that in the discrete metric of [Example 3.4.1](#) every subset is both open and closed, that a sequence converges if and only if it is eventually constant, and that a set is compact if and only if it is finite. Which hypothesis of [Theorem 4.4.2](#) fails for an infinite bounded set in this metric?

Exercise 4.6.3. Let $K \subseteq \mathbb{R}^n$ be compact and non-empty. Show that $\sup K$ and $\inf K$ are attained coordinatewise, that is, for each i there exist $x, y \in K$ with $x_i = \max\{z_i : z \in K\}$ and $y_i = \min\{z_i : z \in K\}$. Give an example of a closed but unbounded $K \subseteq \mathbb{R}$ for which the conclusion fails.

Chapter 5

Continuity, Semicontinuity, and the Weierstrass Theorem

5.1 Motivation: Why Optima Exist

An economic model that begins “the consumer chooses x to maximise $u(x)$ subject to $\langle p, x \rangle \leq m$ ” has presupposed that a maximiser exists. Nothing guarantees this. The supremum of a function over a set need not be attained, and when it is not, the demand function is undefined, comparative statics are vacuous, and equilibrium existence arguments have no starting point.

The Weierstrass theorem gives the standard sufficient conditions: a compact feasible set and enough continuity of the objective. [Chapter 4](#) supplied compactness. This chapter supplies continuity, and then weakens it as far as the existence conclusion permits — to upper semicontinuity for maxima and lower semicontinuity for minima. The weakening is not decoration. Indirect utility functions, value functions of dynamic programs and profit functions arise as suprema of families of continuous functions, and [Proposition 5.3.3](#) shows that such suprema are in general only semicontinuous.

5.2 Continuity

Throughout, $A \subseteq \mathbb{R}^n$ and $f: A \rightarrow \mathbb{R}^m$ unless stated otherwise. Two metrics are in play — the Euclidean metric on the domain and on the codomain — and both are written d .

Definition 5.2.1 (Relative topology). Let $A \subseteq X$ with (X, d) a metric space. A set $U \subseteq A$ is *relatively open set* if $U = G \cap A$ for some open $G \subseteq X$; equivalently, if for every $x \in U$ there is $\varepsilon > 0$ with $B_\varepsilon(x) \cap A \subseteq U$. Relative closedness is defined analogously. This is exactly the topology induced by restricting d to $A \times A$, so “relatively open in A ” and “open in the metric space $(A, d|_{A \times A})$ ” are the same statement.

The distinction matters because $[0, 1]$ is not open in $\mathbb{R}$ but is open relative to $[0, 1]$, and hypotheses about feasible sets are almost always relative hypotheses.

Definition 5.2.2 (Continuity). Let $A \subseteq \mathbb{R}^n$ and $f: A \rightarrow \mathbb{R}^m$. The mapping f is *continuity* $x_0 \in A$ if for every $\varepsilon > 0$ there is $\delta > 0$ such that

$$x \in A \text{ and } d(x_0, x) < \delta \implies d(f(x_0), f(x)) < \varepsilon.$$

It is *continuous mapping* if it is continuous at every point of A .

Two features of the definition are easy to pass over. First, δ may depend on x_0 as well as on ε ; requiring that it not is uniform continuity ([Definition 5.2.3](#)). Second, the implication is restricted to $x \in A$, so continuity is a statement relative to the domain. A function defined only on $\mathbb{Z}$ is continuous at every point of its domain, for trivial reasons.

Example 5.2.1 (Removable and essential discontinuities). Define $f: \mathbb{R} \rightarrow \mathbb{R}$ by $f(x) = \sin(x)/x$ for $x \neq 0$ and $f(0) = 1$. Then f is continuous at 0: since $\cos x \leq \sin(x)/x \leq 1$ for $0 < |x| < \pi/2$ and $\cos$ is

continuous with $\cos 0 = 1$, the value 1 is the only value at 0 that makes f continuous, and it has been assigned. The discontinuity of the function $x \mapsto \sin(x)/x$ on $\mathbb{R} \setminus \{0\}$ is *removable*.

Define $g: \mathbb{R} \rightarrow \mathbb{R}$ by $g(x) = \sin(1/x)$ for $x \neq 0$ and $g(0) = 0$. No assignment at 0 makes g continuous. Taking $x_k = (\pi/2 + 2k\pi)^{-1}$ and $y_k = (-\pi/2 + 2k\pi)^{-1}$ gives $x_k \rightarrow 0$, $y_k \rightarrow 0$ but $g(x_k) = 1$, $g(y_k) = -1$, so no value of $g(0)$ is compatible with [Proposition 5.2.3](#). The discontinuity is *essential* (Fukuda [2026g](#), slides 50–51).

Proposition 5.2.1 (Continuity of linear functionals and of the norm). Let $a \in \mathbb{R}^n$.

- (i) The functional $f(x) = \langle a, x \rangle$ is continuous on $\mathbb{R}^n$.
- (ii) Every linear map $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ is continuous.
- (iii) The norm $\|\cdot\|: \mathbb{R}^n \rightarrow \mathbb{R}$ is continuous.

Proof. (i) If $a = 0$ then $f \equiv 0$ is constant, hence continuous. Let $a \neq 0$, fix $x_0 \in \mathbb{R}^n$ and $\varepsilon > 0$, and set $\delta := \varepsilon / \|a\| > 0$. If $\|x - x_0\| < \delta$ then by Cauchy–Schwarz ([Lemma 3.2.1](#))

$$|f(x) - f(x_0)| = |\langle a, x - x_0 \rangle| \leq \|a\| \|x - x_0\| < \|a\| \delta = \varepsilon.$$

(ii) Write T in the canonical bases as in [Theorem 2.3.1](#), so that the i th component of Tx is $\langle a_i, x \rangle$ with a_i the i th row of the representing matrix. Given $\varepsilon > 0$, apply (i) to each component with tolerance $\varepsilon/\sqrt{m}$ and take the smallest of the resulting δ_i .

(iii) By the reverse triangle inequality ([Proposition 3.3.2](#)), $|\|x\| - \|x_0\|| \leq \|x - x_0\|$, so $\delta = \varepsilon$ works. ■

Part (i) is Proposition 7.17 of Fukuda ([2026d](#)), proved on (Fukuda [2026g](#), slide 52); part (iii) is Proposition 7.18 there, whose hint is Exercise 7.4, recorded here as [Proposition 3.3.2](#). The δ produced in (i) does not depend on x_0 , so linear functionals are in fact uniformly continuous; this is a consequence of linearity, not of the specific bound.

Proposition 5.2.2 (Algebra of continuous functions). Let $A \subseteq \mathbb{R}^n$ and let $f, g: A \rightarrow \mathbb{R}$ be continuous at x_0 .

- (i) $f + g$, fg and αf for $\alpha \in \mathbb{R}$ are continuous at x_0 ; f/g is continuous at x_0 provided $g(x_0) \neq 0$.
- (ii) If $h: B \rightarrow \mathbb{R}^m$ is continuous at $f(x_0)$ and $f(A) \subseteq B$, then $h \circ f$ is continuous at x_0 .
- (iii) If $f_1, \dots, f_m: A \rightarrow \mathbb{R}$ are continuous at x_0 , then so are $\max_{1 \leq i \leq m} f_i$ and $\min_{1 \leq i \leq m} f_i$.

Proof. (i) and (ii) are the standard ε – δ arguments; see Rudin ([1976](#), Thms. 4.4 and 4.7).

(iii) For $m = 2$ use the identity $\max\{s, t\} = \frac{1}{2}(s + t + |s - t|)$ and combine (i), (ii) and the continuity of $|\cdot|$; the general case follows by induction on m , since $\max_{i \leq m} f_i = \max\{\max_{i \leq m-1} f_i, f_m\}$. The corresponding identity for the minimum is $\min\{s, t\} = \frac{1}{2}(s + t - |s - t|)$. ■

Part (iii) is Exercise 7.34 of (Fukuda [2026g](#), slide 56). The restriction to *finitely* many functions is essential and is exactly what fails in [Example 5.3.1](#); the induction consumes one step per function and cannot be run to infinity.

Proposition 5.2.3 (Sequential characterisation). Let $A \subseteq \mathbb{R}^n$ and $f: A \rightarrow \mathbb{R}^m$. Then f is continuous at $x_0 \in A$ if and only if for every sequence $(x_k) \subseteq A$ with $x_k \rightarrow x_0$ one has $f(x_k) \rightarrow f(x_0)$; that is, $f(\lim_k x_k) = \lim_k f(x_k)$.

Proof. ($\Rightarrow$) Given $\varepsilon > 0$, take δ from [Definition 5.2.2](#) and then k_0 with $d(x_0, x_k) < \delta$ for $k \geq k_0$; then $d(f(x_0), f(x_k)) < \varepsilon$ for $k \geq k_0$.

($\Leftarrow$) Suppose f is not continuous at x_0 . Then there is $\varepsilon_0 > 0$ such that for every $\delta > 0$ some $x \in A$ has $d(x_0, x) < \delta$ and $d(f(x_0), f(x)) \geq \varepsilon_0$. Taking $\delta = 1/k$ produces $x_k \in A$ with $x_k \rightarrow x_0$ and $d(f(x_0), f(x_k)) \geq \varepsilon_0$ for every k , so $f(x_k) \not\rightarrow f(x_0)$. ■

This is Proposition 7.22 of Fukuda ([2026d](#)) (Fukuda [2026g](#), slide 57). The contrapositive is the practical tool: to refute continuity, exhibit two sequences approaching the same point along which f has different limits, as in [Example 5.2.2](#).

Theorem 5.2.1 (Topological characterisation). Let $A \subseteq \mathbb{R}^n$ and $f: A \rightarrow \mathbb{R}^m$. The following are equivalent.

- (i) f is continuous on A .
- (ii) $f^{-1}(V)$ is open relative to A for every open $V \subseteq \mathbb{R}^m$.
- (iii) $f^{-1}(F)$ is closed relative to A for every closed $F \subseteq \mathbb{R}^m$.

Proof. (i) $\Rightarrow$ (ii). Let V be open and $x_0 \in f^{-1}(V)$. Since V is open there is $\varepsilon > 0$ with $B_\varepsilon(f(x_0)) \subseteq V$; continuity supplies $\delta > 0$ with $f(B_\delta(x_0) \cap A) \subseteq B_\varepsilon(f(x_0)) \subseteq V$, that is $B_\delta(x_0) \cap A \subseteq f^{-1}(V)$.

(ii) $\Rightarrow$ (i). Fix $x_0 \in A$ and $\varepsilon > 0$. The set $V = B_\varepsilon(f(x_0))$ is open, so $f^{-1}(V)$ is relatively open and contains x_0 ; hence some $\delta > 0$ has $B_\delta(x_0) \cap A \subseteq f^{-1}(V)$, which is the required implication.

(ii) $\Leftrightarrow$ (iii). Preimages commute with complements: $f^{-1}(F^c) = (f^{-1}(F))^c$ relative to A (Proposition 1.2.2). Apply Proposition 4.2.1. ■

Remark 5.2.1 (Continuity, measurability, and the shape of the definition). Theorem 5.2.1 states continuity without reference to any metric: only the two collections of open sets are used. This is the definition of continuity in a general topological space, and it is what makes the parallel of Remark 4.2.1 exact. A map is continuous when preimages of open sets are open; a map is measurable when preimages of measurable sets are measurable (Definition 14.2.1). The same asymmetry appears in both: the condition constrains preimages, never images. Images of open sets under continuous maps need not be open — $x \mapsto x^2$ sends the open set $(-1, 1)$ to $[0, 1)$ — and this is why Proposition 5.2.4, which does concern images, requires compactness rather than openness.

Example 5.2.2 (Separate continuity does not imply joint continuity). Let $f: \mathbb{R}^p \times \mathbb{R}^q \rightarrow \mathbb{R}^m$ be continuous. For fixed y_0 the section $x \mapsto f(x, y_0)$ is continuous, since $x \mapsto (x, y_0)$ is continuous and Proposition 5.2.2(ii) applies; symmetrically in the other argument. The converse fails. Define $f: \mathbb{R}^2 \rightarrow \mathbb{R}$ by

$$f(x_1, x_2) = \begin{cases} \frac{x_1 x_2}{x_1^2 + x_2^2}, & (x_1, x_2) \neq (0, 0), \\ 0, & (x_1, x_2) = (0, 0). \end{cases}$$

Both sections through the origin are identically zero, hence continuous. But along the diagonal $f(\rho, \rho) = \frac{1}{2}$ and along the anti-diagonal $f(\rho, -\rho) = -\frac{1}{2}$ for every $\rho \neq 0$, so by Proposition 5.2.3 f is not continuous at the origin. In fact f is constant on every ray through the origin, and takes every value in $[-\frac{1}{2}, \frac{1}{2}]$ in every ball around it (Fukuda 2026g, slides 54–55).

The example is a standing warning in comparative statics. Verifying that a function of prices and income responds continuously to each argument separately establishes nothing about its joint behaviour, and it is joint continuity that the maximum theorem of Chapter 13 requires.

5.2.1 Continuity, compactness, and connectedness

Proposition 5.2.4 (Continuous images of compact sets). Let $K \subseteq \mathbb{R}^n$ be compact and $f: K \rightarrow \mathbb{R}^m$ continuous. Then $f(K)$ is compact.

Proof. Let $(y_k) \subseteq f(K)$ and choose $x_k \in K$ with $f(x_k) = y_k$. By compactness of K some subsequence $x_{k_j} \rightarrow x^* \in K$, and by Proposition 5.2.3 $y_{k_j} = f(x_{k_j}) \rightarrow f(x^*) \in f(K)$. So $f(K)$ is sequentially compact, hence compact by Theorem 4.4.1. ■

Proposition 5.2.5 (Continuous images of connected sets). Let $C \subseteq \mathbb{R}^n$ be connected and $f: C \rightarrow \mathbb{R}^m$ continuous. Then $f(C)$ is connected.

Proof. Suppose $f(C) = U_1 \cup U_2$ with U_1, U_2 non-empty, disjoint and open relative to $f(C)$. By Theorem 5.2.1 the sets $f^{-1}(U_1)$ and $f^{-1}(U_2)$ are open relative to C , non-empty, disjoint, and their union is C — contradicting connectedness of C . ■

Corollary 5.2.1 (Intermediate value theorem). Let $f: [a, b] \rightarrow \mathbb{R}$ be continuous and let y lie between $f(a)$ and $f(b)$. Then $f(c) = y$ for some $c \in [a, b]$.

Proof. $[a, b]$ is connected by Proposition 4.5.1, so $f([a, b])$ is connected by Proposition 5.2.5, hence an interval by Proposition 4.5.1 again. An interval containing $f(a)$ and $f(b)$ contains everything between them. ■

5.2.2 Uniform and Lipschitz continuity

Definition 5.2.3 (Uniform, Lipschitz and Hölder continuity). Let $A \subseteq \mathbb{R}^n$ and $f: A \rightarrow \mathbb{R}^m$.

- (i) f is *uniform continuity* on A if for every $\varepsilon > 0$ there is $\delta > 0$, depending only on ε , such that $x, y \in A$ and $d(x, y) < \delta$ imply $d(f(x), f(y)) < \varepsilon$.
- (ii) f is *Lipschitz continuity* with constant $L \geq 0$ if $d(f(x), f(y)) \leq L d(x, y)$ for all $x, y \in A$. If $L < 1$ the map is a contraction (Definition 6.4.1).
- (iii) f is *Hölder continuity* of order $\alpha \in (0, 1]$ if $d(f(x), f(y)) \leq L d(x, y)^\alpha$ for all $x, y \in A$.

Lipschitz continuity is Hölder continuity of order 1; Hölder continuity of any order implies uniform continuity, with $\delta = (\varepsilon/L)^{1/\alpha}$; and uniform continuity implies continuity. None of the three implications reverses.

Example 5.2.3 (The implications are strict). On $(0, 1]$, $x \mapsto 1/x$ is continuous but not uniformly continuous: taking $x_k = 1/k$ and $y_k = 1/(k+1)$ gives $|x_k - y_k| \rightarrow 0$ while $|f(x_k) - f(y_k)| = 1$ for every k . On $[0, 1]$, $x \mapsto \sqrt{x}$ is uniformly continuous — it is Hölder of order $\frac{1}{2}$ — but not Lipschitz, since $(\sqrt{x} - \sqrt{0})/(x - 0) = x^{-1/2} \rightarrow \infty$ as $x \downarrow 0$.

Theorem 5.2.2 (Heine–Cantor). A continuous function on a compact set is uniformly continuous.

Proof. Let K be compact, f continuous on K , and suppose f is not uniformly continuous. Then some $\varepsilon_0 > 0$ admits, for each $k \in \mathbb{N}$, points $x_k, y_k \in K$ with $d(x_k, y_k) < 1/k$ and $d(f(x_k), f(y_k)) \geq \varepsilon_0$. By compactness pass to a subsequence with $x_{k_j} \rightarrow x^* \in K$; then $y_{k_j} \rightarrow x^*$ as well, because $d(y_{k_j}, x^*) \leq d(y_{k_j}, x_{k_j}) + d(x_{k_j}, x^*) \rightarrow 0$. Continuity and Proposition 5.2.3 give $f(x_{k_j}) \rightarrow f(x^*)$ and $f(y_{k_j}) \rightarrow f(x^*)$, so $d(f(x_{k_j}), f(y_{k_j})) \rightarrow 0$, contradicting the lower bound ε_0 . ■

Uniform continuity is what allows a single δ to be quoted for an entire domain, and it is the hypothesis under which Riemann sums converge in Chapter 10 and under which numerical approximation error can be bounded uniformly over a state space in Chapter 15.

5.3 Semicontinuity

Continuity is more than existence of optima requires. The following weakening isolates exactly what is needed, and it is the form in which the hypothesis is verifiable for the value functions that arise in economics.

Definition 5.3.1 (Semicontinuity). Let $A \subseteq \mathbb{R}^n$ and $f: A \rightarrow \mathbb{R}$.

- (i) f is *lower semicontinuity* (lsc) if $f^{-1}((\lambda, \infty)) = \{x \in A : f(x) > \lambda\}$ is open relative to A for every $\lambda \in \mathbb{R}$; equivalently, if every lower section $\{x \in A : f(x) \leq \lambda\}$ is closed relative to A .
- (ii) f is *upper semicontinuity* (usc) if $-f$ is lower semicontinuous; equivalently, if every upper section $\{x \in A : f(x) \geq \lambda\}$ is closed relative to A .

Proposition 5.3.1 (Pointwise form). Let $A \subseteq \mathbb{R}^n$ and $f: A \rightarrow \mathbb{R}$.

- (i) f is lsc if and only if for every $x_0 \in A$ and every $\lambda < f(x_0)$ there is $\delta > 0$ such that $x \in B_\delta(x_0) \cap A$ implies $\lambda < f(x)$.
- (ii) f is usc if and only if for every $x_0 \in A$ and every $\lambda > f(x_0)$ there is $\delta > 0$ such that $x \in B_\delta(x_0) \cap A$ implies $f(x) < \lambda$.

Equivalently, in sequential form: f is lsc at x_0 if $x_k \rightarrow x_0$ implies $f(x_0) \leq \liminf_k f(x_k)$, and usc at x_0 if $x_k \rightarrow x_0$ implies $\limsup_k f(x_k) \leq f(x_0)$.

Proof. (i) If f is lsc and $\lambda < f(x_0)$, then x_0 belongs to the relatively open set $\{f > \lambda\}$, which therefore contains $B_\delta(x_0) \cap A$ for some $\delta > 0$. Conversely, given λ and $x_0 \in \{f > \lambda\}$, the hypothesis produces such a δ , so $\{f > \lambda\}$ is relatively open. Part (ii) follows by replacing f with $-f$.

For the sequential form, suppose f is lsc at x_0 and $x_k \rightarrow x_0$. For any $\lambda < f(x_0)$ the ball $B_\delta(x_0)$ eventually contains every x_k , so $f(x_k) > \lambda$ eventually and $\liminf_k f(x_k) \geq \lambda$; letting $\lambda \uparrow f(x_0)$ gives the claim. The converse is the same argument run backwards. ■

This is Proposition 8.10 of Fukuda (2026d, p. 271). The verbal statement is worth fixing: at a point of lower semicontinuity the function cannot *drop* suddenly as one moves away from x_0 , and at a point of upper semicontinuity it cannot *jump up*. A jump downward is compatible with upper semicontinuity, which is why the maximum of a usc function can sit exactly at the top of a jump.

Corollary 5.3.1. f is continuous if and only if it is both lower and upper semicontinuous.

Proof. If f is continuous, then f^{-1} of the open sets (λ, ∞) and $(-\infty, \lambda)$ are relatively open by Theorem 5.2.1, giving both properties. Conversely, if both hold then $f^{-1}((\lambda, \mu)) = f^{-1}((\lambda, \infty)) \cap f^{-1}((-\infty, \mu))$ is relatively open, and every open subset of $\mathbb{R}$ is a countable union of such intervals, so Theorem 4.2.1(i) and Theorem 5.2.1 apply. ■

Proposition 5.3.2 (Epigraph and hypograph). Let $A \subseteq \mathbb{R}^n$ and $f: A \rightarrow \mathbb{R}$. Then f is lsc if and only if its *epigraph*

$$\text{epi } f := \{(x, z) \in A \times \mathbb{R} : f(x) \leq z\}$$

is closed in $A \times \mathbb{R}$, and f is usc if and only if its *hypograph* $\text{hyp } f := \{(x, z) \in A \times \mathbb{R} : f(x) \geq z\}$ is closed.

Proof. Suppose f is lsc and $(x_k, z_k) \rightarrow (x, z)$ with $f(x_k) \leq z_k$. By Proposition 5.3.1 gives $f(x) \leq \liminf_k f(x_k) \leq \liminf_k z_k = z$, so that $(x, z) \in \text{epi } f$; now apply Theorem 4.3.1. Conversely, if $\text{epi } f$ is closed and $x_k \rightarrow x$, put $\ell := \liminf_k f(x_k)$ and pass to a subsequence along which $f(x_{k_j}) \rightarrow \ell$. Then $(x_{k_j}, f(x_{k_j})) \in \text{epi } f$ converges to (x, ℓ) , so $(x, \ell) \in \text{epi } f$, that is $f(x) \leq \ell$. The hypograph statement follows by applying this to $-f$. ■

This is Proposition 8.12 of Fukuda (2026d, p. 273). The epigraph characterisation is the bridge to Chapter 9: convexity of f is convexity of $\text{epi } f$, and lower semicontinuity is closedness of $\text{epi } f$, so “closed convex function” means precisely that the epigraph is a closed convex set.

Proposition 5.3.3 (Suprema and infima). Let $A \subseteq \mathbb{R}^n$ and let $(f_i)_{i \in I}$ be a family of functions $A \rightarrow \mathbb{R}$ indexed by an arbitrary set I .

- (i) If every f_i is lsc and $f := \sup_{i \in I} f_i$ is finite on A , then f is lsc.
- (ii) If every f_i is usc and $f := \inf_{i \in I} f_i$ is finite on A , then f is usc.

Proof. (i) For any λ , $\{x : \sup_i f_i(x) > \lambda\} = \bigcup_{i \in I} \{x : f_i(x) > \lambda\}$: the supremum exceeds λ exactly when some member does. Each set on the right is relatively open, and Theorem 4.2.1(i) permits arbitrary unions. Part (ii) follows by negation. ■

Example 5.3.1 (A supremum of continuous functions that is not usc). For each $a \in (0, 1]$ define the continuous function $f_a: [0, 1] \rightarrow \mathbb{R}$ by $f_a(x) = x/a$ for $0 \leq x \leq a$ and $f_a(x) = 1$ for $a \leq x \leq 1$. Then

$$f(x) := \sup_{a \in (0, 1]} f_a(x) = \begin{cases} 0, & x = 0, \\ 1, & 0 < x \leq 1. \end{cases}$$

Indeed $f_a(0) = 0$ for every a , while for $x > 0$ the choice $a = x$ gives $f_x(x) = 1$. The limit function is lsc — consistent with [Proposition 5.3.3\(i\)](#) — and it is not usc, since $\{f \geq 1\} = (0, 1]$ is not closed in $[0, 1]$. This is Example 8.4 of Fukuda (2026d, p. 272).

The pattern recurs throughout economics. Indirect utility, the profit function and the value function of a dynamic program are all defined as suprema over a feasible set indexed by a parameter, and the general theory delivers only one of the two semicontinuity properties for free. Establishing the other requires additional structure — compactness of the feasible correspondence and continuity of the objective, which is exactly the content of Berge’s theorem in [Theorem 13.5.1](#).

5.4 The Weierstrass Theorem

Theorem 5.4.1 (Weierstrass). Let $A \subseteq \mathbb{R}^n$ be compact and non-empty.

- (i) A lower semicontinuous $f: A \rightarrow \mathbb{R}$ attains its minimum: there is $x_* \in A$ with $f(x_*) = \inf_{x \in A} f(x)$, and this infimum is finite.
- (ii) An upper semicontinuous $f: A \rightarrow \mathbb{R}$ attains its maximum: there is $x^* \in A$ with $f(x^*) = \sup_{x \in A} f(x)$, and this supremum is finite.

Proof. It suffices to prove (i); (ii) follows by applying (i) to $-f$.

Write $\alpha := \inf_{x \in A} f(x) \in [-\infty, \infty)$ and consider the lower sections $L_\lambda := \{x \in A : f(x) \leq \lambda\}$, each closed relative to A by [Definition 5.3.1](#), hence compact by [Proposition 4.4.2\(i\)](#).

First, $\alpha > -\infty$. If not, $L_\lambda \neq \emptyset$ for every $\lambda \in \mathbb{R}$, and the family $(L_\lambda)_{\lambda \in \mathbb{R}}$ is nested decreasing as λ decreases, so every finite subfamily has non-empty intersection. A nested family of non-empty compact sets has non-empty intersection ([Lemma 5.4.1](#)), so some $x \in \bigcap_{\lambda \in \mathbb{R}} L_\lambda$, giving $f(x) \leq \lambda$ for every real λ , which is impossible for a real-valued f . Hence α is finite.

Second, the infimum is attained. The sets L_λ for $\lambda > \alpha$ are non-empty by definition of the infimum, compact, and nested. By [Lemma 5.4.1](#) again, $\bigcap_{\lambda > \alpha} L_\lambda \neq \emptyset$; any x_* in this intersection satisfies $f(x_*) \leq \lambda$ for every $\lambda > \alpha$, hence $f(x_*) \leq \alpha$, and $f(x_*) \geq \alpha$ because α is a lower bound. So $f(x_*) = \alpha = \min_{x \in A} f(x)$. ■

Lemma 5.4.1 (Finite intersection property). Let $(K_i)_{i \in I}$ be a family of non-empty compact subsets of a metric space such that every finite subfamily has non-empty intersection. Then $\bigcap_{i \in I} K_i \neq \emptyset$.

Proof. Fix $i_0 \in I$ and suppose $\bigcap_i K_i = \emptyset$. Then the relatively open sets $K_i^c \cap K_{i_0}$, $i \in I$, cover K_{i_0} : a point of K_{i_0} lying in no K_i^c would lie in every K_i . By compactness finitely many suffice, say for $i_1, \dots, i_m$, which says $K_{i_0} \cap K_{i_1} \cap \dots \cap K_{i_m} = \emptyset$ — contradicting the hypothesis on finite subfamilies. ■

Corollary 5.4.1 (Weierstrass, continuous case). A continuous real-valued function on a non-empty compact set attains both its minimum and its maximum.

Proof. By [Corollary 5.3.1](#) the function is both lsc and usc; apply both parts of [Theorem 5.4.1](#).

A second proof, independent of semicontinuity, is worth recording because it generalises differently. By [Proposition 5.2.4](#) the image $f(A)$ is a compact subset of $\mathbb{R}$, hence closed and bounded by [Theorem 4.4.2](#). Boundedness gives finite $M := \sup f(A)$ and $m := \inf f(A)$, and by [Proposition 1.5.1\(ii\)](#) both are points of closure of $f(A)$; closedness then gives $M, m \in f(A)$, that is, both are attained. ■

This is Corollary 8.6 of Fukuda (2026d, p. 275), with both proofs as given there.

Example 5.4.1 (Each hypothesis is needed). (i) *Semicontinuity of the wrong type.* Let $f: [0, 1] \rightarrow \mathbb{R}$ be $f(0) = \frac{1}{2}$ and $f(x) = 1 - x$ for $x \in (0, 1]$. This function is lsc, so by [Theorem 5.4.1\(i\)](#) it attains its minimum, at $x = 1$. It has no maximum: $\sup f = 1$, approached as $x \downarrow 0$, but $f(0) = \frac{1}{2}$. Symmetrically, $g(0) = \frac{1}{2}$ and $g(x) = x$ on $(0, 1]$ is usc, attains its maximum at $x = 1$, and has no minimum (Fukuda 2026d, Ex. 8.5, p. 274).

(ii) *Non-compact domain, unbounded.* $f(x) = x$ on $[0, \infty)$ is continuous and attains no maximum.

(iii) *Non-compact domain, not closed.* $f(x) = x$ on $[0, 1)$ is continuous and bounded, and attains no maximum: the supremum 1 is not in the image because the point at which it would be attained is not in the domain.

- (iv) *Infinite dimensions.* On the closed unit ball of the space of [Example 4.3.1](#), the continuous functional $x \mapsto \sum_k 2^{-k} x_k$ has supremum not attained on the set of finitely supported points, illustrating that the closed bounded set is not compact and that [Theorem 5.4.1](#) does not apply.

Proposition 5.4.1 (The solution set is compact). Let $A \subseteq \mathbb{R}^n$ be compact and non-empty.

- (i) If $f: A \rightarrow \mathbb{R}$ is usc, then $\arg \max_{x \in A} f(x)$ is a non-empty compact subset of A .
- (ii) If $f: A \rightarrow \mathbb{R}$ is lsc, then $\arg \min_{x \in A} f(x)$ is a non-empty compact subset of A .

Proof. (i) Non-emptiness is [Theorem 5.4.1\(ii\)](#). Writing $M := \max_{x \in A} f(x)$, the solution set is $\{x \in A : f(x) \geq M\}$, which is closed relative to A by upper semicontinuity, hence compact by [Proposition 4.4.2\(i\)](#). Part (ii) is the same argument for $-f$. ■

This is Proposition 8.13 of Fukuda (2026d, p. 275). It is the first appearance of the object that [Chapter 13](#) studies systematically: the solution of an optimisation problem is in general a *set*, and the statement that it varies continuously with the parameters is a statement about correspondences, not functions. Uniqueness — which reduces the set to a singleton and the correspondence to a function — requires strict quasiconcavity of f and convexity of A , and is supplied in [Proposition 9.3.5](#).

5.5 Economic Application: Existence of Consumer Demand

The four questions raised in [Example 1.6.1](#) can now be answered in part.

Environment. There are n goods. A consumption bundle is a vector $x \in \mathbb{R}_+^n$. Prices are $p \in \mathbb{R}_{++}^n$ and wealth is $m > 0$. The consumer's feasible set is the budget set $\mathcal{B}(p, m) = \{x \in \mathbb{R}_+^n : \langle p, x \rangle \leq m\}$, and preferences are represented by a utility function $u: \mathbb{R}_+^n \rightarrow \mathbb{R}$.

Problem. The consumer solves

$$\max_x u(x) \quad \text{subject to} \quad \langle p, x \rangle \leq m, \quad x \geq 0. \quad (5.1)$$

Proposition 5.5.1 (Existence of a solution to the consumer's problem). Let $p \in \mathbb{R}_{++}^n$, $m > 0$, and let $u: \mathbb{R}_+^n \rightarrow \mathbb{R}$ be upper semicontinuous. Then (5.1) has a solution, the set of solutions

$$x(p, m) := \arg \max_{x \in \mathcal{B}(p, m)} u(x)$$

is non-empty and compact, and the indirect utility $v(p, m) := \max_{x \in \mathcal{B}(p, m)} u(x)$ is finite.

Proof. $\mathcal{B}(p, m)$ is non-empty, since 0 belongs to it, and compact by [Example 4.4.1](#). Apply [Theorem 5.4.1\(ii\)](#) and [Proposition 5.4.1\(i\)](#). ■

What the hypotheses mean economically. Strict positivity of every price is what bounds the budget set: a free good can be consumed without limit, and if utility is strictly increasing in it, no optimum exists. Positive wealth guarantees a non-degenerate problem, though $m = 0$ is admissible and yields $x = 0$. Upper semicontinuity of u is weaker than continuity and permits preferences with an upward jump at a threshold — a consumer who strictly prefers reaching a subsistence level to being marginally below it, with the preferred value attained *at* the threshold. A *downward* jump at the threshold violates upper semicontinuity, and then the supremum of utility over the budget set may fail to be attained even though the budget set is compact, exactly as in [Example 5.4.1\(i\)](#).

What the theorem does not deliver. Three things.

First, uniqueness. [Proposition 5.5.1](#) produces a non-empty compact solution *set*. With linear utility $u(x) = \langle a, x \rangle$ and prices proportional to a , every point of the budget hyperplane is optimal. Uniqueness needs strict quasiconcavity ([Chapter 9](#)).

Second, an interior solution. Nothing in the argument prevents $x(p, m)$ from lying on the boundary of $\mathbb{R}_+^n$, and indeed corner solutions are generic when preferences are not strictly monotone in every good.

First-order conditions therefore take the Kuhn–Tucker form of [Chapter 12](#) rather than the gradient-equals-zero form.

Third, continuity of $x(\cdot, \cdot)$ and of $v(\cdot, \cdot)$ in (p, m) . [Proposition 5.5.1](#) is a statement for each fixed (p, m) separately, and [Example 5.2.2](#) is a reminder that pointwise statements do not aggregate into joint continuity. The budget set itself moves with (p, m) ; establishing that the value moves continuously and that the solution set moves upper hemicontinuously is Berge’s maximum theorem ([Theorem 13.5.1](#)), which requires the budget correspondence to be both upper and lower hemicontinuous. That the budget correspondence is lower hemicontinuous fails at $m = 0$, which is the technical reason the Walrasian demand correspondence is well behaved only for $m > 0$.

Remark 5.5.1 (Why upper semicontinuity is the right hypothesis). Preferences, not utility functions, are the primitive. A preference relation $\succsim$ on $\mathbb{R}^n_+$ has *upper contour set* if $\{y : y \succsim x\}$ is closed for every x ; this is a statement about the preference relation alone. If u represents $\succsim$, then the upper contour set at x is $\{y : u(y) \geq u(x)\}$, so closed upper contour sets are exactly upper semicontinuity of u in the sense of [Definition 5.3.1\(ii\)](#). The hypothesis of [Proposition 5.5.1](#) is therefore a restriction on preferences and is invariant to the choice of representing utility function, whereas an assumption such as differentiability of u is not.

5.6 Chapter Summary

Continuity at a point admits three equivalent formulations: the ε – δ condition, preservation of limits of sequences ([Proposition 5.2.3](#)), and relative openness of preimages of open sets ([Theorem 5.2.1](#)). The third is metric-free and is the definition that transfers to general topological spaces and, with “open” replaced by “measurable”, to random variables. Linear maps and norms are continuous; sums, products, quotients with non-vanishing denominator, compositions and finite maxima of continuous functions are continuous; separate continuity in each argument is strictly weaker than joint continuity ([Example 5.2.2](#)). Continuous images of compact sets are compact and of connected sets are connected, which gives the intermediate value theorem and the boundedness half of the Weierstrass theorem. Continuity on a compact set upgrades to uniform continuity ([Theorem 5.2.2](#)).

Splitting continuity into its two halves isolates what existence of an optimum actually needs: upper semicontinuity for a maximum, lower semicontinuity for a minimum, in both cases on a compact domain ([Theorem 5.4.1](#)). The solution set is then non-empty and compact ([Proposition 5.4.1](#)). An arbitrary supremum of lsc functions is lsc, but an arbitrary supremum of continuous functions need not be usc ([Example 5.3.1](#)), which is why the continuity of value functions in their parameters is a separate theorem rather than a corollary. For the consumer’s problem with strictly positive prices, upper semicontinuity of utility delivers existence of demand and finiteness of indirect utility, and nothing more.

Exercise 5.6.1. Prove [Proposition 5.2.2\(i\)](#) directly from [Definition 5.2.2](#), and identify at which step the hypothesis $g(x_0) \neq 0$ is used in the quotient case. Then show by example that continuity of f and g at x_0 is not enough for f/g to have a *removable* discontinuity at a zero of g .

Exercise 5.6.2. Let $A \subseteq \mathbb{R}^n$ and $f : A \rightarrow \mathbb{R}$. Show that f is lsc if and only if $f = \sup_{i \in I} f_i$ for some family of continuous functions f_i when A is compact and f is bounded below. (Hint: for each x_0 and each $\lambda < f(x_0)$, construct a continuous function below f that takes the value λ at x_0 .) Deduce that [Proposition 5.3.3\(i\)](#) is sharp.

Exercise 5.6.3. A firm with technology $Y \subseteq \mathbb{R}^n$ closed and prices $p \in \mathbb{R}^n$ solves $\pi(p) = \sup_{y \in Y} \langle p, y \rangle$. Show that π is convex and lsc on $\mathbb{R}^n$ as a supremum of linear functions, that $\pi(p) = +\infty$ is possible when Y is unbounded, and that if Y is compact then π is finite, continuous, and the supremum is attained. Relate the failure at unbounded Y to [Example 5.4.1\(ii\)](#).

Chapter 6

Completeness, Contractions, and Fixed Points

6.1 Motivation: Proving a Limit Exists Without Knowing It

The definition of convergence names the limit. To use it one must already have a candidate. In most of the arguments that matter — iterating a Bellman operator, solving a differential equation, constructing an equilibrium price — the object whose existence is at issue is precisely the limit, and there is no candidate to name.

Cauchy's condition removes the limit from the statement: it asks only that the terms of the sequence eventually be close *to one another*. In a complete space that internal condition is enough. Completeness is thus the hypothesis that converts a self-referential existence claim into a verifiable one, and it is the hypothesis on which the contraction mapping theorem, and with it the entire recursive method in macroeconomics, rests.

6.2 Cauchy Sequences

Definition 6.2.1 (Cauchy sequence). A sequence $(x_n)_{n \in \mathbb{N}}$ in a metric space (X, d) is a *Cauchy sequence* if for every $\varepsilon > 0$ there is $n_0 \in \mathbb{N}$ such that $m, n \geq n_0$ implies $d(x_m, x_n) < \varepsilon$.

Proposition 6.2.1 (Convergent implies Cauchy). Every convergent sequence in a metric space is Cauchy.

Proof. Let $x_n \rightarrow x^*$ and $\varepsilon > 0$. Choose n_0 with $d(x^*, x_n) < \varepsilon/2$ for $n \geq n_0$. For $m, n \geq n_0$, $d(x_m, x_n) \leq d(x_m, x^*) + d(x^*, x_n) < \varepsilon$. ■

Proposition 6.2.2 (Cauchy implies bounded). Every Cauchy sequence in a metric space is bounded.

Proof. Take $\varepsilon = 1$ and the corresponding n_0 , so $d(x_{n_0}, x_n) < 1$ for $n \geq n_0$. Put

$$M := \max\{1, d(x_{n_0}, x_1), \dots, d(x_{n_0}, x_{n_0-1})\},$$

a maximum over finitely many numbers. Then $(x_n) \subseteq B_{M+1}(x_{n_0})$. ■

These are Propositions 8.14 and 8.15 of Fukuda (2026d) (Fukuda 2026g, slides 58–59).

Lemma 6.2.1 (Cauchy with a convergent subsequence). If a Cauchy sequence has a subsequence converging to x^* , then the whole sequence converges to x^* .

Proof. Let $\varepsilon > 0$. Choose n_0 with $d(x_m, x_n) < \varepsilon/2$ for $m, n \geq n_0$, and choose an index $n_k \geq n_0$ of the subsequence with $d(x^*, x_{n_k}) < \varepsilon/2$. For $n \geq n_0$, $d(x^*, x_n) \leq d(x^*, x_{n_k}) + d(x_{n_k}, x_n) < \varepsilon$. ■

The converse of Proposition 6.2.1 is false in general: the implication “Cauchy $\Rightarrow$ convergent” is not a theorem about metric spaces but a property that some metric spaces have and others lack. That property is completeness.

6.3 Complete Metric Spaces

Definition 6.3.1 (Completeness). A metric space (X, d) is *complete metric space* if every Cauchy sequence in X converges to a limit belonging to X . A subset $A \subseteq X$ is complete if $(A, d|_{A \times A})$ is complete.

In a complete space the two conditions coincide, and the practical consequence is the one stated on (Fukuda 2026g, slide 60): to prove that (x_n) has a limit it suffices to prove that (x_n) is Cauchy, a condition involving only the terms of the sequence.

Example 6.3.1 (An incomplete space). The space $X = (0, 1]$ with the metric inherited from $\mathbb{R}$ is not complete. The sequence $x_n = 1/n$ is Cauchy — given $\varepsilon > 0$, Proposition 1.5.2 supplies n_0 with $1/n_0 < \varepsilon$, and then $|1/m - 1/n| \leq 1/n_0 < \varepsilon$ for $m, n \geq n_0$ — but its limit 0 does not belong to X (Fukuda 2026g, slide 61).

Nothing about the sequence has changed relative to $\mathbb{R}$; what changed is the ambient set. Completeness is not a property of a sequence but of a space, and it is destroyed by deleting a single point.

Theorem 6.3.1 ($\mathbb{R}^n$ is complete). $\mathbb{R}$ is a complete metric space, and so is $\mathbb{R}^n$ for every finite n .

Proof. Let (x_k) be Cauchy in $\mathbb{R}$. By Proposition 6.2.2 it is bounded, so by Theorem 4.3.2 it has a convergent subsequence, and by Lemma 6.2.1 the whole sequence converges. The limit is a real number, so $\mathbb{R}$ is complete. The case of $\mathbb{R}^n$ is identical, using the $\mathbb{R}^n$ form of Theorem 4.3.2. ■

The chain of implications is worth making explicit, because it locates completeness within the logical structure of the real numbers: the least upper bound property (Assumption 1.5.1) gives monotone convergence (Proposition 4.3.3), which with the monotone subsequence lemma gives Bolzano–Weierstrass, which with Lemma 6.2.1 gives metric completeness. Completeness of $\mathbb{R}$ as a metric space is therefore a theorem, not an axiom, once the order-completeness axiom is in place.

Proposition 6.3.1 (Products). If X and Y are complete metric spaces, then $X \times Y$ with the metric $d((x, y), (x', y')) = (d_X(x, x')^2 + d_Y(y, y')^2)^{1/2}$ is complete.

Proof. If (x_k, y_k) is Cauchy in the product, then $d_X(x_j, x_k) \leq d((x_j, y_j), (x_k, y_k))$ shows (x_k) is Cauchy in X , and symmetrically for (y_k) . Their limits x^*, y^* exist by completeness, and $d((x_k, y_k), (x^*, y^*)) \rightarrow 0$. ■

Proposition 6.3.2 (Complete subsets are exactly the closed ones). Let X be a complete metric space and $A \subseteq X$. Then A is complete if and only if A is closed in X .

Proof. ($\Leftarrow$) Let A be closed and $(x_n) \subseteq A$ Cauchy. By completeness of X it converges to some $x^* \in X$, and by Theorem 4.3.1 $x^* \in A$. So the sequence converges in A .

($\Rightarrow$) Let A be complete and $x^* \in \bar{A}$. By the construction in the proof of Theorem 4.3.1 there is $(x_n) \subseteq A$ with $x_n \rightarrow x^*$; this sequence is Cauchy by Proposition 6.2.1, hence converges to a point of A by completeness of A , and by uniqueness of limits (Proposition 4.3.1(i)) that point is x^* . So $\bar{A} \subseteq A$. ■

This is Proposition 8.19 of Fukuda (2026d) (Fukuda 2026g, slide 62). Note the asymmetry of the hypotheses: completeness of the ambient space is used only in one direction, and the equivalence is a statement about subsets of a complete space, not about metric spaces in general.

Corollary 6.3.1. Every vector subspace of $\mathbb{R}^n$ is complete, and hence closed.

Proof. Let $V \subseteq \mathbb{R}^n$ be a subspace with basis $v_1, \dots, v_k$ and let (x_j) be Cauchy in V . Completing to a basis of $\mathbb{R}^n$ and using Theorem 6.3.1, $x_j \rightarrow x^* \in \mathbb{R}^n$. Writing $x_j = \sum_{i=1}^k c_j^i v_i$, the coordinate functionals are linear, hence continuous by Proposition 5.2.1(ii), so the coefficient vectors (c_j) converge in $\mathbb{R}^k$ to some c^* and $x^* = \sum_i c^{i*} v_i \in V$. Closedness follows from Proposition 6.3.2. ■

This is Proposition 8.20 of Fukuda (2026d, p. 279). In infinite dimensions the conclusion fails — the subspace of finitely supported sequences is not closed in ℓ^2 — and this is one of the practical differences that Chapter 11 must accommodate: the projection theorem is stated for closed subspaces because in infinite dimensions “subspace” does not imply “closed”.

6.3.1 Function spaces

The space in which dynamic programming operates is not $\mathbb{R}^n$. It is a space of functions, and its completeness is what the whole method depends on.

Theorem 6.3.2 (The bounded functions are a Banach space). Let S be any non-empty set and let $\mathcal{B}(S, \mathbb{R})$ be the set of bounded functions $f: S \rightarrow \mathbb{R}$, equipped with the supremum norm $\|f\|_\infty = \sup_{s \in S} |f(s)|$. Then $\mathcal{B}(S, \mathbb{R})$ is a Banach space.

Proof. That $\|\cdot\|_\infty$ is a norm is Proposition 3.3.3. For completeness, let (f_k) be Cauchy in $\mathcal{B}(S, \mathbb{R})$.

Step 1: pointwise limit. For each fixed $s \in S$, $|f_j(s) - f_k(s)| \leq \|f_j - f_k\|_\infty$, so $(f_k(s))_k$ is Cauchy in $\mathbb{R}$ and converges by Theorem 6.3.1. Define $f(s) := \lim_k f_k(s)$.

Step 2: the convergence is uniform. Given $\varepsilon > 0$, take k_0 with $\|f_j - f_k\|_\infty < \varepsilon/2$ for $j, k \geq k_0$. Fix s and $k \geq k_0$, and let $j \rightarrow \infty$ in $|f_j(s) - f_k(s)| < \varepsilon/2$; the absolute value is continuous, so $|f(s) - f_k(s)| \leq \varepsilon/2$. The bound is uniform in s , so $\|f - f_k\|_\infty \leq \varepsilon/2 < \varepsilon$ for every $k \geq k_0$.

Step 3: the limit is bounded. With $\varepsilon = 1$ and the corresponding k_0 , $\|f\|_\infty \leq \|f - f_{k_0}\|_\infty + \|f_{k_0}\|_\infty \leq 1 + \|f_{k_0}\|_\infty < \infty$. Hence $f \in \mathcal{B}(S, \mathbb{R})$ and $f_k \rightarrow f$ in the supremum norm. ■

Remark 6.3.1 (Supremum-norm convergence is uniform convergence). Step 2 is the content of the identification: $\|f - f_k\|_\infty \rightarrow 0$ says exactly that $f_k \rightarrow f$ uniformly on S , that is, with a single k_0 serving every s simultaneously. Pointwise convergence alone is a strictly weaker statement and does not preserve continuity: $f_k(s) = s^k$ on $[0, 1]$ converges pointwise to the discontinuous limit that is 0 on $[0, 1)$ and 1 at 1, and $\|f - f_k\|_\infty = 1$ for every k . The supremum norm is chosen for dynamic programming precisely because convergence in it is strong enough to carry continuity, monotonicity and concavity to the limit, as Corollary 6.4.2 exploits.

Proposition 6.3.3 (Bounded continuous functions). Let $S \subseteq \mathbb{R}^n$. The set $\mathcal{C}(S, \mathbb{R}) \cap \mathcal{B}(S, \mathbb{R})$ of bounded continuous functions is a closed subset of $\mathcal{B}(S, \mathbb{R})$, hence a Banach space. If S is compact, every continuous function on S is bounded by Corollary 5.4.1, so $\mathcal{C}(S, \mathbb{R})$ itself is a Banach space in the supremum norm.

Proof. By Proposition 6.3.2 it suffices to show closedness. Let $f_k \rightarrow f$ uniformly with each f_k continuous, fix $s_0 \in S$ and $\varepsilon > 0$. Choose k with $\|f - f_k\|_\infty < \varepsilon/3$ and then $\delta > 0$ with $|f_k(s) - f_k(s_0)| < \varepsilon/3$ for $s \in B_\delta(s_0) \cap S$. For such s ,

$$|f(s) - f(s_0)| \leq |f(s) - f_k(s)| + |f_k(s) - f_k(s_0)| + |f_k(s_0) - f(s_0)| < \varepsilon.$$

The three-term split is the standard device for transferring a property from the approximants to the limit, and each of the three terms is controlled by a different hypothesis: the outer two by uniformity of the convergence, the middle one by continuity of f_k . ■

6.4 The Contraction Mapping Theorem

Definition 6.4.1 (Contraction). Let (X, d) be a metric space. A map $T: X \rightarrow X$ is a *contraction mapping* with *modulus* β if there is $\beta \in [0, 1)$ with

$$d(Tx, Ty) \leq \beta d(x, y) \quad \text{for all } x, y \in X.$$

A point $x^* \in X$ with $Tx^* = x^*$ is a *fixed point* of T .

Two features of the definition deserve emphasis. The modulus must be strictly less than 1 and must be uniform over all pairs x, y . The weaker condition $d(Tx, Ty) < d(x, y)$ for $x \neq y$ — a *contractive* map — does not suffice: on $X = [1, \infty)$ the map $Tx = x + 1/x$ satisfies it, is a self-map of a complete space, and has no fixed point, because the implicit modulus approaches 1 as $x \rightarrow \infty$.

Theorem 6.4.1 (Banach contraction mapping theorem). Let (X, d) be a non-empty complete metric space and let $T: X \rightarrow X$ be a contraction with modulus $\beta \in [0, 1)$. Then T has exactly one fixed point $x^* \in X$. Moreover, for any $x_0 \in X$ the sequence of iterates $x_n := Tx_{n-1} = T^n x_0$ converges to x^* , and

$$d(x_n, x^*) \leq \frac{\beta^n}{1 - \beta} d(x_0, Tx_0) \quad \text{for every } n \in \mathbb{N}_0. \quad (6.1)$$

Proof. Step 1: the iterates are Cauchy. Fix $x_0 \in X$ and define $x_n = Tx_{n-1}$. Applying the contraction property k times, $d(x_k, x_{k+1}) = d(Tx_{k-1}, Tx_k) \leq \beta d(x_{k-1}, x_k) \leq \dots \leq \beta^k d(x_0, x_1)$. For $m > n$, the triangle inequality applied $m - n - 1$ times gives

$$d(x_n, x_m) \leq \sum_{k=n}^{m-1} d(x_k, x_{k+1}) \leq \sum_{k=n}^{m-1} \beta^k d(x_0, x_1) = \beta^n \frac{1 - \beta^{m-n}}{1 - \beta} d(x_0, x_1) \leq \frac{\beta^n}{1 - \beta} d(x_0, x_1),$$

where the geometric sum is finite because $\beta < 1$. Given $\varepsilon > 0$, since $\beta^n \rightarrow 0$ there is n_0 with $\beta^n d(x_0, x_1)/(1 - \beta) < \varepsilon$ for $n \geq n_0$, and then $d(x_n, x_m) < \varepsilon$ for all $m, n \geq n_0$. So (x_n) is Cauchy.

Step 2: the limit exists. X is complete, so $x^* := \lim_n x_n$ exists in X .

Step 3: the limit is a fixed point. T is continuous: given $\varepsilon > 0$, take $\delta = \varepsilon$ (if $\beta = 0$, any δ); then $d(x, y) < \delta$ implies $d(Tx, Ty) \leq \beta d(x, y) < \varepsilon$. Hence, by Proposition 5.2.3,

$$Tx^* = T\left(\lim_{n \rightarrow \infty} x_n\right) = \lim_{n \rightarrow \infty} Tx_n = \lim_{n \rightarrow \infty} x_{n+1} = x^*.$$

Step 4: uniqueness. If $Tx^* = x^*$ and $Ty^* = y^*$, then $d(x^*, y^*) = d(Tx^*, Ty^*) \leq \beta d(x^*, y^*)$. If $d(x^*, y^*) > 0$ this yields $1 \leq \beta$, contradicting $\beta < 1$. Hence $d(x^*, y^*) = 0$ and $x^* = y^*$.

Step 5: the error bound. Let $m \rightarrow \infty$ in the Step 1 estimate $d(x_n, x_m) \leq \beta^n (1 - \beta)^{-1} d(x_0, x_1)$. The metric is continuous in each argument — an immediate consequence of the triangle inequality — so the left-hand side tends to $d(x_n, x^*)$, giving (6.1) with $x_1 = Tx_0$. ■

This is Theorem 8.3 of Fukuda (2026d, p. 284), proved over (Fukuda 2026g, slides 64–72); the proof above follows those steps, with the geometric sum and the passage to the limit in Step 5 written out.

Remark 6.4.1 (What each hypothesis buys). Completeness is used once, in Step 2, and only to produce the limit; it is responsible for *existence*. The strict inequality $\beta < 1$ is used twice: in Step 1 to sum the geometric series, and in Step 4, where it is responsible for *uniqueness*. The self-map property $T(X) \subseteq X$ is used implicitly throughout, in defining the iterates. Dropping any one of the three destroys a different part of the conclusion: on the complete space $\mathbb{R}$ the isometry $Tx = x + 1$ ($\beta = 1$) has no fixed point; on the incomplete space $(0, 1]$ the contraction $Tx = x/2$ has none either, though its iterates are Cauchy.

Remark 6.4.2 (The error bound is what makes the theorem computational). Inequality (6.1) is an *a priori* bound: it is computable from the first iteration alone, before the iteration is run. It converts the theorem from an existence statement into an algorithm with a stopping rule, since $n \geq \log(\varepsilon(1 - \beta)/d(x_0, Tx_0))/\log \beta$ guarantees accuracy ε . Convergence is geometric at rate β , so the discount factor of a dynamic program is simultaneously an economic parameter and the convergence rate of the algorithm that solves it: a model with $\beta = 0.99$ needs roughly $\log(0.5)/\log(0.99) \approx 69$ times as many iterations per digit as one with $\beta = 0.5$.

6.4.1 Three refinements

Corollary 6.4.1 (Iterates of a contraction). Let X be a non-empty complete metric space and $T: X \rightarrow X$ a map such that T^N is a contraction for some $N \in \mathbb{N}$. Then T has a unique fixed point.

Proof. By Theorem 6.4.1, T^N has a unique fixed point x^* . Then $T^N(Tx^*) = T(T^N x^*) = Tx^*$, so Tx^* is also a fixed point of T^N ; by uniqueness $Tx^* = x^*$. Conversely any fixed point of T is a fixed point of T^N , hence equals x^* . ■

This is Exercise 8.30 of Fukuda (2026d, p. 290). It matters because operators that are not contractions can have contracting powers: in a dynamic program with periodic structure, or in a Markov chain whose

transition matrix has zero entries, the one-step operator may fail the modulus condition while the N -step operator satisfies it.

Corollary 6.4.2 (Properties inherited by the fixed point). Let X be a complete metric space, $T: X \rightarrow X$ a contraction with fixed point x^* , and let $Y \subseteq X$ be non-empty and closed with $T(Y) \subseteq Y$. Then $x^* \in T(Y) \subseteq Y$.

Proof. Y is closed in a complete space, hence complete by [Proposition 6.3.2](#), and T restricted to Y is a contraction of Y into itself. By [Theorem 6.4.1](#) applied to $T|_Y$ there is a fixed point in Y ; by uniqueness in X it equals x^* . Finally $x^* = Tx^* \in T(Y)$. ■

Remark 6.4.3 (How to prove that a value function is concave). [Corollary 6.4.2](#) is the standard route to qualitative properties of an object defined only as a fixed point. Let P be a property, and let $Y = \{x \in X : x \text{ has } P\}$. If Y is closed and T preserves P , in the sense that $T(Y) \subseteq Y$, then the fixed point has P . In [Chapter 15](#) this is applied three times over, with $X = \mathcal{C}(S, \mathbb{R})$ and P successively “increasing”, “concave” and “strictly concave”. The closedness requirement is exactly why the supremum norm is the right one: the pointwise limit of increasing functions is increasing and of concave functions is concave, and uniform convergence implies pointwise convergence, so both sets are closed in $\mathcal{B}(S, \mathbb{R})$. Strict concavity is *not* a closed property — $f_k(s) = -s^2/k$ is strictly concave with uniform limit 0 — and must be obtained by a separate argument ([Fukuda 2026d](#), Remark 8.11, p. 290).

Proposition 6.4.1 (Comparison of fixed points). Let X be a complete metric space and let $S, T: X \rightarrow X$ be contractions with a common modulus $\beta \in [0, 1)$ and fixed points x_S, x_T . If $d(Sx, Tx) \leq a$ for all $x \in X$, then

$$d(x_S, x_T) \leq \frac{a}{1 - \beta}.$$

Proof. $d(x_S, x_T) = d(Sx_S, Tx_T) \leq d(Sx_S, Tx_S) + d(Tx_S, Tx_T) \leq a + \beta d(x_S, x_T)$. Rearranging and dividing by $1 - \beta > 0$ gives the bound. ■

This is Exercise 8.29 of [Fukuda \(2026d, p. 290\)](#). It is the stability result behind numerical dynamic programming: if the true Bellman operator and its discretised approximation differ by at most a uniformly, the computed value function differs from the true one by at most $a/(1 - \beta)$. The factor $(1 - \beta)^{-1}$ is an amplification, and it is large exactly when the discount factor is close to 1.

6.5 Blackwell's Sufficient Conditions

Verifying the contraction inequality directly for a Bellman operator requires comparing two suprema, which is awkward. Blackwell's conditions replace that comparison with two structural properties that can be read off the problem.

For $f, g \in \mathcal{B}(S, \mathbb{R})$ write $f \leq g$ to mean $f(s) \leq g(s)$ for every $s \in S$, and for $\alpha \in \mathbb{R}$ let $f + \alpha$ denote the function $s \mapsto f(s) + \alpha$.

Theorem 6.5.1 (Blackwell). Let S be a non-empty set and let $T: \mathcal{B}(S, \mathbb{R}) \rightarrow \mathcal{B}(S, \mathbb{R})$ satisfy

- (i) *monotonicity (Blackwell)*: $f \leq g$ implies $Tf \leq Tg$; and
- (ii) *discounting (Blackwell)*: there is $\beta \in [0, 1)$ such that $T(f + \alpha)(s) \leq (Tf)(s) + \beta\alpha$ for every $f \in \mathcal{B}(S, \mathbb{R})$, every $\alpha \geq 0$ and every $s \in S$.

Then T is a contraction with modulus β on $(\mathcal{B}(S, \mathbb{R}), \|\cdot\|_\infty)$, and therefore has a unique fixed point, reached from any starting point by iteration.

Proof. Let $f, g \in \mathcal{B}(S, \mathbb{R})$. For every s , $g(s) - f(s) \leq |g(s) - f(s)| \leq \|f - g\|_\infty$, so

$$g \leq f + \|f - g\|_\infty.$$

Applying monotonicity and then discounting with $\alpha = \|f - g\|_\infty \geq 0$,

$$Tg \leq T(f + \|f - g\|_\infty) \leq Tf + \beta \|f - g\|_\infty.$$

Interchanging the roles of f and g gives $Tf \leq Tg + \beta \|f - g\|_\infty$. The two inequalities together say that $|(Tf)(s) - (Tg)(s)| \leq \beta \|f - g\|_\infty$ for every $s \in S$, and taking the supremum over s on the left yields $\|Tf - Tg\|_\infty \leq \beta \|f - g\|_\infty$. Since $\mathcal{B}(S, \mathbb{R})$ is complete by [Theorem 6.3.2](#), [Theorem 6.4.1](#) applies. ■

This is Proposition 8.20 of the slides and Proposition 8.21 of Fukuda ([2026d](#), p. 289); the proof is the one sketched on (Fukuda [2026g](#), slide 76).

Remark 6.5.1 (The conditions are sufficient, not necessary). A contraction on $\mathcal{B}(S, \mathbb{R})$ need not be monotone: on $\mathcal{B}(S, \mathbb{R})$ with S a single point, $Tf = -\frac{1}{2}f$ is a contraction with modulus $\frac{1}{2}$ and reverses order. What Blackwell's conditions provide is a criterion adapted to operators that arise from maximisation, for which both properties are transparent. Monotonicity of a Bellman operator says that a pointwise higher continuation value cannot lower today's optimised value, which holds because the same feasible choices remain available. Discounting says that adding a constant α to every continuation value raises today's value by at most $\beta\alpha$, which holds because the constant enters only through the discounted continuation term. Neither statement requires knowing the maximiser.

Example 6.5.1 (Blackwell in $\mathbb{R}^n$). Equip $\mathbb{R}^n$ with $\|x\|_\infty = \max_i |x_i|$ and let $T: \mathbb{R}^n \rightarrow \mathbb{R}^n$ satisfy monotonicity ($x \leq y$ componentwise implies $Tx \leq Ty$) and discounting ($T(x + \alpha \mathbf{1}) \leq Tx + \beta\alpha \mathbf{1}$ for $\alpha \geq 0$, where $\mathbf{1} = (1, \dots, 1)^\top$). The proof of [Theorem 6.5.1](#) applies verbatim with $S = \{1, \dots, n\}$, since $\mathcal{B}(\{1, \dots, n\}, \mathbb{R}) = \mathbb{R}^n$ with the supremum norm. This is Exercise 8.26 of Fukuda ([2026d](#), p. 290), and it is the finite-state case: a Markov decision problem with n states, where the Bellman operator is a map of $\mathbb{R}^n$ into itself and value iteration is a matrix–vector recursion.

6.6 Economic Applications

6.6.1 The Bellman operator

Environment. A planner observes a state $s \in S$, chooses an action a from a feasible set $\Gamma(s) \subseteq A$, receives a period return $r(s, a)$, and moves to the state $g(s, a)$ next period. Future returns are discounted at $\beta \in (0, 1)$. The problem is

$$\sup_{(a_t)_{t \geq 0}} \sum_{t=0}^{\infty} \beta^t r(s_t, a_t) \quad \text{subject to} \quad a_t \in \Gamma(s_t), \quad s_{t+1} = g(s_t, a_t), \quad s_0 \text{ given.} \quad (6.2)$$

The operator. Define T on $\mathcal{B}(S, \mathbb{R})$ by

$$(Tv)(s) := \sup_{a \in \Gamma(s)} \{r(s, a) + \beta v(g(s, a))\}. \quad (6.3)$$

Proposition 6.6.1 (The Bellman operator is a contraction). Suppose r is bounded, $\Gamma(s) \neq \emptyset$ for every s , and $\beta \in [0, 1)$. Then T maps $\mathcal{B}(S, \mathbb{R})$ into itself and is a contraction with modulus β ; consequently (6.3) has a unique bounded solution v^* , and $T^n v_0 \rightarrow v^*$ uniformly from any bounded v_0 .

Proof. Self-map. If $|r| \leq M_r$ and $\|v\|_\infty \leq M_v$, then the expression in braces lies in $[-M_r - \beta M_v, M_r + \beta M_v]$ for every admissible a , so its supremum over the non-empty set $\Gamma(s)$ is finite and bounded by $M_r + \beta M_v$ uniformly in s . Hence $Tv \in \mathcal{B}(S, \mathbb{R})$.

Monotonicity. If $v \leq w$ then for every s and every $a \in \Gamma(s)$, $r(s, a) + \beta v(g(s, a)) \leq r(s, a) + \beta w(g(s, a))$; taking suprema over the same set $\Gamma(s)$ preserves the inequality, so $Tv \leq Tw$.

Discounting. For $\alpha \geq 0$,

$$(T(v + \alpha))(s) = \sup_{a \in \Gamma(s)} \{r(s, a) + \beta v(g(s, a)) + \beta\alpha\} = (Tv)(s) + \beta\alpha,$$

since an additive constant passes through a supremum. The required inequality holds with equality.

Apply [Theorem 6.5.1](#). ■

What this establishes and what it does not. It establishes that the functional equation has exactly one bounded solution and that value iteration converges to it geometrically, with the error after n steps bounded by (6.1). It does not by itself establish that v^* is the value of the sequence problem (6.2) — that equivalence requires a separate argument, given in Chapter 15 — nor that an optimal policy exists, which requires the supremum in (6.3) to be attained and hence the Weierstrass theorem (Theorem 5.4.1) applied to $\Gamma(s)$. Boundedness of r is restrictive: it excludes logarithmic utility on an unbounded state space, for which a weighted supremum norm is used instead (Stokey et al. 1989, Sec. 4.3).

6.6.2 Asset pricing

In the Lucas tree economy the price of a claim to a dividend stream satisfies

$$p(s) = \beta \mathbb{E}[m(s, s')\{d(s') + p(s')\} \mid s], \quad (6.4)$$

where s is the current state, d the dividend and m a positive stochastic discount factor. Define $(Tp)(s)$ as the right-hand side. If $\beta \sup_s \mathbb{E}[m(s, s') \mid s] := \bar{\beta} < 1$ and d is bounded, then for bounded p, q ,

$$|(Tp)(s) - (Tq)(s)| = \beta |\mathbb{E}[m(s, s')\{p(s') - q(s')\} \mid s]| \leq \bar{\beta} \|p - q\|_\infty,$$

using positivity of m and the conditional Jensen inequality of Proposition 14.4.2. So T is a contraction on $\mathcal{B}(S, \mathbb{R})$ and (6.4) has a unique bounded price function. Uniqueness here is an economic statement: it rules out rational bubbles among bounded price functions. Bubbles reappear precisely when the boundedness restriction is dropped, since the homogeneous equation $b(s) = \beta \mathbb{E}[mb(s') \mid s]$ then admits non-zero solutions (Lucas 1978).

6.6.3 The steady-state Kalman filter

Environment. A hidden state follows $\theta_{t+1} = \rho \theta_t + \eta_{t+1}$ with $\rho \in (-1, 1)$ and $\eta_{t+1} \sim N(0, \tau_2)$, and is observed each period through a signal $y_t = \theta_t + \varepsilon_t$ with $\varepsilon_t \sim N(0, \tau_1^{-1})$, all shocks independent. Following the source exercise, τ_2 is the *variance* of the state innovation and τ_1 the *precision* of the signal. An observer forms Bayesian beliefs about θ_t from the signal history; let x_t denote the variance of that belief *before* seeing y_t .

Recursion. Normal-normal updating (Theorem 14.6.1) adds precisions: after observing y_t the posterior variance is $(x_t^{-1} + \tau_1)^{-1}$. The transition then scales the variance by ρ^2 and adds the innovation variance τ_2 , so $x_{t+1} = T(x_t)$ with

$$T(x) = \rho^2(x^{-1} + \tau_1)^{-1} + \tau_2 = \frac{\rho^2 x}{1 + \tau_1 x} + \tau_2, \quad x \in \mathbb{R}_{++}. \quad (6.5)$$

This is the scalar Riccati equation of the Kalman filter.

Proposition 6.6.2 (The Riccati operator is a contraction). Let $\rho \in (-1, 1)$ and $\tau_1, \tau_2 > 0$. Then T in (6.5) maps $\mathbb{R}_{++}$ into itself and is a contraction with modulus $\rho^2 \in [0, 1)$. Consequently the conditional variance converges, from any initial belief, to the unique fixed point

$$x^* = \frac{(\rho^2 + \tau_1 \tau_2 - 1) + \sqrt{(1 - \rho^2 - \tau_1 \tau_2)^2 + 4\tau_1 \tau_2}}{2\tau_1}, \quad (6.6)$$

the unique positive root of $\tau_1 x^2 + (1 - \rho^2 - \tau_1 \tau_2)x - \tau_2 = 0$.

Proof. Self-map. For $x > 0$ both terms of $\rho^2 x / (1 + \tau_1 x) + \tau_2$ are non-negative and the second is strictly positive, so $T(x) \geq \tau_2 > 0$.

Contraction. For $x, y > 0$,

$$T(x) - T(y) = \rho^2 \left(\frac{x}{1 + \tau_1 x} - \frac{y}{1 + \tau_1 y} \right) = \frac{\rho^2(x - y)}{(1 + \tau_1 x)(1 + \tau_1 y)},$$

after clearing the common denominator and cancelling the terms in $\tau_1 xy$. Since $\tau_1 > 0$ and $x, y > 0$, the denominator exceeds 1, so $|T(x) - T(y)| \leq \rho^2 |x - y|$ with $\rho^2 < 1$. The set $[\tau_2, \rho^2 \tau_1^{-1} + \tau_2]$ is closed,

non-empty, and invariant under T , hence complete by [Proposition 6.3.2](#); [Theorem 6.4.1](#) applies on it, and [Corollary 6.4.2](#) places the fixed point there.

Fixed point. Setting $x = \rho^2 x / (1 + \tau_1 x) + \tau_2$ and multiplying by $1 + \tau_1 x > 0$ gives $x + \tau_1 x^2 = \rho^2 x + \tau_2 + \tau_1 \tau_2 x$, that is $\tau_1 x^2 + (1 - \rho^2 - \tau_1 \tau_2)x - \tau_2 = 0$. The product of the two roots is $-\tau_2/\tau_1 < 0$, so exactly one root is positive, and it is [\(6.6\)](#). As a check, $\rho = 0$ gives $\tau_1 x^2 + (1 - \tau_1 \tau_2)x - \tau_2 = (\tau_1 x + 1)(x - \tau_2) = 0$, hence $x^* = \tau_2$: with no persistence, yesterday's signal is worthless and the conditional variance is the innovation variance. ■

Economic content. Three readings. First, uncertainty neither vanishes nor explodes. However long the observer has been learning, the conditional variance settles at $x^* > 0$: new innovations arrive at exactly the rate at which signals resolve old ones. Permanent learning does not produce perfect information, which is why models of learning about a persistent fundamental sustain non-degenerate belief dispersion indefinitely.

Second, the rate of convergence is ρ^2 , the square of the persistence. Beliefs about a near-random-walk state ($\rho \rightarrow 1$) converge to their steady state arbitrarily slowly, so the transitional dynamics of learning are economically relevant over long horizons precisely when the fundamental is persistent.

Third, comparative statics are read off [\(6.6\)](#) through the quadratic. Write $F(x) := \tau_1 x^2 + (1 - \rho^2 - \tau_1 \tau_2)x - \tau_2$, so that $F(x^*) = 0$ and $F'(x^*) > 0$ at the larger root. Since $\partial F / \partial \rho^2 = -x^* < 0$, the implicit function theorem ([Theorem 8.6.1](#)) gives $\partial x^* / \partial \rho^2 = x^* / F'(x^*) > 0$: a more persistent state is harder to track, not easier. Likewise $\partial F / \partial \tau_1 = x^*(x^* - \tau_2)$, which is strictly positive because $x^* = \rho^2 x^* / (1 + \tau_1 x^*) + \tau_2 > \tau_2$; hence $\partial x^* / \partial \tau_1 < 0$ and a more precise signal lowers steady-state uncertainty, without qualification. This is Exercise 8.27 of Fukuda ([2026d](#), p. 290), with the interpretation supplied.

6.6.4 Input–output systems revisited

[Example 2.6.1](#) solved $x = Ax + d$ by a Neumann series under $\rho(A) < 1$. [Theorem 6.4.1](#) gives the same conclusion with a different argument and a computable error bound: if $\|A\| < 1$ in some operator norm, then $Tx = Ax + d$ satisfies $\|Tx - Ty\| = \|A(x - y)\| \leq \|A\| \|x - y\|$, so T is a contraction on the complete space $\mathbb{R}^n$ and the iteration $x_{k+1} = Ax_k + d$ converges to the unique solution from any starting point, with $\|x_k - x^*\| \leq \|A\|^k (1 - \|A\|)^{-1} \|x_1 - x_0\|$. The two hypotheses are not identical: $\rho(A) \leq \|A\|$ always, so the spectral condition is weaker, and by [Corollary 6.4.1](#) the gap is exactly accounted for by passing to a power of T , since $\rho(A) < 1$ implies $\|A^N\| < 1$ for some N .

6.7 Chapter Summary

A Cauchy sequence is one whose terms cluster without reference to any limit; every convergent sequence is Cauchy and every Cauchy sequence is bounded, but the converse of the first implication holds only in complete spaces. $\mathbb{R}^n$ is complete, as a consequence of the least upper bound axiom via Bolzano–Weierstrass; a subset of a complete space is complete exactly when it is closed ([Proposition 6.3.2](#)); the bounded functions on any set form a Banach space under the supremum norm ([Theorem 6.3.2](#)), and the bounded continuous functions form a closed subspace of it.

A contraction on a non-empty complete metric space has exactly one fixed point, reached by iteration from any starting point at geometric rate β with the computable error bound [\(6.1\)](#). Completeness delivers existence and $\beta < 1$ delivers uniqueness; each is indispensable. Three refinements extend the reach of the theorem: a map with a contracting power has a unique fixed point; a closed invariant set contains the fixed point, which is how qualitative properties of value functions are established; and fixed points of uniformly close contractions are within $a/(1 - \beta)$ of each other, which bounds the error of numerical approximation. Blackwell's monotonicity and discounting conditions are sufficient for the contraction property and are verifiable directly on operators defined by maximisation, which is what makes them the standard tool for Bellman operators, asset-pricing operators and Bayesian updating recursions.

Exercise 6.7.1. Let X be a complete normed space and $T: X \rightarrow X$ a contraction with modulus β . Show that for every $y \in X$ there is a unique $x \in X$ with $Tx - x = y$, and that the solution map $y \mapsto x$ is Lipschitz with constant $(1 - \beta)^{-1}$. Interpret the second statement as a stability property of the linear system $x = Ax + d$ with respect to perturbations of d (Fukuda [2026d](#), Ex. 8.31).

Exercise 6.7.2. Show that $Tx = x + 1/x$ on $[1, \infty)$ satisfies $|Tx - Ty| < |x - y|$ for all $x \neq y$ and has no fixed point, and that $Tx = \sqrt{1 + x^2}$ on $\mathbb{R}$ does the same. Identify the supremum of $|Tx - Ty| / |x - y|$ in each case and explain why [Theorem 6.4.1](#) does not apply.

Exercise 6.7.3. Let $S = \{1, \dots, n\}$ and let Q be an $n \times n$ row-stochastic matrix. Show that $Tv = r + \beta Qv$ is a contraction on $(\mathbb{R}^n, \|\cdot\|_\infty)$ with modulus β , that its fixed point is $v^* = (I - \beta Q)^{-1}r$, and that $I - \beta Q$ is invertible for every $\beta \in [0, 1)$. Verify Blackwell's two conditions for T directly, and explain which property of Q delivers each.

Chapter 7

Differential Calculus in One Variable

7.1 Motivation: The Two Firm Problems

Two problems organise this chapter (Fukuda 2026a, slide 3). A price-taking firm solves

$$\max_{q \in [0, \infty)} pq - c(q), \quad (7.1)$$

and a price-setting firm facing inverse demand P solves

$$\max_{q \in [0, \infty)} P(q)q - c(q). \quad (7.2)$$

Assumption 7.1.1 (Standing assumptions on the firm). The cost function $c: [0, \infty) \rightarrow \mathbb{R}$ is differentiable, strictly increasing and strictly convex, with $c(0) = c'(0) = 0$ and $\lim_{q \rightarrow \infty} c(q) = \lim_{q \rightarrow \infty} c'(q) = \infty$. The revenue function $q \mapsto P(q)q$ is concave and differentiable; the leading case is $P(q) = a - bq$ with $a, b > 0$.

Each assumption is used at an identifiable point. Differentiability makes the first-order condition available; strict convexity of c and concavity of revenue make the first-order condition sufficient (Theorem 7.2.3) and the solution unique; $c'(0) = 0$ with $c'(q) \rightarrow \infty$ places the optimum in the interior of $[0, \infty)$, where Theorem 7.2.1 applies; and $c(0) = 0$ makes the value of the problem the profit rather than profit net of a fixed cost.

7.2 The Derivative

Definition 7.2.1 (Derivative). Let $f: (a, b) \rightarrow \mathbb{R}$ with $-\infty \leq a < b \leq \infty$ and let $x_0 \in (a, b)$. The function f is *differentiability* x_0 if

$$f'(x_0) := \lim_{h \rightarrow 0} \frac{f(x_0 + h) - f(x_0)}{h}$$

exists and is finite; the limit is the *derivative* of f at x_0 , also written $Df(x_0)$ or $\frac{df}{dx}(x_0)$. It is the slope of the tangent line to the graph of f at $(x_0, f(x_0))$.

This is Definition 3.1 of Fukuda (2026d) (Fukuda 2026a, slide 4). That x_0 is required to be *interior* to the domain is part of the definition: the two-sided limit must exist, so the difference quotient must be defined for h of both signs.

Proposition 7.2.1 (Equivalent formulation and continuity). Let $f: (a, b) \rightarrow \mathbb{R}$ and $x_0 \in (a, b)$. Then f is differentiable at x_0 with derivative L if and only if

$$f(x_0 + h) = f(x_0) + Lh + r(h), \quad \text{where} \quad \lim_{h \rightarrow 0} \frac{r(h)}{h} = 0. \quad (7.3)$$

In particular, differentiability at x_0 implies continuity at x_0 .

Proof. Define $r(h) := f(x_0 + h) - f(x_0) - Lh$. Then $r(h)/h$ equals the difference quotient minus L , so $r(h)/h \rightarrow 0$ if and only if the difference quotient tends to L . For continuity, (7.3) gives $f(x_0 + h) - f(x_0) = Lh + r(h) \rightarrow 0$ as $h \rightarrow 0$, since $r(h) = h \cdot (r(h)/h) \rightarrow 0 \cdot 0$. ■

Formulation (7.3) is the one that generalises: it says that f is well approximated near x_0 by an affine function, with an error small relative to h . In one variable the approximating linear map is multiplication by a number; in $\mathbb{R}^n$ it is a linear map, and Definition 8.3.1 takes (7.3) verbatim as the definition. The converse of the last sentence fails: $f(x) = |x|$ is continuous at 0 and not differentiable there, since the difference quotient equals $\text{sgn}(h)$.

7.2.1 The first-order condition

Theorem 7.2.1 (Fermat). Let $A \subseteq \mathbb{R}$, let x_0 be an interior point of A at which $f: A \rightarrow \mathbb{R}$ is differentiable, and suppose x_0 is a local maximiser or a local minimiser of f on A . Then $f'(x_0) = 0$.

Proof. Suppose x_0 is a local maximiser, so there is $\delta > 0$ with $B_\delta(x_0) \subseteq A$ and $f(x) \leq f(x_0)$ for $x \in B_\delta(x_0)$. For $0 < h < \delta$,

$$\frac{f(x_0 + h) - f(x_0)}{h} \leq 0,$$

because the numerator is non-positive and the denominator positive; letting $h \downarrow 0$ gives $f'(x_0) \leq 0$. For $-\delta < h < 0$ the denominator is negative and the quotient is non-negative; letting $h \uparrow 0$ gives $f'(x_0) \geq 0$. Since f is differentiable at x_0 the two one-sided limits coincide with $f'(x_0)$, so $f'(x_0) = 0$. The minimiser case follows by applying this to $-f$. ■

A point with $f'(x_0) = 0$ is a *stationary point*, or *critical point*, of f (Fukuda 2026a, slide 5).

Remark 7.2.1 (Three ways the conclusion fails). Each hypothesis of Theorem 7.2.1 is indispensable, and each failure has an economic counterpart.

- (i) *Not interior.* $f(x) = x$ on $[0, 1]$ has its maximum at $x = 1$ with $f'(1) = 1 \neq 0$. The proof used h of both signs, and at a boundary point only one sign is admissible; the conclusion weakens to the inequality $f'(1) \geq 0$. In economics this is the corner solution, and the corresponding first-order conditions are the Kuhn–Tucker inequalities of Chapter 12.
- (ii) *Not differentiable.* $f(x) = -|x|$ has a global maximum at $x = 0$ where no derivative exists. Non-differentiable kinks arise in economics from piecewise-linear tax schedules and from Leontief technology.
- (iii) *The condition is necessary, not sufficient.* $f(x) = x^3$ has $f'(0) = 0$ and no local extremum at 0. Sufficiency requires either a second-order condition or a global structural hypothesis such as concavity.

7.2.2 Concavity and sufficiency

Definition 7.2.2 (Concave and convex functions of one variable). A function $f: I \rightarrow \mathbb{R}$ on an interval $I \subseteq \mathbb{R}$ is *concave function* if

$$f(\lambda x + (1 - \lambda)y) \geq \lambda f(x) + (1 - \lambda)f(y) \quad \text{for all } x, y \in I, \lambda \in [0, 1],$$

strictly concave if the inequality is strict whenever $x \neq y$ and $\lambda \in (0, 1)$, and *convex function* (respectively *strictly convex*) if $-f$ is concave (respectively strictly concave). The general theory, in $\mathbb{R}^n$, is Chapter 9.

Theorem 7.2.2 (Characterisation of differentiable concavity). Let $f: (a, b) \rightarrow \mathbb{R}$ be differentiable. The following are equivalent.

- (i) f is concave on (a, b) .

(ii) $f(x) \leq f(x_0) + f'(x_0)(x - x_0)$ for all $x, x_0 \in (a, b)$.

(iii) f' is non-increasing on (a, b) .

If moreover f is twice differentiable, each is equivalent to $f'' \leq 0$ on (a, b) . The strict versions correspond to strict inequality in (ii) for $x \neq x_0$ and to strictly decreasing f' ; $f'' < 0$ is sufficient but not necessary for strict concavity, as $f(x) = -x^4$ shows at $x = 0$.

Proof. (i) $\Rightarrow$ (ii). For $\lambda \in (0, 1]$, concavity gives $f(x_0 + \lambda(x - x_0)) \geq \lambda f(x) + (1 - \lambda)f(x_0)$, hence

$$\frac{f(x_0 + \lambda(x - x_0)) - f(x_0)}{\lambda} \geq f(x) - f(x_0).$$

Letting $\lambda \downarrow 0$, the left side tends to $f'(x_0)(x - x_0)$ by Definition 7.2.1 applied with $h = \lambda(x - x_0)$.

(ii) $\Rightarrow$ (iii). Apply (ii) twice, at x_0 evaluated at x and at x evaluated at x_0 , and add: $0 \leq (f'(x_0) - f'(x))(x - x_0)$. For $x > x_0$ this forces $f'(x) \leq f'(x_0)$.

(iii) $\Rightarrow$ (i). Let $x < y$ and $z = \lambda x + (1 - \lambda)y$ with $\lambda \in (0, 1)$. By the mean value theorem (Theorem 7.2.5) there are $\xi_1 \in (x, z)$ and $\xi_2 \in (z, y)$ with $f(z) - f(x) = f'(\xi_1)(z - x)$ and $f(y) - f(z) = f'(\xi_2)(y - z)$. Since $\xi_1 < \xi_2$, (iii) gives $f'(\xi_1) \geq f'(\xi_2)$, so

$$\frac{f(z) - f(x)}{z - x} \geq \frac{f(y) - f(z)}{y - z}.$$

Substituting $z - x = (1 - \lambda)(y - x)$ and $y - z = \lambda(y - x)$ and rearranging gives $f(z) \geq \lambda f(x) + (1 - \lambda)f(y)$.

The twice-differentiable statement is (iii) together with the fact that a differentiable function is non-increasing exactly when its derivative is non-positive, itself a consequence of Theorem 7.2.5. ■

Theorem 7.2.3 (The first-order condition is sufficient under concavity). Let $f: (a, b) \rightarrow \mathbb{R}$ be differentiable and concave. Then $x_0 \in (a, b)$ is a global maximiser of f on (a, b) if and only if $f'(x_0) = 0$. If f is convex, the same statement holds with “maximiser” replaced by “minimiser”. If f is strictly concave, the maximiser is unique when it exists.

Proof. Necessity is Theorem 7.2.1. For sufficiency, suppose $f'(x_0) = 0$. By Theorem 7.2.2(ii), $f(x) \leq f(x_0) + f'(x_0)(x - x_0) = f(x_0)$ for every $x \in (a, b)$. Uniqueness under strict concavity: if $x_0 \neq x_1$ were both maximisers with common value M , then $f(\frac{1}{2}x_0 + \frac{1}{2}x_1) > \frac{1}{2}M + \frac{1}{2}M = M$, contradicting maximality. ■

This is the content of (Fukuda 2026a, slides 7–8). The geometric statement — the tangent line at any point lies above the graph, so a horizontal tangent lies above the whole graph — is the proof, once Theorem 7.2.2 makes “lies above” precise.

7.2.3 Mean value theorems

Theorem 7.2.4 (Rolle). Let $f: [a, b] \rightarrow \mathbb{R}$ be continuous on $[a, b]$ and differentiable on (a, b) , with $f(a) = f(b)$. Then $f'(\xi) = 0$ for some $\xi \in (a, b)$.

Proof. $[a, b]$ is compact and f continuous, so by Corollary 5.4.1 f attains a maximum at some $x_{\max}$ and a minimum at some $x_{\min}$ in $[a, b]$. If both are endpoints then f is constant, since $f(a) = f(b)$ forces $\max f = \min f$, and any $\xi \in (a, b)$ works. Otherwise one of them is interior, and Theorem 7.2.1 applies there. ■

Theorem 7.2.5 (Mean value theorem). Let $f: [a, b] \rightarrow \mathbb{R}$ be continuous on $[a, b]$ and differentiable on (a, b) . Then there is $\xi \in (a, b)$ with

$$f(b) - f(a) = f'(\xi)(b - a).$$

Proof. Apply Theorem 7.2.4 to $g(x) := f(x) - f(a) - \frac{f(b) - f(a)}{b - a}(x - a)$, which is continuous on $[a, b]$, differentiable on (a, b) , and satisfies $g(a) = g(b) = 0$. Then $0 = g'(\xi) = f'(\xi) - \frac{f(b) - f(a)}{b - a}$. ■

The mean value theorem is the device by which local information about f' is converted into global information about f : it is what makes “ $f' \geq 0$ on an interval” imply “ f is non-decreasing there”, a step used without comment whenever a comparative-static sign is read off a derivative.

7.3 Rules of Differentiation

Proposition 7.3.1 (Linearity). If $f, g: (a, b) \rightarrow \mathbb{R}$ are differentiable at x and $\alpha, \beta \in \mathbb{R}$, then $\alpha f + \beta g$ is differentiable at x and $(\alpha f + \beta g)'(x) = \alpha f'(x) + \beta g'(x)$.

Proof. The difference quotient of $\alpha f + \beta g$ equals α times that of f plus β times that of g , and limits are linear by [Proposition 5.2.2\(i\)](#). ■

Proposition 7.3.2 (Product rule). If f, g are differentiable at x , so is fg , and $(fg)'(x) = f'(x)g(x) + f(x)g'(x)$.

Proof. Add and subtract $f(x+h)g(x)$:

$$\frac{f(x+h)g(x+h) - f(x)g(x)}{h} = \frac{f(x+h) - f(x)}{h} g(x) + f(x+h) \frac{g(x+h) - g(x)}{h}.$$

As $h \rightarrow 0$ the first quotient tends to $f'(x)$ and the second to $g'(x)$, while $f(x+h) \rightarrow f(x)$ by [Proposition 7.2.1](#). ■

The choice of which factor to hold fixed is the substance of the proof: the decomposition corresponds to splitting the area increment of a rectangle of sides f and g into two strips, and the omitted corner term $(f(x+h) - f(x))(g(x+h) - g(x))$ is of order h^2 (Fukuda 2026a, slide 11).

Proposition 7.3.3 (Quotient rule). If f, g are differentiable at x and $g(x) \neq 0$, then f/g is differentiable at x and

$$\left(\frac{f}{g}\right)'(x) = \frac{f'(x)g(x) - f(x)g'(x)}{g(x)^2}.$$

Proof. Since g is continuous at x and $g(x) \neq 0$, g is non-zero on a neighbourhood of x , so f/g is defined there. Write $h := f/g$, so $f = gh$. By [Proposition 7.3.2](#), $f' = g'h + gh'$, whence

$$h' = \frac{f' - g'h}{g} = \frac{f' - g'f/g}{g} = \frac{f'g - fg'}{g^2}.$$

The step presumes h differentiable; to avoid that, compute directly

$$\frac{1}{h} \left(\frac{1}{g(x+h)} - \frac{1}{g(x)} \right) = -\frac{1}{g(x+h)g(x)} \cdot \frac{g(x+h) - g(x)}{h} \rightarrow -\frac{g'(x)}{g(x)^2},$$

which establishes $(1/g)' = -g'/g^2$, and then apply [Proposition 7.3.2](#) to $f \cdot (1/g)$. ■

Both derivations are those of (Fukuda 2026a, slides 13–14, 16); the second is logically self-contained, whereas the first assumes what it computes and is a mnemonic rather than a proof. The distinction is recorded because the first is frequently presented as though it were the second.

Proposition 7.3.4 (Chain rule). Let $f: (a, b) \rightarrow \mathbb{R}$ and $g: (c, d) \rightarrow \mathbb{R}$ with $f((a, b)) \subseteq (c, d)$. If f is differentiable at x_0 and g is differentiable at $f(x_0)$, then $g \circ f$ is differentiable at x_0 and

$$(g \circ f)'(x_0) = g'(f(x_0))f'(x_0).$$

Proof. Write $y_0 = f(x_0)$ and define

$$\varphi(y) := \begin{cases} \frac{g(y) - g(y_0)}{y - y_0}, & y \neq y_0, \\ g'(y_0), & y = y_0. \end{cases}$$

Then φ is continuous at y_0 by differentiability of g , and $g(y) - g(y_0) = \varphi(y)(y - y_0)$ for all y , including $y = y_0$. Hence for $h \neq 0$,

$$\frac{g(f(x_0+h)) - g(f(x_0))}{h} = \varphi(f(x_0+h)) \cdot \frac{f(x_0+h) - f(x_0)}{h}.$$

As $h \rightarrow 0$, $f(x_0+h) \rightarrow y_0$ by continuity of f , so the first factor tends to $\varphi(y_0) = g'(y_0)$ by [Proposition 5.2.3](#), and the second to $f'(x_0)$. ■

The auxiliary function φ exists to handle the case $f(x_0 + h) = f(x_0)$, at which the naive multiply-and-divide argument would divide by zero. The naive argument is the one behind the suggestive notation $\frac{dz}{dx} = \frac{dz}{dy} \frac{dy}{dx}$ of (Fukuda 2026a, slide 18); it gives the right answer and is not a proof.

Proposition 7.3.5 (Derivative of an inverse function). Let $f: (a, b) \rightarrow \mathbb{R}$ be continuous and strictly monotone, hence injective with continuous inverse f^{-1} on the interval $f((a, b))$. If f is differentiable at x_0 with $f'(x_0) \neq 0$, then f^{-1} is differentiable at $y_0 = f(x_0)$ and

$$(f^{-1})'(y_0) = \frac{1}{f'(f^{-1}(y_0))} = \frac{1}{f'(x_0)}.$$

Proof. Let $y_k \rightarrow y_0$ with $y_k \neq y_0$ and set $x_k := f^{-1}(y_k)$, so $x_k \neq x_0$ and $x_k \rightarrow x_0$ by continuity of f^{-1} . Then

$$\frac{f^{-1}(y_k) - f^{-1}(y_0)}{y_k - y_0} = \frac{x_k - x_0}{f(x_k) - f(x_0)} = \left(\frac{f(x_k) - f(x_0)}{x_k - x_0} \right)^{-1} \rightarrow \frac{1}{f'(x_0)},$$

the last step by $f'(x_0) \neq 0$ and continuity of $t \mapsto 1/t$ away from 0. Proposition 5.2.3 converts the sequential statement into the limit. ■

The hypothesis $f'(x_0) \neq 0$ cannot be dropped: $f(x) = x^3$ is a strictly increasing bijection of $\mathbb{R}$ with $f'(0) = 0$, and $f^{-1}(y) = y^{1/3}$ has a vertical tangent at 0. This is the one-dimensional case of the inverse function theorem, and the condition $f'(x_0) \neq 0$ becomes invertibility of the Jacobian in Corollary 8.6.1. Economically it is the statement that a demand function D and an inverse demand function P have reciprocal slopes, $P'(q) = 1/D'(p)$ at $q = D(p)$ (Fukuda 2026a, slides 19–20).

7.4 Taylor Expansion

The problem is to approximate f near x_0 by a polynomial in h ,

$$f(x_0 + h) \approx \alpha_0 + \alpha_1 h + \alpha_2 h^2 + \cdots + \alpha_n h^n. \quad (7.4)$$

Setting $h = 0$ gives $\alpha_0 = f(x_0)$. Differentiating (7.4) k times with respect to h and again setting $h = 0$ kills every term of degree below k and every term of degree above k , leaving $f^{(k)}(x_0) = k! \alpha_k$, so

$$\alpha_k = \frac{f^{(k)}(x_0)}{k!}, \quad k = 0, 1, \dots, n. \quad (7.5)$$

This is the derivation of (Fukuda 2026a, slides 28–31). It identifies the coefficients on the assumption that an expansion of the stated form exists; it does not establish that one does, nor how large the error is. Both are the content of the following theorem.

Theorem 7.4.1 (Taylor's theorem with Lagrange remainder). Let $n \in \mathbb{N}_0$, let $I \subseteq \mathbb{R}$ be an open interval containing x_0 , and let $f: I \rightarrow \mathbb{R}$ be $n + 1$ times differentiable on I . Define the *Taylor polynomial* of order n ,

$$T_n f(x_0; h) := \sum_{k=0}^n \frac{f^{(k)}(x_0)}{k!} h^k.$$

Then for every h with $x_0 + h \in I$ there exists ξ strictly between x_0 and $x_0 + h$ (or $\xi = x_0$ if $h = 0$) such that

$$f(x_0 + h) = T_n f(x_0; h) + \frac{f^{(n+1)}(\xi)}{(n+1)!} h^{n+1}. \quad (7.6)$$

Proof. Fix $h \neq 0$ and let J be the closed interval with endpoints x_0 and $x_0 + h$. Define M by $f(x_0 + h) = T_n f(x_0; h) + M h^{n+1}$, which determines M uniquely since $h \neq 0$, and set

$$g(t) := f(t) - \sum_{k=0}^n \frac{f^{(k)}(x_0)}{k!} (t - x_0)^k - M(t - x_0)^{n+1}, \quad t \in J.$$

By construction $g^{(k)}(x_0) = 0$ for $k = 0, 1, \dots, n$, and $g(x_0 + h) = 0$ by the choice of M . Since $g(x_0) = g(x_0 + h) = 0$, Theorem 7.2.4 gives ξ_1 strictly between them with $g'(\xi_1) = 0$. Since also $g'(x_0) = 0$, Rolle

applied to g' on the interval with endpoints x_0 and ξ_1 gives ξ_2 with $g''(\xi_2) = 0$. Iterating $n + 1$ times produces $\xi := \xi_{n+1}$ strictly between x_0 and $x_0 + h$ with $g^{(n+1)}(\xi) = 0$. But $g^{(n+1)}(t) = f^{(n+1)}(t) - (n + 1)!M$, so $M = f^{(n+1)}(\xi)/(n + 1)!$, which is (7.6). ■

Corollary 7.4.1 (Peano remainder). If f is n times differentiable at x_0 , then $f(x_0 + h) = T_n f(x_0; h) + o(h^n)$ as $h \rightarrow 0$; that is, the error divided by h^n tends to 0.

The two remainder forms answer different questions. The Lagrange form (7.6) is an exact identity with an unknown evaluation point, and it yields *numerical* error bounds once $f^{(n+1)}$ is bounded on the relevant interval. The Peano form is an asymptotic statement with no evaluation point, and it is the one used when the argument is a limit as a parameter goes to zero — which is exactly the situation in Section 7.6 below.

Example 7.4.1 (Exponential and logarithm). For $f(x) = e^x$ every derivative is e^x and $f^{(k)}(0) = 1$, so

$$e^x = 1 + x + \frac{1}{2}x^2 + \frac{1}{6}x^3 + \cdots + \frac{1}{n!}x^n + \frac{e^\xi}{(n+1)!}x^{n+1}.$$

For $f(x) = \log(1 + x)$ on $(-1, \infty)$, $f'(x) = (1 + x)^{-1}$, $f''(x) = -(1 + x)^{-2}$, $f'''(x) = 2(1 + x)^{-3}$, so $f^{(k)}(0) = (-1)^{k-1}(k-1)!$ and

$$\log(1 + x) = x - \frac{1}{2}x^2 + \frac{1}{3}x^3 - \cdots + \frac{(-1)^{n-1}}{n}x^n + \frac{(-1)^n}{n+1} \frac{x^{n+1}}{(1 + \xi)^{n+1}}.$$

The first-order truncations $e^x \approx 1 + x$ and $\log(1 + x) \approx x$ carry an error of order x^2 , with sign determined by the second derivative: the exponential approximation understates and the logarithmic one overstates (Fukuda 2026a, slides 32–36).

7.4.1 Four uses of the logarithmic approximation

Example 7.4.2 (Growth accounting). Since $(\log f)' = f'/f$ by Proposition 7.3.4, taking logarithms of the Cobb–Douglas aggregate $Y(t) = A(t)K(t)^\alpha L(t)^{1-\alpha}$ and differentiating in t gives the exact identity

$$\frac{Y'(t)}{Y(t)} = \frac{A'(t)}{A(t)} + \alpha \frac{K'(t)}{K(t)} + (1 - \alpha) \frac{L'(t)}{L(t)}.$$

No approximation is involved: the growth rate of a Cobb–Douglas aggregate is exactly the factor-share-weighted sum of the growth rates of its arguments, and the residual A'/A is what growth accounting measures (Fukuda 2026a, slide 37).

Example 7.4.3 (Log differences as growth rates). For a positive sequence (Y_t) with growth rate $r_t = (Y_t - Y_{t-1})/Y_{t-1}$,

$$\Delta \log Y_t = \log Y_t - \log Y_{t-1} = \log\left(\frac{Y_t}{Y_{t-1}}\right) = \log(1 + r_t) \approx r_t.$$

By Example 7.4.1 the error is $-r_t^2/2 + O(r_t^3)$, so log differences *understate* growth, by about half a percentage point at $r_t = 10\%$ and by five at $r_t = 30\%$. The approximation is a good one at quarterly frequencies and a poor one for hyperinflations (Fukuda 2026a, slide 38).

Example 7.4.4 (Fisher equation). The exact relation between the nominal rate i , the inflation rate π and the real rate r is $(1 + i) = (1 + \pi)(1 + r)$. Taking logarithms turns the product into a sum exactly, $\log(1 + i) = \log(1 + \pi) + \log(1 + r)$, and applying $\log(1 + x) \approx x$ three times gives the Fisher equation $i \approx \pi + r$. The neglected term is πr , which is second order in the two rates (Fukuda 2026a, slide 39).

Example 7.4.5 (Log-linearisation). First-order Taylor expansion gives $f(x) \approx f(x_0) + f'(x_0)(x - x_0)$. Writing the absolute change as x_0 times the proportional change and using $\log x - \log x_0 \approx (x - x_0)/x_0$,

$$f(x) \approx f(x_0) + f'(x_0)x_0 \cdot (\log x - \log x_0).$$

The coefficient $f'(x_0)x_0$ is the semi-elasticity of f at x_0 . This is the step by which a nonlinear dynamic model is converted into a linear system in log deviations from steady state, whose stability is then read off the eigenvalues as in Proposition 2.6.1 (Fukuda 2026a, slide 40).

7.5 Economic Application: The Firm's Problem

7.5.1 The price-taking firm

Proposition 7.5.1. Under [Assumption 7.1.1](#) and for every $p > 0$, problem (7.1) has a unique solution $q^*(p) > 0$, characterised by

$$c'(q^*) = p, \quad \text{equivalently} \quad q^*(p) = (c')^{-1}(p). \quad (7.7)$$

The supply function q^* is strictly increasing and differentiable with $(q^*)'(p) = 1/c''(q^*(p)) > 0$ whenever c is twice differentiable with $c'' > 0$.

Proof. The objective $\pi(q) = pq - c(q)$ is strictly concave, being linear minus strictly convex, and differentiable, with $\pi'(q) = p - c'(q)$. Since $c'(0) = 0 < p$ and $c'(q) \rightarrow \infty$, the continuous strictly increasing function c' takes the value p at exactly one point $q^* > 0$, by the intermediate value theorem ([Corollary 5.2.1](#)) and injectivity. At that point $\pi' = 0$, and [Theorem 7.2.3](#) makes it the unique global maximiser on $(0, \infty)$; comparison with $\pi(0) = 0 < \pi(q^*)$ rules out the boundary. The supply function is the inverse of c' , and its derivative is [Proposition 7.3.5](#). ■

The economic content of (7.7) is that the firm equates marginal cost to price. The comparative static $(q^*)' = 1/c'' > 0$ makes precise the sense in which the slope of the supply curve is the reciprocal of the curvature of the cost function: a firm with nearly linear costs responds strongly to price, one with sharply increasing marginal cost hardly at all.

7.5.2 The price-setting firm and the markup

Proposition 7.5.2. Under [Assumption 7.1.1](#), an interior solution $q^* > 0$ of (7.2) satisfies

$$P(q^*) \left(1 + \frac{1}{\eta}\right) = c'(q^*), \quad \text{equivalently} \quad P(q^*) = \underbrace{\frac{1}{1 + \eta^{-1}}}_{\text{markup}} c'(q^*), \quad (7.8)$$

where

$$\eta := \left. \frac{dq}{dp} \frac{p}{q} \right|_{q=q^*} < 0$$

is the price elasticity of demand. If $\eta < -1$ the markup exceeds 1 and $P(q^*) > c'(q^*)$.

Proof. The objective is $\pi(q) = P(q)q - c(q)$, concave by [Assumption 7.1.1](#). Its derivative, by [Proposition 7.3.2](#), is $\pi'(q) = P'(q)q + P(q) - c'(q)$, so [Theorem 7.2.1](#) gives

$$\underbrace{P'(q^*)q^* + P(q^*)}_{\text{marginal revenue}} = \underbrace{c'(q^*)}_{\text{marginal cost}}. \quad (7.9)$$

Factoring $P(q^*) > 0$ out of the left side,

$$P(q^*) \left(1 + \frac{P'(q^*)q^*}{P(q^*)}\right) = c'(q^*).$$

By [Proposition 7.3.5](#) applied to the demand–inverse-demand pair, $P'(q^*) = 1/D'(p^*)$ at $p^* = P(q^*)$, so

$$\frac{P'(q^*)q^*}{P(q^*)} = \frac{dp}{dq} \frac{q}{p} = \left(\frac{dq}{dp} \frac{p}{q}\right)^{-1} = \frac{1}{\eta},$$

which is (7.8). Solving for $P(q^*)$ gives the markup form, and $\eta < -1$ makes $1 + \eta^{-1} \in (0, 1)$, hence the markup exceeds 1. ■

Three observations. First, (7.8) is the Lerner condition, usually written $(P - c')/P = -1/\eta$; the two forms are algebraically identical. Second, the derivation uses the inverse function rule twice over — once to convert P' into $1/D'$ and once implicitly in reading the elasticity in either direction — so [Proposition 7.3.5](#) is not a technicality here but the substance. Third, profit maximisation requires $\eta < -1$: if demand were inelastic, marginal revenue $P(1 + \eta^{-1})$ would be negative while marginal cost is positive, and (7.9) would have no solution. A monopolist never operates on the inelastic portion of its demand curve, and this is a theorem about (7.9) rather than an assumption.

Example 7.5.1 (The dual formulation). Choosing the price rather than the quantity, the firm solves $\max_{p \geq 0} pD(p) - c(D(p))$. The first-order condition is $D(p^*) + p^*D'(p^*) - c'(D(p^*))D'(p^*) = 0$; dividing by $D'(p^*) < 0$ and rearranging gives

$$p^* = \left(\frac{1}{1 + \eta^{-1}} \right) c'(D(p^*)), \quad \eta = \frac{D'(p)p}{D(p)} \Big|_{p=p^*},$$

the same condition as (7.8). That the two formulations agree is a consequence of the chain rule and the strict monotonicity of D , which makes the change of variable $q = D(p)$ a bijection between the two feasible sets. This is Exercise 11.14 of Fukuda (2026d) (Fukuda 2026a, slide 25).

7.6 Economic Application: The Risk Premium

Environment. A decision maker with wealth w and utility $u: \mathbb{R} \rightarrow \mathbb{R}$, strictly increasing and twice continuously differentiable, faces a gamble $\tilde{\varepsilon}$ with $\mathbb{E}[\tilde{\varepsilon}] = 0$ and $\text{Var}[\tilde{\varepsilon}] = \sigma^2$. The *risk premium* π is the sure amount whose surrender leaves the decision maker indifferent to bearing the gamble:

$$u(w - \pi) = \mathbb{E}[u(w + \tilde{\varepsilon})]. \quad (7.10)$$

The slides derive the approximation by expanding each side and matching (Fukuda 2026a, slides 26–27, 41). That derivation is correct in its conclusion and incomplete as an argument: it expands the left side to first order and the right to second order without saying why the orders may differ, and it does not say in what limit the approximation is valid. Both gaps are closed by scaling the gamble.

Theorem 7.6.1 (Arrow–Pratt). Let u be three times continuously differentiable with $u' > 0$ on a neighbourhood of w , and let $\tilde{z}$ be a bounded random variable with $\mathbb{E}[\tilde{z}] = 0$ and $\text{Var}[\tilde{z}] = 1$. For $t > 0$ let $\pi(t)$ solve (7.10) with $\tilde{\varepsilon} = t\tilde{z}$. Then π is twice differentiable at $t = 0$ with $\pi(0) = \pi'(0) = 0$ and

$$\pi(t) = \frac{1}{2} A(w) t^2 + o(t^2) \quad \text{as } t \downarrow 0, \quad \text{where } A(w) := -\frac{u''(w)}{u'(w)}$$

is the *absolute risk aversion*. Writing $\sigma^2 = t^2$ for the variance of the gamble, this is $\pi \approx \frac{1}{2} A(w) \sigma^2$.

Proof. Define $\Phi(\pi, t) := u(w - \pi) - \mathbb{E}[u(w + t\tilde{z})]$. Boundedness of $\tilde{z}$ and continuity of u and its derivatives permit differentiation under the expectation on a neighbourhood of $(0, 0)$, by Lemma 10.5.1. We have $\Phi(0, 0) = 0$ and $\partial\Phi/\partial\pi = -u'(w) \neq 0$ at $(0, 0)$, so by the implicit function theorem (Theorem 8.6.1) there is a twice continuously differentiable $\pi(t)$ near $t = 0$ with $\pi(0) = 0$ and $\Phi(\pi(t), t) = 0$.

Expand the right-hand side of (7.10) with the Lagrange remainder of Theorem 7.4.1 at order 2: for each realisation z ,

$$u(w + tz) = u(w) + u'(w)tz + \frac{1}{2}u''(w)t^2z^2 + \frac{1}{6}u'''(\xi_z)t^3z^3.$$

Taking expectations and using $\mathbb{E}[\tilde{z}] = 0$, $\mathbb{E}[\tilde{z}^2] = 1$ and boundedness of $\tilde{z}$ and of u''' near w ,

$$\mathbb{E}[u(w + t\tilde{z})] = u(w) + \frac{1}{2}u''(w)t^2 + O(t^3).$$

Expanding the left-hand side at order 2 in π , $u(w - \pi) = u(w) - u'(w)\pi + O(\pi^2)$. Equating and using $\pi = O(t^2)$, which makes $O(\pi^2) = O(t^4)$,

$$-u'(w)\pi(t) + O(t^4) = \frac{1}{2}u''(w)t^2 + O(t^3),$$

so $\pi(t) = -\frac{1}{2}(u''(w)/u'(w))t^2 + O(t^3)$, which is the claim. ■

Remark 7.6.1 (Why the two sides are expanded to different orders). The theorem explains the asymmetry that the slide derivation leaves implicit. Because π is itself of order t^2 , the term $-u'(w)\pi$ on the left is already of order t^2 , and the next term $\frac{1}{2}u''(w)\pi^2$ is of order t^4 , hence negligible at the order retained. On the right the linear term vanishes because $\mathbb{E}[\tilde{\varepsilon}] = 0$, so the leading non-trivial term is the quadratic one. Expanding the left side to first order and the right to second is therefore expanding *both* to order t^2 : the orders in π and in t are different bookkeeping for the same expansion.

Remark 7.6.2 (The variance identity). The step $\mathbb{E}[\tilde{\varepsilon}^2] = \sigma^2$ uses $\text{Var}[X] = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$, which follows by expanding $\mathbb{E}[(X - \mathbb{E}X)^2] = \mathbb{E}[X^2] - 2\mathbb{E}[X]\mathbb{E}[X] + (\mathbb{E}X)^2$ and is [Proposition 14.2.2](#). It is exactly here that the mean-zero normalisation of the gamble is used; for a gamble with non-zero mean the first-order term does not vanish and the premium contains a term of order t that reflects the change in expected wealth rather than risk (Fukuda [2026a](#), slide 42).

Economic reading. Three consequences. The premium is proportional to the variance, so risk aversion of this local kind prices only second moments; a gamble's skewness enters at order t^3 and is invisible in the approximation. The factor of proportionality is $A(w)/2$, a property of preferences alone, which is why A is comparable across decision makers and across wealth levels while u'' alone is not — u'' changes under the affine transformations of u that leave preferences unchanged, and the ratio $-u''/u'$ does not. Finally, the premium is second order in the size of the gamble: for small gambles a risk averse agent is approximately risk neutral, which is the formal statement behind the practice of ignoring risk in first-order comparative statics and the reason that observed rejections of small-stakes gambles are hard to reconcile with expected utility.

7.7 Chapter Summary

The derivative is the coefficient of the best affine approximation ([7.3](#)), and differentiability implies continuity but not conversely. At an interior maximiser or minimiser of a differentiable function the derivative vanishes ([Theorem 7.2.1](#)); the conclusion fails at boundaries, at kinks, and as a sufficient condition. Under concavity the first-order condition becomes sufficient and identifies the global maximiser ([Theorem 7.2.3](#)), because a differentiable concave function lies below each of its tangents. Rolle's theorem and the mean value theorem convert information about the derivative into information about the function, and are the engine behind both the monotonicity criterion and Taylor's theorem. The differentiation rules — linearity, product, quotient, chain, inverse — are proved here in the forms actually used later, with the inverse rule carrying the economic content that demand and inverse demand have reciprocal slopes.

Taylor's theorem states that the polynomial with coefficients ([7.5](#)) approximates f with an error given exactly by ([7.6](#)) or asymptotically by [Corollary 7.4.1](#). Applied to $\log(1+x)$ it justifies log differences as growth rates, the Fisher equation, and log-linearisation, in each case with a quantified second-order error. Applied to the certainty-equivalence condition it produces the Arrow–Pratt formula $\pi \approx \frac{1}{2}A(w)\sigma^2$, with the orders of expansion on the two sides reconciled by the observation that the premium is itself second order in the scale of the gamble. For the firm, the first-order condition gives price equal to marginal cost under price taking and the Lerner markup condition under price setting, the latter implying that a monopolist operates only where demand is elastic.

Exercise 7.7.1. Let $c(q) = \frac{1}{2}\gamma q^2$ with $\gamma > 0$ and $P(q) = a - bq$ with $a, b > 0$. Compute q^* and $P(q^*)$ for both firm problems, verify ([7.8](#)) directly, and show that the monopoly quantity is exactly half the competitive quantity when $\gamma = 0$. Identify where [Assumption 7.1.1](#) fails when $\gamma = 0$ and why the conclusion survives anyway.

Exercise 7.7.2. Show that a function may be differentiable everywhere with discontinuous derivative, using $f(x) = x^2 \sin(1/x)$ for $x \neq 0$ and $f(0) = 0$. Compute $f'(0)$ from the definition, compute $f'(x)$ for $x \neq 0$, and show that $\lim_{x \rightarrow 0} f'(x)$ does not exist. Explain why this does not contradict [Theorem 7.2.5](#).

Exercise 7.7.3. For CRRA utility $u(c) = c^{1-\gamma}/(1-\gamma)$ with $\gamma > 0$, $\gamma \neq 1$, compute $A(w)$ and the coefficient of relative risk aversion $wA(w)$. Show that the risk premium for a multiplicative gamble $w(1+t\tilde{z})$ is approximately $\frac{1}{2}\gamma w t^2$, and contrast with [Theorem 7.6.1](#). Verify that $\lim_{\gamma \rightarrow 1} u(c)$, after subtracting the constant $1/(1-\gamma)$, equals $\log c$.

Chapter 8

Differential Calculus in Several Variables

8.1 Motivation: Two Multivariate Problems

Two problems drive the chapter (Fukuda 2026a, slide 43). The representative firm chooses capital and labour to solve

$$\max_{K, L \geq 0} \Pi(K, L), \quad \Pi(K, L) := AK^\alpha L^{1-\alpha} - (rK + wL), \quad (8.1)$$

and the econometrician fits a line by

$$\min_{(\beta_1, \beta_2) \in \mathbb{R}^2} \sum_{i=1}^n (y_i - \beta_1 - \beta_2 x_i)^2. \quad (8.2)$$

Neither is harder than the one-variable problems of Chapter 7 once the right notion of derivative is in place, and finding that notion is the substance of the chapter. Two distinct ideas are involved and are frequently conflated. The first is to vary one coordinate at a time, which produces partial derivatives. The second is to approximate f near a point by an affine function of the whole vector, which produces the total derivative. In one variable the two coincide. In several variables the second is strictly stronger, and it is the second that supports the chain rule, the first-order condition, and the implicit function theorem.

8.2 Partial Derivatives and the Gradient

Definition 8.2.1 (Partial derivative and gradient). Let $f: A \rightarrow \mathbb{R}$ with $A \subseteq \mathbb{R}^n$ and let x be interior to A . For $i \in \{1, \dots, n\}$ the *partial derivative* of f at x is

$$\frac{\partial f}{\partial x_i}(x) := \lim_{h \rightarrow 0} \frac{f(x_1, \dots, x_i + h, \dots, x_n) - f(x_1, \dots, x_i, \dots, x_n)}{h},$$

provided the limit exists and is finite; equivalently, it is the ordinary derivative at x_i of the section $t \mapsto f(x_1, \dots, t, \dots, x_n)$. When all n partial derivatives exist, the *gradient* of f at x is the vector

$$\nabla f(x) := \left(\frac{\partial f}{\partial x_1}(x), \dots, \frac{\partial f}{\partial x_n}(x) \right)^\top \in \mathbb{R}^n.$$

This is Definition 3.5 of Fukuda (2026d) (Fukuda 2026a, slides 46, 48), where the alternatives $\partial f(x)/\partial x_i$, $f_{x_i}(x)$ and $f'(x)$ for the gradient are also recorded. This volume writes the gradient as a column vector, consistently with the front-matter chapter on notation and conventions; the row vector $\nabla f(x)^\top$ is the Jacobian of f viewed as a map into $\mathbb{R}^1$, and conflating the two is the most frequent source of transpose errors in comparative statics.

Example 8.2.1 (Cobb–Douglas). For $f(x, y) = x^\alpha y^{1-\alpha}$ with $\alpha \in (0, 1)$ and $x, y > 0$, fixing y makes f a power function of x , so

$$\frac{\partial f}{\partial x} = \alpha x^{\alpha-1} y^{1-\alpha} = \alpha \left(\frac{y}{x}\right)^{1-\alpha}, \quad \frac{\partial f}{\partial y} = (1-\alpha)x^\alpha y^{-\alpha} = (1-\alpha) \left(\frac{x}{y}\right)^\alpha.$$

Both are positive and each is decreasing in its own argument: marginal products are positive and diminishing. Both are homogeneous of degree zero, so they depend on the two inputs only through the ratio x/y — the property that makes the factor-price equations of [Section 8.7](#) determine the capital–labour ratio and nothing else (Fukuda [2026a](#), slide 47).

Example 8.2.2 (Gradient of a quadratic form). Let $A = (a_{ij}) \in M_{n \times n}(\mathbb{R})$ and $f(x) = x^\top A x = \sum_{i,j} a_{ij} x_i x_j$. Differentiating with respect to x_k , the terms containing x_k are $a_{kk}x_k^2$ and, for $j \neq k$, $a_{kj}x_k x_j$ and $a_{jk}x_j x_k$, so

$$\frac{\partial f}{\partial x_k}(x) = 2a_{kk}x_k + \sum_{j \neq k} (a_{kj} + a_{jk})x_j = \sum_{j=1}^n (a_{kj} + a_{jk})x_j,$$

that is,

$$\nabla(x^\top A x) = (A + A^\top)x. \quad (8.3)$$

For $n = 2$ this is the matrix $\begin{pmatrix} 2a_{11} & a_{12}+a_{21} \\ a_{12}+a_{21} & 2a_{22} \end{pmatrix}$ applied to x , as in Exercise 3.3 and Example 11.2 of Fukuda ([2026d](#)) (Fukuda [2026a](#), slide 49). When A is symmetric (8.3) reduces to the familiar $2Ax$; when it is not, only the symmetric part $\frac{1}{2}(A + A^\top)$ of A affects the quadratic form, which is [Remark 2.5.1](#). The identities that recur in least squares, in the second-order conditions of [Chapter 12](#) and in the Kalman recursion of [subsection 14.6.2](#) are collected in [Appendix B](#).

8.3 Differentiability

Definition 8.3.1 (Differentiability and the total derivative). Let $A \subseteq \mathbb{R}^n$ be open and $f: A \rightarrow \mathbb{R}$. The function f is *differentiable in several variables* $x_0 \in A$ if there is a vector $\nabla f(x_0) \in \mathbb{R}^n$ such that

$$\beta_{x_0}(u) := \frac{f(x_0 + u) - f(x_0) - \langle \nabla f(x_0), u \rangle}{\|u\|} \rightarrow 0 \quad \text{as } u \rightarrow 0, u \neq 0. \quad (8.4)$$

The associated linear functional $Df(x_0): \mathbb{R}^n \rightarrow \mathbb{R}$, $Df(x_0)u := \langle \nabla f(x_0), u \rangle$, is the *total derivative* of f at x_0 . The function is differentiable on A if it is differentiable at every point of A .

This is Definition 11.1 of Fukuda ([2026d](#), p. 369). Condition (8.4) says that $f(x_0) + \langle \nabla f(x_0), u \rangle$ approximates $f(x_0 + u)$ with an error small relative to $\|u\|$; geometrically, the graph has a tangent hyperplane at x_0 (Fukuda [2026a](#), slide 50). The existence of the vector $\nabla f(x_0)$ is part of the requirement, and by [Theorem 3.2.1](#) any linear functional on $\mathbb{R}^n$ has such a representation, so nothing is lost by stating the definition with an inner product rather than an abstract linear map.

Proposition 8.3.1 (Consistency with the one-variable definition). For $n = 1$, [Definition 8.3.1](#) holds at x_0 with $\nabla f(x_0) = L$ if and only if [Definition 7.2.1](#) holds with $f'(x_0) = L$.

Proof. For $n = 1$, (8.4) reads $(f(x_0 + u) - f(x_0) - Lu)/|u| \rightarrow 0$, which since $\|u\| = |u|$ is equivalent to $(f(x_0 + u) - f(x_0))/u \rightarrow L$. This is [Proposition 7.2.1](#), and hence [Definition 7.2.1](#). ■

Proposition 8.3.2 (Differentiability implies continuity and the existence of partials). If f is differentiable at x_0 , then f is continuous at x_0 , every partial derivative of f exists at x_0 , and the vector $\nabla f(x_0)$ in (8.4) is the gradient of [Definition 8.2.1](#). Moreover the directional derivative in any direction $v \in \mathbb{R}^n$ exists and equals $\langle \nabla f(x_0), v \rangle$.

Proof. Continuity: from (8.4), $f(x_0 + u) - f(x_0) = \langle \nabla f(x_0), u \rangle + \|u\| \beta_{x_0}(u)$, and both terms tend to 0 as $u \rightarrow 0$, the first by Cauchy–Schwarz ([Lemma 3.2.1](#)) and the second because $\beta_{x_0}(u) \rightarrow 0$.

Directional derivatives: put $u = tv$ with $t \neq 0$ and $v \neq 0$. Then

$$\frac{f(x_0 + tv) - f(x_0)}{t} = \langle \nabla f(x_0), v \rangle + \frac{|t| \|v\|}{t} \beta_{x_0}(tv) \rightarrow \langle \nabla f(x_0), v \rangle,$$

because $||t|/t| = 1$ and $\beta_{x_0}(tv) \rightarrow 0$. Taking $v = e_i$ gives the i th partial derivative and identifies the i th coordinate of $\nabla f(x_0)$. ■

The converse fails in both of the ways that matter, and the two counterexamples are worth stating because each corresponds to a real modelling error.

Example 8.3.1 (Partials without continuity). Let $f: \mathbb{R}^2 \rightarrow \mathbb{R}$ be $f(x_1, x_2) = 0$ if $x_1 x_2 = 0$ and $f(x_1, x_2) = 1$ otherwise. Along each axis f is identically zero, so $\partial f/\partial x_1(0, 0) = \partial f/\partial x_2(0, 0) = 0$. Yet $f(\varepsilon, \varepsilon) = 1$ for every $\varepsilon \neq 0$, so f is not continuous at the origin. In one variable this cannot happen ([Proposition 7.2.1](#)); in several, the partial derivatives see only n lines through the point and are blind to everything else ([Fukuda 2026d](#), Rem. 11.4, p. 375).

Example 8.3.2 (Continuity and partials without differentiability). Let

$$f(x, y) = \begin{cases} \frac{xy}{\sqrt{x^2 + y^2}}, & (x, y) \neq (0, 0), \\ 0, & (x, y) = (0, 0). \end{cases}$$

Since $|xy| \leq \frac{1}{2}(x^2 + y^2)$, we have $|f(x, y)| \leq \frac{1}{2}\sqrt{x^2 + y^2} \rightarrow 0$, so f is continuous at the origin. Both partial derivatives at the origin vanish, since f is identically zero on each axis. If f were differentiable at the origin, then by [Proposition 8.3.2](#) the gradient would be 0 and (8.4) would require

$$\beta_0(x, y) = \frac{f(x, y)}{\sqrt{x^2 + y^2}} = \frac{xy}{x^2 + y^2} \rightarrow 0.$$

Along $(x, y) = (h, h)$ this ratio equals $\frac{1}{2}$ for every $h \neq 0$, so the limit is not 0 and f is not differentiable at the origin ([Fukuda 2026d](#), Rem. 11.3, p. 374). The obstruction is directional: f has a directional derivative in every direction but they do not fit together into a linear functional.

Proposition 8.3.3 (Continuous partials suffice). Let $A \subseteq \mathbb{R}^n$ be open and let $f: A \rightarrow \mathbb{R}$ have partial derivatives with respect to every variable at every point of A , all of them continuous at $x_0 \in A$. Then f is differentiable at x_0 .

Proof. Write $u = (u_1, \dots, u_n)$ and interpolate one coordinate at a time. Set $v^0 := x_0$ and $v^k := x_0 + \sum_{i \leq k} u_i e_i$, so $v^n = x_0 + u$. Then

$$f(x_0 + u) - f(x_0) = \sum_{k=1}^n (f(v^k) - f(v^{k-1})),$$

and each summand differs only in the k th coordinate. Applying the one-variable mean value theorem ([Theorem 7.2.5](#)) to that section gives ξ^k on the segment from v^{k-1} to v^k with $f(v^k) - f(v^{k-1}) = \frac{\partial f}{\partial x_k}(\xi^k) u_k$. Hence

$$f(x_0 + u) - f(x_0) - \langle \nabla f(x_0), u \rangle = \sum_{k=1}^n \left(\frac{\partial f}{\partial x_k}(\xi^k) - \frac{\partial f}{\partial x_k}(x_0) \right) u_k.$$

Since $|u_k| \leq \|u\|$ and $\xi^k \rightarrow x_0$ as $u \rightarrow 0$, continuity of the partials makes each bracket tend to 0, so the whole expression divided by $\|u\|$ tends to 0. ■

This is Proposition 11.4 of [Fukuda \(2026d, p. 375\)](#). It supplies the practical criterion: a function given by an explicit formula built from elementary operations has continuous partials wherever those operations are defined, hence is differentiable there, and the pathologies of [Example 8.3.1](#) and [Example 8.3.2](#) arise only at points where the formula changes.

Definition 8.3.2 (Continuous differentiability). f is *continuously differentiable* on an open A , written $f \in C^1(A)$, if all partial derivatives exist and are continuous on A . Inductively, $f \in C^k(A)$ if all partial derivatives of order k exist and are continuous.

The implications established so far, none of which reverses, are

$$f \in C^1 \implies f \text{ differentiable} \implies \begin{cases} f \text{ continuous,} \\ \text{all partials exist} \end{cases}$$

and neither of the two consequences on the right, alone or together, implies differentiability.

Remark 8.3.1 (The total differential). Writing dy for the increment of f and dx_i for the increments of the arguments, (8.4) is the statement

$$dy = \frac{\partial f}{\partial x_1} dx_1 + \cdots + \frac{\partial f}{\partial x_n} dx_n$$

up to an error small relative to $\|dx\|$ (Fukuda 2026a, slide 51). The classical notation is convenient and hides the content: the identity is not an algebraic manipulation of infinitesimals but the assertion that the error term in (8.4) vanishes faster than $\|u\|$, which by Example 8.3.2 is a genuine restriction on f and not a consequence of the existence of the partial derivatives appearing in it.

8.3.1 Vector-valued maps and the Jacobian

Definition 8.3.3 (Jacobian). Let $A \subseteq \mathbb{R}^n$ be open and $f = (f_1, \dots, f_m): A \rightarrow \mathbb{R}^m$. Then f is differentiable at x_0 if there is a linear map $Df(x_0): \mathbb{R}^n \rightarrow \mathbb{R}^m$ with $\|f(x_0 + u) - f(x_0) - Df(x_0)u\| / \|u\| \rightarrow 0$ as $u \rightarrow 0$. Equivalently, each component f_k is differentiable at x_0 ; and the matrix of $Df(x_0)$ in the canonical bases is the *Jacobian matrix*

$$Df(x_0) = \begin{pmatrix} \frac{\partial f_1}{\partial x_1}(x_0) & \cdots & \frac{\partial f_1}{\partial x_n}(x_0) \\ \vdots & & \vdots \\ \frac{\partial f_m}{\partial x_1}(x_0) & \cdots & \frac{\partial f_m}{\partial x_n}(x_0) \end{pmatrix} \in M_{m \times n}(\mathbb{R}).$$

The row-wise reading of a matrix flagged in Theorem 2.3.1 appears here: row k of $Df(x_0)$ is $\nabla f_k(x_0)^T$. For $m = 1$ the Jacobian is the *row* vector $\nabla f(x_0)^T$, which is why gradients and Jacobians differ by a transpose and why the chain rule below is stated for matrices rather than vectors.

Theorem 8.3.1 (Chain rule). Let $A \subseteq \mathbb{R}^n$ and $B \subseteq \mathbb{R}^m$ be open, $f: A \rightarrow \mathbb{R}^m$ with $f(A) \subseteq B$, and $\varphi: B \rightarrow \mathbb{R}^p$. If f is differentiable at x_0 and φ is differentiable at $f(x_0)$, then $\varphi \circ f$ is differentiable at x_0 and

$$D(\varphi \circ f)(x_0) = D\varphi(f(x_0)) Df(x_0), \quad (8.5)$$

a product of a $p \times m$ by an $m \times n$ matrix. Componentwise, for $p = 1$,

$$\frac{\partial(\varphi \circ f)}{\partial x_i}(x_0) = \sum_{k=1}^m \frac{\partial \varphi}{\partial y_k}(f(x_0)) \frac{\partial f_k}{\partial x_i}(x_0). \quad (8.6)$$

Proof. Write $y_0 = f(x_0)$, $S := Df(x_0)$ and $T := D\varphi(y_0)$. By differentiability, $f(x_0 + u) = y_0 + Su + \|u\| \varepsilon_1(u)$ with $\varepsilon_1(u) \rightarrow 0$, and $\varphi(y_0 + v) = \varphi(y_0) + Tv + \|v\| \varepsilon_2(v)$ with $\varepsilon_2(v) \rightarrow 0$. Put $v(u) := Su + \|u\| \varepsilon_1(u)$; then

$$\varphi(f(x_0 + u)) - \varphi(y_0) = T(Su) + \|u\| T\varepsilon_1(u) + \|v(u)\| \varepsilon_2(v(u)).$$

Divide by $\|u\|$. The second term tends to 0 because T is linear on a finite-dimensional space, hence continuous by Proposition 5.2.1(ii), and $\varepsilon_1(u) \rightarrow 0$. For the third, $\|v(u)\| \leq (\|S\| + \|\varepsilon_1(u)\|) \|u\|$ stays bounded relative to $\|u\|$, while $v(u) \rightarrow 0$ forces $\varepsilon_2(v(u)) \rightarrow 0$. Hence the quotient tends to 0 and $D(\varphi \circ f)(x_0) = TS$. ■

This is Theorem 11.1 of Fukuda (2026d, p. 380). Identity (8.6) is the rule “sum over all channels through which x_i affects the value”, and it is the formal content of every total-derivative computation in comparative statics: when a policy parameter enters an equilibrium condition both directly and through an endogenous variable, (8.6) is what enumerates the channels.

Proposition 8.3.4 (Gradients are orthogonal to level sets). Let $f: A \rightarrow \mathbb{R}$ be differentiable at x_0 and let $\gamma: (-\delta, \delta) \rightarrow A$ be a differentiable curve with $\gamma(0) = x_0$ lying in the level set $\{x: f(x) = f(x_0)\}$. Then $\langle \nabla f(x_0), \gamma'(0) \rangle = 0$. Moreover, among unit vectors v , the directional derivative $\langle \nabla f(x_0), v \rangle$

is maximised at $v = \nabla f(x_0) / \|\nabla f(x_0)\|$ when $\nabla f(x_0) \neq 0$.

Proof. The function $t \mapsto f(\gamma(t))$ is constant, so by [Theorem 8.3.1](#) its derivative $\langle \nabla f(x_0), \gamma'(0) \rangle$ vanishes. The second claim is the equality case of Cauchy–Schwarz ([Lemma 3.2.1](#)): $\langle \nabla f(x_0), v \rangle \leq \|\nabla f(x_0)\|$ with equality exactly when v is the stated unit vector. ■

The economic reading is that the gradient of a utility function is orthogonal to the indifference curve and points in the direction of steepest preference increase, and that the marginal rate of substitution between goods i and j is the ratio $(\partial u / \partial x_i) / (\partial u / \partial x_j)$ precisely because the level set is orthogonal to the gradient. This is Proposition 11.6 of Fukuda ([2026d](#), p. 385).

8.3.2 Homogeneous functions

Definition 8.3.4 (Homogeneity). Let $X \subseteq \mathbb{R}^n$ be a cone, that is $\lambda x \in X$ whenever $x \in X$ and $\lambda > 0$. A function $f: X \rightarrow \mathbb{R}$ is *homogeneous function* if $f(\lambda x) = \lambda^k f(x)$ for all $x \in X$ and $\lambda > 0$.

Theorem 8.3.2 (Euler). Let $X \subseteq \mathbb{R}^n$ be an open cone and $f: X \rightarrow \mathbb{R}$ differentiable. Then f is homogeneous of degree k if and only if

$$\langle \nabla f(x), x \rangle = \sum_{i=1}^n \frac{\partial f}{\partial x_i}(x) x_i = k f(x) \quad \text{for all } x \in X. \quad (8.7)$$

Moreover, if f is homogeneous of degree k then each partial derivative $\partial f / \partial x_i$ is homogeneous of degree $k - 1$.

Proof. ($\Rightarrow$) Differentiate $f(\lambda x) = \lambda^k f(x)$ with respect to λ using [Theorem 8.3.1](#): $\langle \nabla f(\lambda x), x \rangle = k \lambda^{k-1} f(x)$. Setting $\lambda = 1$ gives (8.7). Differentiating instead with respect to x_i gives $\lambda \partial_i f(\lambda x) = \lambda^k \partial_i f(x)$, which is homogeneity of degree $k - 1$ of $\partial_i f$.

($\Leftarrow$) Fix x and set $g(\lambda) := \lambda^{-k} f(\lambda x)$ for $\lambda > 0$. By [Theorem 8.3.1](#) and (8.7) applied at λx ,

$$g'(\lambda) = -k \lambda^{-k-1} f(\lambda x) + \lambda^{-k} \langle \nabla f(\lambda x), x \rangle = -k \lambda^{-k-1} f(\lambda x) + \lambda^{-k-1} k f(\lambda x) = 0,$$

using $\langle \nabla f(\lambda x), \lambda x \rangle = k f(\lambda x)$. So g is constant on $(0, \infty)$ by [Theorem 7.2.5](#), and $g(1) = f(x)$ gives $f(\lambda x) = \lambda^k f(x)$. ■

Homogeneity of degree 1 of a production function is constant returns to scale, and (8.7) then reads $Y = F_K K + F_L L$: output is exhausted by payments to factors at their marginal products. This is Definition 11.3 of Fukuda ([2026d](#), p. 385), and it is the identity behind the factor shares computed in [Section 8.7](#).

8.4 Optimisation Without Constraints

Theorem 8.4.1 (Fermat, several variables). Let $A \subseteq \mathbb{R}^n$, let x^* be an interior point of A at which $f: A \rightarrow \mathbb{R}$ is differentiable, and suppose x^* is a local maximiser or local minimiser of f on A . Then $\nabla f(x^*) = 0$.

Proof. Fix i and consider the section $g_i(t) := f(x^* + t e_i)$, defined for $|t|$ small because x^* is interior. Then g_i is differentiable at 0 with $g'_i(0) = \partial f / \partial x_i(x^*)$ by [Proposition 8.3.2](#), and 0 is a local extremum of g_i . By [Theorem 7.2.1](#), $g'_i(0) = 0$. As i was arbitrary, $\nabla f(x^*) = 0$. ■

Theorem 8.4.2 (Sufficiency under concavity). Let $C \subseteq \mathbb{R}^n$ be convex, $f: C \rightarrow \mathbb{R}$ continuous on C , differentiable on $\text{Int } C$, and concave. Then $x^* \in \text{Int } C$ is a global maximiser of f on C if and only if $\nabla f(x^*) = 0$. The convex case with minimisers is obtained by replacing f with $-f$.

Proof. Necessity is [Theorem 8.4.1](#). For sufficiency, the multivariate analogue of [Theorem 7.2.2\(ii\)](#), proved as [Theorem 9.3.1](#), gives $f(x) \leq f(x^*) + \langle \nabla f(x^*), x - x^* \rangle = f(x^*)$ for all $x \in C$, first for $x \in \text{Int } C$ and then for $x \in C$ by continuity. ■

This is the content of (Fukuda [2026a](#), slide 52).

Example 8.4.1 (The four canonical surfaces). At the origin, all four functions below have vanishing gradient (Fukuda 2026a, slide 53).

- (i) $f(x, y) = x^2 + y^2$: a strict global minimum. $D^2 f = 2I \succ 0$.
- (ii) $f(x, y) = -x^2 - y^2$: a strict global maximum. $D^2 f = -2I \prec 0$.
- (iii) $f(x, y) = x^2 - y^2$: a saddle. $D^2 f = \text{diag}(2, -2)$ is indefinite; the origin is a minimum along the x -axis and a maximum along the y -axis, so it is neither.
- (iv) $f(x, y) = -x^3$: neither, and the Hessian at the origin is the zero matrix, which is both positive and negative semidefinite. The second-order test is silent, and the third derivative decides.

Cases (iii) and (iv) are the two ways in which a stationary point can fail to be an extremum, and they are distinguished by whether the Hessian is indefinite (the test rules out an extremum) or merely singular (the test is inconclusive).

8.5 Second-Order Theory

Definition 8.5.1 (Hessian). Let $A \subseteq \mathbb{R}^n$ be open and $f: A \rightarrow \mathbb{R}$ twice differentiable at $x \in A$. The *Hessian* of f at x is the matrix of second partial derivatives

$$D^2 f(x) := \left(\frac{\partial^2 f}{\partial x_i \partial x_j}(x) \right)_{i,j=1}^n \in M_{n \times n}(\mathbb{R}),$$

also written $Hf(x)$ or $\nabla^2 f(x)$; it is the Jacobian of the map $x \mapsto \nabla f(x)$.

Theorem 8.5.1 (Schwarz). If $f \in C^2(A)$ then $D^2 f(x)$ is symmetric for every $x \in A$: $\partial^2 f / \partial x_i \partial x_j = \partial^2 f / \partial x_j \partial x_i$.

Symmetry is not automatic without the continuity hypothesis; the standard counterexample is $f(x, y) = xy(x^2 - y^2)/(x^2 + y^2)$ with $f(0, 0) = 0$, for which the two mixed partials at the origin are -1 and $+1$. See Rudin (1976, Thm. 9.41). In economics symmetry of the Hessian is not a technicality: it is the source of the Slutsky symmetry conditions and of the reciprocity relations that make the Hessian of an expenditure function testable.

Theorem 8.5.2 (Second-order Taylor expansion). Let $A \subseteq \mathbb{R}^n$, let $\bar{x} \in \text{Int } A$, and let $f: A \rightarrow \mathbb{R}$ be twice differentiable at $\bar{x}$. Then

$$f(x) = f(\bar{x}) + \langle \nabla f(\bar{x}), x - \bar{x} \rangle + \frac{1}{2}(x - \bar{x})^\top D^2 f(\bar{x})(x - \bar{x}) + \|x - \bar{x}\|^2 \alpha_{\bar{x}}(x - \bar{x}), \quad (8.8)$$

where $\alpha_{\bar{x}}(x - \bar{x}) \rightarrow 0$ as $x \rightarrow \bar{x}$. If $f \in C^2$ on a neighbourhood of the segment $[\bar{x}, x]$, there is $\theta \in (0, 1)$ with the exact expression

$$f(x) = f(\bar{x}) + \langle \nabla f(\bar{x}), x - \bar{x} \rangle + \frac{1}{2}(x - \bar{x})^\top D^2 f(\bar{x} + \theta(x - \bar{x}))(x - \bar{x}).$$

Proof. For the exact form, fix x and set $\varphi(t) := f(\bar{x} + t(x - \bar{x}))$ for $t \in [0, 1]$. By Theorem 8.3.1, $\varphi'(t) = \langle \nabla f(\bar{x} + t(x - \bar{x})), x - \bar{x} \rangle$ and, applying the chain rule again to the gradient map, $\varphi''(t) = (x - \bar{x})^\top D^2 f(\bar{x} + t(x - \bar{x}))(x - \bar{x})$. Applying the one-variable Taylor theorem (Theorem 7.4.1 with $n = 1$) to φ on $[0, 1]$ gives $\varphi(1) = \varphi(0) + \varphi'(0) + \frac{1}{2}\varphi''(\theta)$ for some $\theta \in (0, 1)$, which is the claim. The Peano form (8.8) follows from twice differentiability at the single point $\bar{x}$ by the same argument applied with Corollary 7.4.1. ■

This is Theorem 11.5 of Fukuda (2026d, p. 413).

Theorem 8.5.3 (Second-order conditions). Let $A \subseteq \mathbb{R}^n$ be open, $f \in C^2(A)$, and $x^* \in A$ with $\nabla f(x^*) = 0$.

- (i) If $D^2 f(x^*) \prec 0$, then x^* is a strict local maximiser.

- (ii) If $D^2 f(x^*) \succ 0$, then x^* is a strict local minimiser.
- (iii) If $D^2 f(x^*)$ is indefinite, then x^* is neither.
- (iv) If x^* is a local maximiser then $D^2 f(x^*) \preceq 0$; if a local minimiser then $D^2 f(x^*) \succeq 0$.

Proof. (i) Let $H := D^2 f(x^*) \prec 0$. The function $v \mapsto v^\top H v$ is continuous on the compact unit sphere, so by [Corollary 5.4.1](#) it attains a maximum -2μ there, and $\mu > 0$ by negative definiteness. By homogeneity, $u^\top H u \leq -2\mu \|u\|^2$ for all u . From (8.8) with $u = x - x^*$ and $\nabla f(x^*) = 0$,

$$f(x) - f(x^*) \leq -\mu \|u\|^2 + \|u\|^2 \alpha(u) = \|u\|^2 (\alpha(u) - \mu),$$

which is negative for $u \neq 0$ small enough that $|\alpha(u)| < \mu$. Part (ii) applies (i) to $-f$.

(iii) Indefiniteness supplies v_+ with $v_+^\top H v_+ > 0$ and v_- with $v_-^\top H v_- < 0$. Restricting f to the line $t \mapsto x^* + t v_+$ and applying (8.8) shows $f(x^* + t v_+) > f(x^*)$ for small $t \neq 0$, and symmetrically along v_- ; so x^* is neither a local maximum nor a local minimum.

(iv) If x^* is a local maximiser and some v had $v^\top H v > 0$, the argument of (iii) along v would contradict maximality. ■

The gap between the sufficient conditions (i)–(ii) and the necessary condition (iv) is exactly the semidefinite case, and it cannot be closed: $f(x) = x^3$ and $f(x) = x^4$ both have $f''(0) = 0$ and differ in their behaviour. Definiteness is checked by the eigenvalue criterion ([Proposition 2.5.1](#)) or by Sylvester's criterion ([Proposition 2.5.2](#)), with the warning of [Remark 2.5.2](#) about the semidefinite case.

Theorem 8.5.4 (Hessian characterisation of convexity). Let $A \subseteq \mathbb{R}^n$ be open and convex and $f \in C^2(A)$. Then f is convex on A if and only if $D^2 f(x) \succeq 0$ for every $x \in A$, and concave if and only if $D^2 f(x) \preceq 0$ for every $x \in A$. If $D^2 f(x) \succ 0$ everywhere then f is strictly convex; the converse fails, as $f(x) = x^4$ shows.

Proof. Convexity of f on the convex set A is equivalent to convexity of $\varphi_{x,y}(t) := f(x + t(y - x))$ on $[0, 1]$ for every $x, y \in A$, since the two statements are the same inequality. By [Theorem 7.2.2](#) applied to $-\varphi_{x,y}$, that holds if and only if $\varphi_{x,y}''(t) \geq 0$, and $\varphi_{x,y}''(t) = (y - x)^\top D^2 f(x + t(y - x))(y - x)$ as computed in the proof of [Theorem 8.5.2](#). Ranging over $x, y \in A$ makes $y - x$ an arbitrary direction and $x + t(y - x)$ an arbitrary point of A , which is exactly $D^2 f \succeq 0$ on A . ■

This is Theorem 11.6 of Fukuda ([2026d](#), p. 415).

8.6 The Implicit and Inverse Function Theorems

Theorem 8.6.1 (Implicit function theorem). Let $\Lambda \subseteq \mathbb{R}^n \times \mathbb{R}^m$ be open and $f: \Lambda \rightarrow \mathbb{R}^m$ continuously differentiable. Let $(\bar{x}, \bar{y}) \in \Lambda$ satisfy $f(\bar{x}, \bar{y}) = 0$ and suppose the $m \times m$ matrix $\nabla_y f(\bar{x}, \bar{y})$ is nonsingular. Then there are an open neighbourhood $\Gamma \subseteq \mathbb{R}^n$ of $\bar{x}$ and a continuously differentiable map $e: \Gamma \rightarrow \mathbb{R}^m$ such that

- (i) $(x, e(x)) \in \Lambda$ and $f(x, e(x)) = 0$ for all $x \in \Gamma$;
- (ii) $e(\bar{x}) = \bar{y}$;
- (iii) $\nabla_y f(x, e(x))$ is nonsingular for all $x \in \Gamma$; and
- (iv) for all $x \in \Gamma$,

$$De(x) = -[\nabla_y f(x, e(x))]^{-1} \nabla_x f(x, e(x)). \quad (8.9)$$

Moreover e is the unique such function on a possibly smaller neighbourhood: the solutions of $f(x, y) = 0$ near $(\bar{x}, \bar{y})$ are exactly the points $(x, e(x))$.

This is Theorem 11.7 of Fukuda ([2026d](#), p. 419). The proof constructs e as the fixed point of a contraction: for each x near $\bar{x}$ the map $y \mapsto y - [\nabla_y f(\bar{x}, \bar{y})]^{-1} f(x, y)$ is a contraction on a small closed ball, by continuity of Df , and [Theorem 6.4.1](#) applies on that complete set, with [Corollary 6.4.2](#) keeping the fixed point inside the ball; the regularity of e then follows from the regularity of f . It is carried out in full in [Appendix A](#), where it is placed because it is long and purely technical, and because it is

exactly the use of the contraction mapping theorem advertised on (Fukuda 2026g, slide 65); the classical treatment is Rudin (1976, Thm. 9.28).

Formula (8.9) is not an extra: it is obtained by differentiating the identity $f(x, e(x)) = 0$ with respect to x using Theorem 8.3.1,

$$\nabla_x f(x, e(x)) + \nabla_y f(x, e(x)) De(x) = 0,$$

and solving, which is legitimate precisely because (iii) guarantees the inverse exists. The differentiation is valid only once e is known to be differentiable, which is why (iv) is a conclusion of the theorem and not a derivation of it.

Corollary 8.6.1 (Inverse function theorem). Let $A \subseteq \mathbb{R}^n$ be open, $g: A \rightarrow \mathbb{R}^n$ continuously differentiable, and $\bar{x} \in A$ with $Dg(\bar{x})$ nonsingular. Then there are open neighbourhoods $U \ni \bar{x}$ and $V \ni g(\bar{x})$ such that $g|_U: U \rightarrow V$ is a bijection with continuously differentiable inverse, and

$$D(g^{-1})(y) = [Dg(g^{-1}(y))]^{-1}, \quad y \in V.$$

Proof. Apply Theorem 8.6.1 to $f(y, x) := g(x) - y$ on $\mathbb{R}^n \times A$, at the point $(g(\bar{x}), \bar{x})$. Here the roles are exchanged: the “parameter” is y and the “unknown” is x , and $\nabla_x f = Dg(\bar{x})$ is nonsingular by hypothesis. The implicit function e is the local inverse, and (8.9) gives $De(y) = -[Dg]^{-1}(-I) = [Dg]^{-1}$. ■

For $n = 1$ this is Proposition 7.3.5, with $Dg(\bar{x}) \neq 0$ in place of $f'(x_0) \neq 0$. The theorem is local and cannot be made global: the map $g(x_1, x_2) = (e^{x_1} \cos x_2, e^{x_1} \sin x_2)$ has everywhere nonsingular Jacobian and is not injective, being 2π -periodic in x_2 .

Remark 8.6.1 (Comparative statics is the implicit function theorem). The standard comparative-statics exercise has this form. An endogenous vector $y \in \mathbb{R}^m$ and an exogenous vector $x \in \mathbb{R}^n$ satisfy m equilibrium conditions $f(x, y) = 0$ — first-order conditions, market-clearing conditions, or both. At a reference equilibrium $(\bar{x}, \bar{y})$ one asks how y responds to x . The theorem answers three questions at once, and it is worth separating them.

Does an equilibrium exist nearby? Yes, and it varies continuously: this is conclusion (i)–(ii).

Is it locally unique? Yes: this is the uniqueness clause, and it is what makes “the” comparative static well posed. Uniqueness is local only; the theorem says nothing about other equilibria far away.

What is the response? It is (8.9). The economic content is in the two factors. The matrix $\nabla_x f$ collects the direct effects of the parameters on the equilibrium conditions; the matrix $[\nabla_y f]^{-1}$ propagates them through the simultaneous system. When $f = \nabla_y \Pi$ for an objective Π , the propagating matrix is the inverse Hessian, and Theorem 8.5.3 makes it negative definite at a maximum — which is the source of the standard sign restrictions on comparative statics.

The condition that can fail is nonsingularity of $\nabla_y f$. At such a point the equilibrium is not locally unique, comparative statics are undefined, and the model is at a bifurcation.

8.7 Economic Application: The Representative Firm

Environment. A firm rents capital $K \geq 0$ at rate $r > 0$ and hires labour $L \geq 0$ at wage $w > 0$, producing $Y = F(K, L) = AK^\alpha L^{1-\alpha}$ with $A > 0$ and $\alpha \in (0, 1)$. Output is the numeraire. The firm solves (8.1).

Proposition 8.7.1 (First-order conditions and factor shares). Let (K^*, L^*) be an interior solution of (8.1). Then

$$r = \frac{\partial Y}{\partial K} = \alpha A \left(\frac{K^*}{L^*} \right)^{\alpha-1}, \quad w = \frac{\partial Y}{\partial L} = (1 - \alpha) A \left(\frac{K^*}{L^*} \right)^{\alpha}, \quad (8.10)$$

and consequently

$$\alpha = \frac{rK^*}{Y^*}, \quad 1 - \alpha = \frac{wL^*}{Y^*}, \quad \text{and} \quad \Pi(K^*, L^*) = 0. \quad (8.11)$$

Proof. Π is differentiable on $\mathbb{R}_{++}^2$ and (K^*, L^*) is interior, so Theorem 8.4.1 gives $\nabla \Pi = 0$, which is (8.10) after computing the partials as in Example 8.2.1. Multiplying the first equation by K^* gives

$rK^* = \alpha A(K^*)^\alpha (L^*)^{1-\alpha} = \alpha Y^*$, and similarly for labour. Adding, $rK^* + wL^* = Y^*$, so profit is zero. ■

Reading the result. Three points, each with a mathematical source.

The zero-profit conclusion is [Theorem 8.3.2](#) with $k = 1$: F is homogeneous of degree 1, so $\langle \nabla F, (K, L) \rangle = F(K, L)$, and payment of each factor at its marginal product exhausts output exactly. Nothing about Cobb–Douglas is used beyond constant returns.

The factor shares in (8.11) are constants, independent of r, w and A . This is special to Cobb–Douglas, and it is the reason the function is used: it is the production function whose elasticity of substitution between capital and labour equals 1, and constancy of shares is equivalent to that.

The scale of the firm is indeterminate. Equations (8.10) involve K^* and L^* only through their ratio, because the marginal products are homogeneous of degree 0 by the second part of [Theorem 8.3.2](#). Given (r, w) , either no ratio satisfies both equations, or one does and then every scalar multiple of (K^*, L^*) is also optimal. Formally, the Hessian

$$D^2 \Pi = D^2 F = \alpha(1 - \alpha) \frac{Y}{KL} \begin{pmatrix} -L/K & 1 \\ 1 & -K/L \end{pmatrix}$$

has determinant 0 and trace $-\alpha(1 - \alpha) \frac{Y}{KL} (L/K + K/L) < 0$, so it is negative *semidefinite* with a one-dimensional null space spanned by (K, L) — the direction of proportional scaling. The maximum is therefore not strict, and [Theorem 8.5.3\(i\)](#) does not apply. This is not a defect of the mathematics but the correct statement that a constant-returns firm in a competitive market has an indeterminate size: only the aggregate is pinned down, by demand. This is the optional exercise of (Fukuda [2026a](#), slides 43, 54).

8.8 Economic Application: Least Squares by Calculus

Problem. Given data $(x_i, y_i)_{i=1}^n$ with at least two distinct x_i , minimise the sum of squared residuals (8.2).

Proposition 8.8.1. Let $S(\beta_1, \beta_2) := \sum_{i=1}^n (y_i - \beta_1 - \beta_2 x_i)^2$. If the x_i are not all equal, then S has a unique global minimiser, given by

$$\hat{\beta}_2 = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sum_i (x_i - \bar{x})^2}, \quad \hat{\beta}_1 = \bar{y} - \hat{\beta}_2 \bar{x}, \quad (8.12)$$

where $\bar{x} = n^{-1} \sum_i x_i$ and $\bar{y} = n^{-1} \sum_i y_i$.

Proof. S is a polynomial, hence C^∞ . Its partial derivatives are

$$\frac{\partial S}{\partial \beta_1} = -2 \sum_i (y_i - \beta_1 - \beta_2 x_i), \quad \frac{\partial S}{\partial \beta_2} = -2 \sum_i x_i (y_i - \beta_1 - \beta_2 x_i),$$

so $\nabla S = 0$ is the pair of *normal equations*

$$n\beta_1 + \left(\sum_i x_i\right)\beta_2 = \sum_i y_i, \quad \left(\sum_i x_i\right)\beta_1 + \left(\sum_i x_i^2\right)\beta_2 = \sum_i x_i y_i. \quad (8.13)$$

The Hessian is

$$D^2 S = 2 \begin{pmatrix} n & \sum_i x_i \\ \sum_i x_i & \sum_i x_i^2 \end{pmatrix},$$

constant in β , with $\det D^2 S = 4(n \sum_i x_i^2 - (\sum_i x_i)^2) = 4n \sum_i (x_i - \bar{x})^2 > 0$ when the x_i are not all equal, and leading entry $2n > 0$. By [Proposition 2.5.2](#), $D^2 S \succ 0$, so S is strictly convex by [Theorem 8.5.4](#) and the unique stationary point is the global minimiser by [Theorem 8.4.2](#). Solving (8.13) by Cramer's rule ([Example 2.4.2](#)) yields (8.12). ■

What is and is not established. The calculus route delivers existence, uniqueness and a formula, and it makes the role of the data condition explicit: $\sum_i (x_i - \bar{x})^2 > 0$ is simultaneously the nondegeneracy condition on the regressors, the invertibility condition for the normal equations, and the positive definiteness condition for the Hessian. It is the two-variable case of the no-perfect-multicollinearity condition of [Remark 2.3.1](#).

What the calculus route does not deliver is the interpretation. Formula (8.12) says that $\hat{\beta}_2$ is the ratio of a sample covariance to a sample variance, and [Example 3.2.2](#) identified that ratio as a cosine rescaled by norms. The geometry behind it — that the fitted values are the orthogonal projection of y onto the span of the regressors, that the residual is orthogonal to that span, and that the normal equations (8.13) are the orthogonality conditions — is developed in [Chapter 11](#), where the same estimator is obtained without differentiating anything and in a form that survives to infinite dimensions (Fukuda 2026a, slides 55–64).

8.9 Chapter Summary

Partial derivatives differentiate along the coordinate axes; the total derivative requires a single linear map to approximate the increment in every direction at once. In one variable the two notions coincide; in several they do not, and the gap is genuine: a function can have all partial derivatives at a point and be discontinuous there ([Example 8.3.1](#)), or be continuous with all partials and still fail to be differentiable ([Example 8.3.2](#)). Continuity of the partials is sufficient ([Proposition 8.3.3](#)), which is why C^1 is the standard working hypothesis. Differentiability delivers continuity, all directional derivatives, the chain rule in matrix form (8.5), orthogonality of the gradient to level sets, and Euler’s identity for homogeneous functions.

At an interior local extremum of a differentiable function the gradient vanishes; under concavity the condition is also sufficient and global. The second-order theory rests on the Hessian: symmetric under C^2 by [Theorem 8.5.1](#), appearing as the quadratic term of the Taylor expansion (8.8), and determining local behaviour at a stationary point through definiteness, with the semidefinite case inconclusive. Global convexity or concavity is exactly semidefiniteness of the Hessian everywhere.

The implicit function theorem converts a system of equilibrium conditions into a locally unique, continuously differentiable solution mapping with derivative (8.9); it is the theorem underlying essentially all differentiable comparative statics, and its hypothesis — nonsingularity of the Jacobian in the endogenous variables — is exactly what fails at a bifurcation. Applied to the constant-returns firm, the first-order conditions determine the capital–labour ratio, exhaust output through Euler’s identity, and leave scale indeterminate, the last fact readable from the singular Hessian.

Exercise 8.9.1. Let $f(x, y) = x^3y/(x^4 + y^2)$ for $(x, y) \neq (0, 0)$ and $f(0, 0) = 0$. Show that the directional derivative of f at the origin exists and equals 0 in every direction, yet f is not continuous at the origin. Locate the failure relative to [Proposition 8.3.2](#), and explain why this is not a counterexample to it.

Exercise 8.9.2. Let F be C^2 , homogeneous of degree 1, and strictly quasiconcave on $\mathbb{R}_{++}^2$. Show that $D^2 F(K, L)$ is singular with (K, L) in its kernel, and deduce that no interior strict local maximum of $F(K, L) - rK - wL$ exists. Then show that the cost-minimisation problem $\min\{rK + wL : F(K, L) \geq \bar{Y}\}$ does have a unique solution, and explain the difference.

Exercise 8.9.3. A monopolist chooses q facing inverse demand $P(q, \theta)$ with a demand shifter θ , and cost $c(q)$. Write the first-order condition as $f(\theta, q) = 0$ and use (8.9) to compute $dq^*/d\theta$. State the exact sign restriction on $\partial^2 \Pi / \partial q \partial \theta$ that makes the response positive, and verify that the second-order condition supplies the sign of the denominator. Relate the result to [Remark 8.6.1](#).

Chapter 9

Convexity and Separation

9.1 Motivation: What Convexity Buys

Convexity does three things that nothing else in this volume does.

It converts local information into global information. A stationary point of a concave function is a global maximiser ([Theorem 8.4.2](#)), so the first-order conditions that [Chapter 8](#) derived as necessary become sufficient, and the search for an optimum reduces to solving equations.

It delivers uniqueness. Strict concavity of the objective on a convex feasible set makes the solution a point rather than a set ([Proposition 9.3.5](#)), which is what turns a demand correspondence into a demand function and makes comparative statics well posed.

It produces prices. The separation theorems say that two disjoint convex sets can be told apart by a linear functional — a vector of coefficients, one per coordinate. When the coordinates are commodities, that vector is a price system, and the existence of supporting prices in the second welfare theorem, of state prices under no arbitrage, and of Lagrange multipliers in [Chapter 12](#) are three instances of one theorem.

9.2 Convex Sets

Definition 9.2.1 (Convex set). A subset C of a real vector space X is *convex set* if

$$\lambda x + (1 - \lambda)y \in C \quad \text{for all } x, y \in C \text{ and } \lambda \in [0, 1].$$

Equivalently, C contains the segment joining any two of its points. The empty set and every singleton are convex, vacuously and trivially.

Example 9.2.1 (Convex sets in economics). (i) The budget set $\mathcal{B}(p, m) = \{x \in \mathbb{R}_+^n : \langle p, x \rangle \leq m\}$ is convex, being the intersection of the half-space $\{\langle p, x \rangle \leq m\}$ with the cone $\mathbb{R}_+^n$.

(ii) The unit simplex $\Delta_n = \{p \in \mathbb{R}^n : \sum_i p_i = 1, p_i \geq 0\}$, the set of probability distributions on n states, is convex and compact.

(iii) An upper contour set $\{x : u(x) \geq u(x_0)\}$ is convex precisely when u is quasiconcave ([Definition 9.3.2](#)); this is the formal content of “convex preferences”.

(iv) A production set $Y \subseteq \mathbb{R}^n$ is convex when the technology admits time-sharing between any two feasible plans, and convexity of Y is exactly what excludes increasing returns.

Proposition 9.2.1 (Operations preserving convexity). Let $C, D \subseteq \mathbb{R}^n$ be convex, $\lambda, \mu \in \mathbb{R}$, and $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ linear. Then

- (i) $\bigcap_{i \in I} C_i$ is convex for any family of convex sets;
- (ii) $\lambda C + \mu D := \{\lambda x + \mu y : x \in C, y \in D\}$ is convex;

- (iii) $T(C) \subseteq \mathbb{R}^m$ and $T^{-1}(E) \subseteq \mathbb{R}^n$ are convex, the latter for any convex $E \subseteq \mathbb{R}^m$;
- (iv) $\text{Int } C$ and $\overline{C}$ are convex.

Proof. (i) If $x, y \in \bigcap_i C_i$ then the segment lies in each C_i .

(ii) Let $z_k = \lambda x_k + \mu y_k$ for $k = 1, 2$ and $\theta \in [0, 1]$. Then $\theta z_1 + (1 - \theta)z_2 = \lambda(\theta x_1 + (1 - \theta)x_2) + \mu(\theta y_1 + (1 - \theta)y_2)$, and both brackets lie in the respective sets.

(iii) Immediate from linearity of T , which commutes with convex combinations.

(iv) For the closure: let $x, y \in \overline{C}$ and take $(x_k), (y_k) \subseteq C$ with $x_k \rightarrow x, y_k \rightarrow y$ (Theorem 4.3.1). Then $\lambda x_k + (1 - \lambda)y_k \in C$ converges to $\lambda x + (1 - \lambda)y$, which is therefore in $\overline{C}$. For the interior: let $x, y \in \text{Int } C$ with $B_\varepsilon(x), B_\varepsilon(y) \subseteq C$ and set $z = \lambda x + (1 - \lambda)y$. For $\|u\| < \varepsilon$, $z + u = \lambda(x + u) + (1 - \lambda)(y + u) \in C$, so $B_\varepsilon(z) \subseteq C$. ■

This is Proposition 6.3 and the results of Section 7.6 of Fukuda (2026d). Part (i) is the one used most often: a feasible set described by a list of convex constraints is convex, however long the list, and part (ii) is what makes the aggregate of convex production sets convex.

Definition 9.2.2 (Convex combination and convex hull). A point $a \in \mathbb{R}^n$ is a *convex combination* of $x_1, \dots, x_k$ if $a = \sum_{i=1}^k \lambda_i x_i$ with $\lambda_i \geq 0$ and $\sum_{i=1}^k \lambda_i = 1$. The *convex hull* $\text{co } A$ of $A \subseteq \mathbb{R}^n$ is the set of all convex combinations of finitely many points of A ; equivalently, by Proposition 9.2.1(i), the intersection of all convex sets containing A , hence the smallest such set.

Theorem 9.2.1 (Carathéodory). Let $A \subseteq \mathbb{R}^n$. Every point of $\text{co } A$ is a convex combination of at most $n + 1$ points of A .

Proof. Let $x = \sum_{i=1}^k \lambda_i x_i$ with $x_i \in A$, $\lambda_i > 0$, $\sum_i \lambda_i = 1$, and suppose $k \geq n + 2$. The $k - 1 \geq n + 1$ vectors $x_2 - x_1, \dots, x_k - x_1$ lie in $\mathbb{R}^n$ and are therefore linearly dependent: there are $\mu_2, \dots, \mu_k$, not all zero, with $\sum_{i=2}^k \mu_i (x_i - x_1) = 0$. Setting $\mu_1 := -\sum_{i=2}^k \mu_i$ gives $\sum_{i=1}^k \mu_i x_i = 0$ and $\sum_{i=1}^k \mu_i = 0$, with some $\mu_i > 0$ because they sum to zero and are not all zero.

For $t \geq 0$ the coefficients $\lambda_i - t\mu_i$ still sum to 1 and still represent x . Let $t^* := \min\{\lambda_i/\mu_i : \mu_i > 0\}$, attained at some index j . Then $\lambda_i - t^*\mu_i \geq 0$ for every i and equals 0 at $i = j$, so x is a convex combination of the remaining $k - 1$ points. Iterate until $k \leq n + 1$. ■

This is Theorem 6.1 of Fukuda (2026d, p. 190). The bound $n + 1$ is sharp — a point in the interior of a simplex in $\mathbb{R}^n$ needs all $n + 1$ vertices — and the theorem is the reason that finitely many agents suffice to generate any aggregate consumption bundle in an n -commodity economy.

Lemma 9.2.1 (Nearest point in a closed convex set). Let $\emptyset \neq D \subseteq \mathbb{R}^n$ be closed and convex and let $x \in \mathbb{R}^n$. Then there is exactly one $P_D x \in D$ with

$$\|x - P_D x\| = \text{dist}(x, D) := \inf_{y \in D} \|x - y\|,$$

and $P_D x$ is characterised by the variational inequality

$$\langle x - P_D x, y - P_D x \rangle \leq 0 \quad \text{for all } y \in D. \quad (9.1)$$

Proof. Existence. Fix $y_0 \in D$ and set $R := \|x - y_0\|$. The set $D \cap \overline{B_R(x)}$ is non-empty, closed and bounded, hence compact by Theorem 4.4.2, and the continuous function $y \mapsto \|x - y\|$ attains its minimum there by Corollary 5.4.1. That minimum is the infimum over all of D , since points outside the ball are further away than y_0 .

Uniqueness. Let $d := \text{dist}(x, D)$ and suppose $\|x - y_1\| = \|x - y_2\| = d$ with $y_1 \neq y_2$. The midpoint $\bar{y} = \frac{1}{2}(y_1 + y_2)$ lies in D by convexity, and the parallelogram law (Remark 3.3.1) gives

$$\|x - \bar{y}\|^2 = \frac{1}{2} \|x - y_1\|^2 + \frac{1}{2} \|x - y_2\|^2 - \frac{1}{4} \|y_1 - y_2\|^2 = d^2 - \frac{1}{4} \|y_1 - y_2\|^2 < d^2,$$

contradicting minimality of d .

Characterisation. Let $z := P_D x$, $y \in D$ and $t \in (0, 1]$. Then $z + t(y - z) \in D$, so

$$\|x - z\|^2 \leq \|x - z - t(y - z)\|^2 = \|x - z\|^2 - 2t \langle x - z, y - z \rangle + t^2 \|y - z\|^2.$$

Cancelling, dividing by $t > 0$ and letting $t \downarrow 0$ gives (9.1). Conversely, if $z \in D$ satisfies (9.1), then for any $y \in D$,

$$\|x - y\|^2 = \|x - z\|^2 - 2\langle x - z, y - z \rangle + \|y - z\|^2 \geq \|x - z\|^2,$$

so z is the minimiser. ■

This is the finite-dimensional case of Theorem 9.1 of Fukuda (2026d, p. 295). The Hilbert-space version, in which compactness is unavailable and completeness does the work instead, is Theorem 11.3.1; the uniqueness and variational arguments above carry over verbatim, and only the existence argument changes.

9.3 Convex and Concave Functions

Definition 9.3.1 (Convexity and concavity). Let $C \subseteq \mathbb{R}^n$ be convex. A function $f: C \rightarrow \mathbb{R}$ is *convex function* on $\mathbb{R}^n$ if

$$f(\lambda x + (1 - \lambda)y) \leq \lambda f(x) + (1 - \lambda)f(y) \quad \text{for all } x, y \in C, \lambda \in [0, 1],$$

strictly convex if the inequality is strict for $x \neq y$ and $\lambda \in (0, 1)$, and *concave function* on $\mathbb{R}^n$ (respectively *strictly concave*) if $-f$ is convex (respectively strictly convex).

Proposition 9.3.1 (Epigraph and Jensen). Let $C \subseteq \mathbb{R}^n$ be convex and $f: C \rightarrow \mathbb{R}$.

- (i) f is convex if and only if $\text{epi } f = \{(x, z) \in C \times \mathbb{R} : f(x) \leq z\}$ is a convex subset of $\mathbb{R}^{n+1}$.
- (ii) If f is convex, then for all $x_1, \dots, x_k \in C$ and weights $\lambda_i \geq 0$ with $\sum_i \lambda_i = 1$,

$$f\left(\sum_{i=1}^k \lambda_i x_i\right) \leq \sum_{i=1}^k \lambda_i f(x_i). \quad (9.2)$$

Proof. (i) Suppose f convex and $(x_1, z_1), (x_2, z_2) \in \text{epi } f$. For $\lambda \in [0, 1]$, $f(\lambda x_1 + (1 - \lambda)x_2) \leq \lambda f(x_1) + (1 - \lambda)f(x_2) \leq \lambda z_1 + (1 - \lambda)z_2$, so the convex combination is in $\text{epi } f$. Conversely, apply convexity of $\text{epi } f$ to the points $(x_i, f(x_i))$.

(ii) Induction on k . The case $k = 2$ is the definition. For the step, if $\lambda_k < 1$ write $\sum_{i \leq k} \lambda_i x_i = (1 - \lambda_k) \sum_{i < k} \frac{\lambda_i}{1 - \lambda_k} x_i + \lambda_k x_k$, apply the definition, and then the inductive hypothesis to the inner combination, whose weights sum to 1. ■

Combining Proposition 9.3.1(i) with Proposition 5.3.2 gives the identification announced in Chapter 5: f is convex and lower semicontinuous exactly when $\text{epi } f$ is a closed convex set. Inequality (9.2) is Jensen's inequality for a finitely supported distribution; the general version, for an arbitrary distribution and for conditional expectations, is Proposition 14.4.2.

Proposition 9.3.2 (Operations). Let $C \subseteq \mathbb{R}^n$ be convex.

- (i) If f, g are convex and $\alpha, \beta \geq 0$, then $\alpha f + \beta g$ is convex.
- (ii) If $(f_i)_{i \in I}$ are convex and $f = \sup_{i \in I} f_i$ is finite on C , then f is convex. The corresponding statement for infima of concave functions holds; suprema of concave functions need not be concave.
- (iii) If f is convex and $\varphi: \mathbb{R} \rightarrow \mathbb{R}$ is convex and non-decreasing, then $\varphi \circ f$ is convex.
- (iv) If T is affine, $f \circ T$ is convex whenever f is.

Proof. (i) and (iv) are immediate from the definition. (ii) is $\text{epi}(\sup_i f_i) = \bigcap_i \text{epi } f_i$ together with Proposition 9.2.1(i) and Proposition 9.3.1(i). (iii): $\varphi(f(\lambda x + (1 - \lambda)y)) \leq \varphi(\lambda f(x) + (1 - \lambda)f(y)) \leq \lambda \varphi(f(x)) + (1 - \lambda)\varphi(f(y))$, the first step by monotonicity of φ and convexity of f , the second by convexity of φ . Monotonicity cannot be dropped: $\varphi(t) = t^2$ is convex, $f(x) = -x$ is convex, and $\varphi \circ f = x^2$ happens to be convex, but with $f(x) = |x| - 1$ and $\varphi(t) = -t$ the composition is concave. ■

Part (ii) is the source of the profit function's convexity in prices, as [Exercise 5.6.3](#) noted: $\pi(p) = \sup_{y \in Y} \langle p, y \rangle$ is a supremum of linear, hence convex, functions of p , and the conclusion holds for any production set whatsoever, convex or not.

Proposition 9.3.3 (Continuity). A convex function on an open convex set $C \subseteq \mathbb{R}^n$ is continuous on C , and Lipschitz on every compact subset of C .

The proof is a standard argument bounding a convex function on a simplex by its values at the vertices; see Rockafellar (1970, Thm. 10.4). The openness hypothesis is essential: $f(x) = 0$ on $[0, 1)$ and $f(1) = 1$ is convex on $[0, 1]$ and discontinuous at the endpoint. In economics this is why boundary behaviour of convex cost or value functions requires separate attention even when the interior behaviour is smooth.

Theorem 9.3.1 (Differentiable characterisation). Let $C \subseteq \mathbb{R}^n$ be open and convex and $f: C \rightarrow \mathbb{R}$ differentiable. The following are equivalent.

- (i) f is concave on C .
- (ii) $f(x) \leq f(x_0) + \langle \nabla f(x_0), x - x_0 \rangle$ for all $x, x_0 \in C$.
- (iii) $\langle \nabla f(x) - \nabla f(x_0), x - x_0 \rangle \leq 0$ for all $x, x_0 \in C$, that is, ∇f is monotone decreasing.

If f is twice differentiable, each is equivalent to $D^2 f \preceq 0$ on C ([Theorem 8.5.4](#)). The convex case is obtained by reversing every inequality.

Proof. Fix $x, x_0 \in C$ and set $\varphi(t) := f(x_0 + t(x - x_0))$ on an open interval containing $[0, 1]$. By [Theorem 8.3.1](#), $\varphi'(t) = \langle \nabla f(x_0 + t(x - x_0)), x - x_0 \rangle$.

f is concave on C if and only if every such φ is concave, since the defining inequality for f at the pair (x, x_0) is the inequality for φ at the pair $(1, 0)$ and conversely. By [Theorem 7.2.2](#), φ concave is equivalent to $\varphi(1) \leq \varphi(0) + \varphi'(0)$, which is (ii), and to φ' non-increasing, which evaluated at $t = 0$ and $t = 1$ is (iii). ■

This is Theorem 11.3 of Fukuda (2026d, p. 388). Statement (ii) is the tangent-plane picture of (Fukuda 2026a, slides 8, 52): a concave function lies below each of its tangent hyperplanes, which is why a horizontal tangent certifies a global maximum.

9.3.1 Quasiconcavity

Definition 9.3.2 (Quasiconcavity). Let $C \subseteq \mathbb{R}^n$ be convex. A function $f: C \rightarrow \mathbb{R}$ is *quasiconcave function* if every upper contour set $\{x \in C : f(x) \geq \alpha\}$ is convex, equivalently if

$$f(\lambda x + (1 - \lambda)y) \geq \min\{f(x), f(y)\} \quad \text{for all } x, y \in C, \lambda \in [0, 1].$$

It is *strictly quasiconcave* if the inequality is strict whenever $x \neq y$ and $\lambda \in (0, 1)$. Quasiconvexity is the same with lower contour sets and max.

Proposition 9.3.4. Every concave function is quasiconcave, and quasiconcavity is preserved by strictly increasing transformations of the values, whereas concavity is not.

Proof. If f is concave then $f(\lambda x + (1 - \lambda)y) \geq \lambda f(x) + (1 - \lambda)f(y) \geq \min\{f(x), f(y)\}$. For the second claim, let ψ be strictly increasing. The upper contour sets of $\psi \circ f$ are upper contour sets of f , since $\psi(f(x)) \geq \alpha$ if and only if $f(x) \geq \psi^{-1}(\alpha)$ on the range; convexity of the latter is quasiconcavity of f . That concavity is not preserved is shown by $f(x) = x$ on $\mathbb{R}$, concave, and $\psi(t) = t^3$, giving the non-concave x^3 . ■

The asymmetry in [Proposition 9.3.4](#) is the reason quasiconcavity, not concavity, is the right hypothesis on a utility function. Utility is an ordinal representation of a preference relation, unique only up to strictly increasing transformation, and convexity of preferences — convexity of the sets $\{y : y \succ x\}$ — is exactly quasiconcavity of every representation. Concavity, by contrast, is a property of a particular representation and has no preference-level meaning.

Proposition 9.3.5 (Uniqueness of the maximiser). Let $C \subseteq \mathbb{R}^n$ be convex and $f: C \rightarrow \mathbb{R}$ strictly quasiconcave. Then f has at most one maximiser on C . If in addition C is compact and non-empty and f is upper semicontinuous, the maximiser exists and is unique.

Proof. Suppose $x \neq y$ both maximise f on C , with common value M . Then $\frac{1}{2}x + \frac{1}{2}y \in C$ and strict quasiconcavity gives $f(\frac{1}{2}x + \frac{1}{2}y) > \min\{f(x), f(y)\} = M$, contradicting maximality. Existence is [Theorem 5.4.1\(ii\)](#). ■

Proposition 9.3.6 (Differentiable quasiconcavity). Let $C \subseteq \mathbb{R}^n$ be open and convex and $f: C \rightarrow \mathbb{R}$ differentiable. Then f is quasiconcave if and only if

$$f(x) \geq f(x_0) \implies \langle \nabla f(x_0), x - x_0 \rangle \geq 0 \quad \text{for all } x, x_0 \in C.$$

This is Proposition 11.7 of Fukuda (2026d, p. 396). The condition says that moving from x_0 towards any weakly preferred point does not initially decrease f ; geometrically the gradient at x_0 supports the upper contour set at x_0 . It is weaker than [Theorem 9.3.1\(ii\)](#), which bounds f globally, and correspondingly it does *not* make a stationary point a global maximiser: $f(x) = x^3$ is quasiconcave on $\mathbb{R}$ with $f'(0) = 0$ and 0 is not a maximiser. Under quasiconcavity the first-order condition remains only necessary, which is why [Chapter 12](#) states its sufficiency results under concavity and treats quasiconcavity separately.

9.4 Hyperplanes and Separation

Definition 9.4.1 (Hyperplane and half-spaces). A *hyperplane* in $\mathbb{R}^n$ is a set of the form

$$H_{a,\alpha} := \{x \in \mathbb{R}^n : \langle a, x \rangle = \alpha\}, \quad a \in \mathbb{R}^n \setminus \{0\}, \alpha \in \mathbb{R},$$

equivalently an affine subspace of dimension $n - 1$. It determines the two closed *half-space* $\{\langle a, x \rangle \geq \alpha\}$ and $\{\langle a, x \rangle \leq \alpha\}$. The hyperplane *separating* A and B if

$$\langle a, x \rangle \geq \alpha \geq \langle a, y \rangle \quad \text{for all } x \in A, y \in B,$$

and *strict separation* them if both inequalities can be taken strict with a gap, that is if $\sup_{y \in B} \langle a, y \rangle < \alpha < \inf_{x \in A} \langle a, x \rangle$.

That every hyperplane has the stated form is Proposition 10.1 of Fukuda (2026d, p. 334), and it is [Theorem 3.2.1](#) in disguise: an affine subspace of codimension 1 is a level set of a non-zero linear functional, and every such functional is an inner product with a fixed vector.

Lemma 9.4.1 (Separating a point from a closed convex set). Let $D \subseteq \mathbb{R}^n$ be non-empty, closed and convex and let $x \notin D$. Then there are $a \in \mathbb{R}^n \setminus \{0\}$ and $\alpha \in \mathbb{R}$ with

$$\langle a, x \rangle > \alpha > \langle a, y \rangle \quad \text{for all } y \in D.$$

Proof. Let $z := P_D x$ from [Lemma 9.2.1](#) and put $a := x - z \neq 0$, since $x \notin D$ and $z \in D$. By (9.1), $\langle a, y - z \rangle \leq 0$, that is $\langle a, y \rangle \leq \langle a, z \rangle$ for every $y \in D$. Also

$$\langle a, x \rangle - \langle a, z \rangle = \langle a, x - z \rangle = \|a\|^2 > 0.$$

Any α strictly between $\langle a, z \rangle$ and $\langle a, x \rangle$ — for instance $\alpha = \langle a, z \rangle + \frac{1}{2} \|a\|^2$ — works. ■

Theorem 9.4.1 (Strict separation). Let $C, D \subseteq \mathbb{R}^n$ be non-empty, convex and disjoint, with C compact and D closed. Then there is a hyperplane strictly separating them.

Proof. The set $E := C + (-D) = \{x - y : x \in C, y \in D\}$ is convex by [Proposition 9.2.1\(ii\)](#), and $0 \notin E$ because $C \cap D = \emptyset$. It is closed: if $x_k - y_k \rightarrow w$ with $x_k \in C$ and $y_k \in D$, compactness of C gives a subsequence $x_{k_j} \rightarrow x \in C$, whence $y_{k_j} = x_{k_j} - (x_{k_j} - y_{k_j}) \rightarrow x - w$, which lies in D because D is closed; so $w = x - (x - w) \in E$.

Apply [Lemma 9.4.1](#) to the closed convex set E and the point $0 \notin E$: there are $b \neq 0$ and α_0 with $0 = \langle b, 0 \rangle > \alpha_0 > \langle b, w \rangle$ for every $w \in E$, that is

$$\langle b, x \rangle - \langle b, y \rangle < \alpha_0 < 0 \quad \text{for all } x \in C, y \in D.$$

Taking the supremum over $x \in C$ and the infimum over $y \in D$,

$$\sup_{x \in C} \langle b, x \rangle \leq \inf_{y \in D} \langle b, y \rangle + \alpha_0 < \inf_{y \in D} \langle b, y \rangle,$$

so the two quantities are separated by the strictly positive gap $|\alpha_0|$. Setting $a := -b$ and taking α to be -1 times the midpoint of the two bounds gives $\inf_{x \in C} \langle a, x \rangle > \alpha > \sup_{y \in D} \langle a, y \rangle$, which is strict separation in the orientation of [Definition 9.4.1](#). ■

Theorem 9.4.2 (Supporting hyperplane). Let $C \subseteq \mathbb{R}^n$ be non-empty and convex, not necessarily closed, and let $x_0 \in \partial C$. Then there is $a \in \mathbb{R}^n \setminus \{0\}$ with

$$\langle a, x \rangle \leq \langle a, x_0 \rangle \quad \text{for all } x \in C.$$

Proof. Since $x_0 \in \partial C = \overline{C} \setminus \text{Int } C$, there is a sequence $(x_k) \subseteq (\overline{C})^c$ with $x_k \rightarrow x_0$. For each k , [Lemma 9.4.1](#) applied to the closed convex set $\overline{C}$ and the point x_k supplies $a_k \neq 0$ with $\langle a_k, x \rangle < \langle a_k, x_k \rangle$ for all $x \in \overline{C}$; normalising, $\|a_k\| = 1$. The unit sphere in $\mathbb{R}^n$ is compact by [Theorem 4.4.2](#), so a subsequence converges, $a_{k_j} \rightarrow a$ with $\|a\| = 1$, hence $a \neq 0$. For fixed $x \in C$, $\langle a_{k_j}, x \rangle < \langle a_{k_j}, x_{k_j} \rangle$ for every j ; letting $j \rightarrow \infty$ and using continuity of the inner product gives $\langle a, x \rangle \leq \langle a, x_0 \rangle$. ■

This is Theorem 10.3 of Fukuda (2026d, p. 343). The proof is an instance of a recurring pattern: obtain the result at nearby points where a stronger theorem applies, normalise, and extract a convergent subsequence by compactness of the sphere — a step available only in finite dimensions, which is why the infinite-dimensional supporting hyperplane theorem requires non-empty interior or a different argument (Aliprantis and Border 2006, Ch. 7). [Figure 9.1](#) contrasts the two conclusions: strict separation puts a gap on each side of the hyperplane, support allows the hyperplane to touch.

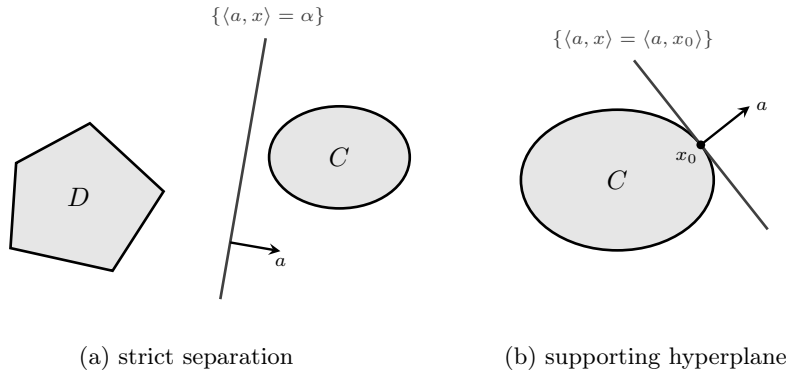


Figure 9.1: The two geometries. In (a) the sets are disjoint, one of them compact, and the hyperplane passes strictly between them with a gap on each side, so $\sup_{y \in D} \langle a, y \rangle < \alpha < \inf_{x \in C} \langle a, x \rangle$ ([Theorem 9.4.1](#)). In (b) the hyperplane touches C at a boundary point and leaves the whole set in one closed half-space, so $\langle a, x \rangle \leq \langle a, x_0 \rangle$ for every $x \in C$ ([Theorem 9.4.2](#)). In both panels a points into the half-space on which $\langle a, \cdot \rangle$ is larger, which is toward C in (a) and away from C in (b): the two statements orient the normal oppositely, and replacing a by $-a$ converts either into the other.

Theorem 9.4.3 (Separation). Let $C, D \subseteq \mathbb{R}^n$ be non-empty, convex and disjoint. Then there is a hyperplane separating them; moreover the separating inequality is strict on $\text{Int } C$ whenever $\text{Int } C \neq \emptyset$.

Proof. As above, $E = C + (-D)$ is convex with $0 \notin E$. If $0 \notin \overline{E}$, apply [Lemma 9.4.1](#) to $\overline{E}$, which is convex by [Proposition 9.2.1\(iv\)](#). If $0 \in \overline{E}$, then $0 \in \partial E$, since $0 \notin E$ excludes $0 \in \text{Int } E$, and [Theorem 9.4.2](#) applied to E at its boundary point 0 supplies $a \neq 0$ with $\langle a, w \rangle \leq \langle a, 0 \rangle = 0$ for all $w \in E$. In both

cases $\langle a, x \rangle \leq \langle a, y \rangle$ for all $x \in C$ and $y \in D$, so with $\alpha := \sup_{x \in C} \langle a, x \rangle$, which is finite and at most $\inf_{y \in D} \langle a, y \rangle$,

$$\langle a, x \rangle \leq \alpha \leq \langle a, y \rangle \quad \text{for all } x \in C, y \in D,$$

which is separation in the sense of [Definition 9.4.1](#), with the roles of the two sets interchanged.

Strictness on $\text{Int } C$: suppose $\langle a, x_0 \rangle = \alpha$ for some $x_0 \in \text{Int } C$. Then $x_0 + \varepsilon a \in C$ for small $\varepsilon > 0$, and $\langle a, x_0 + \varepsilon a \rangle = \alpha + \varepsilon \|a\|^2 > \alpha$, contradicting the inequality just established. ■

These are Theorems 10.1 and 10.2 of Fukuda (2026d, pp. 337, 339), and Boyd and Vandenberghe (2004, Sec. 2.5, p. 46) states the same pair in the form used by the optimisation literature; the compactness hypothesis in [Theorem 9.4.1](#) is what upgrades separation to strict separation and cannot be dropped. The standard witness is $C = \{(x_1, x_2) : x_2 \geq e^{x_1}\}$ and $D = \{(x_1, 0) : x_1 \in \mathbb{R}\}$: both are closed and convex and disjoint, the separating hyperplane $x_2 = 0$ is unique, and no strict separation exists because $\text{dist}(C, D) = 0$.

9.5 The Farkas–Minkowski Theorem

Theorem 9.5.1 (Farkas–Minkowski). Let $f, g_1, \dots, g_m : \mathbb{R}^n \rightarrow \mathbb{R}$ be concave, and let $q \in \{0, 1, \dots, m\}$ be such that g_i is affine for $i \in \{q+1, \dots, m\}$. Assume the system

$$g_i(x) > 0 \quad (i \leq q), \quad g_i(x) \geq 0 \quad (i > q) \quad (9.3)$$

has a solution. Then the system

$$f(x) > 0 \quad \text{and} \quad g_i(x) \geq 0 \quad (i = 1, \dots, m) \quad (9.4)$$

has no solution in $\mathbb{R}^n$ if and only if there exists $y = (y_1, \dots, y_m) \in \mathbb{R}_+^m$ such that

$$\sup_{x \in \mathbb{R}^n} \left\{ f(x) + \sum_{i=1}^m y_i g_i(x) \right\} \leq 0.$$

This is Theorem 10.9 of Fukuda (2026d, p. 353), proved there from [Theorem 9.4.3](#) applied to the convex set $\{(f(x), g_1(x), \dots, g_m(x)) + \mathbb{R}_{-}^{m+1} : x \in \mathbb{R}^n\}$ and the non-negative orthant. The proof is not reproduced here; it is a page of careful bookkeeping with no additional economic content, and the statement is used only through [Chapter 12](#).

The theorem is the mechanism by which multipliers appear. Read (9.4) as “no feasible point strictly improves on the candidate”: f is the improvement in the objective and the g_i are the constraints. The conclusion produces non-negative numbers y_i — the multipliers — such that the aggregate $f + \sum_i y_i g_i$ is nowhere positive, which is the Lagrangian inequality of the Kuhn–Tucker theorem. Condition (9.3) is Slater’s constraint qualification, and [Example 9.5.1](#) shows what goes wrong without it.

Example 9.5.1 (Slater’s condition is not decorative). Take $n = 1$, $m = 1$, $g_1(x) = -x^2$, and $f(x) = x$. The constraint set is $\{x : -x^2 \geq 0\} = \{0\}$, and (9.3) fails: no x has $-x^2 > 0$. The system (9.4) indeed has no solution, since it requires $x > 0$ and $x = 0$. But no $y \geq 0$ makes $\sup_x \{x - yx^2\} \leq 0$: for $y > 0$ the supremum is $1/(4y) > 0$, and for $y = 0$ it is $+\infty$. The conclusion of [Theorem 9.5.1](#) fails, and with it the existence of a multiplier at the constrained optimum $x = 0$.

9.6 Economic Applications

9.6.1 Supporting prices

Environment. An economy has n commodities, a single consumer with convex preferences represented by a quasiconcave, locally non-satiated utility $u : \mathbb{R}_+^n \rightarrow \mathbb{R}$, and an aggregate production set $Y \subseteq \mathbb{R}^n$ that is convex and contains 0. An allocation is a pair (x, y) with $x \in \mathbb{R}_+^n$, $y \in Y$ and $x \leq \omega + y$, where $\omega \in \mathbb{R}_+^n$ is the endowment.

Proposition 9.6.1 (Second welfare theorem, one-consumer case). Let (x^*, y^*) be a Pareto efficient allocation with $x^* \in \mathbb{R}_{++}^n$. Suppose u is quasiconcave and continuous and Y is convex. Then there

is a price vector $p \in \mathbb{R}^n \setminus \{0\}$ such that

- (i) $u(x) > u(x^*)$ implies $\langle p, x \rangle \geq \langle p, x^* \rangle$; and
- (ii) $\langle p, y \rangle \leq \langle p, y^* \rangle$ for all $y \in Y$.

Proof. Let $B := \{x \in \mathbb{R}_+^n : u(x) > u(x^*)\}$, convex by quasiconcavity of u and [Definition 9.3.2](#), and let $A := \omega + Y$, convex because Y is. Efficiency of (x^*, y^*) says exactly that no point of A dominates a point of B componentwise; setting $C := A - \mathbb{R}_+^n$, which is convex by [Proposition 9.2.1\(ii\)](#), efficiency gives $B \cap \text{Int } C = \emptyset$. Since B and $\text{Int } C$ are disjoint convex sets, [Theorem 9.4.3](#) supplies $p \neq 0$ and α with $\langle p, x \rangle \geq \alpha \geq \langle p, z \rangle$ for $x \in B$ and $z \in \text{Int } C$. Continuity of u makes x^* a limit of points of B , so $\langle p, x^* \rangle \geq \alpha$; and $x^* \in \bar{C}$ gives $\langle p, x^* \rangle \leq \alpha$. Hence $\alpha = \langle p, x^* \rangle$, which is (i). For (ii), $\omega + y - x^* \in C$ for every $y \in Y$ after subtracting a non-negative vector, and the same inequality applies. ■

Reading the hypotheses. Convexity of Y is what makes the theorem true, and it fails under increasing returns: with a non-convex production set an efficient allocation need not be supportable by any price vector, which is the formal basis of the case for regulation of natural monopoly. Quasiconcavity of u — convexity of preferences — plays the same role on the consumption side; without it the separating hyperplane at x^* may cut through the preferred set, and the consumer facing p would choose a different bundle. Neither convexity assumption can be weakened to something merely topological: [Theorem 9.4.3](#) is a theorem about convex sets and there is no version of it for arbitrary disjoint sets.

Note also what [Proposition 9.6.1](#) does *not* say. It gives $\langle p, x \rangle \geq \langle p, x^* \rangle$ for strictly preferred x , which is cost-minimisation, not utility maximisation. Upgrading it to “ x^* maximises u subject to $\langle p, x \rangle \leq \langle p, x^* \rangle$ ” requires that the budget set have a cheaper point, which is the standard cheaper-point condition and fails when $\langle p, x^* \rangle = \min_{x \geq 0} \langle p, x \rangle = 0$.

9.6.2 No arbitrage and state prices

Environment. There are S states tomorrow and J assets. Asset j pays R_{sj} in state s ; write $R \in M_{S \times J}(\mathbb{R})$. A portfolio is $\theta \in \mathbb{R}^J$, costing $\langle q, \theta \rangle$ today at prices $q \in \mathbb{R}^J$ and paying $R\theta \in \mathbb{R}^S$ tomorrow.

Definition 9.6.1 (Arbitrage). An *arbitrage* is a portfolio θ with $\langle q, \theta \rangle \leq 0$ and $R\theta \geq 0$, at least one of the $S + 1$ inequalities being strict.

Theorem 9.6.1 (Fundamental theorem of asset pricing). There is no arbitrage if and only if there exists a strictly positive state price vector $\psi \in \mathbb{R}_{++}^S$ with

$$q = R^\top \psi, \quad \text{that is} \quad q_j = \sum_{s=1}^S \psi_s R_{sj} \quad \text{for every } j. \quad (9.5)$$

Proof. ($\Leftarrow$) Suppose (9.5) holds and θ satisfies $R\theta \geq 0$. Then $\langle q, \theta \rangle = \langle R^\top \psi, \theta \rangle = \langle \psi, R\theta \rangle \geq 0$, with strict inequality if $R\theta \neq 0$ because $\psi \gg 0$. So no arbitrage exists.

($\Rightarrow$) Consider in $\mathbb{R}^{1+S}$ the subspace $M := \{(-\langle q, \theta \rangle, R\theta) : \theta \in \mathbb{R}^J\}$, which is a linear subspace hence convex and closed ([Corollary 6.3.1](#)), and the set $K := \{v \in \mathbb{R}_+^{1+S} : \sum_i v_i = 1\}$, which is convex and compact. No arbitrage says $M \cap K = \emptyset$: a point of the intersection would be a portfolio with $-\langle q, \theta \rangle \geq 0$, $R\theta \geq 0$ and not all components zero. By [Theorem 9.4.1](#) there are $\phi = (\phi_0, \psi) \in \mathbb{R}^{1+S} \setminus \{0\}$ and α with

$$\langle \phi, v \rangle > \alpha > \langle \phi, w \rangle \quad \text{for all } v \in K, w \in M.$$

M is a subspace, so $\langle \phi, w \rangle$ is bounded above on it only if it vanishes identically; hence $\langle \phi, w \rangle = 0$ for all $w \in M$ and $\alpha < 0 < \langle \phi, v \rangle$ for all $v \in K$. Taking v to be each standard basis vector of $\mathbb{R}^{1+S}$, which lies in K , gives $\phi_0 > 0$ and $\psi \gg 0$. Applying $\langle \phi, w \rangle = 0$ to w generated by $\theta = e_j$ gives $-\phi_0 q_j + \langle \psi, R_{\cdot j} \rangle = 0$, that is $q = R^\top(\psi/\phi_0)$. Replacing ψ by $\psi/\phi_0 > 0$ completes the proof. ■

Economic content. The state prices ψ_s are the prices today of one unit of consumption in state s tomorrow, and (9.5) says every asset is priced by its payoff-weighted sum of state prices. Strict positivity of ψ is the whole content: it is equivalent to the absence of arbitrage, and it is what permits the normalisation $\psi_s = \beta \pi_s m_s$ into a discount factor, a probability and a stochastic discount factor, giving

the Euler equation form used in [subsection 6.6.2](#). If the market is complete — rank $R = S$ — then ψ is unique, since $R^\top$ is then injective on $\mathbb{R}^S$ by [Theorem 2.3.2](#); if incomplete, the set of consistent state price vectors is the non-empty relatively open convex set $\{\psi \gg 0 : R^\top \psi = q\}$, and asset prices outside the span of R are not determined by no arbitrage alone. This is the connection to [Example 3.2.4](#), where the same vector appeared as the Riesz representation of the pricing functional.

9.7 Chapter Summary

Convex sets are closed under intersection, sums, scalar multiples, linear images and preimages, and interiors and closures; convex hulls in $\mathbb{R}^n$ need at most $n + 1$ points per element ([Theorem 9.2.1](#)). A non-empty closed convex set contains a unique nearest point to any given point, characterised by the variational inequality (9.1), and that lemma is the engine of every separation result: a point outside a closed convex set is strictly separated from it, disjoint convex sets with one compact and one closed are strictly separated, disjoint convex sets are separated, and a boundary point of a convex set carries a supporting hyperplane.

Convex functions are those with convex epigraph, are stable under non-negative combinations, arbitrary suprema, and composition with increasing convex maps, and are continuous on the interior of their domain. For differentiable functions, concavity is the tangent-plane inequality, is monotonicity of the gradient, and is negative semidefiniteness of the Hessian. Quasiconcavity — convexity of upper contour sets — is the ordinal weakening appropriate to utility, is preserved by increasing transformations, and yields uniqueness of the maximiser in its strict form but not sufficiency of the first-order condition.

The Farkas–Minkowski theorem converts the non-existence of an improving feasible direction into the existence of non-negative multipliers, under a Slater condition that cannot be dropped. Separation delivers supporting prices for a Pareto efficient allocation when both preferences and technology are convex, and delivers strictly positive state prices exactly when no arbitrage exists.

Exercise 9.7.1. Show that the set of positive semidefinite symmetric $n \times n$ matrices is a closed convex cone, and that $A \mapsto \lambda_{\min}(A)$ is concave on it. Deduce that $\{A : A \succeq \varepsilon I\}$ is convex for each $\varepsilon \geq 0$, and relate this to [Proposition 2.5.1](#).

Exercise 9.7.2. Let $u : \mathbb{R}_+^n \rightarrow \mathbb{R}$ be continuous and quasiconcave, and let $v(p, m) = \max_{x \in \mathcal{B}(p, m)} u(x)$. Show that v is quasiconvex in (p, m) : that is, $\{(p, m) : v(p, m) \leq \alpha\}$ is convex. (Hint: show that $\mathcal{B}(\lambda p + (1-\lambda)p', \lambda m + (1-\lambda)m') \subseteq \mathcal{B}(p, m) \cup \mathcal{B}(p', m')$.) Explain why the conclusion is quasiconvexity rather than convexity, and contrast with the profit function of [Exercise 5.6.3](#).

Exercise 9.7.3. In the setting of [Theorem 9.6.1](#), suppose $S = 3$, $J = 2$, and $R = \begin{pmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{pmatrix}$ with $q = (1, 2)^\top$. Determine the set of state price vectors consistent with no arbitrage, verify that it is a non-empty relatively open convex set, and compute the interval of prices at which a claim paying $(0, 1, 0)^\top$ can trade without creating an arbitrage. Identify where completeness fails.

Chapter 10

Integration

10.1 Motivation: Moments of a Distribution

Expectations are integrals. The question that organises this chapter is the one posed on (Fukuda 2026a, slide 65): if X is normally distributed with density

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right), \quad (10.1)$$

confirm that $\mathbb{E}[X] = \mu$ and $\text{Var}[X] = \sigma^2$. Doing so requires three tools — integration by parts, integration by substitution, and a change of variables in the plane — and, before any of them, a definition of the integral precise enough that the manipulations are licensed rather than merely plausible.

10.2 The Riemann Integral

Definition 10.2.1 (Partitions and Darboux sums). Let $f: [a, b] \rightarrow \mathbb{R}$ be bounded. A *partition* of $[a, b]$ is a finite set $P = \{a = t_0 < t_1 < \cdots < t_k = b\}$. Writing $m_i := \inf_{t \in [t_{i-1}, t_i]} f(t)$ and $M_i := \sup_{t \in [t_{i-1}, t_i]} f(t)$, the *lower* and *upper Darboux sums* are

$$L(f, P) := \sum_{i=1}^k m_i(t_i - t_{i-1}), \quad U(f, P) := \sum_{i=1}^k M_i(t_i - t_{i-1}).$$

The function is *Riemann integrable* on $[a, b]$ if $\sup_P L(f, P) = \inf_P U(f, P)$, and the common value is $\int_a^b f(x) dx$.

Boundedness is part of the definition: an unbounded f has $M_i = +\infty$ on some subinterval of every partition. Integrals over unbounded domains, or of unbounded functions, are defined as limits in Definition 10.3.2.

Proposition 10.2.1 (Riemann's criterion). A bounded $f: [a, b] \rightarrow \mathbb{R}$ is Riemann integrable if and only if for every $\varepsilon > 0$ there is a partition P with $U(f, P) - L(f, P) < \varepsilon$.

Proof. Refining a partition cannot increase U or decrease L , and any two partitions have a common refinement, so $L(f, P) \leq U(f, Q)$ for all P, Q and hence $\sup_P L \leq \inf_Q U$. If the criterion holds, the two are within ε of each other for every ε , hence equal. Conversely, if they are equal to I , choose P_1 with $L(f, P_1) > I - \varepsilon/2$ and P_2 with $U(f, P_2) < I + \varepsilon/2$ and take their common refinement. ■

Proposition 10.2.2 (Sufficient conditions for integrability). Let $f: [a, b] \rightarrow \mathbb{R}$.

- (i) If f is continuous, it is Riemann integrable.
- (ii) If f is bounded with finitely many discontinuities, it is Riemann integrable.

(iii) If f is monotone, it is Riemann integrable.

Proof. (i) $[a, b]$ is compact, so f is bounded by [Corollary 5.4.1](#) and uniformly continuous by [Theorem 5.2.2](#). Given $\varepsilon > 0$ choose $\delta > 0$ with $|f(s) - f(t)| < \varepsilon/(b - a)$ whenever $|s - t| < \delta$, and take any partition of mesh less than δ . On each subinterval the supremum and infimum are attained ([Corollary 5.4.1](#)) at points less than δ apart, so $M_i - m_i \leq \varepsilon/(b - a)$ and

$$U(f, P) - L(f, P) = \sum_i (M_i - m_i)(t_i - t_{i-1}) \leq \frac{\varepsilon}{b - a} \sum_i (t_i - t_{i-1}) = \varepsilon.$$

Apply [Proposition 10.2.1](#).

(ii) Let D be the finite set of discontinuities and $|f| \leq M$. Cover D by finitely many intervals of total length less than $\varepsilon/(4M)$; on those intervals the contribution to $U - L$ is at most $2M \cdot \varepsilon/(4M) = \varepsilon/2$. On the complement, a finite union of closed intervals on which f is continuous, apply (i) to get a further $\varepsilon/2$.

(iii) Let f be non-decreasing and take the uniform partition of mesh $(b - a)/k$. Then $M_i = f(t_i)$ and $m_i = f(t_{i-1})$, so the sum telescopes:

$$U(f, P) - L(f, P) = \frac{b - a}{k} \sum_{i=1}^k (f(t_i) - f(t_{i-1})) = \frac{b - a}{k} (f(b) - f(a)) \rightarrow 0.$$

This is Proposition 3.11 of Fukuda ([2026d](#)) (Fukuda [2026a](#), slide 66), with the proofs supplied. Part (iii) is worth noting: a monotone function may have countably many discontinuities and is still integrable, so (ii) and (iii) are genuinely different criteria. The exact characterisation — integrable if and only if the set of discontinuities has Lebesgue measure zero — is Lebesgue's criterion, for which see Rudin ([1976](#), Thm. 11.33); it is the point at which the Riemann theory is superseded by the Lebesgue theory of [Chapter 14](#). ■

Proposition 10.2.3 (Basic properties). Let f, g be Riemann integrable on $[a, b]$ and $\alpha, \beta \in \mathbb{R}$. Then

- (i) $\alpha f + \beta g$ is integrable with $\int_a^b (\alpha f + \beta g) = \alpha \int_a^b f + \beta \int_a^b g$;
- (ii) for $c \in (a, b)$, $\int_a^b f = \int_a^c f + \int_c^b f$;
- (iii) $f \leq g$ implies $\int_a^b f \leq \int_a^b g$; in particular $|\int_a^b f| \leq \int_a^b |f|$;
- (iv) if f is continuous there is $\xi \in [a, b]$ with $\int_a^b f = f(\xi)(b - a)$.

Proof. (i)–(iii) follow from the corresponding statements for Darboux sums and [Proposition 10.2.1](#); see Rudin ([1976](#), Thm. 6.12). For (iv), let $m = \min_{[a, b]} f$ and $M = \max_{[a, b]} f$, both attained by [Corollary 5.4.1](#). By (iii), $m(b - a) \leq \int_a^b f \leq M(b - a)$, so $\frac{1}{b - a} \int_a^b f \in [m, M]$, and the intermediate value theorem ([Corollary 5.2.1](#)) supplies ξ with $f(\xi)$ equal to it. ■

10.3 The Fundamental Theorem of Calculus

Definition 10.3.1 (Indefinite integral and antiderivative). Let $f: [a, b] \rightarrow \mathbb{R}$ be integrable. The *indefinite integral* of f is $F(x) := \int_a^x f(t) dt$ for $x \in [a, b]$. A function $G: [a, b] \rightarrow \mathbb{R}$ is an *antiderivative* (a primitive) of f if $G'(x) = f(x)$ for every $x \in (a, b)$. Any two antiderivatives differ by a constant, by [Theorem 7.2.5](#), which is why the generic antiderivative is written $\int f(x) dx = G(x) + c$.

Theorem 10.3.1 (Fundamental theorem of calculus). Let $f: [a, b] \rightarrow \mathbb{R}$ be Riemann integrable and $F(x) = \int_a^x f(t) dt$.

- (i) F is Lipschitz, hence continuous, on $[a, b]$; and F is differentiable with $F'(x_0) = f(x_0)$ at every point x_0 at which f is continuous. In particular, if f is continuous on $[a, b]$ then $F' = f$ throughout.

(ii) If G is any antiderivative of f on $[a, b]$ and f is integrable, then

$$\int_a^b f(t) dt = G(b) - G(a) := [G(t)]_a^b.$$

Proof. (i) With $|f| \leq M$, [Proposition 10.2.3\(ii\)–\(iii\)](#) gives $|F(y) - F(x)| = \left| \int_x^y f \right| \leq M|y - x|$, which is the Lipschitz bound. Let f be continuous at x_0 and $\varepsilon > 0$; choose $\delta > 0$ with $|f(t) - f(x_0)| < \varepsilon$ for $|t - x_0| < \delta$. For $0 < |h| < \delta$,

$$\left| \frac{F(x_0 + h) - F(x_0)}{h} - f(x_0) \right| = \left| \frac{1}{h} \int_{x_0}^{x_0+h} (f(t) - f(x_0)) dt \right| \leq \varepsilon,$$

using [Proposition 10.2.3\(iii\)](#) and $\frac{1}{h} \int_{x_0}^{x_0+h} dt = 1$.

(ii) Let $P = \{a = t_0 < \dots < t_k = b\}$ be any partition. By [Theorem 7.2.5](#) applied to G on each $[t_{i-1}, t_i]$ there is $\xi_i \in (t_{i-1}, t_i)$ with $G(t_i) - G(t_{i-1}) = G'(\xi_i)(t_i - t_{i-1}) = f(\xi_i)(t_i - t_{i-1})$. Summing, the left side telescopes to $G(b) - G(a)$, and since $m_i \leq f(\xi_i) \leq M_i$,

$$L(f, P) \leq G(b) - G(a) \leq U(f, P)$$

for every P . Integrability makes the two outer quantities have the common supremum and infimum $\int_a^b f$, so $G(b) - G(a) = \int_a^b f$. ■

The two halves say different things, and the slides' formulation $\frac{d}{dx} \int f(x) dx = f$ and $\int f'(x) dx = f + c$ (Fukuda [2026a](#), slide 68) compresses them. Part (i) constructs an antiderivative and requires continuity of f at the point in question; part (ii) evaluates an integral given an antiderivative and requires only integrability of f together with the existence of G . Neither implies the other: there are differentiable G whose derivative is not Riemann integrable, for which (ii) does not apply, and integrable f with no antiderivative, for which (i) fails at the discontinuities.

10.3.1 Integration by parts

Theorem 10.3.2 (Integration by parts). Let $u, v: [a, b] \rightarrow \mathbb{R}$ be differentiable with u' and v' Riemann integrable. Then

$$\int_a^b u'(x)v(x) dx = [u(x)v(x)]_a^b - \int_a^b u(x)v'(x) dx. \quad (10.2)$$

Proof. By the product rule ([Proposition 7.3.2](#)), uv is differentiable with $(uv)' = u'v + uv'$, and the right side is integrable as a sum of products of integrable functions. Apply [Theorem 10.3.1\(ii\)](#) to uv and then [Proposition 10.2.3\(i\)](#). ■

Identity (10.2) is usually applied in the form of (Fukuda [2026a](#), slide 71): given a product fg to integrate, take an antiderivative F of f and differentiate g ,

$$\int_a^b f(x)g(x) dx = [F(x)g(x)]_a^b - \int_a^b F(x)g'(x) dx,$$

choosing the split so that Fg' is simpler than fg . Two standard choices: $g = \log x$ with $f = 1$ reduces $\int \log x dx$ to $\int 1 dx$, giving $x(\log x - 1) + c$ (Fukuda [2026a](#), slide 78); and g a power with f an exponential reduces the power by one at each application, which is the recursion used in [subsection 10.7.1](#).

10.3.2 Substitution

Theorem 10.3.3 (Change of variable in one dimension). Let $\varphi: [a, b] \rightarrow \mathbb{R}$ be continuously differentiable and let f be continuous on an interval containing $\varphi([a, b])$. Then

$$\int_{\varphi(a)}^{\varphi(b)} f(x) dx = \int_a^b f(\varphi(t))\varphi'(t) dt. \quad (10.3)$$

Proof. Let F be an antiderivative of f , which exists by [Theorem 10.3.1\(i\)](#) since f is continuous. By the chain rule ([Proposition 7.3.4](#)), $(F \circ \varphi)'(t) = F'(\varphi(t))\varphi'(t) = f(\varphi(t))\varphi'(t)$, which is continuous hence integrable. Applying [Theorem 10.3.1\(ii\)](#) twice,

$$\int_a^b f(\varphi(t))\varphi'(t) dt = [F(\varphi(t))]_a^b = F(\varphi(b)) - F(\varphi(a)) = \int_{\varphi(a)}^{\varphi(b)} f(x) dx.$$

■

This is Proposition 3.16 of Fukuda ([2026d](#)) (Fukuda [2026a](#), slide 82). Note what the theorem does *not* require: φ need not be injective, and the limits of integration are $\varphi(a)$ and $\varphi(b)$ in that order, which may be reversed relative to $a < b$. The mnemonic $x = \varphi(t)$, $dx = \varphi'(t) dt$ (Fukuda [2026a](#), slide 81) is the identity (10.3) written in differentials; the substance is the chain rule, and the differentials are notation for it.

Example 10.3.1 (Two substitutions). With $\varphi(t) = 2t + 3$, $\varphi' = 2$, and $f(x) = x^4$,

$$\int_1^2 (2t + 3)^4 dt = \frac{1}{2} \int_5^7 x^4 dx = \frac{1}{10} [x^5]_5^7 = \frac{7^5 - 5^5}{10} = \frac{6841}{10}.$$

With $\varphi(x) = -x^2$, $\varphi'(x) = -2x$, and $f(y) = e^y$,

$$\int xe^{-x^2} dx = -\frac{1}{2} \int e^y dy = -\frac{1}{2} e^{-x^2} + c.$$

These are the computations of (Fukuda [2026a](#), slides 83–84). The second is the pattern used for the normal density in [subsection 10.7.3](#): an integrand of the form $g'(x)h(g(x))$ integrates to $H(g(x))$ with $H' = h$.

10.3.3 Improper integrals

Definition 10.3.2 (Improper integral). Let f be Riemann integrable on $[a, c]$ for every $c > a$. Then

$$\int_a^\infty f(x) dx := \lim_{c \rightarrow \infty} \int_a^c f(x) dx$$

when the limit exists in $\mathbb{R}$, and analogously at $-\infty$. The integral over $\mathbb{R}$ is defined as $\int_{-\infty}^0 f + \int_0^\infty f$, requiring *both* limits to exist separately. The integral *absolute convergence* if $\int |f| < \infty$, and absolute convergence implies convergence.

The requirement that the two halves converge separately is not pedantry. For $f(x) = x$ the symmetric limit $\lim_{c \rightarrow \infty} \int_{-c}^c x dx = 0$ exists while $\int_{-\infty}^\infty x dx$ does not, and a distribution with such a density tail has no mean — the Cauchy distribution being the standard example. The distinction is exactly the one between the principal value and the integral, and it is what makes “the expectation does not exist” a meaningful statement.

10.4 Multivariate Integration

Two results are needed for the Gaussian computation. Both are stated without proof and used only where cited; complete treatments are in Rudin ([1976](#), Ch. 10) and, in the measure-theoretic form, Le Gall ([2022](#), Ch. 5).

Theorem 10.4.1 (Fubini–Tonelli). Let $f: \mathbb{R}^m \times \mathbb{R}^n \rightarrow \mathbb{R}$ be measurable. Suppose either that $f \geq 0$, or that the iterated integral of $|f|$ is finite. Then

$$\int_{\mathbb{R}^{m+n}} f = \int_{\mathbb{R}^m} \left(\int_{\mathbb{R}^n} f(x, y) dy \right) dx = \int_{\mathbb{R}^n} \left(\int_{\mathbb{R}^m} f(x, y) dx \right) dy,$$

all three integrals existing and being equal.

The non-negativity or absolute integrability hypothesis is essential; without it the two iterated integrals can exist and differ. In economics the theorem is what licenses interchanging an expectation over states with an integral over time, or summing a double sum in either order — routine steps that are valid under conditions worth stating rather than assuming.

Theorem 10.4.2 (Change of variables). Let $U, V \subseteq \mathbb{R}^n$ be open and $\Phi: U \rightarrow V$ a continuously differentiable bijection with $\det D\Phi(u) \neq 0$ for every $u \in U$. Then for every integrable $f: V \rightarrow \mathbb{R}$,

$$\int_V f(x) dx = \int_U f(\Phi(u)) |\det D\Phi(u)| du. \quad (10.4)$$

The factor $|\det D\Phi|$ is the local volume magnification of Φ , which is the interpretation of the determinant established in [Example 2.4.1](#): Φ maps an infinitesimal box of volume du to a parallelepiped of volume $|\det D\Phi(u)| du$ (Fukuda 2026a, slide 93). For $n = 1$, $|\det D\Phi| = |\varphi'|$ and (10.4) is (10.3) with the absolute value accounting for orientation, which the one-dimensional statement instead encodes in the order of the limits.

10.5 Differentiating Under the Integral Sign

Lemma 10.5.1 (Differentiation under the integral). Let $f: [a, b] \times [\underline{r}, \bar{r}] \rightarrow \mathbb{R}$ be such that $f(\cdot, r)$ is continuous on $[a, b]$ for each r , and $\partial f / \partial r$ exists and is continuous on the whole rectangle. Then $r \mapsto \int_a^b f(x, r) dx$ is differentiable on $(\underline{r}, \bar{r})$ with

$$\frac{d}{dr} \int_a^b f(x, r) dx = \int_a^b \frac{\partial f}{\partial r}(x, r) dx.$$

Proof. Fix r and let $h \neq 0$ be small enough that $r + h$ stays in the interval. By [Theorem 7.2.5](#) applied to $s \mapsto f(x, s)$, for each x there is $\theta_x \in (0, 1)$ with $(f(x, r + h) - f(x, r)) / h = \frac{\partial f}{\partial r}(x, r + \theta_x h)$. Hence

$$\left| \frac{1}{h} \int_a^b (f(x, r + h) - f(x, r)) dx - \int_a^b \frac{\partial f}{\partial r}(x, r) dx \right| \leq (b - a) \sup_{x \in [a, b]} \left| \frac{\partial f}{\partial r}(x, r + \theta_x h) - \frac{\partial f}{\partial r}(x, r) \right|.$$

The rectangle is compact, so $\partial f / \partial r$ is uniformly continuous on it by [Theorem 5.2.2](#). The two arguments differ by at most $|h|$, uniformly in x , so the supremum tends to 0 as $h \rightarrow 0$. ■

Theorem 10.5.1 (Leibniz formula). Under the hypotheses of [Lemma 10.5.1](#), let $A, B: [\underline{r}, \bar{r}] \rightarrow [a, b]$ be continuously differentiable and set $V(r) := \int_{A(r)}^{B(r)} f(x, r) dx$. Then V is differentiable with

$$V'(r) = f(B(r), r)B'(r) - f(A(r), r)A'(r) + \int_{A(r)}^{B(r)} \frac{\partial f}{\partial r}(x, r) dx. \quad (10.5)$$

Proof. Let $g(x, r) := \int_a^x f(u, r) du$, so that

$$\frac{\partial g}{\partial x} = f(x, r) \text{ by Theorem 10.3.1(i),} \quad \frac{\partial g}{\partial r} = \int_a^x \frac{\partial f}{\partial r}(u, r) du \text{ by Lemma 10.5.1;}$$

both are continuous, so g is differentiable in the sense of [Definition 8.3.1](#) by [Proposition 8.3.3](#). By [Proposition 10.2.3\(ii\)](#), $V(r) = g(B(r), r) - g(A(r), r)$, and the chain rule ([Theorem 8.3.1](#)) gives

$$V'(r) = \frac{\partial g}{\partial x}(B(r), r)B'(r) + \frac{\partial g}{\partial r}(B(r), r) - \frac{\partial g}{\partial x}(A(r), r)A'(r) - \frac{\partial g}{\partial r}(A(r), r).$$

Substituting the two partial derivatives of g and combining the two $\partial g / \partial r$ terms into a single integral from $A(r)$ to $B(r)$ yields (10.5). ■

This is Theorem 3.3 of Fukuda (2026d), and the proof is the route mapped out in Exercise 3.6 there (Fukuda 2026a, slides 95–96), carried out. The intuitive derivation on (Fukuda 2026a, slides 97–98) identifies the three terms correctly — the value gained at the moving upper endpoint, the value lost at the moving lower endpoint, and the change in the integrand over the fixed region — and the proof above supplies the hypotheses under which the approximations become identities.

10.6 Economic Application: Continuation Values

Proposition 10.6.1 (Differential form of a continuation value). Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be bounded and continuous and $\rho > 0$, and define the continuation value

$$V(t) := \int_t^\infty e^{-\rho(s-t)} f(s) ds.$$

Then the integral converges absolutely for every t , V is differentiable, and

$$V'(t) = \rho V(t) - f(t), \quad \text{equivalently} \quad \rho V(t) = f(t) + V'(t). \quad (10.6)$$

Proof. Let $|f| \leq M$. Then $\int_t^\infty e^{-\rho(s-t)} |f(s)| ds \leq M \int_t^\infty e^{-\rho(s-t)} ds = M/\rho < \infty$, so the integral converges absolutely.

Write $W(t) := \int_t^\infty e^{-\rho s} f(s) ds$, which converges absolutely by the same bound, so that $V(t) = e^{\rho t} W(t)$. Fixing any t_0 , $W(t) = W(t_0) - \int_{t_0}^t e^{-\rho s} f(s) ds$, and the integrand is continuous, so [Theorem 10.3.1\(i\)](#) gives $W'(t) = -e^{-\rho t} f(t)$. By the product rule ([Proposition 7.3.2](#)),

$$V'(t) = \rho e^{\rho t} W(t) + e^{\rho t} W'(t) = \rho V(t) - f(t).$$

■

Reading (10.6). In the form $\rho V = f + V'$ the identity is an asset-pricing equation: the required return ρV on holding the “asset” whose value is V equals the flow payoff f plus the capital gain V' . It is the deterministic, uncontrolled special case of the Hamilton–Jacobi–Bellman equation of [Chapter 16](#), and the same three terms reappear there with a maximisation over controls inserted in front of f . This is Example 3.22 of Fukuda ([2026d](#)) (Fukuda [2026a](#), slide 94).

The proof deliberately avoids differentiating under an improper integral. The Leibniz formula ([10.5](#)) applies on compact rectangles, and $[t, \infty)$ is not one; a direct application would require a domination argument. Rewriting $V = e^{\rho t} W$ moves the t -dependence out of the integrand entirely, leaving only a moving endpoint, and [Theorem 10.3.1\(i\)](#) handles that with no additional hypotheses.

10.7 Economic Application: Moments of Two Distributions

10.7.1 The exponential distribution

Proposition 10.7.1. Let X have density $f(x) = \lambda e^{-\lambda x}$ for $x \geq 0$ and $f(x) = 0$ for $x < 0$, with $\lambda > 0$. Then its distribution function is $F(x) = 1 - e^{-\lambda x}$ for $x > 0$, and

$$\mathbb{E}[X] = \frac{1}{\lambda}, \quad \mathbb{E}[X^2] = \frac{2}{\lambda^2}, \quad \text{Var}[X] = \frac{1}{\lambda^2}.$$

Proof. For $x > 0$, $F(x) = \int_0^x \lambda e^{-\lambda t} dt = [-e^{-\lambda t}]_0^x = 1 - e^{-\lambda x}$ by [Theorem 10.3.1\(ii\)](#).

For the first moment, write $\lambda e^{-\lambda x} = (-e^{-\lambda x})'$ and apply [Theorem 10.3.2](#) on $[0, c]$:

$$\int_0^c x \lambda e^{-\lambda x} dx = [-x e^{-\lambda x}]_0^c + \int_0^c e^{-\lambda x} dx = -c e^{-\lambda c} + \frac{1 - e^{-\lambda c}}{\lambda}.$$

Letting $c \rightarrow \infty$ and using $c e^{-\lambda c} \rightarrow 0$ gives $\mathbb{E}[X] = 1/\lambda$.

For the second moment, the same split with x^2 in place of x gives

$$\int_0^c x^2 \lambda e^{-\lambda x} dx = [-x^2 e^{-\lambda x}]_0^c + 2 \int_0^c x e^{-\lambda x} dx = -c^2 e^{-\lambda c} + \frac{2}{\lambda} \int_0^c x \lambda e^{-\lambda x} dx,$$

so letting $c \rightarrow \infty$ and substituting the first moment yields $\mathbb{E}[X^2] = 2/\lambda^2$. Finally $\text{Var}[X] = \mathbb{E}[X^2] - (\mathbb{E}[X])^2 = 2/\lambda^2 - 1/\lambda^2 = 1/\lambda^2$ by [Proposition 14.2.2](#). ■

These are the computations of (Fukuda [2026a](#), slides 73–77), with the improper integrals handled as limits of integrals over $[0, c]$ in accordance with [Definition 10.3.2](#) rather than by evaluating boundary terms at ∞ directly. The exponential distribution is the continuous-time counterpart of a constant hazard rate: $\mathbb{P}(X > s + t \mid X > s) = e^{-\lambda t}$ does not depend on s , which is why it is the standard model of the arrival of a price-adjustment opportunity in Calvo pricing and of job separation in search models.

10.7.2 The Gaussian integral

Proposition 10.7.2 (Gaussian integral).

$$\int_{-\infty}^{\infty} e^{-x^2} dx = \sqrt{\pi}, \quad \text{and more generally} \quad \int_{-\infty}^{\infty} e^{-a(x+b)^2} dx = \sqrt{\frac{\pi}{a}}$$

for every $a > 0$ and $b \in \mathbb{R}$. Consequently the function (10.1) integrates to 1 and is a probability density.

Proof. Let $I := \int_{-\infty}^{\infty} e^{-x^2} dx$, which converges because $e^{-x^2} \leq e^{-|x|}$ for $|x| \geq 1$. The integrand of the product I^2 is non-negative, so Theorem 10.4.1 applies and

$$I^2 = \left(\int_{\mathbb{R}} e^{-x^2} dx \right) \left(\int_{\mathbb{R}} e^{-y^2} dy \right) = \int_{\mathbb{R}} \int_{\mathbb{R}} e^{-(x^2+y^2)} dy dx.$$

Apply Theorem 10.4.2 with polar coordinates $\Phi(r, \theta) = (r \cos \theta, r \sin \theta)$, a continuously differentiable bijection from $U = (0, \infty) \times (0, 2\pi)$ onto $\mathbb{R}^2$ minus a half-line, a set of measure zero which does not affect the integral. Its Jacobian is

$$D\Phi(r, \theta) = \begin{pmatrix} \cos \theta & -r \sin \theta \\ \sin \theta & r \cos \theta \end{pmatrix}, \quad \det D\Phi = r(\cos^2 \theta + \sin^2 \theta) = r > 0.$$

Since $x^2 + y^2 = r^2$,

$$I^2 = \int_0^{2\pi} \int_0^{\infty} e^{-r^2} r dr d\theta = 2\pi \left[-\frac{1}{2} e^{-r^2} \right]_0^{\infty} = 2\pi \cdot \frac{1}{2} = \pi.$$

As $I > 0$, $I = \sqrt{\pi}$. For the general case substitute $x = y/\sqrt{a} - b$ in Theorem 10.3.3, so that $dx = dy/\sqrt{a}$ and $a(x+b)^2 = y^2$, giving $\int e^{-a(x+b)^2} dx = I/\sqrt{a} = \sqrt{\pi/a}$. Taking $a = 1/(2\sigma^2)$ and $b = -\mu$ gives $\int_{\mathbb{R}} \exp(-(x-\mu)^2/(2\sigma^2)) dx = \sqrt{2\pi\sigma^2}$, so (10.1) integrates to 1. ■

This is the derivation of (Fukuda 2026a, slides 89–93), with the interchange of the two integrals justified by Theorem 10.4.1 and the polar substitution by Theorem 10.4.2 rather than by the differential mnemonic $dy dx = |\det J| dr d\theta$. The trick is that e^{-x^2} has no elementary antiderivative, so Theorem 10.3.1(ii) is unavailable; squaring converts the problem to two dimensions, where the radial symmetry of $e^{-(x^2+y^2)}$ makes the extra factor r from the Jacobian exactly the derivative needed for a substitution.

10.7.3 Mean and variance of the normal

Proposition 10.7.3. Let X have the density (10.1) with $\sigma > 0$. Then $\mathbb{E}[X] = \mu$ and $\text{Var}[X] = \sigma^2$.

Proof. Mean. Since the density integrates to 1 (Proposition 10.7.2),

$$\mathbb{E}[X] = \int_{\mathbb{R}} x f(x) dx = \int_{\mathbb{R}} (x - \mu) f(x) dx + \mu.$$

The remaining integral vanishes. Substituting $t = (x - \mu)/\sigma$, so $dx = \sigma dt$ by Theorem 10.3.3, turns it into $\sigma \int_{\mathbb{R}} t \phi(t) dt$ with $\phi(t) = (2\pi)^{-1/2} e^{-t^2/2}$; the integrand is odd and absolutely integrable, so each half contributes the negative of the other and the integral is 0. Equivalently, and without appealing to symmetry, note that

$$(x - \mu) f(x) = -\sigma^2 \frac{d}{dx} f(x), \quad \text{so} \quad \int_{-c}^c (x - \mu) f(x) dx = -\sigma^2 [f(x)]_{-c}^c \rightarrow 0$$

as $c \rightarrow \infty$, since $f(x) \rightarrow 0$ at both ends.

Variance. Using the same identity $f'(x) = -(x - \mu)\sigma^{-2} f(x)$ and integrating by parts (Theorem 10.3.2) on $[-c, c]$,

$$\int_{-c}^c (x - \mu)^2 f(x) dx = \int_{-c}^c (x - \mu)(-\sigma^2 f'(x)) dx = [-\sigma^2 (x - \mu) f(x)]_{-c}^c + \sigma^2 \int_{-c}^c f(x) dx.$$

The boundary term tends to 0 because f decays faster than any polynomial grows, and the remaining integral tends to 1. Hence $\text{Var}[X] = \mathbb{E}[(X - \mu)^2] = \sigma^2$. ■

These are the computations of (Fukuda 2026a, slides 85–88). Two points of technique are worth isolating. First, the identity $f' = -(x - \mu)\sigma^{-2}f$ is what makes the normal tractable: the density is its own antiderivative up to the linear factor, so every moment computation is an integration by parts that reduces the power by one. Second, both boundary terms are evaluated as limits over $[-c, c]$; writing $[\cdot]_{-\infty}^{\infty}$ directly, as the slides do, presupposes the convergence that Definition 10.3.2 requires be checked, and here it holds because of the Gaussian decay.

10.8 Chapter Summary

The Riemann integral is defined by squeezing a bounded function between step functions; continuity, boundedness with finitely many discontinuities, and monotonicity each suffice for integrability, by arguments that use uniform continuity, a covering of the bad set, and telescoping respectively. The fundamental theorem has two halves with different hypotheses: the indefinite integral of an integrable function is Lipschitz and differentiates back to the integrand wherever the integrand is continuous; and any antiderivative evaluates the definite integral. Integration by parts is the product rule integrated, and substitution is the chain rule integrated; in several variables the substitution rule acquires the factor $|\det D\Phi|$, the local volume magnification established in Chapter 2. Improper integrals are limits and require both tails to converge separately, which is what makes non-existence of a mean a meaningful statement.

Differentiation under the integral sign holds when the parameter derivative is continuous on a compact rectangle, by the mean value theorem and uniform continuity; with moving endpoints it becomes the Leibniz formula (10.5), whose three terms are the flows at the two endpoints and the change in the integrand. Applied to a discounted flow it gives $\rho V = f + V'$, the deterministic Hamilton–Jacobi–Bellman equation.

The exponential distribution’s moments follow from repeated integration by parts; the normal density integrates to 1 by the polar-coordinate evaluation of the Gaussian integral, which needs both Fubini’s theorem and the multivariate change of variables; and its mean and variance follow from the identity $f' = -(x - \mu)\sigma^{-2}f$ together with one integration by parts each.

Exercise 10.8.1. Show that $f(x) = 1$ for $x \in \mathbb{Q} \cap [0, 1]$ and $f(x) = 0$ otherwise is bounded and not Riemann integrable, by computing $L(f, P)$ and $U(f, P)$ for an arbitrary partition. Then show that it is Lebesgue integrable with integral 0, and identify which hypothesis of Proposition 10.2.2 fails.

Exercise 10.8.2. Compute the moment generating function $M(s) = \mathbb{E}[e^{sX}]$ for X normal with mean μ and variance σ^2 , by completing the square in the exponent and applying Proposition 10.7.2. Verify $M'(0) = \mu$ and $M''(0) = \mu^2 + \sigma^2$, stating the hypothesis under which the two differentiations under the integral sign are legitimate.

Exercise 10.8.3. A firm’s value is $V(t) = \int_t^{T(t)} e^{-\rho(s-t)} \pi(s, t) ds$, where the exit date T and the flow profit π both depend on t . Apply (10.5) to compute $V'(t)$, identify the three terms economically, and show that when T is chosen optimally the term involving $T'(t)$ vanishes. Relate the last observation to the envelope theorem of Chapter 12.

Chapter 11

Projection, Orthogonality, and Least Squares

11.1 Motivation: Least Squares Is Not About Calculus

[Proposition 8.8.1](#) solved the least-squares problem by setting a gradient to zero. The solution was correct and the method was, in a precise sense, beside the point. Writing $\mathbf{1} = (1, \dots, 1)^\top$, $x = (x_1, \dots, x_n)^\top$ and $y = (y_1, \dots, y_n)^\top$ in $\mathbb{R}^n$, the objective is

$$S(\beta_1, \beta_2) = \sum_{i=1}^n (y_i - \beta_1 - \beta_2 x_i)^2 = \|y - \beta_1 \mathbf{1} - \beta_2 x\|^2, \quad (11.1)$$

so the problem is to find the point of the plane $\text{span}\{\mathbf{1}, x\}$ closest to y ([Fukuda 2026a](#), slide 61). That is a question about distance in an inner product space, and its answer — drop a perpendicular — does not involve differentiation, holds in infinite dimensions where the calculus argument would not, and explains why the first-order conditions look the way they do: the normal equations *are* the statement that the residual is orthogonal to the regressors.

This chapter develops that geometry. Its three payoffs are the orthogonal decomposition theorem, the projection formula $P = X(X^\top X)^{-1}X^\top$, and the identification of conditional expectation as a projection, which [Chapter 14](#) uses to characterise the best predictor.

11.2 Orthogonality

Throughout, H is a real inner product space with inner product $\langle \cdot, \cdot \rangle$ and induced norm $\|\cdot\|$ ([Proposition 3.3.1](#)), and it is a Hilbert space when complete ([Definition 3.5.1](#)). The leading cases are $H = \mathbb{R}^n$ and $H = L^2$.

Definition 11.2.1 (Orthogonal complement). For $\emptyset \neq A \subseteq H$, the *orthogonal complement* is

$$A^\perp := \{x \in H : \langle x, a \rangle = 0 \text{ for all } a \in A\}.$$

A set A is *orthogonal set* if $\langle x, y \rangle = 0$ for all distinct $x, y \in A$, and *orthonormal set* if in addition $\|x\| = 1$ for every $x \in A$.

Proposition 11.2.1 (Elementary properties). Let $A, B \subseteq H$ be non-empty.

- (i) $A^\perp$ is a closed vector subspace of H , whether or not A is.
- (ii) $A \subseteq B$ implies $B^\perp \subseteq A^\perp$, and $A \subseteq A^{\perp\perp}$, and $A^\perp = A^{\perp\perp\perp}$.
- (iii) $x \in (\text{span}\{x_1, \dots, x_m\})^\perp$ if and only if $\langle x, x_i \rangle = 0$ for every i .
- (iv) An orthogonal set not containing 0 is linearly independent.

Proof. (i) Linearity of the inner product in its first argument makes $A^\perp$ a subspace. For closedness, let $x_k \rightarrow x$ with $x_k \in A^\perp$; for each $a \in A$, $\langle x, a \rangle = \lim_k \langle x_k, a \rangle = 0$ by continuity of the inner product, which follows from Cauchy–Schwarz (Lemma 3.2.1).

(ii) The first claim is immediate. For $A \subseteq A^{\perp\perp}$: if $a \in A$ then $\langle a, z \rangle = 0$ for every $z \in A^\perp$ by definition of $A^\perp$, so $a \in A^{\perp\perp}$. Applying the first claim to $A \subseteq A^{\perp\perp}$ gives $A^{\perp\perp\perp} \subseteq A^\perp$, and applying $A \subseteq A^{\perp\perp}$ with $A^\perp$ in place of A gives the reverse inclusion.

(iii) If $\langle x, x_i \rangle = 0$ for every i , then $\langle x, \sum_i \lambda_i x_i \rangle = 0$ by linearity. The converse is the special case of singleton combinations.

(iv) Let $x_1, \dots, x_m$ be orthogonal and non-zero with $\sum_i \lambda_i x_i = 0$. Taking the inner product with x_j kills every term but the j th, giving $\lambda_j \|x_j\|^2 = 0$, hence $\lambda_j = 0$. ■

These are Propositions 9.4 and 9.6 of Fukuda (2026d, pp. 299–301). Part (i) contains a fact used repeatedly: orthogonal complements are automatically closed, which is why the decomposition theorem below can be stated for an arbitrary closed subspace M without any hypothesis on $M^\perp$.

Proposition 11.2.2 (Pythagoras). If $x \perp y$ then $\|x + y\|^2 = \|x\|^2 + \|y\|^2$. More generally, for a finite orthogonal set $\{x_1, \dots, x_m\}$, $\|\sum_i x_i\|^2 = \sum_i \|x_i\|^2$.

Proof. Expand $\|x + y\|^2 = \|x\|^2 + 2\langle x, y \rangle + \|y\|^2$ and use $\langle x, y \rangle = 0$; iterate for the general case. ■

11.3 The Projection Theorem

Theorem 11.3.1 (Projection onto a closed convex set). Let H be a Hilbert space, $\emptyset \neq M \subseteq H$ closed and convex, and $x \in H$. Then there is a unique $P_M x \in M$ with

$$\|x - P_M x\| = \inf_{y \in M} \|x - y\|,$$

and $y_0 = P_M x$ is characterised by

$$\langle x - y_0, z - y_0 \rangle \leq 0 \quad \text{for all } z \in M. \quad (11.2)$$

Proof. Existence. Let $d := \inf_{y \in M} \|x - y\|$ and take $(y_k) \subseteq M$ with $\|x - y_k\| \rightarrow d$. The parallelogram law (Remark 3.3.1) applied to $x - y_j$ and $x - y_k$ gives

$$\|y_j - y_k\|^2 = 2\|x - y_j\|^2 + 2\|x - y_k\|^2 - 4\left\|x - \frac{y_j + y_k}{2}\right\|^2 \leq 2\|x - y_j\|^2 + 2\|x - y_k\|^2 - 4d^2,$$

where the inequality uses $\frac{1}{2}(y_j + y_k) \in M$ by convexity. As $j, k \rightarrow \infty$ the right side tends to $2d^2 + 2d^2 - 4d^2 = 0$, so (y_k) is Cauchy. Completeness of H and closedness of M (Proposition 6.3.2) give a limit $y_0 \in M$, and $\|x - y_0\| = \lim_k \|x - y_k\| = d$ by continuity of the norm.

Uniqueness and characterisation. Identical to the corresponding parts of the proof of Lemma 9.2.1, which used only the parallelogram law and convexity. ■

This is Theorem 9.1 with Proposition 9.1 of Fukuda (2026d, pp. 295–298). The comparison with Lemma 9.2.1 is instructive. There, existence came from compactness: a closed bounded subset of $\mathbb{R}^n$ is compact and a continuous function attains its minimum. Here compactness is unavailable — the closed unit ball of an infinite-dimensional space is not compact (Remark 4.4.1) — and the parallelogram law does the work instead, converting a minimising sequence into a Cauchy sequence, which completeness then converges. Convexity is used exactly once, to place the midpoint $\frac{1}{2}(y_j + y_k)$ in M , and the theorem is false without it: in $\mathbb{R}^2$ the point 0 has no unique nearest point in the closed non-convex set $\{y : \|y\| = 1\}$.

Proposition 11.3.1 (Characterisation for subspaces). Let $M \subseteq H$ be a closed subspace and $x \in H$. Then $y_0 \in M$ minimises $\|x - y\|$ over M if and only if

$$\langle x - y_0, z \rangle = 0 \quad \text{for all } z \in M, \quad (11.3)$$

that is, if and only if $x - y_0 \in M^\perp$.

Proof. ($\Rightarrow$) Let $u \in M$ and $t \in \mathbb{R}$. Then $y_0 + tu \in M$, so

$$\|x - y_0\|^2 \leq \|x - y_0 - tu\|^2 = \|x - y_0\|^2 - 2t \langle x - y_0, u \rangle + t^2 \|u\|^2.$$

Hence $t^2 \|u\|^2 - 2t \langle x - y_0, u \rangle \geq 0$ for *every* real t , including t of both signs. A quadratic in t with a non-negative value everywhere and a zero at $t = 0$ must have vanishing linear coefficient, so $\langle x - y_0, u \rangle = 0$.

($\Leftarrow$) If $x - y_0 \perp M$ then for any $y \in M$, $y_0 - y \in M$ and Pythagoras (Proposition 11.2.2) gives

$$\|x - y\|^2 = \|(x - y_0) + (y_0 - y)\|^2 = \|x - y_0\|^2 + \|y_0 - y\|^2 \geq \|x - y_0\|^2.$$

■

This is Proposition 9.8 of Fukuda (2026d, p. 301). Comparing (11.3) with (11.2): on a subspace the variational inequality holds for z and for $-z$ simultaneously, so it collapses to an equality. The last display also proves more than was asked: it quantifies the loss from using any other $y \in M$, and that identity is the decomposition of total variation used in Equation 11.7 below.

Theorem 11.3.2 (Orthogonal decomposition). Let M be a closed subspace of a Hilbert space H . Then every $x \in H$ has a unique decomposition

$$x = y + z, \quad y = P_M x \in M, \quad z \in M^\perp,$$

so that $H = M \oplus M^\perp$, and $M^{\perp\perp} = M$.

Proof. Existence. If $x \in M$ take $(y, z) = (x, 0)$. Otherwise let $y = P_M x$ from Theorem 11.3.1, which applies since a closed subspace is closed and convex, and let $z := x - y$; then $z \in M^\perp$ by Proposition 11.3.1.

Uniqueness. If $y_1 + z_1 = y_2 + z_2$ with $y_i \in M$, $z_i \in M^\perp$, then $y_1 - y_2 = z_2 - z_1 \in M \cap M^\perp$. Any w in that intersection satisfies $\langle w, w \rangle = 0$, hence $w = 0$.

$M^{\perp\perp} = M$. The inclusion $M \subseteq M^{\perp\perp}$ is Proposition 11.2.1(ii). Conversely, let $x \in M^{\perp\perp}$ and decompose $x = y + z$ as above. Then $z = x - y \in M^\perp$, and also $z \in M^{\perp\perp}$ since both x and $y \in M \subseteq M^{\perp\perp}$ lie there. So $\langle z, z \rangle = 0$ and $x = y \in M$. ■

This is Theorem 9.2 of Fukuda (2026d, p. 302). The hypothesis that M be *closed* cannot be dropped, and in infinite dimensions it is not automatic: by Corollary 6.3.1 every subspace of $\mathbb{R}^n$ is closed, but the subspace of finitely supported sequences in L^2 is dense and proper, so its orthogonal complement is $\{0\}$ and no decomposition of the stated form exists.

Proposition 11.3.2 (The projection operator). Let $M \subseteq H$ be a closed subspace. The map $P_M: H \rightarrow M$ is

- (i) linear and surjective onto M ;
- (ii) idempotent, $P_M^2 = P_M$, with $\text{Ker } P_M = M^\perp$ and $\text{Im } P_M = M$;
- (iii) self-adjoint, $\langle P_M x, w \rangle = \langle x, P_M w \rangle$ for all $x, w \in H$;
- (iv) norm-reducing, $\|P_M x\| \leq \|x\|$, with equality if and only if $x \in M$.

Proof. (i) Let $x = y + z$ and $x' = y' + z'$ be the decompositions of Theorem 11.3.2 and $\alpha, \beta \in \mathbb{R}$. Then $\alpha x + \beta x' = (\alpha y + \beta y') + (\alpha z + \beta z')$ with the first bracket in M and the second in $M^\perp$, both being subspaces; by uniqueness this is *the* decomposition, so $P_M(\alpha x + \beta x') = \alpha P_M x + \beta P_M x'$. Surjectivity: $P_M y = y$ for $y \in M$.

(ii) Immediate from $P_M y = y$ on M and $P_M z = 0$ on $M^\perp$.

(iii) Write $x = y + z$, $w = y' + z'$. Then $\langle P_M x, w \rangle = \langle y, y' + z' \rangle = \langle y, y' \rangle$ since $y \in M$ and $z' \in M^\perp$, and symmetrically $\langle x, P_M w \rangle = \langle y + z, y' \rangle = \langle y, y' \rangle$.

(iv) $\|x\|^2 = \|y\|^2 + \|z\|^2 \geq \|y\|^2$ by Proposition 11.2.2, with equality exactly when $z = 0$. ■

This is Proposition 9.9 of Fukuda (2026d, p. 302), with the self-adjointness added because it is what identifies the projection matrix in Proposition 11.4.1 and what makes the residual maker symmetric.

Theorem 11.3.3 (Riesz representation). Let H be a Hilbert space and $f: H \rightarrow \mathbb{R}$ a continuous linear

functional. Then there is a unique $a \in H$ with $f(x) = \langle a, x \rangle$ for all $x \in H$, and $\|a\|$ equals the operator norm of f .

Proof. If $f \equiv 0$ take $a = 0$. Otherwise $M := \text{Ker } f$ is a closed proper subspace — closed because f is continuous and $M = f^{-1}(\{0\})$ (Theorem 5.2.1) — so by Theorem 11.3.2 there is $u \in M^\perp$ with $\|u\| = 1$. For any $x \in H$ the vector $f(x)u - f(u)x$ lies in M , since f of it is $f(x)f(u) - f(u)f(x) = 0$; taking the inner product with $u \in M^\perp$ gives $f(x) - f(u)\langle u, x \rangle = 0$, that is $f(x) = \langle f(u)u, x \rangle$. So $a = f(u)u$ works. Uniqueness: if $\langle a, x \rangle = \langle a', x \rangle$ for all x , take $x = a - a'$. ■

This is Theorem 9.3 of Fukuda (2026d, p. 304) and the promised generalisation of Theorem 3.2.1; in $\mathbb{R}^n$ continuity was automatic by Proposition 5.2.1(ii), and in infinite dimensions it must be assumed, since discontinuous linear functionals exist there.

11.4 Projection in $\mathbb{R}^n$ and the Projection Matrix

Proposition 11.4.1 (Projection matrix). Let $X \in M_{n \times k}(\mathbb{R})$ have linearly independent columns, and let $M := \text{Im } X = \text{span}\{X_1, \dots, X_k\} \subseteq \mathbb{R}^n$. Then $X^\top X$ is invertible, and for every $y \in \mathbb{R}^n$,

$$P_M y = X(X^\top X)^{-1} X^\top y, \quad P := X(X^\top X)^{-1} X^\top. \quad (11.4)$$

The matrix P is symmetric and idempotent with $\text{rank } P = k$, and $Q := I_n - P$ is the projection onto $M^\perp$, symmetric and idempotent with $\text{rank } Q = n - k$.

Proof. Invertibility. Suppose $X^\top X v = 0$ for some $v \in \mathbb{R}^k$. Then $0 = v^\top X^\top X v = \|Xv\|^2$, so $Xv = 0$, and linear independence of the columns forces $v = 0$. Hence $\text{Ker}(X^\top X) = \{0\}$ and Theorem 2.3.2 makes the square matrix $X^\top X$ invertible.

The formula. Set $\hat{y} := X(X^\top X)^{-1} X^\top y$, which lies in M because it is X times a vector. By Proposition 11.3.1 it suffices to check $y - \hat{y} \perp M$, and by Proposition 11.2.1(iii) it suffices to check orthogonality to each column of X , that is $X^\top(y - \hat{y}) = 0$:

$$X^\top y - X^\top X(X^\top X)^{-1} X^\top y = X^\top y - X^\top y = 0.$$

Properties. Symmetry: $P^\top = X((X^\top X)^{-1})^\top X^\top = P$, since $X^\top X$ is symmetric and so is its inverse. Idempotence: $P^2 = X(X^\top X)^{-1}(X^\top X)(X^\top X)^{-1} X^\top = P$. Rank: $\text{Im } P = M$ has dimension k . The statements for Q follow from Theorem 11.3.2, $Q = I - P$ being the projection onto $M^\perp$, whose dimension is $n - k$ by Theorem 2.3.2. ■

This is Theorem 9.4 of Fukuda (2026d, p. 305), and it is the formula displayed on (Fukuda 2026a, slide 62). Both hypotheses on X carry economic content: linear independence of the columns is the no-perfect-multicollinearity condition of Remark 2.3.1, and $k \leq n$ is the requirement of at least as many observations as parameters, which is forced because k independent vectors cannot live in $\mathbb{R}^n$ with $k > n$ by Theorem 2.2.1.

11.4.1 Gram–Schmidt and QR

Proposition 11.4.2 (Gram–Schmidt). Let $v_1, \dots, v_k \in H$ be linearly independent. Define recursively

$$u_1 := v_1, \quad u_j := v_j - \sum_{i=1}^{j-1} \frac{\langle v_j, u_i \rangle}{\|u_i\|^2} u_i \quad (j = 2, \dots, k), \quad e_j := \frac{u_j}{\|u_j\|}.$$

Then $\{e_1, \dots, e_k\}$ is orthonormal and $\text{span}\{e_1, \dots, e_j\} = \text{span}\{v_1, \dots, v_j\}$ for every j .

Proof. Induction on j . Each u_j is v_j minus its projection onto $\text{span}\{u_1, \dots, u_{j-1}\}$, expressed through the orthogonal basis by Proposition 11.2.2; by Proposition 11.3.1 the difference is orthogonal to that span. It is non-zero because $v_j \notin \text{span}\{v_1, \dots, v_{j-1}\} = \text{span}\{u_1, \dots, u_{j-1}\}$ by independence, so the normalisation is legitimate, and the span statement follows since the transformation is triangular with unit diagonal. ■

Corollary 11.4.1 (QR decomposition). Every $X \in M_{n \times k}(\mathbb{R})$ with independent columns factors as $X = QR$ with $Q \in M_{n \times k}(\mathbb{R})$ having orthonormal columns and $R \in M_{k \times k}(\mathbb{R})$ upper triangular with positive diagonal. Then $P = QQ^\top$ and the least-squares coefficient solves $R\hat{\beta} = Q^\top y$.

Proof. Apply Proposition 11.4.2 to the columns of X and collect the coefficients of v_j in terms of $e_1, \dots, e_j$ into the j th column of R , which is upper triangular by construction. Then $X^\top X = R^\top Q^\top QR = R^\top R$ and (11.4) gives $P = QR(R^\top R)^{-1}R^\top Q^\top = QQ^\top$; substituting into $X^\top X\hat{\beta} = X^\top y$ gives $R^\top R\hat{\beta} = R^\top Q^\top y$, and $R^\top$ is invertible. ■

This is Section 9.5 of Fukuda (2026d, pp. 320–323). The practical point is numerical: forming $X^\top X$ squares the condition number of X , so the explicit inverse in (11.4) is a formula for analysis and a poor algorithm, whereas solving the triangular system $R\hat{\beta} = Q^\top y$ is stable. Statistical software computes the QR factorisation, not the inverse.

11.5 Econometric Application: Least Squares

Setting. Observations $(x_i, y_i)_{i=1}^n$ are stacked as $y \in \mathbb{R}^n$ and $X = [\mathbf{1} \ x] \in M_{n \times 2}(\mathbb{R})$. The least-squares problem is (11.1).

Proposition 11.5.1 (Least squares as projection). Suppose the columns of X are linearly independent, equivalently that the x_i are not all equal. Then (11.1) has the unique solution

$$\hat{\beta} = (X^\top X)^{-1} X^\top y, \quad (11.5)$$

the fitted vector is $\hat{y} = X\hat{\beta} = Py$, the residual is $\hat{e} = y - \hat{y} = Qy$, and

$$X^\top \hat{e} = 0, \quad \text{that is} \quad \sum_{i=1}^n \hat{e}_i = 0 \quad \text{and} \quad \sum_{i=1}^n x_i \hat{e}_i = 0. \quad (11.6)$$

Moreover

$$\|y - \bar{y}\mathbf{1}\|^2 = \|\hat{y} - \bar{y}\mathbf{1}\|^2 + \|\hat{e}\|^2. \quad (11.7)$$

Proof. Minimising (11.1) is minimising $\|y - v\|$ over $v \in M = \text{Im } X$, so the solution is $\hat{y} = P_M y$, unique by Theorem 11.3.1, and given by (11.4). Since X has independent columns, the representation $\hat{y} = X\hat{\beta}$ has a unique coefficient vector, namely (11.5). Orthogonality (11.6) is Proposition 11.3.1 combined with Proposition 11.2.1(iii), the two scalar equations being orthogonality to $\mathbf{1}$ and to x .

For (11.7): the first equation of (11.6) gives $\bar{\hat{e}} = 0$, hence $\bar{\hat{y}} = \bar{y}$, so $\hat{y} - \bar{y}\mathbf{1} \in M$ and $\hat{e} \in M^\perp$ are orthogonal; $y - \bar{y}\mathbf{1} = (\hat{y} - \bar{y}\mathbf{1}) + \hat{e}$ and Proposition 11.2.2 apply. ■

What the geometry explains. Four things that the calculus derivation leaves opaque.

First, the normal equations (11.6) are not computational artefacts. They say the residual is orthogonal to every regressor, which is the defining property of the projection, and they are the sample analogue of the population moment conditions $\mathbb{E}[\hat{e}] = 0$, $\mathbb{E}[x\hat{e}] = 0$ that identify the estimand. The first equation holds *because* $\mathbf{1}$ is a regressor: a regression without an intercept has residuals that need not sum to zero, and then $\hat{y} \neq \bar{y}\mathbf{1}$ and (11.7) fails.

Second, (11.7) is the analysis-of-variance decomposition, total sum of squares equals explained plus residual, and it is Pythagoras' theorem. The coefficient of determination $R^2 = \|\hat{y} - \bar{y}\mathbf{1}\|^2 / \|y - \bar{y}\mathbf{1}\|^2$ is therefore $\cos^2$ of the angle between the demeaned data vector and the demeaned fit, which by Example 3.2.2 is the squared sample correlation between y and $\hat{y}$. That $R^2 \in [0, 1]$ is Cauchy–Schwarz, not a convention.

Third, adding a regressor cannot raise the residual sum of squares, because it enlarges M and Theorem 11.3.1 minimises over a larger set. This is a geometric fact about nested subspaces, which is why R^2 is non-decreasing in the number of regressors and why model selection needs a penalty rather than a fit criterion.

Fourth, the estimator is linear in y : $\hat{\beta} = (X^\top X)^{-1} X^\top y$ applies a fixed matrix to the data. Linearity is Proposition 11.3.2(i), and it is the property that makes the sampling distribution of $\hat{\beta}$ computable from that of y — normal if y is normal, by the linear-image property of the Gaussian family.

Example 11.5.1 (The simulation). Example 8.1 of Fukuda (2026d) (Fukuda 2026a, slides 63–64) draws $n = 500$ independent standard normal regressors x_i and errors ε_i , sets $y_i = 1 + 3x_i + \varepsilon_i$, and computes $\hat{\beta}$ from (11.5). The accompanying programs — `regression.m` for MATLAB, `regression_Octave.m`, and `regression_Python.py` — fix the random seed, build $X = [\mathbf{1} \ x]$, and evaluate `inv(X'*X)*X'*y`, comparing the result with the built-in `regress`. The exercise is worth performing once for the sake of a single observation: $\hat{\beta}$ is a random vector, being a fixed linear map applied to the random y , and its realisation differs from the true $(1, 3)^\top$ by an amount that is itself the projection of the error vector ε ,

$$\hat{\beta} - \beta = (X^\top X)^{-1} X^\top \varepsilon.$$

Every result in the asymptotic theory of least squares is a statement about that expression.

11.5.1 The best linear predictor

Proposition 11.5.2 (Population least squares). Let (X, Y) be square-integrable random variables with $\text{Var}[X] > 0$. The minimiser of $\mathbb{E}[(Y - a - bX)^2]$ over $(a, b) \in \mathbb{R}^2$ is

$$b^* = \frac{\text{Cov}(X, Y)}{\text{Var}[X]}, \quad a^* = \mathbb{E}[Y] - b^* \mathbb{E}[X],$$

and the prediction error $Y - a^* - b^*X$ has mean zero and is uncorrelated with X .

Proof. Work in the Hilbert space L^2 with $\langle U, V \rangle = \mathbb{E}[UV]$ (Example 3.2.1) and let $M := \text{span}\{1, X\}$, a two-dimensional, hence closed, subspace by Corollary 6.3.1 applied inside the finite-dimensional span. Minimising $\mathbb{E}[(Y - a - bX)^2] = \|Y - a - bX\|^2$ is projecting Y onto M , so by Proposition 11.3.1 the solution is characterised by $\mathbb{E}[(Y - a - bX) \cdot 1] = 0$ and $\mathbb{E}[(Y - a - bX)X] = 0$. The first gives $a = \mathbb{E}[Y] - b\mathbb{E}[X]$; substituting into the second gives $\mathbb{E}[(Y - \mathbb{E}Y)X] - b\mathbb{E}[(X - \mathbb{E}X)X] = 0$, that is $\text{Cov}(X, Y) = b \text{Var}[X]$. ■

The formulas are the population counterparts of (8.12), and the argument is the same projection applied in L^2 rather than in $\mathbb{R}^n$: the sample version replaces $\mathbb{E}$ by the sample average, that is, replaces the L^2 inner product by $\langle u, v \rangle = n^{-1} \sum_i u_i v_i$. Enlarging M from $\text{span}\{1, X\}$ to the set of all square-integrable functions of X turns the best *linear* predictor into the best predictor, and Theorem 14.5.1 identifies it as $\mathbb{E}[Y | X]$.

11.6 Chapter Summary

Orthogonal complements are always closed subspaces; orthogonal sets of non-zero vectors are independent; and orthogonality makes norms add, which is Pythagoras. In a Hilbert space, every non-empty closed convex set contains a unique nearest point to any given point, with existence obtained from the parallelogram law and completeness rather than from compactness; on a subspace the characterising variational inequality becomes the orthogonality condition (11.3). Hence $H = M \oplus M^\perp$ for every closed subspace M , the projection operator is linear, idempotent, self-adjoint and norm-reducing, and every continuous linear functional on a Hilbert space is an inner product with a fixed vector.

In $\mathbb{R}^n$ the projection onto the column space of a full-column-rank X is $X(X^\top X)^{-1}X^\top$, and invertibility of $X^\top X$ is exactly independence of the columns. Least squares is that projection: the normal equations are orthogonality of the residual to the regressors, the analysis-of-variance identity is Pythagoras, R^2 is a squared cosine, adding regressors cannot worsen the fit because it enlarges the subspace, and the estimator is linear in the data. The same projection performed in L^2 gives the best linear predictor, with covariance over variance replacing the sample formula.

Exercise 11.6.1. Let $M_1 \subseteq M_2$ be closed subspaces of a Hilbert space. Show that $P_{M_1} P_{M_2} = P_{M_2} P_{M_1} = P_{M_1}$, and deduce that $\|y - P_{M_2} y\| \leq \|y - P_{M_1} y\|$. Give the econometric reading, and explain why the inequality is generally strict even when the added regressor is independent of y in the population.

Exercise 11.6.2. (Frisch–Waugh–Lovell.) Partition $X = [X_1 \ X_2]$ and let $Q_1 = I - X_1(X_1^\top X_1)^{-1}X_1^\top$. Show that the sub-vector $\hat{\beta}_2$ of (11.5) equals the least-squares coefficient from regressing $Q_1 y$ on $Q_1 X_2$. Identify at which step Proposition 11.3.2(ii) and (iii) are used, and interpret the result as “controlling for X_1 ”.

Exercise 11.6.3. Let $H = L^2([0, 1])$ and M the subspace of polynomials of degree at most 2. Compute $P_M f$ for $f(t) = e^t$ by applying Proposition 11.4.2 to $\{1, t, t^2\}$, and compare the resulting quadratic with the second-order Taylor polynomial of e^t at $t = 0$. Explain in terms of the two different optimality criteria why they differ.

Chapter 12

Constrained Optimisation: Lagrange and Kuhn–Tucker

12.1 The Programming Problem

Definition 12.1.1 (The programming problem). Let $f, g_1, \dots, g_n: \mathbb{R}^m \rightarrow \mathbb{R}$. The *programming problem*, henceforth *Problem 1*, is

$$\max_{x \in \mathbb{R}^m} f(x) \quad \text{subject to} \quad g_j(x) \geq 0 \quad \text{for each } j \in \{1, \dots, n\}. \quad (12.1)$$

The inequalities are the *constraint*, f is the *objective function*, and

$$D := \{x \in \mathbb{R}^m : g_j(x) \geq 0 \text{ for each } j\}$$

is the *domain of a program* of the program. The program is *bounded program* if f is bounded above on D .

The sign convention $g_j \geq 0$ is not restrictive. A constraint $h(x) \leq 0$ is $-h(x) \geq 0$, and an equality constraint $h(x) = 0$ is the pair $h(x) \geq 0$ and $-h(x) \geq 0$; nothing else is needed (Fukuda 2026c, slide 4). The convention is chosen so that the multipliers below are non-negative, and it must be applied consistently: writing a budget constraint as $m - \langle p, x \rangle \geq 0$ rather than $\langle p, x \rangle - m \geq 0$ is what makes the multiplier on wealth positive rather than negative.

Definition 12.1.2 (Binding and slack). The constraint $g_j(x) \geq 0$ is *binding constraint* at $x^* \in D$ if $g_j(x^*) = 0$, and *slack constraint* if $g_j(x^*) > 0$.

Remark 12.1.1 (Identify the binding constraints first). Before differentiating anything, determine which constraints bind at the optimum. The recommendation of (Fukuda 2026c, slides 6–9) is not a matter of convenience; it is where the economics of a problem is located, and three cases recur.

Non-negativity. With $\lim_{c \downarrow 0} u'(c) = \infty$ — an Inada condition — the constraint $c \geq 0$ cannot bind at an optimum, since moving to $c = \varepsilon$ raises utility without bound relative to the cost. The first-order conditions are then equalities and the analysis is standard. When instead a non-negativity-type constraint does bind — a borrowing constraint, a short-sale constraint, limited liability, the zero lower bound on nominal rates — the character of the solution changes, and solving the unconstrained first-order condition gives the wrong answer rather than an approximation to the right one.

Incentive constraints. A mechanism-design or agency problem may carry dozens of incentive-compatibility constraints. Establishing which of them bind — typically the local downward constraints under a single-crossing condition — reduces the problem to a tractable one and is the substantive step.

Direction of an equality. If a constraint is known to hold with equality at the optimum, it still matters which inequality is imposed. Writing the budget set as $\langle p, x \rangle \leq m$ rather than $\langle p, x \rangle \geq m$ yields a multiplier with the sign that makes it the marginal utility of wealth.

12.2 Five Formulations

The Kuhn–Tucker theorem is the statement that five apparently different problems have the same solutions under suitable hypotheses. Stating them separately makes visible exactly which hypothesis is responsible for which implication (Fukuda 2026c, slides 10–20).

Definition 12.2.1 (Problem 2: local maximum). Find $x \in D$ such that for every $u \in \mathbb{R}^m$ there is $\lambda_0 > 0$ with

$$0 \leq \lambda \leq \lambda_0 \text{ and } x + \lambda u \in D \implies f(x + \lambda u) \leq f(x).$$

The quantifier structure matters: λ_0 may depend on the direction u . This is weaker than requiring a single ball around x on which $f(x)$ dominates, and it is the formulation that matches the way local optimality is verified in economics — by considering one-dimensional deviations. The one-shot deviation principle in repeated games and the perturbation of a consumption path in a growth model are both instances of checking Definition 12.2.1 along a particular family of directions u .

Definition 12.2.2 (Problem 3: no improving feasible direction). Suppose $f, g_1, \dots, g_n$ are differentiable. Find $x \in D$ such that

$$\langle \nabla f(x), u \rangle \leq 0 \quad \text{for every } u \in \mathbb{R}^m \text{ satisfying } [g_j(x) = 0 \implies \langle \nabla g_j(x), u \rangle \geq 0].$$

In words: every direction that does not immediately violate a binding constraint fails to increase the objective to first order. The geometry of (Fukuda 2026c, slides 12–13) is that $\nabla f(x)$ makes an angle of at least 90° with every such u , while each binding $\nabla g_j(x)$ makes an angle of at most 90° . With a single binding constraint this forces $\nabla f(x)$ and $\nabla g_1(x)$ to be antiparallel, that is $\nabla f(x) + y_1 \nabla g_1(x) = 0$ for some $y_1 \geq 0$, which is the Lagrange condition.

Definition 12.2.3 (Problem 4: Lagrange’s problem). Suppose $f, g_1, \dots, g_n$ are differentiable. Find $x \in D$ and $y = (y_1, \dots, y_n) \in \mathbb{R}_+^n$ such that

$$\frac{\partial f}{\partial x_i}(x) + \sum_{j=1}^n y_j \frac{\partial g_j}{\partial x_i}(x) = 0 \quad \text{for each } i \in \{1, \dots, m\}, \quad (12.2)$$

$$\sum_{j=1}^n y_j g_j(x) = 0. \quad (12.3)$$

Definition 12.2.4 (Problem 5: saddle point). Define the *Lagrangian* $F: \mathbb{R}^m \times \mathbb{R}_+^n \rightarrow \mathbb{R}$ by

$$F(x, y) := f(x) + \sum_{j=1}^n y_j g_j(x). \quad (12.4)$$

Find $(\bar{x}, \bar{y}) \in \mathbb{R}^m \times \mathbb{R}_+^n$ that is a *saddle point* of F :

$$F(x, \bar{y}) \leq F(\bar{x}, \bar{y}) \leq F(\bar{x}, y) \quad \text{for all } x \in \mathbb{R}^m \text{ and } y \in \mathbb{R}_+^n. \quad (12.5)$$

Condition (12.5) has two halves: $\bar{x}$ maximises $F(\cdot, \bar{y})$ over the *whole* of $\mathbb{R}^m$, with no constraint, and $\bar{y}$ minimises $F(\bar{x}, \cdot)$ over $\mathbb{R}_+^n$. This is the saddle-point interpretation of Boyd and Vandenberghe (2004, Sec. 5.4, p. 237), whose Section 5.5 (p. 241) states the optimality conditions (12.2)–(12.3) in the same form and whose Section 5.6 (p. 249) develops the sensitivity reading of the multiplier recovered below in Theorem 12.3.2. The first is what makes the Lagrangian method a method: an unconstrained problem replaces a constrained one. The second encodes feasibility, as the next proposition shows.

Proposition 12.2.1 (Complementary slackness at a saddle point). Let $(\bar{x}, \bar{y})$ solve Problem 5. Then $\bar{x} \in D$ and $\bar{y}_j g_j(\bar{x}) = 0$ for every j ; in particular (12.3) holds.

Proof. Feasibility. Suppose $g_{j_0}(\bar{x}) < 0$ for some j_0 . Taking $y = \bar{y} + te_{j_0}$ with $t > 0$ gives $F(\bar{x}, y) = F(\bar{x}, \bar{y}) + tg_{j_0}(\bar{x}) \rightarrow -\infty$ as $t \rightarrow \infty$, contradicting the right-hand inequality of (12.5). So $\bar{x} \in D$.

Complementary slackness. Taking $y = 0 \in \mathbb{R}_+^n$ in (12.5) gives $F(\bar{x}, \bar{y}) \leq F(\bar{x}, 0) = f(\bar{x})$, that is $\sum_j \bar{y}_j g_j(\bar{x}) \leq 0$. Every summand is a product of the non-negative numbers $\bar{y}_j$ and $g_j(\bar{x})$, so every summand is non-negative; a non-negative sum that is at most zero has all terms zero. ■

This is Exercise 12.3 of Fukuda (2026d) (Fukuda 2026c, slide 19). The economic reading of $\bar{y}_j g_j(\bar{x}) = 0$ is exactly the dichotomy on (Fukuda 2026c, slide 16): either the constraint binds, or its marginal valuation is zero. A resource in excess supply has price zero, and a resource with a positive price is exhausted.

Proposition 12.2.2 (The unconditional implications). (i) If $(\bar{x}, \bar{y})$ solves Problem 5, then $\bar{x}$ solves Problem 1.

(ii) If $\bar{x}$ solves Problem 1, then $\bar{x}$ solves Problem 2.

Neither implication requires any hypothesis on f or the g_j .

Proof. (i) By Proposition 12.2.1, $\bar{x} \in D$ and $F(\bar{x}, \bar{y}) = f(\bar{x})$. For any $x \in D$, the left inequality of (12.5) gives

$$f(x) \leq f(x) + \sum_j \bar{y}_j g_j(x) = F(x, \bar{y}) \leq F(\bar{x}, \bar{y}) = f(\bar{x}),$$

the first step because $\bar{y}_j \geq 0$ and $g_j(x) \geq 0$ for feasible x .

(ii) A global maximiser over D satisfies the local condition of Definition 12.2.1 with λ_0 arbitrary. ■

This is Proposition 12.1 of Fukuda (2026d) (Fukuda 2026c, slide 19). Part (i) is the part used in practice: exhibiting a saddle point *proves* global optimality with no concavity, no differentiability and no constraint qualification. The converse implications are where hypotheses are needed.

12.3 The Kuhn–Tucker Theorem

Assumption 12.3.1 (Slater’s condition). There exists $\hat{x} \in \mathbb{R}^m$ such that $g_j(\hat{x}) \geq 0$ for every j , and $g_j(\hat{x}) > 0$ for every j for which g_j is not affine.

Theorem 12.3.1 (Kuhn–Tucker). Suppose $f, g_1, \dots, g_n$ are concave and Assumption 12.3.1 holds.

(i) Problems 1, 2 and 5 are equivalent: $\bar{x}$ solves one if and only if it solves the others, where for Problem 5 this means that some $\bar{y} \in \mathbb{R}_+^n$ makes $(\bar{x}, \bar{y})$ a saddle point.

(ii) If in addition $f, g_1, \dots, g_n$ are differentiable, all five problems are equivalent.

Proof. By Proposition 12.2.2 it remains to prove that a solution of Problem 2 is a solution of Problem 5, and, under differentiability, to relate Problems 3 and 4 to the rest.

Problem 2 $\Rightarrow$ Problem 1 under concavity. Let $\bar{x}$ solve Problem 2 and let $x \in D$. Concavity of each g_j makes D convex (it is an intersection of upper contour sets of concave, hence quasiconcave, functions), so $\bar{x} + \lambda(x - \bar{x}) \in D$ for $\lambda \in [0, 1]$. Applying Definition 12.2.1 with $u = x - \bar{x}$ gives, for small $\lambda > 0$,

$$f(\bar{x}) \geq f(\bar{x} + \lambda(x - \bar{x})) \geq (1 - \lambda)f(\bar{x}) + \lambda f(x),$$

the second inequality by concavity of f . Rearranging, $\lambda f(\bar{x}) \geq \lambda f(x)$, so $f(\bar{x}) \geq f(x)$.

Problem 1 $\Rightarrow$ Problem 5. Let $\bar{x}$ solve Problem 1 and set $\tilde{f}(x) := f(x) - f(\bar{x})$, which is concave. Optimality of $\bar{x}$ says the system

$$\tilde{f}(x) > 0 \quad \text{and} \quad g_j(x) \geq 0 \quad (j = 1, \dots, n)$$

has no solution. Assumption 12.3.1 is exactly the solvability hypothesis (9.3) of the Farkas–Minkowski theorem, so Theorem 9.5.1 supplies $\bar{y} \in \mathbb{R}_+^n$ with

$$\sup_{x \in \mathbb{R}^m} \left\{ \tilde{f}(x) + \sum_j \bar{y}_j g_j(x) \right\} \leq 0, \quad \text{that is} \quad F(x, \bar{y}) \leq f(\bar{x}) \quad \text{for all } x \in \mathbb{R}^m.$$

Evaluating at $x = \bar{x}$ gives $\sum_j \bar{y}_j g_j(\bar{x}) \leq 0$; every term is non-negative, so all vanish and $F(\bar{x}, \bar{y}) = f(\bar{x})$. Hence $F(x, \bar{y}) \leq F(\bar{x}, \bar{y})$ for all x , the left half of (12.5). For the right half, $y \in \mathbb{R}_+^n$ gives $F(\bar{x}, y) = f(\bar{x}) + \sum_j y_j g_j(\bar{x}) \geq f(\bar{x}) = F(\bar{x}, \bar{y})$, since each $g_j(\bar{x}) \geq 0$.

Problems 3, 4 and 5 under differentiability. If $(\bar{x}, \bar{y})$ is a saddle point, then $\bar{x}$ maximises the differentiable function $F(\cdot, \bar{y})$ over the open set $\mathbb{R}^m$, so Theorem 8.4.1 gives (12.2), and (12.3) is Proposition 12.2.1: Problem 5 implies Problem 4. Conversely, if (12.2)–(12.3) hold, then $F(\cdot, \bar{y})$ is concave — a non-negative combination of concave functions, by Proposition 9.3.2(i) applied to $-F$ — with vanishing gradient at $\bar{x}$, so Theorem 8.4.2 makes $\bar{x}$ a global maximiser of $F(\cdot, \bar{y})$, and the right half of (12.5) follows as above: Problem 4 implies Problem 5. That Problem 4 implies Problem 3 is immediate on taking the inner product of (12.2) with an admissible u ; the converse uses Theorem 9.5.1 applied to the linearised system at $\bar{x}$. ■

This is Theorem 12.1 of Fukuda (2026d, p. 436) (Fukuda 2026c, slide 20). The proof locates each hypothesis. Concavity of the g_j makes D convex, which is what turns local into global. Concavity of f is used in the same step and again in Theorem 8.4.2. Slater’s condition is used once, to invoke Theorem 9.5.1, and it is what guarantees a multiplier *exists*; Example 9.5.1 exhibits a program in which it does not.

Remark 12.3.1 (Constraint qualifications). Assumption 12.3.1 is one of several conditions under which multipliers exist; they are collectively the *constraint qualification*. Three points about them.

First, they are conditions on the *constraint functions*, not on the feasible set. The set $\{x \in \mathbb{R} : -x^2 \geq 0\} = \{0\}$ and the set $\{x \in \mathbb{R} : x \geq 0, -x \geq 0\} = \{0\}$ are identical, but the second description satisfies Assumption 12.3.1 — both constraints are affine — while the first does not, as Example 9.5.1 showed. A change of description changes whether multipliers exist.

Second, the affine exemption in Assumption 12.3.1 is not cosmetic. Linear programs and problems with only budget and non-negativity constraints satisfy the qualification automatically, which is why the constraint qualification is almost never mentioned in consumer theory and is essential in problems with nonlinear constraints such as incentive compatibility.

Third, when the objective and constraints are not concave the relevant qualification is local and differential rather than global. The standard condition is that the gradients of the binding constraints at $\bar{x}$ be linearly independent, and under it the conclusion is weaker: the first-order conditions (12.2)–(12.3) are necessary at a local maximum, but not sufficient, and Problem 5 is not equivalent to Problem 1. See Mangasarian (1994, Ch. 7).

Example 12.3.1 (Failure of the constraint qualification). Maximise $f(x_1, x_2) = x_1$ subject to $g_1(x) = -x_2 \geq 0$ and $g_2(x) = x_2 - (1 - x_1)^3 \geq 0$. The feasible set is $\{(x_1, 0) : x_1 \geq 1\}$ together with the region below the cubic, and $\bar{x} = (1, 0)$ is a local maximiser on the curve. At $\bar{x}$ both constraints bind, and $\nabla g_1(\bar{x}) = (0, -1)^\top$, $\nabla g_2(\bar{x}) = (0, 1)^\top$ are linearly dependent. No $y \in \mathbb{R}_+^2$ satisfies $\nabla f(\bar{x}) + y_1 \nabla g_1 + y_2 \nabla g_2 = 0$, since the first component of ∇f is 1 and both constraint gradients have first component 0. The first-order conditions therefore have no solution at a point that is optimal, and a search over stationary points would miss it.

12.3.1 Interpretation of the multipliers

Theorem 12.3.2 (Envelope theorem). Consider the parametrised program

$$V(b, \alpha) := \max_{x \in \mathbb{R}^m} f(x, \alpha) \quad \text{subject to} \quad g(x, \alpha) = b,$$

with f, g continuously differentiable, and suppose that on a neighbourhood of $(\bar{b}, \bar{\alpha})$ the solution $x^*(b, \alpha)$ and the multiplier $\lambda^*(b, \alpha)$ are unique and continuously differentiable. Then

$$\frac{\partial V}{\partial \alpha} = \frac{\partial f}{\partial \alpha}(x^*, \alpha) - \lambda^* \frac{\partial g}{\partial \alpha}(x^*, \alpha), \quad \frac{\partial V}{\partial b} = \lambda^*. \quad (12.6)$$

Proof. Write $V(b, \alpha) = f(x^*, \alpha) + \lambda^*(b - g(x^*, \alpha))$, which is legitimate because the bracket vanishes at the optimum. Differentiate with respect to α using Theorem 8.3.1:

$$\frac{\partial V}{\partial \alpha} = \underbrace{(f_x - \lambda^* g_x)}_{=0 \text{ by (12.2)}} \cdot \frac{\partial x^*}{\partial \alpha} + \underbrace{(b - g(x^*, \alpha))}_{=0 \text{ by feasibility}} \cdot \frac{\partial \lambda^*}{\partial \alpha} + f_\alpha - \lambda^* g_\alpha.$$

The first bracket vanishes by the first-order condition and the second by the constraint, leaving the last two terms. Differentiating instead with respect to b leaves only the term $\lambda^* \cdot \partial b / \partial b = \lambda^*$. ■

This is Proposition 13.9 and the constrained envelope theorem of Fukuda (2026d, pp. 479–481), and the computation of (Fukuda 2026c, slide 28). The content is that the indirect effects, operating through the response $\partial x^* / \partial \alpha$ of the optimal choice, cancel: at an optimum the marginal value of moving x is zero, so a marginal change in the parameter has only its direct effect. This is what makes λ^* the *shadow price* of the constraint: it is the rate at which the maximised value rises when the constraint is relaxed, and it is computable without differentiating the solution.

Remark 12.3.2 (Differentiability of V is an assumption, not a conclusion). **Theorem 12.3.2** presumes that x^* and λ^* are differentiable in the parameters, which is itself a conclusion of the implicit function theorem (**Theorem 8.6.1**) applied to the first-order conditions, and requires the bordered Hessian to be nonsingular at the optimum. When the solution set is not a singleton or when the active set changes, V can have a kink and (12.6) fails as stated. The general form due to Milgrom and Segal (2002) replaces the derivative by one-sided derivatives and requires only that $f_\alpha(x^*, \alpha)$ exist at a single maximiser: $V'(\alpha^-) \leq f_\alpha(x^*, \alpha) \leq V'(\alpha^+)$ whenever the one-sided derivatives exist, with equality when V is differentiable. That version is what applies to mechanism design, where X may be a discrete set with no differentiable structure at all.

12.4 Economic Application: The Consumer's Problem

Problem. With $f(x_1, x_2) = x_1^\alpha x_2^{1-\alpha}$, $\alpha \in (0, 1)$, and constraints

$$g_1(x) = m - p_1 x_1 - p_2 x_2 \geq 0, \quad g_2(x) = x_1 \geq 0, \quad g_3(x) = x_2 \geq 0,$$

with $p_1, p_2, m > 0$, solve (12.1).

Proposition 12.4.1. The problem has the unique solution

$$x_1^* = \frac{\alpha m}{p_1}, \quad x_2^* = \frac{(1-\alpha)m}{p_2}, \quad (12.7)$$

with multipliers $y_2^* = y_3^* = 0$ and

$$y_1^* = \frac{(x_1^*)^\alpha (x_2^*)^{1-\alpha}}{m} = \frac{v(p, m)}{m}, \quad (12.8)$$

where $v(p, m)$ is the indirect utility. Expenditure shares are constant: $p_1 x_1^* / m = \alpha$ and $p_2 x_2^* / m = 1 - \alpha$.

Proof. Existence and uniqueness. The budget set is compact (**Example 4.4.1**) and non-empty, and f is continuous, so a solution exists by **Corollary 5.4.1**. On $\mathbb{R}_{++}^2$ the function f is strictly quasiconcave, so the solution is unique by **Proposition 9.3.5** (Fukuda 2026c, slide 21).

Which constraints bind. If $x_1^* = 0$ then $f = 0$, while the feasible point $(m/(2p_1), m/(2p_2))$ gives $f > 0$; so g_2 is slack, and symmetrically g_3 . The budget constraint binds: f is strictly increasing in each argument on $\mathbb{R}_{++}^2$, so an unspent unit of wealth could be spent to raise utility.

First-order conditions. With $y_2 = y_3 = 0$ by **Proposition 12.2.1**, (12.2) reads

$$\alpha(x_1^*)^{\alpha-1}(x_2^*)^{1-\alpha} = y_1 p_1, \quad (1-\alpha)(x_1^*)^\alpha (x_2^*)^{-\alpha} = y_1 p_2.$$

Dividing eliminates y_1 and gives $\frac{x_2^*}{x_1^*} = \frac{1-\alpha}{\alpha} \cdot \frac{p_1}{p_2}$, that is $p_2 x_2^* = \frac{1-\alpha}{\alpha} p_1 x_1^*$. Substituting into the binding budget constraint $p_1 x_1^* + p_2 x_2^* = m$ yields $p_1 x_1^* (1 + \frac{1-\alpha}{\alpha}) = m$, hence (12.7).

Multiplier. From the first condition,

$$y_1^* = \frac{\alpha(x_1^*)^{\alpha-1}(x_2^*)^{1-\alpha}}{p_1} = \frac{\alpha(x_1^*)^\alpha (x_2^*)^{1-\alpha}}{p_1 x_1^*} = \frac{(x_1^*)^\alpha (x_2^*)^{1-\alpha}}{m},$$

using $p_1 x_1^* = \alpha m$. The second condition gives the same value, using $p_2 x_2^* = (1-\alpha)m$.

Verification via Problem 5. f is concave on $\mathbb{R}_+^2$ and the g_j are affine, so [Assumption 12.3.1](#) holds automatically and [Theorem 12.3.1](#) applies; alternatively, the candidate with its multipliers satisfies (12.2)–(12.3), which by the last part of that proof makes it a saddle point, and [Proposition 12.2.2\(i\)](#) then certifies global optimality without any further argument. ■

Reading the multiplier. Formula (12.8) says $y_1^* = v(p, m)/m$, and since Cobb–Douglas indirect utility is $v(p, m) = m \cdot (\alpha/p_1)^\alpha ((1-\alpha)/p_2)^{1-\alpha}$, the multiplier is independent of m : the marginal utility of wealth is constant, because the utility function is homogeneous of degree one in the bundle and hence in wealth. [Theorem 12.3.2](#) confirms this directly, $\partial v/\partial m = y_1^*$, which is the computation of (Fukuda 2026c, slides 28–29) in which every term involving dx_i^*/dm cancels against a first-order condition.

The constant expenditure shares are the economic signature of Cobb–Douglas preferences and are the reason the functional form is used: demand for each good is independent of the price of the other, income elasticities are unity, and the own-price elasticity is exactly -1 . Each of these is a strong restriction, and all three follow from the single fact that the elasticity of substitution is 1.

Numerical counterpart. Three of the accompanying programs solve the same problem by sequential quadratic programming (Fukuda 2026c, slides 33–36): `consumer_choice_fmincon.m`, `consumer_choice_sqp_Octave.m` and `consumer_choice_python.py`. Two observations are worth making about the comparison. The solver returns a multiplier alongside the solution, and it is the same y_1^* ; a numerical multiplier that differs from $\partial V/\partial m$ computed by finite differences signals either a non-binding constraint or a failure of the constraint qualification. And the solver requires the problem in minimisation form with constraints written as $c(x) \leq 0$, which is the sign flip of [Definition 12.1.1](#): the multiplier returned is then non-negative for the reversed convention, and reading its sign without checking the convention is the most common error in applied work.

12.5 Economic Application: Intertemporal Choice

Problem. A consumer with wealth $W > 0$ chooses consumption in two periods, facing gross interest rate $R > 0$ and discount factor $\beta \in (0, 1)$:

$$\max_{c_1, c_2 \geq 0} u(c_1) + \beta u(c_2) \quad \text{subject to} \quad c_1 + \frac{c_2}{R} \leq W, \quad (12.9)$$

with u strictly increasing, strictly concave, differentiable and satisfying $\lim_{c \downarrow 0} u'(c) = \infty$.

Proposition 12.5.1 (Euler equation). Problem (12.9) has a unique solution, it is interior, the budget constraint binds, and the solution satisfies

$$u'(c_1^*) = \beta R u'(c_2^*), \quad (12.10)$$

with multiplier $\lambda^* = u'(c_1^*)$. For $u(c) = c^{1-\gamma}/(1-\gamma)$ this gives $c_2^*/c_1^* = (\beta R)^{1/\gamma}$ and

$$c_1^* = \frac{W}{1 + \beta^{1/\gamma} R^{(1-\gamma)/\gamma}}.$$

Proof. The feasible set is compact and convex and the objective continuous and strictly concave, so a unique solution exists by [Corollary 5.4.1](#) and [Proposition 9.3.5](#). The Inada condition rules out $c_i^* = 0$: at such a point the derivative of the objective in c_i is $+\infty$ while the marginal cost is finite, so a small increase is feasible and improving. Monotonicity makes the budget constraint bind. The constraints are affine, so [Assumption 12.3.1](#) holds and [Theorem 12.3.1](#) applies. The Lagrangian $F = u(c_1) + \beta u(c_2) + \lambda(W - c_1 - c_2/R)$ gives $u'(c_1^*) = \lambda^*$ and $\beta u'(c_2^*) = \lambda^*/R$; eliminating λ^* yields (12.10). With CRRA utility, $u'(c) = c^{-\gamma}$ and (12.10) reads $(c_1^*)^{-\gamma} = \beta R (c_2^*)^{-\gamma}$, that is $c_2^* = (\beta R)^{1/\gamma} c_1^*$; substituting into the binding budget constraint gives the stated c_1^* . ■

Content of (12.10). The marginal utility cost of saving one more unit today equals its discounted marginal utility benefit tomorrow, scaled by the gross return. Three comparative statics follow, each requiring a different piece of the machinery. The consumption growth rate $(\beta R)^{1/\gamma}$ is increasing in R with elasticity $1/\gamma$, the elasticity of intertemporal substitution. The effect of R on c_1^* is ambiguous in sign

and is positive exactly when $\gamma > 1$, the income effect dominating the substitution effect. And $\lambda^* = u'(c_1^*)$ is the marginal utility of wealth, so by [Theorem 12.3.2](#) $\partial V/\partial W = u'(c_1^*)$ — an identity that is trivial here and is the Benveniste–Scheinkman formula of [Chapter 15](#) in the infinite-horizon case. The corresponding programs are `two_periods_consumption_fmincon.m`, `two_periods_consumption_sqp_Octave.m` and `two_periods_consumption_python.py` (Fukuda 2026c, slides 37–40).

12.6 Economic Application: Nash Bargaining

Problem. Two players divide a surplus. The set of feasible utility pairs is $S \subseteq \mathbb{R}^2$, compact and convex, and the disagreement point is $d \in S$. The *Nash bargaining solution* solves

$$\max_{(u_1, u_2) \in S} (u_1 - d_1)(u_2 - d_2) \quad \text{subject to} \quad u_1 \geq d_1, u_2 \geq d_2. \quad (12.11)$$

Proposition 12.6.1. Suppose S is compact and convex and there is $u \in S$ with $u \gg d$. Then (12.11) has a unique solution u^* , it satisfies $u^* \gg d$, and if $S = \{u \in \mathbb{R}_+^2 : u_1/a_1 + u_2/a_2 \leq 1\}$ with $a_1, a_2 > 0$ and $d = 0$, then $u^* = (a_1/2, a_2/2)$: the surplus is split equally in normalised units.

Proof. Existence follows from compactness of the feasible set, which is the intersection of S with two closed half-spaces, and continuity of the objective ([Corollary 5.4.1](#)). The objective is not concave, but its logarithm $\log(u_1 - d_1) + \log(u_2 - d_2)$ is strictly concave on the region where both factors are positive, and by [Proposition 9.3.4](#) the strictly increasing transformation preserves the maximiser; uniqueness is [Proposition 9.3.5](#). The hypothesis $u \gg d$ makes the maximum positive, so $u^* \gg d$ and the transformation is legitimate at the optimum.

For the linear frontier with $d = 0$, maximise $\log u_1 + \log u_2$ subject to $u_1/a_1 + u_2/a_2 \leq 1$. This is [Proposition 12.4.1](#) with $\alpha = \frac{1}{2}$, $p_i = 1/a_i$ and $m = 1$, giving $u_i^* = a_i/2$. ■

The identification with the Cobb–Douglas consumer problem is not a coincidence: maximising a product subject to a linear constraint is the same problem in both cases, and the equal split of the Nash solution corresponds to the equal expenditure shares of symmetric Cobb–Douglas preferences. The numerical counterparts are `Nash_bargaining_fmincon.m`, `Nash_bargaining_sqp_Octave.m` and `Nash_bargaining_python.py` (Fukuda 2026c, slides 41–44). Two practical points appear there. Solving (12.11) directly is numerically worse conditioned than solving the logarithmic version, because the product objective is flat near the axes; and a solver started at the disagreement point returns it, since the gradient of the product vanishes there, so the initial guess must lie in the interior.

12.7 Chapter Summary

A programming problem in the standard form (12.1) admits five formulations: global maximum, local maximum, absence of an improving feasible direction, the Lagrange conditions with complementary slackness, and a saddle point of the Lagrangian. Two implications hold with no hypotheses at all: a saddle point yields a global maximum, and a global maximum is a local maximum. The converses require concavity of the objective and constraints and a constraint qualification, and under those hypotheses the Kuhn–Tucker theorem makes all five equivalent, with differentiability needed only for the two formulations that mention derivatives. The proof of the essential implication — from global optimality to the existence of multipliers — is an application of the Farkas–Minkowski theorem, itself a consequence of the separating hyperplane theorem, so the multipliers are the coefficients of a separating hyperplane.

Complementary slackness says a constraint either binds or has zero multiplier. The multiplier is the shadow price of the constraint, and the envelope theorem makes this precise: at an optimum, the derivative of the value function with respect to a parameter is the partial derivative of the Lagrangian, with all indirect effects through the optimal choice cancelling against the first-order conditions. Applied to the Cobb–Douglas consumer, the machinery yields constant expenditure shares and a multiplier equal to indirect utility divided by wealth; applied to two-period consumption, the Euler equation $u'(c_1) = \beta R u'(c_2)$ with the multiplier equal to the marginal utility of current consumption; and applied to bargaining, the Nash solution, which is formally the same optimisation as symmetric Cobb–Douglas demand.

Exercise 12.7.1. A firm maximises $pF(K, L) - rK - wL$ subject to $K \leq \bar{K}$, with F concave and increasing. Write the Kuhn–Tucker conditions, and show that the multiplier on the capacity constraint equals $pF_K - r$ when the constraint binds and 0 otherwise. Interpret it as the shadow rental rate, and use [Theorem 12.3.2](#) to compute the value of an extra unit of capacity.

Exercise 12.7.2. Consider $\max x_1$ subject to $g_1(x) = x_2 - x_1^3 \geq 0$ and $g_2(x) = -x_2 \geq 0$. Show that $\bar{x} = (0, 0)$ is the unique feasible point with $x_1 \geq 0$ maximal, that both constraints bind there, and that no multipliers satisfy (12.2). Identify which hypothesis of [Theorem 12.3.1](#) fails, and verify that adding the affine constraint $x_1 \leq 0$ makes the program satisfy [Assumption 12.3.1](#).

Exercise 12.7.3. Derive the expenditure-minimisation dual of [Proposition 12.4.1](#): minimise $p_1x_1 + p_2x_2$ subject to $x_1^\alpha x_2^{1-\alpha} \geq \bar{u}$. Show that the multiplier on the utility constraint is the reciprocal of y_1^* from (12.8) evaluated at the corresponding wealth, and that the expenditure function $e(p, \bar{u})$ satisfies $\partial e / \partial p_i = x_i^h$, the Hicksian demand. State which part of [Theorem 12.3.2](#) delivers the last identity.

Chapter 13

Correspondences, the Maximum Theorem and Fixed Points

13.1 Why Set-Valued Maps

The optimisation of [Chapter 12](#) produced a maximiser and treated it as a point. Two features of economic models make that treatment inadequate.

The maximiser need not be unique. Consumer demand is single-valued only when preferences are strictly convex; with linear indifference curves the consumer is indifferent over a whole face of the budget set, and the demand *set* moves discontinuously as relative prices cross the ratio at which the linear indifference curves are parallel to the budget line. In game theory the best response to a mixed strategy is generically a vertex of the simplex but is a whole face whenever the player is indifferent, and indifference is exactly what characterises equilibrium in mixed strategies. Nothing can be said about the existence of equilibrium if the objects being compared are required to be single-valued.

The feasible set itself moves with the parameters. In the consumer's problem the budget set depends on prices and income; in a dynamic programme the set of feasible continuations depends on the current state. Asking whether the value of the problem varies continuously with the parameters is therefore a question about the joint behaviour of a function and a moving domain, and it cannot be posed until the movement of the domain has been given a topology.

This chapter supplies both. It defines continuity for set-valued maps, proves Berge's theorem that the value function of a parametrised programme is continuous and the maximiser correspondence upper hemicontinuous, and states the two fixed-point theorems — Brouwer's for functions and Kakutani's for correspondences — on which the existence of Walrasian and Nash equilibrium rests. [Theorem 6.4.1](#) gave a fixed-point theorem of a different kind, and [Proposition 6.4.1](#) recorded the trade: contraction buys uniqueness and an algorithm at the price of a Lipschitz hypothesis, whereas Brouwer and Kakutani buy existence alone, from compactness and convexity, with no uniqueness and no constructive procedure.

13.2 Correspondences

Definition 13.2.1 (Correspondence). A *correspondence* $\varphi: X \rightrightarrows Y$ from a set X into a set Y is a function $\varphi: X \rightarrow \mathcal{P}(\cdot)Y$ into the power set of Y ; with each $x \in X$ it associates a subset $\varphi(x)$ of Y . Its *graph* of a correspondence is

$$\text{gr } \varphi := \{(x, y) \in X \times Y : y \in \varphi(x)\} \subseteq X \times Y,$$

and φ is recovered from it by $\varphi(x) = \{y \in Y : (x, y) \in \text{gr } \varphi\}$.

The definition permits $\varphi(x) = \emptyset$; φ is *non-empty-valued correspondence* if $\varphi(x) \neq \emptyset$ for every x . A function $f: X \rightarrow Y$ is the single-valued correspondence $x \mapsto \{f(x)\}$, so every statement below specialises to a statement about functions, and the specialisation is worth checking each time. Identifying φ with $\text{gr } \varphi$ also identifies it with a binary relation from X to Y in the sense of [Chapter 1](#), which is the reason the graph, rather than the map into $\mathcal{P}(\cdot)Y$, is the object one topologises.

Definition 13.2.2 (Value restrictions). Let X, Y be topological spaces. A correspondence $\varphi: X \rightrightarrows Y$ is *open-valued*, *closed-valued*, *compact-valued* or *convex-valued* if $\varphi(x)$ is respectively open, closed, compact or convex in Y for every $x \in X$ (convexity requiring Y to be a vector space).

Example 13.2.1 (Budget and demand correspondences). Define $B: \mathbb{R}_{++}^2 \times \mathbb{R}_+ \rightrightarrows \mathbb{R}_+^2$ by

$$B(p, m) := \{x \in \mathbb{R}_+^2 : \langle p, x \rangle \leq m\}. \quad (13.1)$$

By [Example 4.4.1](#) the budget set is compact when $p \gg 0$, and it is convex as the intersection of $\mathbb{R}_+^2$ with a half-space, so B is compact-valued, closed-valued and convex-valued. The associated *demand correspondence*

$$D(p, m) := \arg \max_{x \in B(p, m)} u(x) \quad (13.2)$$

is the leading example of an *argmax* correspondence, which associates with each parameter the set of maximisers. It is non-empty-valued whenever u is continuous, by [Corollary 5.4.1](#), and it is single-valued whenever u is strictly quasiconcave, by [Proposition 9.3.5](#); without strict quasiconcavity it need be neither.

Example 13.2.2 (A correspondence with a closed graph). $\varphi: \mathbb{R} \rightrightarrows \mathbb{R}$, $\varphi(x) = [x - 1, x + 1]$, has graph $\text{gr } \varphi = \{(x, y) : |y - x| \leq 1\}$, a closed band about the diagonal. It is compact-valued and convex-valued.

13.2.1 Two inverses

A function has one inverse image; a correspondence has two, according to whether the whole value or merely part of it is required to land in the target set. The distinction is the source of the two distinct notions of continuity.

Definition 13.2.3 (Upper and lower inverse). Let $\varphi: X \rightrightarrows Y$ and $A \subseteq Y$. The *upper inverse* (or strong inverse) and *lower inverse* (or weak inverse) of A are

$$\varphi^u(A) := \{x \in X : \varphi(x) \subseteq A\}, \quad \varphi^\ell(A) := \{x \in X : \varphi(x) \cap A \neq \emptyset\}. \quad (13.3)$$

If φ is single-valued, $\varphi^u(A) = f^{-1}(A) = \varphi^\ell(A)$, so the two notions coincide precisely on functions.

Proposition 13.2.1 (Calculus of the two inverses). Let $\varphi: X \rightrightarrows Y$ and let $A, B \subseteq Y$ and $(A_i)_{i \in I}$ be subsets of Y .

- (i) If φ is non-empty-valued then $\varphi^u(A) \subseteq \varphi^\ell(A)$.
- (ii) $\varphi^u(A) = (\varphi^\ell(A^c))^c$ and $\varphi^\ell(A) = (\varphi^u(A^c))^c$.
- (iii) $\varphi^u(\bigcap_i A_i) = \bigcap_i \varphi^u(A_i)$ and $\varphi^\ell(\bigcup_i A_i) = \bigcup_i \varphi^\ell(A_i)$.
- (iv) $A \subseteq B$ implies $\varphi^u(A) \subseteq \varphi^u(B)$ and $\varphi^\ell(A) \subseteq \varphi^\ell(B)$; consequently $\bigcup_i \varphi^u(A_i) \subseteq \varphi^u(\bigcup_i A_i)$ and $\varphi^\ell(\bigcap_i A_i) \subseteq \bigcap_i \varphi^\ell(A_i)$, both inclusions being strict in general.

Proof. (i) If $\varphi(x) \subseteq A$ and $\varphi(x) \neq \emptyset$ then $\varphi(x) \cap A = \varphi(x) \neq \emptyset$.

(ii) $x \notin \varphi^\ell(A^c)$ says $\varphi(x) \cap A^c = \emptyset$, that is $\varphi(x) \subseteq A$, which is $x \in \varphi^u(A)$. The second identity is the first applied to A^c and complemented.

(iii) $\varphi(x) \subseteq \bigcap_i A_i$ if and only if $\varphi(x) \subseteq A_i$ for every i ; and $\varphi(x) \cap \bigcup_i A_i \neq \emptyset$ if and only if $\varphi(x) \cap A_i \neq \emptyset$ for some i .

(iv) Monotonicity is immediate from (13.3), and the two inclusions follow. For strictness take $X = \{x\}$, $Y = \{1, 2\}$, $\varphi(x) = \{1, 2\}$, $A_1 = \{1\}$, $A_2 = \{2\}$: then $\varphi^u(A_1) \cup \varphi^u(A_2) = \emptyset$ while $\varphi^u(A_1 \cup A_2) = \{x\}$, and dually $\varphi^\ell(A_1 \cap A_2) = \emptyset$ while $\varphi^\ell(A_1) \cap \varphi^\ell(A_2) = \{x\}$. ■

Part (iii) is the reason the two inverses divide the work of [Proposition 1.2.2](#), where a single preimage commuted with both unions and intersections. Set-valuedness breaks that symmetry, and the break is what forces two continuity concepts rather than one.

13.3 Hemicontinuity

Definition 13.3.1 (Upper and lower hemicontinuity). Let X, Y be topological spaces and $\varphi: X \rightrightarrows Y$.

- (i) φ is *upper hemicontinuity* at $x \in X$ if for every open $U \supseteq \varphi(x)$ there is an open neighbourhood V of x with $\varphi(z) \subseteq U$ for every $z \in V$.
- (ii) φ is *lower hemicontinuity* at $x \in X$ if for every open U with $\varphi(x) \cap U \neq \emptyset$ there is an open neighbourhood V of x with $\varphi(z) \cap U \neq \emptyset$ for every $z \in V$.
- (iii) φ is *continuous* at x if it is both; and φ is upper hemicontinuous, lower hemicontinuous or continuous if it is so at every point of X .

Read the two conditions as restrictions on how the value set may move. Upper hemicontinuity forbids the value from suddenly *expanding*: an open set that contains $\varphi(x)$ still contains $\varphi(z)$ for z near x . Lower hemicontinuity forbids it from suddenly *contracting*: any part of $\varphi(x)$ that an open set reaches is still reached by $\varphi(z)$ for z near x . Neither implies the other.

Proposition 13.3.1 (Global characterisations). Let $\varphi: X \rightrightarrows Y$ between topological spaces.

- (i) φ is upper hemicontinuous if and only if $\varphi^u(V)$ is open in X for every open $V \subseteq Y$, if and only if $\varphi^\ell(F)$ is closed in X for every closed $F \subseteq Y$.
- (ii) φ is lower hemicontinuous if and only if $\varphi^\ell(V)$ is open in X for every open $V \subseteq Y$, if and only if $\varphi^u(F)$ is closed in X for every closed $F \subseteq Y$.

Consequently a single-valued correspondence is upper hemicontinuous if and only if it is lower hemicontinuous, if and only if the underlying function is continuous.

Proof. (i) Suppose φ is upper hemicontinuous and $V \subseteq Y$ is open. If $x \in \varphi^u(V)$ then V is an open set containing $\varphi(x)$, so there is an open $W \ni x$ with $\varphi(z) \subseteq V$ for all $z \in W$, that is $W \subseteq \varphi^u(V)$; hence $\varphi^u(V)$ is open. Conversely, if $\varphi^u(V)$ is open for every open V and $U \supseteq \varphi(x)$ is open, then $\varphi^u(U)$ is an open set containing x and serves as the required neighbourhood. The equivalence with the closed-set statement is Proposition 13.2.1(ii): $\varphi^u(V)$ open for all open V means $\varphi^\ell(V^c) = \varphi^u(V)^c$ closed for all such V , and $V \mapsto V^c$ is a bijection between open and closed sets.

(ii) The same argument with the roles of the two inverses exchanged.

The final claim follows because for a single-valued φ the two inverses agree with f^{-1} , and Theorem 5.2.1 identifies continuity of f with openness of $f^{-1}(V)$ for open V . ■

Example 13.3.1 (The two notions are independent). Define $\varphi, \psi, \gamma: [0, 1] \rightrightarrows [0, 1]$ by

$$\varphi(x) = \begin{cases} \{0\}, & x \in [0, 1), \\ [0, 1], & x = 1, \end{cases} \quad \psi(x) = \begin{cases} [0, 1], & x \in [0, 1), \\ \{0\}, & x = 1, \end{cases} \quad \gamma(x) = [0, 1].$$

Then φ is upper hemicontinuous everywhere but not lower hemicontinuous at $x = 1$: the open set $U = (\frac{1}{2}, 1]$ meets $\varphi(1)$ but misses $\varphi(z) = \{0\}$ for every $z < 1$. Symmetrically ψ is lower hemicontinuous everywhere but not upper hemicontinuous at $x = 1$: the open set $U = [0, \frac{1}{2})$ contains $\psi(1)$ but contains no $\psi(z)$ with $z < 1$. The constant correspondence γ is continuous. The value set of φ explodes at $x = 1$ in the sense forbidden by lower hemicontinuity only in the limit from the left, and the value set of ψ implodes there. The three graphs are drawn in Figure 13.1.

Proposition 13.3.2 (Compactness is preserved). Let $\varphi: X \rightrightarrows Y$ be compact-valued and upper hemicontinuous, and let $A \subseteq X$ be compact. Then $\varphi(A) := \bigcup_{x \in A} \varphi(x)$ is compact in Y .

Proof. Let $\mathcal{V}$ be an open cover of $\varphi(A)$. Fix $x \in A$. The compact set $\varphi(x)$ is covered by finitely many members of $\mathcal{V}$; let U_x be their union, an open set containing $\varphi(x)$. Upper hemicontinuity supplies an open $W_x \ni x$ with $\varphi(z) \subseteq U_x$ for $z \in W_x$. The family $(W_x)_{x \in A}$ covers the compact set A , so finitely many $W_{x_1}, \dots, W_{x_k}$ suffice, and then $\varphi(A) \subseteq \bigcup_{i=1}^k U_{x_i}$, a union of finitely many finite subfamilies of $\mathcal{V}$. ■

This generalises the statement that a continuous image of a compact set is compact (Proposition 5.2.4), and both hypotheses are needed: the upper hemicontinuous but non-compact-valued $\varphi(x) =$

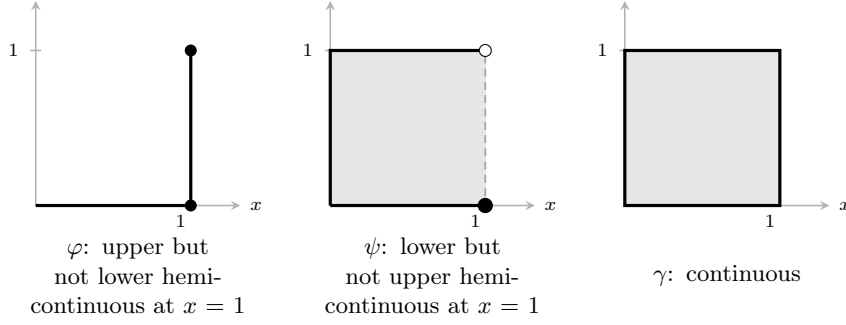


Figure 13.1: The graphs of the three correspondences of [Example 13.3.1](#). Shading marks the values; a dashed edge is not part of the graph. Upper hemicontinuity fails when the value set explodes in the limit, lower hemicontinuity when it implodes.

$\mathbb{R}$ has non-compact image, and the compact-valued but not upper hemicontinuous ψ of [Example 13.3.1](#) would fail if $[0, 1]$ were replaced by an unbounded set.

13.3.1 Closed graphs

Definition 13.3.2 (Closed graph). A correspondence $\varphi: X \rightrightarrows Y$ between topological spaces *closed graph* if $\text{gr } \varphi$ is closed in $X \times Y$ with the product topology.

Theorem 13.3.1 (Closed graph theorem). Let X be a metric space and Y a compact metric space. A correspondence $\varphi: X \rightrightarrows Y$ has a closed graph if and only if it is upper hemicontinuous and closed-valued.

Proof. ($\Leftarrow$) A closed subset of a compact metric space is compact, so a closed-valued φ into Y is compact-valued. Let $(x, y) \notin \text{gr } \varphi$. Since $\varphi(x)$ is compact and $y \notin \varphi(x)$, the number $\varepsilon := \text{dist}(y, \varphi(x))$ is strictly positive, and $U := \bigcup_{z \in \varphi(x)} B_{\varepsilon/2}(z)$ is an open set containing $\varphi(x)$ and disjoint from $B_{\varepsilon/2}(y)$. Upper hemicontinuity gives an open $W \ni x$ with $\varphi(w) \subseteq U$ for $w \in W$, so $(W \times B_{\varepsilon/2}(y)) \cap \text{gr } \varphi = \emptyset$ and $(\text{gr } \varphi)^c$ is open.

($\Rightarrow$) A closed graph gives closed values, since $\varphi(x) = \{y : (x, y) \in \text{gr } \varphi\}$ is the slice of a closed set. Suppose φ fails to be upper hemicontinuous at some x : there is an open $V \supseteq \varphi(x)$ such that every neighbourhood of x contains a point whose value is not contained in V . Taking the neighbourhoods $B_{1/n}(x)$ produces $x_n \rightarrow x$ and $y_n \in \varphi(x_n) \setminus V$. Since Y is compact, [Theorem 4.4.1](#) gives a subsequence $y_{n_k} \rightarrow y$, and $y \in V^c$ because V^c is closed, so $y \notin \varphi(x)$. Then $(x_{n_k}, y_{n_k}) \in \text{gr } \varphi$ converges to $(x, y) \notin \text{gr } \varphi$, and by [Theorem 4.3.1](#) the graph is not closed. ■

Remark 13.3.1 (Compactness of Y is not decoration). Without compactness of Y a closed graph does not give upper hemicontinuity. Take $X = Y = \mathbb{R}$ and $\varphi(x) = \{1/x\}$ for $x \neq 0$, $\varphi(0) = \{0\}$. The graph is the hyperbola together with the origin, which is closed in $\mathbb{R}^2$, but φ is not upper hemicontinuous at 0: the open set $(-1, 1) \supseteq \varphi(0)$ contains no $\varphi(z)$ with z small. Economic models in which quantities are unbounded require the closed-graph criterion to be supplemented by a compactness or boundedness argument before upper hemicontinuity may be inferred.

13.3.2 Sequential criteria

In finite-dimensional applications the definitions are almost always verified through sequences.

Theorem 13.3.2 (Sequential criterion for upper hemicontinuity). Let $X \subseteq \mathbb{R}^k$ and $Y \subseteq \mathbb{R}^d$ and let $\varphi: X \rightrightarrows Y$ be compact-valued. Then φ is upper hemicontinuous at $x \in X$ if and only if for every sequence $(x_n, y_n) \in \text{gr } \varphi$ with $x_n \rightarrow x$ the sequence (y_n) has a subsequence converging to a point of $\varphi(x)$.

Proof. ($\Rightarrow$) Let $x_n \rightarrow x$ with $y_n \in \varphi(x_n)$. Since $\varphi(x)$ is compact it is bounded, so it is contained in some open ball B ; upper hemicontinuity gives a neighbourhood W of x with $\varphi(z) \subseteq B$ for $z \in W$, and $x_n \in W$ for $n \geq n_0$, whence $y_n \in B$ for such n and (y_n) is bounded. By [Theorem 4.3.2](#) a subsequence $y_{n_k} \rightarrow y$ exists. Suppose $y \notin \varphi(x)$. As $\varphi(x)$ is closed and non-empty, $\varepsilon := \text{dist}(y, \varphi(x)) > 0$, and the open set $U := \bigcup_{z \in \varphi(x)} B_{\varepsilon/2}(z)$ contains $\varphi(x)$ but not y ; indeed $\text{dist}(y, U) \geq \varepsilon/2$. Upper hemicontinuity gives n_1 with $y_n \in \varphi(x_n) \subseteq U$ for $n \geq n_1$, so $y = \lim_k y_{n_k}$ lies in $\overline{U}$, contradicting $\text{dist}(y, U) \geq \varepsilon/2$. Hence $y \in \varphi(x)$.

($\Leftarrow$) Suppose φ is not upper hemicontinuous at x : there is an open $V \supseteq \varphi(x)$ such that for every n some $x_n \in B_{1/n}(x)$ has $\varphi(x_n) \not\subseteq V$. Pick $y_n \in \varphi(x_n) \cap V^c$. Then $x_n \rightarrow x$, and any convergent subsequence of (y_n) has its limit in the closed set V^c , hence outside $\varphi(x) \subseteq V$. So no subsequence converges into $\varphi(x)$. ■

Theorem 13.3.3 (Sequential criterion for lower hemicontinuity). Let $X \subseteq \mathbb{R}^k$, $Y \subseteq \mathbb{R}^d$ and $\varphi: X \rightrightarrows Y$. Then φ is lower hemicontinuous at x if and only if for every sequence $x_n \rightarrow x$ in X and every $y \in \varphi(x)$ there exist $y_n \in \varphi(x_n)$ with $y_n \rightarrow y$.

Proof. ($\Rightarrow$) Let $x_n \rightarrow x$ and $y \in \varphi(x)$. For each $k \in \mathbb{N}$ the open ball $B_{1/k}(y)$ meets $\varphi(x)$, so lower hemicontinuity gives a neighbourhood U_k of x with $\varphi(z) \cap B_{1/k}(y) \neq \emptyset$ for $z \in U_k$. Choose $n_1 < n_2 < \dots$ with $x_n \in U_k$ for all $n \geq n_k$. For $n_k \leq n < n_{k+1}$ pick $y_n \in \varphi(x_n) \cap B_{1/k}(y)$, and for $n < n_1$ pick $y_n \in \varphi(x_n)$ arbitrarily. Then $\|y_n - y\| < 1/k$ once $n \geq n_k$, so $y_n \rightarrow y$.

($\Leftarrow$) Suppose φ is not lower hemicontinuous at x : there is an open V with $\varphi(x) \cap V \neq \emptyset$ such that every neighbourhood of x contains a point z with $\varphi(z) \cap V = \emptyset$. Taking $B_{1/n}(x)$ produces $x_n \rightarrow x$ with $\varphi(x_n) \cap V = \emptyset$. Fix $y \in \varphi(x) \cap V$. Any $y_n \in \varphi(x_n)$ lies in the closed set V^c , so no such sequence can converge to $y \in V$. ■

Remark 13.3.2 (The asymmetry of the two criteria). The two criteria are not mirror images, and the difference is exactly the difference between the two inverses. Upper hemicontinuity asks that limits of *arbitrary* selections stay inside the value at the limit; it is a statement about the graph and, with compact values, is equivalent to closedness of the graph by [Theorem 13.3.1](#). Lower hemicontinuity asks that *every* point of the value at the limit be approachable by a selection; it is a statement about the existence of continuous selections and has no closed-graph reformulation. In economic terms: upper hemicontinuity rules out the sudden appearance of new optima, lower hemicontinuity rules out the sudden disappearance of existing ones.

Corollary 13.3.1 (Closed sub-correspondences). Let $\varphi, \psi: X \rightrightarrows Y$ between subsets of finite-dimensional Euclidean spaces with $\psi(x) \subseteq \varphi(x)$ for all x . If φ is compact-valued and upper hemicontinuous and $\text{gr } \psi$ is closed, then ψ is upper hemicontinuous and compact-valued.

Proof. $\psi(x)$ is a closed subset of the compact set $\varphi(x)$, hence compact. Let $(x_n, y_n) \in \text{gr } \psi$ with $x_n \rightarrow x$. Then $(x_n, y_n) \in \text{gr } \varphi$, so by [Theorem 13.3.2](#) a subsequence $y_{n_k} \rightarrow y \in \varphi(x)$. Since $(x_{n_k}, y_{n_k}) \rightarrow (x, y)$ and $\text{gr } \psi$ is closed, $(x, y) \in \text{gr } \psi$, that is $y \in \psi(x)$. Apply [Theorem 13.3.2](#) again. ■

13.3.3 Operations

Proposition 13.3.3 (Composition, union, intersection, product). Let X, Y, Z be topological spaces.

- (i) For $\varphi: X \rightrightarrows Y$ and $\psi: Y \rightrightarrows Z$ define $(\psi \circ \varphi)(x) := \bigcup_{y \in \varphi(x)} \psi(y)$. Then $(\psi \circ \varphi)^u(A) = \varphi^u(\psi^u(A))$ and $(\psi \circ \varphi)^\ell(A) = \varphi^\ell(\psi^\ell(A))$, and the composition inherits upper, respectively lower, hemicontinuity from both factors.
- (ii) A union of lower hemicontinuous correspondences is lower hemicontinuous; a union of two upper hemicontinuous correspondences is upper hemicontinuous.
- (iii) If φ has a closed graph and ψ is upper hemicontinuous and compact-valued, then $\varphi \cap \psi$ is upper hemicontinuous.
- (iv) A product of upper hemicontinuous compact-valued correspondences is upper hemicontinuous

and compact-valued; a product of lower hemicontinuous correspondences is lower hemicontinuous.

Proof. (i) For the upper inverse, $x \in (\psi \circ \varphi)^u(A)$ says $\psi(y) \subseteq A$ for every $y \in \varphi(x)$, that is $\varphi(x) \subseteq \psi^u(A)$; the lower inverse is analogous with “some” in place of “every”. Hemicontinuity then follows from [Proposition 13.3.1](#): openness of $\psi^u(V)$ and of φ^u of an open set compose.

(ii) For unions, $(\bigcup_i \varphi_i)^\ell(V) = \bigcup_i \varphi_i^\ell(V)$ is open when each summand is; for two upper hemicontinuous correspondences, $(\varphi \cup \psi)^u(V) = \varphi^u(V) \cap \psi^u(V)$ is a finite intersection of open sets.

(iii) Let $V \subseteq Y$ be open with $(\varphi \cap \psi)(x) \subseteq V$, and put $K := \psi(x) \setminus V$, a compact set. If $K = \emptyset$ take $N := \psi^u(V)$. Otherwise each $y \in K$ lies outside $\varphi(x)$, so $(x, y) \notin \text{gr } \varphi$, and closedness of the graph supplies open $U_y \ni x$ and $W_y \ni y$ with $(U_y \times W_y) \cap \text{gr } \varphi = \emptyset$. Finitely many $W_{y_1}, \dots, W_{y_r}$ cover K ; set $W := \bigcup_i W_{y_i}$ and $N := \psi^u(V \cup W) \cap \bigcap_i U_{y_i}$. Since $\psi(x) \subseteq V \cup W$, upper hemicontinuity of ψ makes N an open neighbourhood of x . For $z \in N$ we have $\psi(z) \subseteq V \cup W$ and $\varphi(z) \cap W = \emptyset$, hence $(\varphi \cap \psi)(z) \subseteq V$.

(iv) See Aliprantis and Border (2006, Thm. 17.28); the compact-valued case in a finite product follows from [Theorem 13.3.2](#) applied coordinatewise together with [Proposition 4.3.4](#). ■

13.4 Economic Application: Continuity of the Budget Correspondence

Proposition 13.4.1. Let $B: \mathbb{R}_{++}^n \times \mathbb{R}_{++} \rightrightarrows \mathbb{R}_+^n$ be the budget correspondence (13.1). Then B is non-empty-valued, compact-valued, convex-valued and continuous.

Proof. Non-emptiness, compactness and convexity are [Example 13.2.1](#).

Upper hemicontinuity. Let $(p_k, m_k) \rightarrow (p, m)$ in $\mathbb{R}_{++}^n \times \mathbb{R}_{++}$ and $x_k \in B(p_k, m_k)$. Since $p \gg 0$, there are $\underline{p} > 0$ and $\bar{m} < \infty$ with $p_k \geq \underline{p}$ and $m_k \leq \bar{m}$ for large k , so $\|x_k\|_1 \leq \bar{m}/\underline{p}$ and (x_k) is bounded. By [Theorem 4.3.2](#) a subsequence converges, say $x_{k_j} \rightarrow x \geq 0$, and passing to the limit in $\langle p_{k_j}, x_{k_j} \rangle \leq m_{k_j}$ gives $\langle p, x \rangle \leq m$, so $x \in B(p, m)$. Apply [Theorem 13.3.2](#).

Lower hemicontinuity. Let $(p_k, m_k) \rightarrow (p, m)$ and $x \in B(p, m)$. If $\langle p, x \rangle < m$, then $\langle p_k, x \rangle < m_k$ for large k and the constant sequence $x_k := x$ works for those k ; for the finitely many remaining indices take $x_k := 0$. If instead $\langle p, x \rangle = m$, set

$$\lambda_k := \min\left\{1, \frac{m_k}{\langle p_k, x \rangle}\right\}, \quad x_k := \lambda_k x.$$

Then $x_k \in \mathbb{R}_+^n$ and $\langle p_k, x_k \rangle \leq m_k$, so $x_k \in B(p_k, m_k)$; and $\langle p_k, x \rangle \rightarrow \langle p, x \rangle = m > 0$, so $\lambda_k \rightarrow \min\{1, m/m\} = 1$ and $x_k \rightarrow x$. Apply [Theorem 13.3.3](#). ■

Example 13.4.1 (Both hypotheses on the parameters are needed). Lower hemicontinuity fails at $m = 0$. Take $n = 1$, $p = 1$ fixed, and $m_k = 0$ for all k with $m = 0$: here $B(1, 0) = \{0\}$ and the correspondence is continuous. The genuine failure is at a zero price. Take $n = 2$ and $p^k = (1, 1/k)$, $m = 1$. Then $B(p^k, 1)$ is the triangle with vertices $(0, 0)$, $(1, 0)$, $(0, k)$, which grows without bound, while $B((1, 0), 1) = \{x \in \mathbb{R}_+^2 : x_1 \leq 1\}$ is unbounded and not compact. Upper hemicontinuity in the sense of [Theorem 13.3.2](#) fails because the criterion presupposes compact values. Economically: when a good is free, the budget set ceases to be compact and Weierstrass no longer delivers a demand, which is why general equilibrium existence proofs either bound consumption sets artificially or work only on the interior of the price simplex and control the boundary separately.

13.5 Berge’s Theorem of the Maximum

Recall from [Definition 5.3.1](#) that $m: X \rightarrow \mathbb{R}$ is lower semicontinuous if $\{x : m(x) > \alpha\}$ is open for every $\alpha \in \mathbb{R}$, and upper semicontinuous if $-m$ is lower semicontinuous. The maximum theorem is proved by establishing the two halves separately, each from one half of the continuity of the constraint correspondence.

Lemma 13.5.1 (Lower semicontinuity of the value). Let $\varphi: X \rightrightarrows Y$ be lower hemicontinuous and let $f: \text{gr } \varphi \rightarrow \mathbb{R}$ be lower semicontinuous. Then $m(x) := \sup_{y \in \varphi(x)} f(x, y)$, with $m(x) = -\infty$ when $\varphi(x) = \emptyset$, is lower semicontinuous.

Proof. Fix $\alpha \in \mathbb{R}$ and x_0 with $m(x_0) > \alpha$. Then $f(x_0, y_0) > \alpha$ for some $y_0 \in \varphi(x_0)$, so $\varphi(x_0) \neq \emptyset$. The set $W := \{(x, y) \in \text{gr } \varphi : f(x, y) > \alpha\}$ is open in $\text{gr } \varphi$ by lower semicontinuity of f and contains (x_0, y_0) , so there are open $U \ni x_0$ in X and $V \ni y_0$ in Y with $(U \times V) \cap \text{gr } \varphi \subseteq W$. Put $N := U \cap \varphi^\ell(V)$, which is open by [Proposition 13.3.1\(ii\)](#) and contains x_0 because $y_0 \in \varphi(x_0) \cap V$. For $x \in N$ pick $y \in \varphi(x) \cap V$; then $(x, y) \in (U \times V) \cap \text{gr } \varphi \subseteq W$, so $f(x, y) > \alpha$ and $m(x) > \alpha$. Hence $\{m > \alpha\}$ is open. ■

Lemma 13.5.2 (Upper semicontinuity of the value). Let $\varphi: X \rightrightarrows Y$ be upper hemicontinuous, non-empty- and compact-valued, and let $f: \text{gr } \varphi \rightarrow \mathbb{R}$ be upper semicontinuous. Then $m(x) := \max_{y \in \varphi(x)} f(x, y)$ is well defined and upper semicontinuous.

Proof. For fixed x the function $f(x, \cdot)$ is upper semicontinuous on the non-empty compact set $\varphi(x)$, so [Theorem 5.4.1](#) gives the maximum. Fix $\alpha \in \mathbb{R}$ and x_0 with $m(x_0) < \alpha$, and put $W := \{(x, y) \in \text{gr } \varphi : f(x, y) < \alpha\}$, open in $\text{gr } \varphi$ and containing $\{x_0\} \times \varphi(x_0)$. For each $y \in \varphi(x_0)$ choose open $U_y \ni x_0$ and $V_y \ni y$ with $(U_y \times V_y) \cap \text{gr } \varphi \subseteq W$. Compactness of $\varphi(x_0)$ gives $y_1, \dots, y_r$ with $\varphi(x_0) \subseteq V := \bigcup_i V_{y_i}$; set $U := \bigcap_i U_{y_i}$ and $N := U \cap \varphi^u(V)$, open by [Proposition 13.3.1\(i\)](#) and containing x_0 . For $x \in N$ and $y \in \varphi(x)$ we have $(x, y) \in (U \times V) \cap \text{gr } \varphi \subseteq W$, so $f(x, y) < \alpha$ and therefore $m(x) < \alpha$. ■

Theorem 13.5.1 (Berge's theorem of the maximum). Let X, Y be topological spaces, let $\varphi: X \rightrightarrows Y$ be continuous with non-empty compact values, and let $f: \text{gr } \varphi \rightarrow \mathbb{R}$ be continuous. Define the *value function* and the *argmax correspondence*

$$m(x) := \max_{y \in \varphi(x)} f(x, y), \quad \mu(x) := \{y \in \varphi(x) : f(x, y) = m(x)\}. \quad (13.4)$$

Then

- (i) m is continuous on X ;
- (ii) μ is non-empty- and compact-valued;
- (iii) if Y is Hausdorff, or if f extends continuously to $X \times Y$, then μ is upper hemicontinuous.

Proof. (i) f is both upper and lower semicontinuous and φ is both upper and lower hemicontinuous, so [Lemma 13.5.1](#) and [Lemma 13.5.2](#) make m both lower and upper semicontinuous, hence continuous by [Corollary 5.3.1](#).

(ii) This is [Proposition 5.4.1](#) applied on the non-empty compact set $\varphi(x)$ to the continuous function $f(x, \cdot)$.

(iii) Suppose first that f extends continuously to $X \times Y$. Then $\psi(x) := \{y \in Y : f(x, y) = m(x)\}$ has closed graph, being the zero set of the continuous function $(x, y) \mapsto f(x, y) - m(x)$ on $X \times Y$, and $\mu = \varphi \cap \psi$, so [Proposition 13.3.3\(iii\)](#) applies.

Suppose instead that Y is Hausdorff. By [Corollary 13.3.1](#) it suffices to show that $\text{gr } \mu$ is closed. Let $(x_n, y_n) \in \text{gr } \mu$ with $(x_n, y_n) \rightarrow (x, y)$. First, $y \in \varphi(x)$: otherwise, since Y is Hausdorff and $\varphi(x)$ is compact, there are disjoint open sets $V \ni y$ and $W \supseteq \varphi(x)$, and upper hemicontinuity puts $y_n \in \varphi(x_n) \subseteq W$ for large n while $y_n \in V$ for large n , contradicting $V \cap W = \emptyset$. Then, using continuity of m from (i) and of f ,

$$m(x) = \lim_{n \rightarrow \infty} m(x_n) = \lim_{n \rightarrow \infty} f(x_n, y_n) = f(x, y),$$

so $y \in \mu(x)$. ■

Remark 13.5.1 (μ is not lower hemicontinuous). Upper hemicontinuity is the strongest conclusion available. Let $X = [0, 1]$, $Y = [1, 2]$, $\varphi \equiv Y$ and $f(x, y) = y^x$. Then $\mu(x) = \{2\}$ for $x > 0$ and $\mu(0) = Y$, since $f(0, \cdot) \equiv 1$ is constant. The correspondence μ is upper hemicontinuous at 0, as the theorem asserts, and is not lower hemicontinuous there: every $y \in [1, 2)$ belongs to $\mu(0)$ and is approached by no selection from $\mu(x) = \{2\}$. Ties in the objective produce large optimal sets that collapse under an arbitrarily small perturbation, and no hypothesis of the theorem prevents this. In economics the phenomenon is the discontinuous collapse of demand when an indifference curve is tangent to a whole segment of the budget line.

Example 13.5.1 (Every hypothesis is used). Let $X = Y = [0, 1]$, $f(x, y) = y$, and

$$\varphi(x) = \begin{cases} [0, 1], & x = 0, \\ \{0\}, & x > 0. \end{cases}$$

Then φ is upper hemicontinuous and compact-valued but not lower hemicontinuous at 0. Consequently $m(0) = 1$ and $m(x) = 0$ for $x > 0$, so m is not continuous but merely upper semicontinuous, as [Lemma 13.5.2](#) predicts; and $\mu(0) = \{1\}$, $\mu(x) = \{0\}$ for $x > 0$, so μ is not upper hemicontinuous. Losing lower hemicontinuity of the constraint correspondence loses both conclusions at once.

Corollary 13.5.1 (Continuity of Marshallian demand). Let $u: \mathbb{R}_+^n \rightarrow \mathbb{R}$ be continuous. Then the indirect utility function $v(p, m) = \max_{x \in B(p, m)} u(x)$ is continuous on $\mathbb{R}_{++}^n \times \mathbb{R}_{++}$ and the demand correspondence D of (13.2) is non-empty- and compact-valued and upper hemicontinuous there. If in addition u is strictly quasiconcave, D is single-valued and the demand function is continuous.

Proof. [Proposition 13.4.1](#) and [Theorem 13.5.1](#) give the first two claims, $\mathbb{R}_+^n$ being Hausdorff as a metric space. If u is strictly quasiconcave, [Proposition 9.3.5](#) makes D single-valued, and by [Proposition 13.3.1](#) a single-valued upper hemicontinuous correspondence is a continuous function. ■

The corollary is the theoretical warrant for the comparative statics performed throughout demand theory. Without it, an expression such as $\partial x^*/\partial p$ could not be assumed even to exist, and the observation of [Remark 8.6.1](#) — that the implicit function theorem delivers differentiability from a nonsingular bordered Hessian — would have nothing to be applied to. [Corollary 13.5.1](#) also shows what is available without any differentiability at all: continuity of v and upper hemicontinuity of D come from compactness and continuity alone.

Proposition 13.5.1 (Concavity of the value function). Let $f: \mathbb{R}^n \times \mathbb{R}^k \rightarrow \mathbb{R}$ be concave and suppose $f^*(\alpha) := \max_{x \in \mathbb{R}^n} f(x, \alpha)$ exists for every $\alpha \in \mathbb{R}^k$. Then f^* is concave.

Proof. Fix $\alpha, \alpha' \in \mathbb{R}^k$ and $\lambda \in [0, 1]$, and choose maximisers $x^*(\alpha), x^*(\alpha')$. Then

$$\begin{aligned} f^*((1-\lambda)\alpha + \lambda\alpha') &\geq f((1-\lambda)x^*(\alpha) + \lambda x^*(\alpha'), (1-\lambda)\alpha + \lambda\alpha') \\ &\geq (1-\lambda)f(x^*(\alpha), \alpha) + \lambda f(x^*(\alpha'), \alpha') = (1-\lambda)f^*(\alpha) + \lambda f^*(\alpha'), \end{aligned}$$

the first inequality because the displayed point is feasible and the second by concavity of f on $\mathbb{R}^n \times \mathbb{R}^k$. ■

Maximisation over an unconstrained variable therefore preserves concavity. This is what makes the value function of a concave dynamic programme concave, and hence — by [Proposition 9.3.3](#) and [Theorem 9.3.1](#) — continuous on the interior and differentiable wherever the maximiser is unique. The corresponding statement for constrained problems requires the constraint correspondence to have a convex graph, not merely convex values; the budget correspondence (13.1) has a convex graph in (x, m) for fixed p , which is why the indirect utility function is convex in p and concave in m in the appropriate senses.

13.6 Fixed-Point Theorems

Theorem 13.6.1 (Brouwer). Let $C \subseteq \mathbb{R}^n$ be non-empty, compact and convex. Every continuous $f: C \rightarrow C$ has a fixed point.

The proof is topological — it rests on the fact that the n -sphere is not a retract of the closed ball — and is not reproduced here; see [Ok \(2007, Ch. E\)](#) for a full treatment and [Fukuda \(2026d, p. 270\)](#) for the statement used in this course. What is reproduced is the exact role of each hypothesis, since the failures are what one must check in applications.

Example 13.6.1 (The three hypotheses of Brouwer). *Compactness.* $C = \mathbb{R}$ and $f(x) = x + 1$: continuous, C convex, no fixed point. Closedness alone is not enough, and neither is boundedness alone: on $C = (0, 1)$ the map $f(x) = x/2$ is continuous with no fixed point.

Convexity. $C = \{x \in \mathbb{R}^2 : \|x\| = 1\}$, the unit circle, is compact but not convex, and the rotation by π , $f(x) = -x$, is continuous with no fixed point.

Continuity. $C = [0, 1]$ and $f(x) = 1$ for $x < \frac{1}{2}$, $f(x) = 0$ for $x \geq \frac{1}{2}$: compact convex domain, no fixed point.

Corollary 13.6.1 (Brouwer on the simplex). Let $\Delta^{n-1} := \{p \in \mathbb{R}_+^n : \sum_{i=1}^n p_i = 1\}$. Every continuous $f: \Delta^{n-1} \rightarrow \Delta^{n-1}$ has a fixed point.

Proof. Δ^{n-1} is convex as the intersection of $\mathbb{R}_+^n$ with a hyperplane, and compact by Theorem 4.4.2, being closed and contained in the unit ball of $\|\cdot\|_1$. ■

Theorem 13.6.2 (Kakutani). Let $X \subseteq \mathbb{R}^n$ be non-empty, compact and convex, and let $\varphi: X \rightrightarrows X$ be non-empty-valued, convex-valued and have a closed graph. Then φ has a *fixed point of a correspondence*: some $x \in X$ with $x \in \varphi(x)$.

By Theorem 13.3.1 the closed-graph hypothesis is, on a compact X , the same as requiring φ upper hemicontinuous with closed values. Kakutani's theorem contains Brouwer's: a continuous function is a single-valued correspondence with convex values and, by Proposition 13.3.1 and Theorem 13.3.1, a closed graph. The statement is Fukuda (2026d, Thm. 13.2, p. 478); the proof, which approximates φ by continuous functions on successively finer simplicial subdivisions and applies Brouwer to each, is in Aliprantis and Border (2006, Ch. 17).

Example 13.6.2 (The four hypotheses of Kakutani). *Compactness.* $X = \mathbb{R}$, $\varphi(x) = \{x + 1\}$.

Convexity of X . X the unit circle, $\varphi(x) = \{-x\}$.

Convex values. $X = [0, 1]$ and $\varphi(x) = \{0, 1\} \setminus \{x\}$ for $x \in \{0, 1\}$, $\varphi(x) = \{0, 1\}$ otherwise. This has a closed graph and compact values, but $x \in \varphi(x)$ never holds. A cleaner instance is $X = [0, 1]$ with $\varphi(x) = \{1\}$ for $x < \frac{1}{2}$, $\varphi(1/2) = \{0, 1\}$ and $\varphi(x) = \{0\}$ for $x > \frac{1}{2}$: the graph is closed, the values are compact and non-empty, and only convexity at $x = \frac{1}{2}$ fails, which is exactly where a fixed point would have to be.

Closed graph. $X = [0, 1]$ with $\varphi(x) = \{1\}$ for $x < \frac{1}{2}$ and $\varphi(x) = \{0\}$ for $x \geq \frac{1}{2}$: convex-valued, non-empty-valued, no fixed point, and the graph is not closed at $x = \frac{1}{2}$.

13.7 Economic Application: Existence of Nash Equilibrium

Environment. A finite game has players indexed by $i \in \{1, \dots, I\}$, finite action sets A_i , and payoffs $u_i: \prod_j A_j \rightarrow \mathbb{R}$. A mixed strategy for player i is an element σ_i of the simplex $\Delta(A_i)$ over A_i . Write $\Sigma := \prod_i \Delta(A_i)$ and extend u_i to Σ by expectation,

$$u_i(\sigma) := \sum_{a \in \prod_j A_j} \left(\prod_{j=1}^I \sigma_j(a_j) \right) u_i(a). \quad (13.5)$$

A *Nash equilibrium* is $\sigma^* \in \Sigma$ with $u_i(\sigma^*) \geq u_i(\sigma_i, \sigma_{-i}^*)$ for every i and every $\sigma_i \in \Delta(A_i)$.

Theorem 13.7.1 (Nash). Every finite game has a Nash equilibrium in mixed strategies.

Proof. Define the best-response correspondence $\beta_i: \Sigma \rightrightarrows \Delta(A_i)$ by $\beta_i(\sigma) := \arg \max_{\sigma_i \in \Delta(A_i)} u_i(\sigma_i, \sigma_{-i})$ and set $\beta := \prod_i \beta_i: \Sigma \rightrightarrows \Sigma$. A fixed point of β is precisely a Nash equilibrium, by the definition of β_i . The hypotheses of Theorem 13.6.2 are verified in turn.

Σ is non-empty, compact and convex. Each $\Delta(A_i)$ is a simplex, compact and convex by Corollary 13.6.1's argument, and a finite product of compact convex sets is compact by Proposition 4.4.2 and convex by Proposition 9.2.1.

β is non-empty-valued. $u_i(\cdot, \sigma_{-i})$ is a polynomial, hence continuous, on the compact set $\Delta(A_i)$, so Corollary 5.4.1 applies.

β is convex-valued. By (13.5), u_i is affine in σ_i for fixed σ_{-i} . The set of maximisers of an affine function on a convex set is convex: if $u_i(\sigma_i, \sigma_{-i}) = u_i(\sigma'_i, \sigma_{-i}) = \max$ then affinity gives the same value at every convex combination. A product of convex sets is convex.

β has a closed graph. The constraint correspondence $\sigma \mapsto \Delta(A_i)$ is constant, hence continuous with non-empty compact values, and u_i is continuous on $\Sigma \times \Delta(A_i)$; so Theorem 13.5.1(iii) makes β_i upper hemicontinuous with compact values, and Theorem 13.3.1 converts this into a closed graph on the compact space Σ . Closed graphs are preserved by finite products.

Theorem 13.6.2 now supplies $\sigma^* \in \beta(\sigma^*)$. ■

Two features of the argument deserve emphasis. Convex-valuedness rests entirely on the *linearity* of expected utility in own mixed strategy; it is what mixing is for, and it is why an equilibrium in pure strategies need not exist while one in mixed strategies always does. And upper hemicontinuity of the best response is an instance of [Theorem 13.5.1](#) in the degenerate case of a constant constraint correspondence, so the maximum theorem is doing the work even where no constraint moves.

13.8 Economic Application: Existence of Walrasian Equilibrium

Environment. There are n goods and a price vector p normalised to lie in Δ^{n-1} . Let $Z: \Delta^{n-1} \rightarrow \mathbb{R}^n$ denote aggregate excess demand: $Z_i(p)$ is the quantity of good i demanded in aggregate, net of the aggregate endowment, at prices p .

Assumption 13.8.1 (Excess demand). Z is continuous on Δ^{n-1} and satisfies *Walras' law* $\langle p, Z(p) \rangle = 0$ for every $p \in \Delta^{n-1}$.

Walras' law is the aggregate budget identity: each consumer spends exactly their wealth, so the value of aggregate excess demand is zero at every price vector, whether or not markets clear. Continuity of Z is not primitive; it is the conclusion of [Corollary 13.5.1](#) applied to each consumer, requiring strictly quasiconcave utility for single-valuedness and interior prices for compactness of the budget set ([Example 13.4.1](#)).

Theorem 13.8.1 (Existence of equilibrium prices). Under [Assumption 13.8.1](#) there exists $p^* \in \Delta^{n-1}$ with $Z(p^*) \leq 0$; and $Z_i(p^*) < 0$ only if $p_i^* = 0$.

Proof. Define $g: \Delta^{n-1} \rightarrow \Delta^{n-1}$ by

$$g_i(p) := \frac{p_i + \max\{0, Z_i(p)\}}{1 + \sum_{k=1}^n \max\{0, Z_k(p)\}}. \quad (13.6)$$

The denominator is at least 1, the numerators are non-negative and sum to the denominator, so g maps Δ^{n-1} into itself; g is continuous because Z is and $t \mapsto \max\{0, t\}$ is. By [Corollary 13.6.1](#) there is p^* with $g(p^*) = p^*$. Writing $S := \sum_k \max\{0, Z_k(p^*)\}$, the fixed-point equation is $p_i^*(1+S) = p_i^* + \max\{0, Z_i(p^*)\}$, that is

$$p_i^* S = \max\{0, Z_i(p^*)\} \quad \text{for each } i. \quad (13.7)$$

Multiply (13.7) by $Z_i(p^*)$ and sum over i :

$$\underbrace{S \langle p^*, Z(p^*) \rangle}_{=0 \text{ (Walras)}} = \sum_{i=1}^n Z_i(p^*) \max\{0, Z_i(p^*)\} = \sum_{i=1}^n (\max\{0, Z_i(p^*)\})^2.$$

A sum of squares equal to zero forces $\max\{0, Z_i(p^*)\} = 0$ for every i , that is $Z(p^*) \leq 0$. Finally, if $Z_i(p^*) < 0$ for some i with $p_i^* > 0$, then $\langle p^*, Z(p^*) \rangle < 0$ because every term is at most zero and this one is strictly negative, contradicting Walras' law. ■

The map (13.6) is a normalised *tâtonnement*: prices of goods in excess demand are raised in proportion to the excess, prices of the remaining goods fall through the normalisation. The theorem says the process has a rest point; it says nothing about whether the process converges to it, and in general it does not. The last sentence of the statement is free disposal in equilibrium: a good in strict excess supply must be free.

Remark 13.8.1 (What has and has not been proved). [Theorem 13.8.1](#) is a statement about a function Z that has been *assumed* continuous on the whole simplex, boundary included. Deriving that hypothesis from preferences and endowments is the substance of the existence problem, and it is where the difficulties of [Example 13.4.1](#) bite: as some $p_i \rightarrow 0$ the budget set ceases to be compact and, under monotone preferences, demand for good i diverges, so Z is unbounded near the boundary. The standard resolutions — truncating consumption sets to a large compact box and showing the truncation does not bind, or replacing Z by a correspondence and invoking [Theorem 13.6.2](#) in the form of the Gale–Nikaidô–Debreu lemma — are treated in Aliprantis and Border (2006, Ch. 17). What the argument above establishes cleanly is that once continuity and Walras' law are available,

existence is a one-page consequence of Brouwer's theorem.

13.9 Chapter Summary

A correspondence is a map into subsets, identified with its graph. Because a correspondence has two inverse images — the upper inverse demanding containment of the whole value, the lower inverse demanding only non-empty intersection — there are two continuity notions. Upper hemicontinuity, equivalent to openness of upper inverses of open sets and, on compact ranges, to closedness of the graph, forbids the value from exploding; lower hemicontinuity forbids it from imploding. In finite dimensions both are checked by sequences: upper hemicontinuity says every selection along a convergent sequence of parameters has a subsequence converging into the limit value, lower hemicontinuity says every point of the limit value is the limit of some selection.

The budget correspondence is continuous with compact convex values at strictly positive prices and positive income, and fails to be so when a price is zero. Berge's theorem of the maximum then yields, for a continuous objective on the graph of a continuous compact-valued constraint correspondence, a continuous value function and an upper hemicontinuous compact-valued argmax; lower hemicontinuity of the argmax is false in general, and the counterexample is ties in the objective. Applied to the consumer, this gives continuity of indirect utility and upper hemicontinuity of Marshallian demand, single-valued and continuous under strict quasiconcavity. Maximising out a variable preserves concavity of the objective in the remaining parameters.

Existence of equilibrium comes from fixed points. Brouwer's theorem gives a fixed point of a continuous self-map of a non-empty compact convex set; Kakutani's extends this to correspondences that are non-empty- and convex-valued with a closed graph. Kakutani applied to the best-response correspondence gives Nash equilibrium in mixed strategies, convex-valuedness being exactly the linearity of expected utility in own strategy. Brouwer applied to a normalised tâtonnement map gives Walrasian equilibrium prices from continuity of excess demand and Walras' law. Neither theorem gives uniqueness, and neither gives an algorithm — in contrast to [Theorem 6.4.1](#), which gives both at the cost of a contraction hypothesis.

Exercise 13.9.1. Prove [Proposition 13.3.1](#) in the pointwise form: φ is upper hemicontinuous at x if and only if $\varphi^u(U)$ is a neighbourhood of x for every neighbourhood U of $\varphi(x)$; and dually for lower hemicontinuity. Then show directly from the definitions that a correspondence is upper hemicontinuous and lower hemicontinuous at x if and only if it is continuous there.

Exercise 13.9.2. Let $\varphi: X \rightrightarrows Y$ be a correspondence and $A \subseteq X$. Prove that $A \subseteq \varphi^u(\varphi(A))$, and that if φ is non-empty-valued then $A \subseteq \varphi^\ell(\varphi(A))$. Give an example in which both inclusions are strict, and reconcile the result with [Proposition 1.2.2](#).

Exercise 13.9.3. Let $\Gamma: S \rightrightarrows S$ be the feasible-continuation correspondence of a dynamic programme on a compact state space S , assumed continuous with non-empty compact values, and let $F: \text{gr } \Gamma \rightarrow \mathbb{R}$ be continuous and $\beta \in (0, 1)$. Using [Theorem 13.5.1](#), show that the Bellman operator $(Tv)(s) = \max_{s' \in \Gamma(s)} \{F(s, s') + \beta v(s')\}$ maps $\mathcal{C}(S, \mathbb{R})$ into itself, and combine this with [Proposition 6.3.3](#) and [Theorem 6.5.1](#) to conclude that the value function is the unique continuous fixed point and that the policy correspondence is upper hemicontinuous. This is the argument completed in [Chapter 15](#).

Chapter 14

Probability, Conditional Expectation and Bayesian Updating

14.1 Probability Spaces

Uncertainty enters an economic model as a set of states, a rule assigning likelihoods to statements about the state, and functions of the state that the agent cares about. The three objects are the sample space, the probability measure and the random variables, and the first difficulty is that the second cannot in general be defined on all statements.

Definition 14.1.1 (σ -algebra). Let Ω be a set. A collection $\mathcal{F}$ of subsets of Ω is a *sigma-algebra* on Ω if

- (i) $\Omega \in \mathcal{F}$;
- (ii) $E \in \mathcal{F}$ implies $E^c \in \mathcal{F}$;
- (iii) $E_n \in \mathcal{F}$ for every $n \in \mathbb{N}$ implies $\bigcap_{n \in \mathbb{N}} E_n \in \mathcal{F}$.

The pair $(\Omega, \mathcal{F})$ is then a *measurable space* and the members of $\mathcal{F}$ are *event*.

Proposition 14.1.1 (Closure properties). Let $\mathcal{F}$ be a σ -algebra on Ω . Then $\emptyset \in \mathcal{F}$ and $\bigcup_{n \in \mathbb{N}} E_n \in \mathcal{F}$ whenever every $E_n \in \mathcal{F}$; consequently $\mathcal{F}$ is closed under finite unions, finite intersections and set differences.

Proof. $\emptyset = \Omega^c \in \mathcal{F}$ by (i) and (ii). For unions, De Morgan's law (Proposition 1.2.1) gives $\bigcup_n E_n = (\bigcap_n E_n^c)^c$, and each $E_n^c \in \mathcal{F}$ by (ii), so the intersection lies in $\mathcal{F}$ by (iii) and its complement by (ii). Finite operations follow by padding the sequence with Ω or $\emptyset$, and $E \setminus F = E \cap F^c$. ■

An event is a statement about the experiment admitting a yes-or-no answer, and the three axioms are the closure of such statements under negation and countable conjunction. The restriction to *countable* conjunction is what makes the definition non-trivial: uncountable operations are excluded because they would force $\mathcal{F}$ to be the whole power set, and on an uncountable Ω no probability measure with the required additivity can be defined on the whole power set. This is a theorem of measure theory rather than a technical convenience; see Le Gall (2022, Ch. 1). Chapter 4 recorded the formal parallel with Definition 4.2.2: a topology is closed under arbitrary unions and finite intersections, a σ -algebra under countable operations of both kinds and under complementation, and neither contains the other.

Definition 14.1.2 (Probability measure and probability space). Let $(\Omega, \mathcal{F})$ be a measurable space. A function $\mathbb{P}: \mathcal{F} \rightarrow [0, 1]$ is a *probability measure* if

- (i) $\mathbb{P}(\Omega) = 1$;
- (ii) (σ -additivity) $\mathbb{P}(\bigcup_{n \in \mathbb{N}} E_n) = \sum_{n=1}^{\infty} \mathbb{P}(E_n)$ whenever $(E_n)_{n \in \mathbb{N}}$ is pairwise disjoint.

The triple $(\Omega, \mathcal{F}, \mathbb{P})$ is a *probability space*.

Proposition 14.1.2 (Elementary properties). Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and $E, F \in \mathcal{F}$.

- (i) $\mathbb{P}(\emptyset) = 0$.
- (ii) (Monotonicity) $E \subseteq F$ implies $\mathbb{P}(E) \leq \mathbb{P}(F)$.
- (iii) (Finite additivity) $E \cap F = \emptyset$ implies $\mathbb{P}(E \cup F) = \mathbb{P}(E) + \mathbb{P}(F)$; in particular $\mathbb{P}(E) + \mathbb{P}(E^c) = 1$.
- (iv) (Inclusion–exclusion) $\mathbb{P}(E \cup F) = \mathbb{P}(E) + \mathbb{P}(F) - \mathbb{P}(E \cap F)$, hence $\mathbb{P}(E \cup F) \leq \mathbb{P}(E) + \mathbb{P}(F)$.

Proof. (i) Suppose $\mathbb{P}(\emptyset) = a > 0$ and take $E_n = \emptyset$ for every n . These are pairwise disjoint, so σ -additivity gives $a = \mathbb{P}(\emptyset) = \sum_{n=1}^{\infty} a = \infty$, contradicting $\mathbb{P} \leq 1$.

(ii) Put $E_1 = E$, $E_2 = F \setminus E$ and $E_n = \emptyset$ for $n \geq 3$; these are disjoint with union F , so $\mathbb{P}(F) = \mathbb{P}(E) + \mathbb{P}(F \setminus E) \geq \mathbb{P}(E)$ by (i) and non-negativity.

(iii) Put $E_1 = E$, $E_2 = F$, $E_n = \emptyset$ for $n \geq 3$. Taking $F = E^c$ gives $\mathbb{P}(E) + \mathbb{P}(E^c) = \mathbb{P}(\Omega) = 1$.

(iv) Decompose $E \cup F$ into the three disjoint events $E \setminus F$, $E \cap F$, $F \setminus E$ and apply (iii) three times: $\mathbb{P}(E \cup F) = \mathbb{P}(E \setminus F) + \mathbb{P}(E \cap F) + \mathbb{P}(F \setminus E)$, while $\mathbb{P}(E) = \mathbb{P}(E \setminus F) + \mathbb{P}(E \cap F)$ and similarly for F ; adding the last two and subtracting $\mathbb{P}(E \cap F)$ gives the identity. The inequality follows since $\mathbb{P}(E \cap F) \geq 0$. ■

Remark 14.1.1 (The countable case). When Ω is at most countable one may take $\mathcal{F} = \mathcal{P}(\Omega)$ and specify $\mathbb{P}$ by the numbers $\mathbb{P}(\{\omega\})$, $\omega \in \Omega$, which are non-negative and sum to 1; every event then has $\mathbb{P}(E) = \sum_{\omega \in E} \mathbb{P}(\{\omega\})$ and σ -additivity is automatic. The four properties listed on (Fukuda 2026f, slide 4) — $\mathbb{P}(\emptyset) = 0$, $\mathbb{P}(\Omega) = 1$, monotonicity and additivity on disjoint events — are exactly Definition 14.1.2 together with Proposition 14.1.2 specialised to this case. Every statement in Section 14.3 and Section 14.4 below is proved in the countable setting and holds verbatim in general once the sums are read as integrals.

14.2 Random Variables and Expectation

Definition 14.2.1 (Random variable). Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space. A *random variable* is a function $X: \Omega \rightarrow \mathbb{R}$ that is *measurable function*: $X^{-1}(B) \in \mathcal{F}$ for every Borel set $B \subseteq \mathbb{R}$. When Ω is at most countable and $\mathcal{F} = \mathcal{P}(\Omega)$ the measurability requirement is vacuous. Writing $X(\Omega) = \{x_i\}_i$ for the set of realisations, the *probability mass function* of X is

$$f(x) := \mathbb{P}(\{\omega \in \Omega : X(\omega) = x\}) = \mathbb{P}(X^{-1}(\{x\})), \quad x \in X(\Omega), \quad (14.1)$$

and one writes $p_i := f(x_i)$ and $[X = x] := X^{-1}(\{x\})$.

Measurability is the requirement that every statement about the value of X be an event, so that its probability is defined. It is the probabilistic counterpart of continuity: Theorem 5.2.1 characterised continuity by $f^{-1}(\text{open})$ open, and Remark 5.2.1 recorded that measurability replaces “open” by “Borel” and “open” by “in $\mathcal{F}$ ”. On a countable sample space the condition is empty, which is why the slides define a random variable simply as a function (Fukuda 2026f, slide 15).

Definition 14.2.2 (Expectation). For a random variable X on an at most countable probability space with $X(\Omega) = \{x_i\}_{i=1}^n$, the *expectation* is

$$\mathbb{E}[X] := \sum_{i=1}^n p_i x_i = \sum_{\omega \in \Omega} \mathbb{P}(\{\omega\}) X(\omega), \quad (14.2)$$

the two expressions being equal because the events $[X = x_i]$ partition Ω .

Proof of the second equality in (14.2). Group the sum over Ω by the value of X :

$$\sum_{\omega \in \Omega} \mathbb{P}(\{\omega\})X(\omega) = \sum_{i=1}^n \sum_{\omega: X(\omega)=x_i} \mathbb{P}(\{\omega\})X(\omega) = \sum_{i=1}^n x_i \sum_{\omega: X(\omega)=x_i} \mathbb{P}(\{\omega\}) = \sum_{i=1}^n x_i p_i,$$

using (14.1) in the last step. Rearrangement is legitimate because the family $([X = x_i])_i$ partitions Ω and, for a countable Ω , absolute convergence of $\sum_{\omega} \mathbb{P}(\{\omega\})|X(\omega)|$ is exactly the integrability of X . ■

Proposition 14.2.1 (Properties of expectation). Let X, Y be integrable random variables and $a, b \in \mathbb{R}$.

- (i) (Linearity) $\mathbb{E}[aX + bY] = a\mathbb{E}[X] + b\mathbb{E}[Y]$.
- (ii) (Monotonicity) $X \leq Y$ pointwise implies $\mathbb{E}[X] \leq \mathbb{E}[Y]$; and $X \geq 0$ with $\mathbb{E}[X] = 0$ implies $\mathbb{P}(X = 0) = 1$.
- (iii) (Change of variables) $\mathbb{E}[g(X)] = \sum_{x \in X(\Omega)} g(x)f(x)$ for any $g: \mathbb{R} \rightarrow \mathbb{R}$ for which the sum converges absolutely.
- (iv) (Independence) If X and Y are independent, meaning $\mathbb{P}([X \in A] \cap [Y \in B]) = \mathbb{P}(X \in A)\mathbb{P}(Y \in B)$ for all Borel A, B , then $\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y]$.

Proof. (i)–(iii) are rearrangements of absolutely convergent sums, using (14.2) in its second form for (i) and its first form for (iii). For the second half of (ii), if $\mathbb{P}(X > 0) > 0$ then $\mathbb{P}(X > 1/k) > 0$ for some k by countable additivity applied to $[X > 0] = \bigcup_k [X > 1/k]$, and then $\mathbb{E}[X] \geq (1/k)\mathbb{P}(X > 1/k) > 0$. For (iv), independence makes the joint mass function factor, $f_{X,Y}(x, y) = f_X(x)f_Y(y)$, and $\mathbb{E}[XY] = \sum_{x,y} xyf_X(x)f_Y(y) = (\sum_x xf_X(x))(\sum_y yf_Y(y))$. ■

Definition 14.2.3 (Variance and covariance). For square-integrable X, Y ,

$$\begin{aligned} \text{Var}(X) &:= \mathbb{E}[(X - \mathbb{E}[X])^2], & \text{Cov}(X, Y) &:= \mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])], \\ \text{Corr}(X, Y) &:= \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X)\text{Var}(Y)}}, & \text{defined when } \text{Var}(X)\text{Var}(Y) > 0. \end{aligned}$$

Proposition 14.2.2 (The variance identity). For square-integrable X and Y ,

$$\text{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2, \quad \text{Cov}(X, Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y]. \quad (14.3)$$

More generally, for any $c \in \mathbb{R}$, $\mathbb{E}[(X - c)^2] = \text{Var}(X) + (\mathbb{E}[X] - c)^2$, so $c \mapsto \mathbb{E}[(X - c)^2]$ is minimised uniquely at $c = \mathbb{E}[X]$.

Proof. Write $\mu := \mathbb{E}[X]$ and expand, using Proposition 14.2.1(i) and the fact that μ is a constant:

$$\mathbb{E}[(X - c)^2] = \mathbb{E}[(X - \mu)^2] + 2(\mu - c)\underbrace{\mathbb{E}[X - \mu]}_{=0} + (\mu - c)^2 = \text{Var}(X) + (\mu - c)^2.$$

Taking $c = 0$ gives the first identity in (14.3); the covariance identity is the same expansion applied to the product. The minimisation claim is immediate since $(\mu - c)^2 \geq 0$ with equality only at $c = \mu$. ■

The last sentence is the reason the mean, and not some other centre, is the object that appears throughout asset pricing and econometrics: it is the constant that best predicts X in mean square. Theorem 14.5.1 extends the statement from constants to functions of a conditioning variable, and Proposition 11.5.2 to affine functions.

Theorem 14.2.1 (Jensen's inequality). Let $I \subseteq \mathbb{R}$ be an interval, $\varphi: I \rightarrow \mathbb{R}$ convex, and X an integrable random variable with $\mathbb{P}(X \in I) = 1$ and $\mathbb{E}[X]$ in the interior of I . Then $\mathbb{E}[\varphi(X)] \geq$

$\varphi(\mathbb{E}[X])$, with equality if and only if φ is affine on the smallest interval carrying all the mass of X . The inequality reverses for concave φ .

Proof. Put $\mu := \mathbb{E}[X]$. By Proposition 9.3.1 the set $\text{epi } \varphi$ is convex, and $(\mu, \varphi(\mu))$ lies on its boundary, so Theorem 9.4.2 supplies a supporting hyperplane: there is $(a, b) \neq 0$ with $ax + bt \geq a\mu + b\varphi(\mu)$ for every $(x, t) \in \text{epi } \varphi$. Since t may be taken arbitrarily large, $b \geq 0$; and $b = 0$ would force $ax \geq a\mu$ for all $x \in I$ with $a \neq 0$, impossible because μ is interior to I . Dividing by $b > 0$ and setting $s := -a/b$ gives the supporting line

$$\varphi(x) \geq \varphi(\mu) + s(x - \mu) \quad \text{for all } x \in I. \quad (14.4)$$

Substituting X and taking expectations, monotonicity and linearity (Proposition 14.2.1) give $\mathbb{E}[\varphi(X)] \geq \varphi(\mu) + s\mathbb{E}[X - \mu] = \varphi(\mu)$. Equality forces $\mathbb{E}[\varphi(X) - \varphi(\mu) - s(X - \mu)] = 0$ with a non-negative integrand, hence by Proposition 14.2.1(ii) the integrand vanishes almost surely: φ agrees with the affine function (14.4) on the support of X . ■

Example 14.2.1 (Risk aversion is Jensen). If u is concave, Theorem 14.2.1 gives $\mathbb{E}[u(W)] \leq u(\mathbb{E}[W])$: the agent prefers the expected value of the gamble to the gamble. The certainty equivalent c defined by $u(c) = \mathbb{E}[u(W)]$ therefore satisfies $c \leq \mathbb{E}[W]$, and the risk premium $\mathbb{E}[W] - c$ is non-negative. The second-order expansion of Theorem 7.6.1 quantified this premium as $\frac{1}{2}A(\mathbb{E}[W])\text{Var}(W)$ for small risks; Jensen gives the sign for risks of any size, with no differentiability and no smallness. Equality holds only if u is affine on the support of W , which is the definition of risk neutrality there.

14.3 Conditional Probability and Bayes' Rule

Definition 14.3.1 (Conditional probability). For events D, H with $\mathbb{P}(H) > 0$,

$$\mathbb{P}(D \mid H) := \frac{\mathbb{P}(D \cap H)}{\mathbb{P}(H)}. \quad (14.5)$$

Events D and H are *independent events* if $\mathbb{P}(D \cap H) = \mathbb{P}(D)\mathbb{P}(H)$, equivalently $\mathbb{P}(D \mid H) = \mathbb{P}(D)$ when $\mathbb{P}(H) > 0$.

For fixed H with $\mathbb{P}(H) > 0$ the map $E \mapsto \mathbb{P}(E \mid H)$ is itself a probability measure on $(\Omega, \mathcal{F})$: it assigns 1 to Ω and is σ -additive, both immediately from (14.5) and σ -additivity of $\mathbb{P}$. Everything proved for probability measures therefore applies to conditional probabilities without further argument.

Theorem 14.3.1 (Bayes' rule). Let D, H be events with $\mathbb{P}(D) > 0$ and $\mathbb{P}(H) > 0$. Then

$$\mathbb{P}(H \mid D) = \frac{\mathbb{P}(H)\mathbb{P}(D \mid H)}{\mathbb{P}(D)}. \quad (14.6)$$

If moreover $0 < \mathbb{P}(H) < 1$,

$$\mathbb{P}(H \mid D) = \frac{\mathbb{P}(H)\mathbb{P}(D \mid H)}{\mathbb{P}(H)\mathbb{P}(D \mid H) + \mathbb{P}(H^c)\mathbb{P}(D \mid H^c)}. \quad (14.7)$$

More generally, if $(H_j)_{j=1}^n$ partitions Ω with $\mathbb{P}(H_j) > 0$ for every j , then for each i

$$\mathbb{P}(H_i \mid D) = \frac{\mathbb{P}(D \mid H_i)\mathbb{P}(H_i)}{\sum_{j=1}^n \mathbb{P}(D \mid H_j)\mathbb{P}(H_j)}. \quad (14.8)$$

Proof. By (14.5) applied twice, $\mathbb{P}(H \mid D)\mathbb{P}(D) = \mathbb{P}(D \cap H) = \mathbb{P}(D \mid H)\mathbb{P}(H)$; dividing by $\mathbb{P}(D) > 0$ gives (14.6). For the denominators, the events $D \cap H_j$ are pairwise disjoint with union D because (H_j) partitions Ω , so finite additivity (Proposition 14.1.2(iii)) gives the *law of total probability*

$$\mathbb{P}(D) = \sum_{j=1}^n \mathbb{P}(D \cap H_j) = \sum_{j=1}^n \mathbb{P}(D \mid H_j)\mathbb{P}(H_j). \quad (14.9)$$

Substituting (14.9) into (14.6) gives (14.8), and (14.7) is the case $n = 2$ with $H_1 = H$, $H_2 = H^c$. ■

The content of (14.6) is a reversal of the direction of conditioning. The likelihood $\mathbb{P}(D | H)$ runs from hypothesis to data, from cause to effect; the posterior $\mathbb{P}(H | D)$ runs from data back to hypothesis, and the theorem says the two are related by the ratio of the prior to the marginal likelihood of the data (Fukuda 2026f, slide 6).

Example 14.3.1 (Base rates dominate). A disease affects 1% of the population. The test is positive with probability 0.95 on the diseased and negative with probability 0.9 on the healthy. Let H be “diseased” and D “tests positive”. Then

$$\mathbb{P}(H | D) = \frac{0.01 \times 0.95}{0.01 \times 0.95 + 0.99 \times 0.10} = \frac{0.0095}{0.1085} = \frac{19}{217} \approx 0.0876.$$

A test with 95% sensitivity and 90% specificity leaves the posterior probability of disease below 9%. The reason is arithmetic rather than statistical: the 99% healthy generate 9.9 false positives per hundred while the 1% diseased generate 0.95 true positives, and no reading of the likelihoods alone can reveal this. Neglect of the denominator in (14.7) is the base-rate fallacy, and it is the mechanism behind overconfidence in models of belief updating.

Example 14.3.2 (The airport carousel). Bags are correctly loaded with probability 0.9. Conditional on the bag having been loaded, it fails to appear after nine minutes with probability 0.1; if it was not loaded it certainly does not appear. With H “not loaded” and D “bag missing”,

$$\mathbb{P}(H | D) = \frac{0.1 \times 1}{0.1 \times 1 + 0.9 \times 0.1} = \frac{0.1}{0.19} = \frac{10}{19} \approx 0.5263.$$

Waiting turns a prior of 0.1 into a posterior above one half. The magnitude of the revision is governed by the likelihood ratio $\mathbb{P}(D | H)/\mathbb{P}(D | H^c) = 10$, which is the sufficient statistic for the evidence: (14.7) in odds form reads

$$\frac{\mathbb{P}(H | D)}{\mathbb{P}(H^c | D)} = \frac{\mathbb{P}(H)}{\mathbb{P}(H^c)} \cdot \frac{\mathbb{P}(D | H)}{\mathbb{P}(D | H^c)}, \quad (14.10)$$

posterior odds equal prior odds times the likelihood ratio.

Example 14.3.3 (A poll). Party A has 30% support, party B has 40%, and 30% support neither. The cabinet is supported by 80% of A 's supporters, 20% of B 's and 50% of the rest. By (14.8) with the three-element partition, a randomly chosen cabinet supporter belongs to party A with probability

$$\frac{0.3 \times 0.8}{0.3 \times 0.8 + 0.4 \times 0.2 + 0.3 \times 0.5} = \frac{0.24}{0.47} = \frac{24}{47} \approx 0.5106.$$

14.3.1 Economic application: order flow and the market maker's beliefs

Environment. An asset has value $V \in \{H, L\}$ with $H > L$. A market maker holds the prior $p := \mathbb{P}(V = H) \in (0, 1)$ and quotes prices at which they will trade one unit. A trader arrives; with probability $q \in (0, 1)$ the trader is *informed* and knows V , and with probability $1 - q$ the trader is *uninformed*. The informed trader buys when $V = H$ and sells when $V = L$; the uninformed trader buys or sells with probability $\frac{1}{2}$ each, independently of V .

Proposition 14.3.1 (Posterior after a sale). Let S denote the event that the first trade is a sale. Then

$$\mathbb{P}(V = H | S) = \frac{p(1 - q)}{p(1 - q) + (1 - p)(1 + q)}, \quad (14.11)$$

which is strictly less than p for every $q \in (0, 1)$, is strictly decreasing in q , equals p at $q = 0$ and tends to 0 as $q \rightarrow 1$.

Proof. Conditioning on the trader's type and using (14.9) within each state,

$$\mathbb{P}(S | V = H) = q \cdot 0 + (1 - q) \cdot \frac{1}{2} = \frac{1 - q}{2}, \quad \mathbb{P}(S | V = L) = q \cdot 1 + (1 - q) \cdot \frac{1}{2} = \frac{1 + q}{2},$$

because an informed trader never sells when the value is high and always sells when it is low. Substituting into (14.7) and cancelling the factor $\frac{1}{2}$ gives (14.11). Writing the posterior as $(1 + \frac{1-p}{p} \cdot \frac{1+q}{1-q})^{-1}$ exhibits it as strictly decreasing in q , since $q \mapsto (1 + q)/(1 - q)$ is strictly increasing on $(0, 1)$; the boundary values follow by inspection. ■

The comparative static in q is the informational content of order flow. When $q = 0$ every trade is noise and prices do not move; as $q \rightarrow 1$ a single sale reveals the state. The market maker who sets the bid at $\mathbb{E}[V | S] = L + (H - L)\mathbb{P}(V = H | S)$ and the ask at $\mathbb{E}[V | B]$ makes zero expected profit on each trade and covers the losses to informed traders with the gains from uninformed ones; the difference between the two quotes is the bid–ask spread, and (14.11) shows it widens in q . This is the adverse-selection account of the spread, and the calculation above is its entire mechanism (Fukuda 2026f, slide 12); see O’Hara (1998, Sec. 3.A.1) for the sequential-trade model built on it and Kyle (1985) for the continuous version in which the informed trader chooses order size strategically.

14.4 Conditional Expectation

Definition 14.4.1 (Conditional expectation given an event). For an event H with $\mathbb{P}(H) > 0$ and a random variable X ,

$$\mathbb{E}[X | H] := \sum_{x \in X(H)} x \mathbb{P}([X = x] | H) = \sum_{\omega \in H} X(\omega) \mathbb{P}(\{\omega\} | H) = \frac{\mathbb{E}[X \mathbb{1}_H]}{\mathbb{P}(H)}, \quad (14.12)$$

where $X(H) := \{X(\omega) : \omega \in H\}$ and $\mathbb{1}_H$ is the indicator of H .

Proof of the equalities in (14.12). The first equals the second by grouping $\omega \in H$ according to the value of $X(\omega)$, exactly as in (14.2). For the third, expand $\mathbb{P}(\{\omega\} | H) = \mathbb{P}(\{\omega\})/\mathbb{P}(H)$ for $\omega \in H$ and note that $X(\omega)\mathbb{1}_H(\omega) = 0$ for $\omega \notin H$, so $\sum_{\omega \in H} X(\omega)\mathbb{P}(\{\omega\}) = \sum_{\omega \in \Omega} X(\omega)\mathbb{1}_H(\omega)\mathbb{P}(\{\omega\}) = \mathbb{E}[X \mathbb{1}_H]$. ■

Definition 14.4.2 (Conditional expectation given a random variable). Let X, Y be random variables. The *conditional expectation* $\mathbb{E}[Y | X]$ is the random variable $\omega \mapsto \mathbb{E}[Y | [X = X(\omega)]]$, that is, the function of X whose value at $x \in X(\Omega)$ is

$$\mathbb{E}[Y | X = x] := \sum_{y \in Y(\Omega)} y f_{Y|X}(y | x), \quad f_{Y|X}(y | x) := \mathbb{P}([Y = y] | [X = x]). \quad (14.13)$$

The distinction between Definition 14.4.1 and Definition 14.4.2 is that the first is a number and the second a random variable — a function of X , hence itself measurable with respect to the information generated by X . Everything that follows turns on this: $\mathbb{E}[Y | X]$ is the best summary of Y available to someone who observes X .

Proposition 14.4.1 (Taking out what is known). Let X, Y be random variables and $g: \mathbb{R} \rightarrow \mathbb{R}$ with $g(X)Y$ integrable. Then

$$\mathbb{E}[g(X)Y | X] = g(X) \mathbb{E}[Y | X]. \quad (14.14)$$

In particular $\mathbb{E}[g(X) | X] = g(X)$ and $\mathbb{E}[a | X] = a$ for constant a .

Proof. Fix $x_0 \in X(\Omega)$ and put $W := g(X)Y$. If $g(x_0) = 0$ then W is identically zero on $[X = x_0]$ and both sides of (14.14) vanish there. If $g(x_0) \neq 0$ then on the event $[X = x_0]$ we have $W = w$ if and only if $Y = w/g(x_0)$, so

$$\mathbb{P}([W = w] | [X = x_0]) = \mathbb{P}\left([Y = \frac{w}{g(x_0)}] | [X = x_0]\right),$$

and substituting $y = w/g(x_0)$ in (14.13),

$$\mathbb{E}[W | X = x_0] = \sum_w w \mathbb{P}([W = w] | [X = x_0]) = g(x_0) \sum_y y \mathbb{P}([Y = y] | [X = x_0]) = g(x_0) \mathbb{E}[Y | X = x_0].$$

Since x_0 was arbitrary, (14.14) holds as an identity between random variables. Taking $Y \equiv 1$ gives $\mathbb{E}[g(X) | X] = g(X)$, and g constant gives the last claim. ■

Theorem 14.4.1 (Law of iterated expectations). (i) Let $(H_j)_{j=1}^m$ partition Ω with $\mathbb{P}(H_j) > 0$ for

every j , and let X be integrable. Then

$$\mathbb{E}[X] = \sum_{j=1}^m \mathbb{E}[X | H_j] \mathbb{P}(H_j). \quad (14.15)$$

(ii) For integrable X and any random variable Y ,

$$\mathbb{E}[X] = \mathbb{E}[\mathbb{E}[X | Y]]. \quad (14.16)$$

Proof. (i) By the third form of (14.12), $\mathbb{E}[X | H_j] \mathbb{P}(H_j) = \mathbb{E}[X \mathbb{1}_{H_j}]$. Since (H_j) partitions Ω , $\sum_j \mathbb{1}_{H_j} = 1$ pointwise, so linearity (Proposition 14.2.1(i)) gives $\sum_j \mathbb{E}[X \mathbb{1}_{H_j}] = \mathbb{E}[X \sum_j \mathbb{1}_{H_j}] = \mathbb{E}[X]$.

(ii) Apply (i) to the partition $H_j := [Y = y_j]$, $y_j \in Y(\Omega)$, and observe that $\sum_j \mathbb{E}[X | Y = y_j] \mathbb{P}(Y = y_j)$ is by (14.2) and (14.13) precisely the expectation of the random variable $\mathbb{E}[X | Y]$, whose value on $[Y = y_j]$ is $\mathbb{E}[X | Y = y_j]$ and which takes that value with probability $\mathbb{P}(Y = y_j)$. ■

These are Theorems 2.4 and 2.5 of Fukuda (2026d, pp. 74–75) (Fukuda 2026f, slides 18–19). The statement to hold on to is that conditioning on less information cannot change the average: an expectation computed in two stages, first within each cell of a partition and then across cells, equals the expectation computed at once. Almost every argument in dynamic economics that replaces a many-period expectation by a one-period expectation of a continuation value is an application of (14.16).

Proposition 14.4.2 (Conditional Jensen). Let $\varphi: I \rightarrow \mathbb{R}$ be convex on an interval I , and let Y be integrable with $\mathbb{P}(Y \in I) = 1$ and $\mathbb{E}[Y | X]$ almost surely interior to I . Then

$$\mathbb{E}[\varphi(Y) | X] \geq \varphi(\mathbb{E}[Y | X]) \quad \text{almost surely,} \quad (14.17)$$

and the inequality reverses for concave φ .

Proof. Fix $x \in X(\Omega)$. Conditional on $[X = x]$ the map $E \mapsto \mathbb{P}(E | [X = x])$ is a probability measure, as noted after Definition 14.3.1, and $\mathbb{E}[\cdot | X = x]$ is the expectation under it. Theorem 14.2.1 applied under that measure, with $\mu := \mathbb{E}[Y | X = x]$ interior to I , gives $\mathbb{E}[\varphi(Y) | X = x] \geq \varphi(\mu)$. As x was arbitrary, the inequality holds pointwise as an inequality between random variables. ■

Taking expectations in (14.17) and applying (14.16) recovers Theorem 14.2.1, so the conditional version is strictly stronger. It is what makes the value function of a dynamic programme concave in the state when the return function is concave, and it is what forces the martingale $(\mathbb{E}[Y | X_t])_t$ to have a convex-increasing transform.

14.5 Conditional Expectation as a Projection

Let $L^2 := L^2(\Omega, \mathcal{F}, \mathbb{P})$ be the space of square-integrable random variables with the inner product $\langle U, V \rangle := \mathbb{E}[UV]$; it is a Hilbert space, and the geometry of Chapter 11 applies verbatim with $\|U\|^2 = \mathbb{E}[U^2]$.

Theorem 14.5.1 (Conditional expectation is the best predictor). Let $X, Y \in L^2$ and let $M(X) := \{\phi(X) : \phi \text{ Borel}, \phi(X) \in L^2\}$, a closed subspace of L^2 . Then for every ϕ with $\phi(X) \in L^2$,

$$\mathbb{E}[(Y - \phi(X))^2] = \mathbb{E}[(Y - \mathbb{E}[Y | X])^2] + \mathbb{E}[(\mathbb{E}[Y | X] - \phi(X))^2] \geq \mathbb{E}[(Y - \mathbb{E}[Y | X])^2], \quad (14.18)$$

with equality if and only if $\phi(X) = \mathbb{E}[Y | X]$ almost surely. Equivalently, $\mathbb{E}[Y | X]$ is the orthogonal projection of Y onto $M(X)$: $P_{M(X)} Y = \mathbb{E}[Y | X]$.

Proof. Write $\hat{Y} := \mathbb{E}[Y | X]$ and decompose $Y - \phi(X) = (Y - \hat{Y}) + (\hat{Y} - \phi(X))$. Expanding the square,

$$\mathbb{E}[(Y - \phi(X))^2] = \mathbb{E}[(Y - \hat{Y})^2] + 2\mathbb{E}[(Y - \hat{Y})(\hat{Y} - \phi(X))] + \mathbb{E}[(\hat{Y} - \phi(X))^2].$$

The cross term vanishes. Indeed $\hat{Y} - \phi(X)$ is a function of X , so Proposition 14.4.1 and then (14.16) give

$$\mathbb{E}[(Y - \hat{Y})(\hat{Y} - \phi(X))] = \mathbb{E}[\mathbb{E}[(Y - \hat{Y})(\hat{Y} - \phi(X)) | X]] = \mathbb{E}[(\hat{Y} - \phi(X)) \underbrace{(\mathbb{E}[Y | X] - \hat{Y})}_{=0}] = 0,$$

using $\mathbb{E}[\hat{Y} | X] = \hat{Y}$ from [Proposition 14.4.1](#). This proves the identity in (14.18); the inequality and its equality case follow because the second term is non-negative and vanishes only when $\hat{Y} = \phi(X)$ almost surely, by [Proposition 14.2.1\(ii\)](#).

The vanishing of the cross term for *every* ϕ says $Y - \hat{Y} \perp M(X)$, and $\hat{Y} \in M(X)$; by [Proposition 11.3.1](#) this characterises $\hat{Y}$ as the orthogonal projection of Y onto $M(X)$, whose existence and uniqueness are [Theorem 11.3.2](#). ■

Three readings of (14.18) are worth separating. As a minimisation, it says the conditional expectation is the mean-square-optimal forecast among all functions of the conditioning variable, extending the constant-forecast statement of [Proposition 14.2.2](#). As a geometric statement, it identifies conditioning with orthogonal projection, so that all the structure of [Chapter 11](#) — idempotence, the Pythagorean decomposition, contraction in norm — transfers to conditioning; the tower property (14.16) becomes the composition of projections onto nested subspaces. As a variance decomposition, taking $\phi \equiv \mathbb{E}[Y]$ in (14.18) and using (14.16) yields

$$\text{Var}(Y) = \mathbb{E}[\text{Var}(Y | X)] + \text{Var}(\mathbb{E}[Y | X]), \quad (14.19)$$

the law of total variance: unexplained variation plus explained variation. In a regression context the second term over the first is the population R^2 .

Remark 14.5.1 (Linear versus general prediction). [Proposition 11.5.2](#) solved the same problem over the strictly smaller subspace $\text{span}\{1, X\} \subseteq M(X)$ and produced

$$\mathbb{E}^*[Y | X] = \mathbb{E}[Y] + \frac{\text{Cov}(X, Y)}{\text{Var}(X)}(X - \mathbb{E}[X]). \quad (14.20)$$

Since the subspace is smaller, the linear predictor is weakly worse, and strictly worse whenever $\mathbb{E}[Y | X]$ is not affine in X . The two coincide exactly when $\mathbb{E}[Y | X]$ happens to be affine, which is the content of the next proposition and the reason the bivariate normal is the workhorse of applied work: it is the case in which linear regression estimates the conditional expectation rather than merely its best affine approximation.

Proposition 14.5.1 (The Gaussian case). Let (X, Y) be jointly normal with $\text{Var}(X) > 0$. Then

$$\mathbb{E}[Y | X] = \mathbb{E}[Y] + \frac{\text{Cov}(X, Y)}{\text{Var}(X)}(X - \mathbb{E}[X]), \quad \text{Var}(Y | X) = (1 - \text{Corr}(X, Y)^2) \text{Var}(Y), \quad (14.21)$$

the conditional variance being a constant, not a function of X .

The multivariate statement, in which X and Y are vectors and the ratio $\text{Cov}(X, Y)/\text{Var}(X)$ becomes $\Sigma_{12}\Sigma_{22}^{-1}$, is [Theorem C.2.1](#) of [Appendix C](#), together with the affine stability of the family that makes the argument work.

Proof. Let $g(X)$ denote the right-hand side of (14.20). By construction $Y - g(X)$ is orthogonal to 1 and to X , that is $\mathbb{E}[Y - g(X)] = 0$ and $\text{Cov}(X, Y - g(X)) = 0$. Now $Y - g(X)$ and X are jointly normal, being affine images of the jointly normal pair (X, Y) , and for jointly normal variables zero correlation implies independence, since the joint density then factors. Hence $\mathbb{E}[Y - g(X) | X] = \mathbb{E}[Y - g(X)] = 0$, which is the first identity in (14.21).

For the variance, $Y - \mathbb{E}[Y] = \frac{\text{Cov}(X, Y)}{\text{Var}(X)}(X - \mathbb{E}[X]) + (Y - g(X))$ is an orthogonal decomposition, so [Proposition 11.2.2](#) gives

$$\text{Var}(Y) = \frac{\text{Cov}(X, Y)^2}{\text{Var}(X)^2} \text{Var}(X) + \text{Var}(Y - g(X)) = \frac{\text{Cov}(X, Y)^2}{\text{Var}(X)} + \text{Var}(Y | X),$$

and rearranging gives the stated expression with $\text{Corr}(X, Y)^2 = \text{Cov}(X, Y)^2/(\text{Var}(X) \text{Var}(Y))$. ■

14.6 Normal–Normal Updating

Theorem 14.6.1 (Gaussian updating, one observation). Let $X \sim \mathcal{N}(\mu_x, \sigma_x^2)$ with $\sigma_x^2 > 0$ and let $Y = X + \varepsilon$ with $\varepsilon \sim \mathcal{N}(0, \sigma_\varepsilon^2)$, $\sigma_\varepsilon^2 > 0$, independent of X . Then

$$X | Y = y \sim \mathcal{N}\left(\frac{\sigma_\varepsilon^2 \mu_x + \sigma_x^2 y}{\sigma_x^2 + \sigma_\varepsilon^2}, \frac{\sigma_x^2 \sigma_\varepsilon^2}{\sigma_x^2 + \sigma_\varepsilon^2}\right). \quad (14.22)$$

Writing $\tau_x := 1/\sigma_x^2$ and $\tau_\varepsilon := 1/\sigma_\varepsilon^2$ for the *precision* and $\tau := \tau_x + \tau_\varepsilon$, this reads

$$X | Y = y \sim \mathcal{N}\left(\frac{\tau_x \mu_x + \tau_\varepsilon y}{\tau}, \tau^{-1}\right). \quad (14.23)$$

Proof. The prior density and the likelihood are

$$\pi(x) = \frac{1}{\sqrt{2\pi}\sigma_x} \exp\left(-\frac{(x - \mu_x)^2}{2\sigma_x^2}\right), \quad L(y | x) = \frac{1}{\sqrt{2\pi}\sigma_\varepsilon} \exp\left(-\frac{(y - x)^2}{2\sigma_\varepsilon^2}\right),$$

the second because $Y - X = \varepsilon$ is independent of X . By Bayes' rule in density form, $f(x | Y = y) = \pi(x)L(y | x) / \int_{-\infty}^{\infty} \pi(\tilde{x})L(y | \tilde{x}) d\tilde{x}$, so as a function of x ,

$$f(x | Y = y) \propto \exp\left(-\frac{(x - \mu_x)^2}{2\sigma_x^2} - \frac{(y - x)^2}{2\sigma_\varepsilon^2}\right).$$

Complete the square in the exponent. Multiplying out,

$$\begin{aligned} -\frac{(x - \mu_x)^2}{2\sigma_x^2} - \frac{(y - x)^2}{2\sigma_\varepsilon^2} &= -\frac{1}{2\sigma_x^2 \sigma_\varepsilon^2} \left[\sigma_\varepsilon^2 (x - \mu_x)^2 + \sigma_x^2 (y - x)^2 \right] \\ &= -\frac{1}{2\sigma_x^2 \sigma_\varepsilon^2} \left[(\sigma_x^2 + \sigma_\varepsilon^2) x^2 - 2(\sigma_\varepsilon^2 \mu_x + \sigma_x^2 y) x + (\sigma_\varepsilon^2 \mu_x^2 + \sigma_x^2 y^2) \right] \\ &= -\frac{\sigma_x^2 + \sigma_\varepsilon^2}{2\sigma_x^2 \sigma_\varepsilon^2} \left(x - \frac{\sigma_\varepsilon^2 \mu_x + \sigma_x^2 y}{\sigma_x^2 + \sigma_\varepsilon^2} \right)^2 + c(y), \end{aligned}$$

where $c(y)$ collects all terms free of x . Exponentiating, the posterior density is proportional to a Gaussian kernel in x with mean $(\sigma_\varepsilon^2 \mu_x + \sigma_x^2 y) / (\sigma_x^2 + \sigma_\varepsilon^2)$ and variance $\sigma_x^2 \sigma_\varepsilon^2 / (\sigma_x^2 + \sigma_\varepsilon^2)$; since a density is determined by its kernel up to the normalising constant, and by [Proposition 10.7.2](#) that constant is $(2\pi \sigma_x^2 \sigma_\varepsilon^2 / (\sigma_x^2 + \sigma_\varepsilon^2))^{-1/2}$, (14.22) follows. The precision form is the substitution $\sigma_x^2 = 1/\tau_x$, $\sigma_\varepsilon^2 = 1/\tau_\varepsilon$, under which $\sigma_x^2 + \sigma_\varepsilon^2 = \tau / (\tau_x \tau_\varepsilon)$. ■

This is the conjugate normal model of Gelman et al. (2013, Sec. 2.5, pp. 39–42), whose multivariate counterpart with known variance is Section 3.5 of the same work; the precision form (14.23) is the statement there that the posterior precision is the prior precision plus the data precision. Three of its properties are the reason the Gaussian model is used wherever information is aggregated.

Precisions add. The posterior precision $\tau = \tau_x + \tau_\varepsilon$ does not depend on the realisation y : the informativeness of an observation is known before it is made. Consequently the posterior variance $\tau^{-1} < \sigma_x^2$ falls with every observation, deterministically.

The posterior mean is a precision-weighted average. It is linear in y ,

$$\mathbb{E}[X | Y = y] = \mu_x + \frac{\sigma_x^2}{\sigma_x^2 + \sigma_\varepsilon^2} (y - \mu_x),$$

so the response to the surprise $y - \mu_x$ is the ratio of prior variance to total variance, between 0 and 1. A precise prior or a noisy signal produces sluggish updating; the converse produces aggressive updating. Linearity in y is also the statement that the best predictor is the best *linear* predictor, which is [Proposition 14.5.1](#) applied to the jointly normal pair (X, Y) .

Conjugacy. The posterior is again normal, so the operation can be iterated without leaving the family — and this is exactly what makes recursive filtering tractable.

Theorem 14.6.2 (Gaussian updating, many observations). Let $X \sim \mathcal{N}(\mu_x, \tau_x^{-1})$ and let $Y_j = X + \varepsilon_j$ for $j = 1, \dots, n$, where the $\varepsilon_j \sim \mathcal{N}(0, \tau_j^{-1})$ are mutually independent and independent of X . Then

with $\tau := \tau_x + \sum_{j=1}^n \tau_j$,

$$X \mid (Y_j = y_j)_{j=1}^n \sim \mathcal{N}\left(\frac{\tau_x}{\tau} \mu_x + \sum_{j=1}^n \frac{\tau_j}{\tau} y_j, \tau^{-1}\right). \quad (14.24)$$

Proof. Independence of the errors makes the joint likelihood the product $\prod_j L(y_j \mid x)$, so

$$f(x \mid (Y_j = y_j)_j) \propto \exp\left(-\frac{1}{2}\left[\tau_x(x - \mu_x)^2 + \sum_{j=1}^n \tau_j(x - y_j)^2\right]\right).$$

The bracket is a quadratic in x with leading coefficient $\tau_x + \sum_j \tau_j = \tau$ and linear coefficient $-2(\tau_x \mu_x + \sum_j \tau_j y_j)$, so completing the square as in [Theorem 14.6.1](#) gives $\tau(x - (\tau_x \mu_x + \sum_j \tau_j y_j)/\tau)^2$ plus terms free of x . Identification of the Gaussian kernel gives (14.24). ■

The posterior mean is a weighted average of the prior mean and all the observations, with weights proportional to precisions summing to one, and the posterior precision is the sum of all precisions. Applying [Theorem 14.6.1](#) sequentially, using the posterior after j observations as the prior for observation $j + 1$, gives the same answer — the order of arrival is irrelevant, which is a substantive restriction and not an accounting identity.

14.6.1 Economic application: CARA–normal portfolio choice

Environment. An investor with initial wealth W_0 and constant absolute risk aversion $a > 0$, $u(W) = -e^{-aW}$, chooses a quantity $\theta \in \mathbb{R}$ of a risky asset with payoff $V \sim \mathcal{N}(\mu, \sigma^2)$ at price p ; the riskless rate is normalised to zero, so terminal wealth is $W = W_0 + \theta(V - p)$.

Proposition 14.6.1 (Gaussian demand). The investor’s optimal holding is

$$\theta^* = \frac{\mu - p}{a\sigma^2}, \quad (14.25)$$

and the certainty equivalent of the optimal portfolio is $W_0 + (\mu - p)^2/(2a\sigma^2)$. If instead the investor observes a signal $Y = V + \varepsilon$ with $\varepsilon \sim \mathcal{N}(0, \tau_\varepsilon^{-1})$ independent of V , then θ^* is (14.25) with μ and σ^2 replaced by the posterior mean and variance of [Theorem 14.6.1](#).

Proof. Terminal wealth is normal with mean $W_0 + \theta(\mu - p)$ and variance $\theta^2 \sigma^2$, so by [Proposition 10.7.1](#)

$$\mathbb{E}[u(W)] = -\exp\left(-a(W_0 + \theta(\mu - p)) + \frac{1}{2}a^2\theta^2\sigma^2\right).$$

Maximising this is minimising the exponent, that is maximising $\theta(\mu - p) - \frac{1}{2}a\theta^2\sigma^2$, a strictly concave quadratic in θ ; [Theorem 7.2.3](#) gives the unique maximiser (14.25), and substituting back gives an exponent of $-a(W_0 + (\mu - p)^2/(2a\sigma^2))$, whose certainty equivalent is the stated value. Conditioning on $Y = y$ replaces the distribution of V by its posterior, which is again normal by [Theorem 14.6.1](#), and the same computation applies. ■

Demand is linear in the price with slope $-1/(a\sigma^2)$, and initial wealth does not appear — the wealth effect is exactly zero under CARA, which is what makes the model aggregate. Two consequences follow. Because demand is linear in the posterior mean and the posterior mean is linear in the signal by [Theorem 14.6.1](#), individual demands are linear in private signals, so aggregate demand is linear in the average signal and the market-clearing price is a linear statistic of the population’s information. This is the mechanism by which prices aggregate dispersed information; adding noise traders to prevent the price from being fully revealing gives the noisy rational-expectations equilibrium of Grossman and Stiglitz (1980), in which the price is informative but not sufficient, so that costly information acquisition remains worthwhile. Second, the precision arithmetic of (14.23) means an investor with a more precise signal takes a larger position on the same price, which is the source of the informed trader’s profit in [subsection 14.3.1](#).

14.6.2 Economic application: recursive filtering

Conjugacy makes updating recursive. Let the state follow $X_{t+1} = \rho X_t + \eta_{t+1}$ with $\eta_{t+1} \sim \mathcal{N}(0, \tau_2)$ and let the observation be $Y_t = X_t + \varepsilon_t$ with $\varepsilon_t \sim \mathcal{N}(0, \tau_1^{-1})$, all errors independent. If the prior on X_t given the history is $\mathcal{N}(\hat{x}_t, P_t)$, then [Theorem 14.6.1](#) gives the posterior precision $P_t^{-1} + \tau_1$ after observing Y_t , and the transition inflates the variance by τ_2 after scaling by ρ^2 :

$$P_{t+1} = \rho^2 (P_t^{-1} + \tau_1)^{-1} + \tau_2. \quad (14.26)$$

The right-hand side is exactly the operator T of (6.5), shown in [Proposition 6.6.2](#) to be a contraction of modulus ρ^2 on $\mathbb{R}_{++}$. [Theorem 6.4.1](#) therefore gives a unique steady-state variance x^* satisfying (6.6), reached from every initial uncertainty at the geometric rate ρ^2 . The economics of the recursion is that the filter never learns the state exactly, no matter how long it observes: each period's transition injects fresh variance τ_2 , and the steady state balances that injection against the precision τ_1 acquired from the new observation. Setting $\rho = 0$ makes the state i.i.d. and gives $x^* = \tau_2$, the variance of a state about which yesterday's data say nothing.

Equation (14.26) is the scalar case of the Kalman recursions of Brockwell and Davis (2016, Sec. 9.4, p. 270), whose general form replaces the variances by covariance matrices and the scalar ρ by a transition matrix; the fixed point x^* is then the steady-state error covariance of the filter. The contraction argument survives verbatim in the matrix case with the operator acting on positive definite matrices, which is why the filter forgets its initialisation at a geometric rate there too.

14.7 Probability Inequalities

Theorem 14.7.1 (Markov and Chebyshev). Let $X \geq 0$ be a random variable and $a > 0$. Then

$$\mathbb{P}(X \geq a) \leq \frac{\mathbb{E}[X]}{a}. \quad (14.27)$$

Consequently, for any square-integrable Z and $k > 0$,

$$\mathbb{P}(|Z - \mathbb{E}[Z]| \geq k) \leq \frac{\text{Var}(Z)}{k^2}. \quad (14.28)$$

Proof. Since $X \geq 0$, the pointwise inequality $X \geq a \mathbb{1}_{[X \geq a]}$ holds: on $[X \geq a]$ the left side is at least a , and elsewhere the right side is zero. Taking expectations and using monotonicity and linearity ([Proposition 14.2.1](#)) gives $\mathbb{E}[X] \geq a \mathbb{E}[\mathbb{1}_{[X \geq a]}] = a \mathbb{P}(X \geq a)$, which is (14.27). For (14.28), apply (14.27) to the non-negative variable $X := (Z - \mathbb{E}[Z])^2$ with threshold k^2 and use $\{|Z - \mathbb{E}[Z]| \geq k\} = \{X \geq k^2\}$ together with $\mathbb{E}[X] = \text{Var}(Z)$. ■

Remark 14.7.1 (Sharpness and use). Markov's inequality is attained: if $X = a$ with probability $\mathbb{E}[X]/a$ and $X = 0$ otherwise, (14.27) holds with equality, so no distribution-free improvement is possible. Its role is to convert a statement about a mean into a statement about a tail using no distributional assumption, and this is what drives the weak law of large numbers: for i.i.d. square-integrable (Z_i) with mean μ and variance σ^2 , the sample mean $\bar{Z}_n$ has $\mathbb{E}[\bar{Z}_n] = \mu$ and, by independence, $\text{Var}(\bar{Z}_n) = \sigma^2/n$, so (14.28) gives $\mathbb{P}(|\bar{Z}_n - \mu| \geq k) \leq \sigma^2/(nk^2) \rightarrow 0$ for every $k > 0$. Convergence in probability of sample averages is therefore a two-line consequence of the variance identity and Markov's inequality, requiring no central limit theorem and no normality.

14.8 Chapter Summary

A probability space consists of a sample space, a σ -algebra of events closed under complementation and countable intersection, and a countably additive measure of total mass one; monotonicity, finite additivity and inclusion–exclusion follow from the two axioms. A random variable is a measurable real function on the sample space, its expectation the mass-weighted sum of its values, and the variance identity $\text{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$ is the special case, at $c = 0$, of the decomposition $\mathbb{E}[(X - c)^2] = \text{Var}(X) + (\mathbb{E}[X] - c)^2$ that identifies the mean as the best constant predictor. Jensen's inequality follows

from the existence of a supporting line to a convex function, and yields the sign of the risk premium without differentiability.

Conditional probability makes $\mathbb{P}(\cdot | H)$ a probability measure in its own right; Bayes' rule reverses the direction of conditioning, and in odds form states that posterior odds are prior odds times the likelihood ratio. The denominator, computed by the law of total probability, is what the base-rate fallacy neglects. Conditional expectation given a random variable is itself a random variable, satisfies the pull-out property $\mathbb{E}[g(X)Y | X] = g(X)\mathbb{E}[Y | X]$ and the law of iterated expectations $\mathbb{E}[\mathbb{E}[X | Y]] = \mathbb{E}[X]$, and, viewed in L^2 , is the orthogonal projection of Y onto the functions of X . That identification supplies at once the best-predictor property, the law of total variance, and the fact that the best linear predictor is a strictly worse projection onto a strictly smaller subspace, the two coinciding exactly in the jointly normal case.

Gaussian updating is conjugate: posterior precision is the sum of prior and signal precisions, independent of the realisation, and the posterior mean is the corresponding precision-weighted average, linear in the signal. With many independent signals the same arithmetic applies to their sum, and the sequential and simultaneous updates agree. Applied to a CARA investor the linearity yields demand $(\mu - p)/(a\sigma^2)$ with no wealth effect, hence prices that are linear statistics of dispersed information; applied recursively it yields the Kalman recursion, whose variance obeys the Riccati equation shown in Chapter 6 to be a contraction. Markov's inequality bounds a tail by a mean with no distributional assumption, and through Chebyshev delivers the weak law of large numbers.

Exercise 14.8.1. Prove the general Bayes formula (14.8) directly from Definition 14.3.1 without invoking (14.6), and verify it against Example 14.3.3. Then show that the odds form (14.10) extends to a partition of size n as the statement that $\mathbb{P}(H_i | D)/\mathbb{P}(H_j | D)$ equals the prior odds ratio times the likelihood ratio, so that the normalising constant never has to be computed to compare two hypotheses.

Exercise 14.8.2. Prove the law of total variance (14.19) from Theorem 14.5.1 and Theorem 14.4.1. Then show that $\text{Var}(\mathbb{E}[Y | X]) \leq \text{Var}(Y)$, with equality if and only if Y is almost surely a function of X , and interpret the ratio of the two as the population R^2 of the regression of Y on X .

Exercise 14.8.3. In the setting of Proposition 14.3.1, compute $\mathbb{P}(V = H | B)$ after a buy, and show that the market maker's bid $\mathbb{E}[V | S]$ and ask $\mathbb{E}[V | B]$ satisfy $\mathbb{E}[V | S] < \mathbb{E}[V] < \mathbb{E}[V | B]$ with the spread strictly increasing in q and vanishing at $q = 0$. Then verify that the martingale property $\mathbb{E}[\mathbb{E}[V | \text{trade}]] = \mathbb{E}[V]$ holds, and explain why this is Theorem 14.4.1 rather than an assumption about the market maker's behaviour.

Chapter 15

Discrete-Time Dynamic Programming

15.1 The Sequential Problem

Definition 15.1.1 (Sequential problem). Let $r: \mathbb{R}^n \times \mathbb{R}^k \rightarrow \mathbb{R}$ be the *return function*, $g: \mathbb{R}^n \times \mathbb{R}^k \rightarrow \mathbb{R}^n$ the *transition function* and $\beta \in (0, 1)$ the *discount factor*. The *sequential problem* from an initial state $x_0 \in \mathbb{R}^n$ is

$$\sup_{(u_t)_{t=0}^{\infty}} \sum_{t=0}^{\infty} \beta^t r(x_t, u_t) \quad \text{subject to} \quad x_{t+1} = g(x_t, u_t) \quad \text{for every } t \geq 0, \quad x_0 \text{ given.} \quad (15.1)$$

Its value is the *value function* $V(x_0)$. The variable $x_t \in \mathbb{R}^n$ is the *state*, $u_t \in \mathbb{R}^k$ the *control*, and a *policy function* is a map $h: \mathbb{R}^n \rightarrow \mathbb{R}^k$ prescribing $u = h(x)$.

Assumption 15.1.1 (Standing hypotheses). Throughout this chapter:

- (i) the state lies in a set $X \subseteq \mathbb{R}^n$ and the feasible controls at x form a set $\Gamma(x) \subseteq \mathbb{R}^k$, defining a correspondence $\Gamma: X \rightrightarrows \mathbb{R}^k$ that is non-empty- and compact-valued and continuous;
- (ii) $g(x, u) \in X$ for every $x \in X$ and $u \in \Gamma(x)$, so the problem is well defined;
- (iii) r is continuous and bounded on $\text{gr } \Gamma$;
- (iv) $\beta \in (0, 1)$.

Boundedness in (iii) is what makes the infinite sum in (15.1) converge absolutely: $|r| \leq \bar{r}$ gives $|\sum_t \beta^t r(x_t, u_t)| \leq \bar{r}/(1 - \beta)$ for every feasible sequence, so the supremum is finite and no ordering or summability question arises. It is a genuine restriction — it excludes log utility on an unbounded state space, and the cake-eating problem of Section 15.5 violates it as $w \downarrow 0$ — and Remark 15.3.2 records what survives without it.

Remark 15.1.1 (Stationarity and the prime notation). The problem is *stationary problem*: neither r , g nor Γ depends on t , and the horizon is infinite at every date. Therefore

$$V(x_t) = \beta^{-t} \sup_{(u_\tau)_{\tau \geq t}} \sum_{\tau=t}^{\infty} \beta^\tau r(x_\tau, u_\tau)$$

subject to the same constraints: the problem faced at date t in state x_t is the same problem faced at date 0 in state $x_0 = x_t$. Only the state matters, not the date at which it is reached, and one may therefore drop time subscripts and write x for today's state and x' for tomorrow's (Fukuda 2026b, slide 8). Every statement about V and h below is a statement about functions of the state alone; the reduction from a sequence problem in infinitely many variables to a functional equation in one is the entire content of the method.

15.2 Bellman's Principle of Optimality

"An optimal policy has the property that whatever the initial state and initial decision are, the remaining decisions must constitute an optimal policy with regard to the state resulting from the first decision." (Bellman 1957, p. 83)

Definition 15.2.1 (Bellman operator). Under [Assumption 15.1.1](#), define T on the space $\mathcal{B}(X, \mathbb{R})$ of bounded continuous functions $X \rightarrow \mathbb{R}$ by

$$(TW)(x) := \max_{u \in \Gamma(x)} \{r(x, u) + \beta W(g(x, u))\}, \quad x \in X. \quad (15.2)$$

A function W with $W = TW$ solves the *Bellman equation*.

Definition (15.2) is (6.3) of [Chapter 6](#) with the supremum replaced by a maximum. The replacement is a theorem, not a convention: [Assumption 15.1.1](#) makes the constraint correspondence compact-valued and the objective continuous, so [Proposition 15.2.1](#) shows the supremum is attained. Where it is not, every statement below about the policy correspondence fails while the contraction argument of [Proposition 6.6.1](#) survives unchanged.

Proposition 15.2.1 (T is well defined). Under [Assumption 15.1.1](#), T maps $\mathcal{B}(X, \mathbb{R})$ into itself, and the associated *policy correspondence*

$$H_W(x) := \arg \max_{u \in \Gamma(x)} \{r(x, u) + \beta W(g(x, u))\} \quad (15.3)$$

is non-empty- and compact-valued and upper hemicontinuous.

Proof. Fix $W \in \mathcal{B}(X, \mathbb{R})$. The objective $(x, u) \mapsto r(x, u) + \beta W(g(x, u))$ is continuous on $\text{gr } \Gamma$, being a sum of a continuous function and the composition of the continuous W with the continuous g . [Assumption 15.1.1](#)(i) makes Γ continuous with non-empty compact values, so [Theorem 13.5.1](#) applies: the maximum is attained, TW is continuous, and H_W is non-empty- and compact-valued and upper hemicontinuous, $\mathbb{R}^k$ being Hausdorff. Boundedness is immediate from $|TW| \leq \bar{r} + \beta \|W\|_\infty$. ■

Theorem 15.2.1 (Principle of optimality). Under [Assumption 15.1.1](#):

- (i) the value function V of (15.1) satisfies the Bellman equation $V = TV$;
- (ii) conversely, if $W \in \mathcal{B}(X, \mathbb{R})$ satisfies $W = TW$, then $W = V$;
- (iii) a feasible plan (u_t^*) from x_0 is optimal for (15.1) if and only if $u_t^* \in H_V(x_t^*)$ for every t , where $x_{t+1}^* = g(x_t^*, u_t^*)$.

Proof. (i) Fix $x \in X$ and write $S(x)$ for the value of (15.1). Any feasible plan from x decomposes into a first control $u \in \Gamma(x)$ and a continuation plan feasible from $g(x, u)$, and its objective is $r(x, u) + \beta$ times the objective of the continuation plan. Taking suprema over continuation plans first and then over u , and using that the supremum over a product decomposes,

$$S(x) = \sup_{u \in \Gamma(x)} \{r(x, u) + \beta S(g(x, u))\} = (TS)(x),$$

the interchange being legitimate because the inner supremum is finite, uniformly bounded by $\bar{r}/(1 - \beta)$.

(ii) Let $W = TW$ with W bounded. By (i) $V = TV$, and by [Theorem 15.3.1](#) below T has at most one fixed point in $\mathcal{B}(X, \mathbb{R})$, so $W = V$ provided $V \in \mathcal{B}(X, \mathbb{R})$; that V is bounded is the estimate $|V| \leq \bar{r}/(1 - \beta)$, and that it is continuous follows from (i) and [Proposition 15.2.1](#) once V is known to be the limit of $T^m W_0$, which is [Theorem 15.3.1](#).

(iii) ($\Leftarrow$) Suppose $u_t^* \in H_V(x_t^*)$ for all t . Then $V(x_t^*) = r(x_t^*, u_t^*) + \beta V(x_{t+1}^*)$ for every t ; iterating m times,

$$V(x_0) = \sum_{t=0}^{m-1} \beta^t r(x_t^*, u_t^*) + \beta^m V(x_m^*). \quad (15.4)$$

Since $|V| \leq \bar{r}/(1 - \beta)$, the last term vanishes as $m \rightarrow \infty$, so the plan attains $V(x_0)$ and is optimal.

($\Rightarrow$) Suppose the plan is optimal but $u_{t_0}^* \notin H_V(x_{t_0}^*)$ for some t_0 . Then $r(x_{t_0}^*, u_{t_0}^*) + \beta V(x_{t_0+1}^*) < V(x_{t_0}^*)$ by the definition of the argmax, so replacing the continuation from t_0 by an optimal one — which exists by [Proposition 15.2.1](#) and the argument just given — raises the objective by β^{t_0} times a strictly positive amount, contradicting optimality. ■

Remark 15.2.1 (Where the transversality condition hides). The step $\beta^m V(x_m^*) \rightarrow 0$ in (15.4) is the transversality condition, and boundedness of V is what supplies it. Without [Assumption 15.1.1\(iii\)](#) it must be imposed separately: a plan satisfying the Bellman equation at every date but with $\limsup_m \beta^m V(x_m) > 0$ need not be optimal, and the standard example is a rational asset-price bubble, in which the price satisfies the pricing equation period by period while violating the terminal condition. The converse failure — a solution of the Bellman equation that is not the value function — is the same phenomenon read in the other direction.

15.3 The Contraction Argument

Theorem 15.3.1 (The Bellman operator is a contraction). Under [Assumption 15.1.1](#), $T: \mathcal{B}(X, \mathbb{R}) \rightarrow \mathcal{B}(X, \mathbb{R})$ is a contraction of modulus β in the supremum norm. Consequently:

- (i) T has a unique fixed point $V \in \mathcal{B}(X, \mathbb{R})$, so the value function exists, is unique among bounded continuous functions, and is continuous;
- (ii) for every $W_0 \in \mathcal{B}(X, \mathbb{R})$ the iterates $W_m := T^m W_0$ converge uniformly to V , with

$$\|W_m - V\|_\infty \leq \frac{\beta^m}{1 - \beta} \|W_1 - W_0\|_\infty \quad \text{and} \quad \|W_m - V\|_\infty \leq \frac{\beta}{1 - \beta} \|W_m - W_{m-1}\|_\infty; \quad (15.5)$$

- (iii) the policy correspondence H_V of (15.3) is non-empty- and compact-valued and upper hemicontinuous, and is a continuous function whenever the maximiser is unique.

Proof. $\mathcal{B}(X, \mathbb{R})$ is complete in the supremum norm by [Theorem 6.3.2](#) and [Proposition 6.3.3](#), and T maps it into itself by [Proposition 15.2.1](#). We verify Blackwell's two sufficient conditions of [Theorem 6.5.1](#).

Monotonicity. If $W \leq W'$ pointwise then for every x and every $u \in \Gamma(x)$, $r(x, u) + \beta W(g(x, u)) \leq r(x, u) + \beta W'(g(x, u))$, and taking maxima over the same set $\Gamma(x)$ preserves the inequality: $TW \leq TW'$.

Discounting. For $\alpha \geq 0$ constant,

$$(T(W + \alpha))(x) = \max_{u \in \Gamma(x)} \{r(x, u) + \beta W(g(x, u)) + \beta \alpha\} = (TW)(x) + \beta \alpha,$$

the constant $\beta \alpha$ passing outside the maximum because it does not depend on u .

[Theorem 6.5.1](#) therefore makes T a contraction of modulus β , and [Theorem 6.4.1](#) gives (i) and the first bound in (ii); the second bound is (6.1). Continuity of V holds because $\mathcal{B}(X, \mathbb{R})$ consists of continuous functions and uniform limits of continuous functions are continuous. Claim (iii) is [Proposition 15.2.1](#) applied to $W = V$, together with [Proposition 13.3.1](#) for the single-valued case. ■

The proof separates two things that are easily conflated. Monotonicity is a property of maximisation — raising the continuation value cannot lower today's optimum — and holds regardless of discounting. Discounting is a property of $\beta < 1$ and holds regardless of the maximisation. Neither alone gives a contraction, and the failure of the second is exactly what makes undiscounted infinite-horizon problems hard.

Remark 15.3.1 (Value function iteration). The two bounds in (15.5) play different roles. The first is *a priori*: before running anything, it says $m \geq \log(\varepsilon(1 - \beta)/\|W_1 - W_0\|_\infty)/\log \beta$ iterations suffice for accuracy ε . The second is *a posteriori*: it converts the observed change $\|W_m - W_{m-1}\|_\infty$, which the algorithm computes anyway, into a certified bound on the distance to the unknown V . Stopping when $\|W_m - W_{m-1}\|_\infty < \varepsilon$, as on (Fukuda 2026b, slides 14–16), therefore guarantees an error of at most $\beta\varepsilon/(1 - \beta)$, not ε — a distinction that matters when $\beta = 0.99$, where the inflation factor is 99.

Both bounds are uniform in the initial guess, so a bad W_0 costs iterations but never convergence.

Both degrade as $\beta \rightarrow 1$, and this is the practical reason patient problems are computationally hard: the number of iterations needed for fixed accuracy grows like $1/(1 - \beta)$ up to logarithms.

Implemented on a finite grid, W is a vector of values at nodes and the maximum in (15.2) is taken over grid points, which introduces a discretisation error that (15.5) does not control. The programs `DP_cake_eating.m` and `DP_cake_eating_CRRA.m` choose the state and control so that the transition maps grid points to grid points, avoiding interpolation entirely: consumption is read off as $c_{ij} = w_i - w_j$ from the matrix of all pairs of nodes, and the maximisation is a row-wise maximum. The Lucas program of Section 15.7 cannot do this, because the transition $y \mapsto y^\alpha r$ carries nodes off the grid, and it interpolates.

Remark 15.3.2 (Unbounded returns). [Assumption 15.1.1\(iii\)](#) fails for log utility near zero consumption and for CRRA with $\gamma > 1$. Two repairs are standard. If the state space can be restricted to a compact set that the optimal path never leaves — as in [Section 15.6](#), where capital stays in $[k_{\min}, k_{\max}]$ — then r is bounded there and the theory applies verbatim. Otherwise one replaces the supremum norm by a weighted norm $\|W\|_\phi = \sup_x |W(x)| / \phi(x)$ for a suitable ϕ and re-establishes the contraction property in that norm; see Stokey et al. (1989, Ch. 4). The cake-eating problem of [Section 15.5](#) is solved below by guess-and-verify precisely because it falls outside [Assumption 15.1.1](#), and its verification consists in checking (15.4) directly rather than invoking [Theorem 15.3.1](#).

15.4 First-Order Conditions and the Envelope Formula

Contraction gives existence, uniqueness and an algorithm, but no characterisation. Characterisation comes from differentiating the Bellman equation, and that requires knowing that V is differentiable — which is a conclusion, not an assumption, since V is defined as a supremum.

Lemma 15.4.1 (Supergradients of a concave function). Let $V: X \rightarrow \mathbb{R}$ be concave on a convex $X \subseteq \mathbb{R}^n$ and let $x_0 \in \text{Int } X$. Then there exists $p \in \mathbb{R}^n$ with

$$V(x) \leq V(x_0) + \langle p, x - x_0 \rangle \quad \text{for every } x \in X, \quad (15.6)$$

and V is differentiable at x_0 if and only if such a p is unique, in which case $p = \nabla V(x_0)$.

Proof. The hypograph $\text{hyp } V = \{(x, t) \in X \times \mathbb{R} : t \leq V(x)\}$ is convex, by the concave analogue of [Proposition 9.3.1](#), and $(x_0, V(x_0))$ is a boundary point of it. [Theorem 9.4.2](#) supplies $(a, b) \neq 0$ with $\langle a, x \rangle + bt \leq \langle a, x_0 \rangle + bV(x_0)$ for all $(x, t) \in \text{hyp } V$. Letting $t \rightarrow -\infty$ forces $b \geq 0$; and $b = 0$ would give $\langle a, x - x_0 \rangle \leq 0$ for all $x \in X$ with $a \neq 0$, impossible for x_0 interior. Dividing by $b > 0$ and setting $p := -a/b$ gives (15.6) on taking $t = V(x)$. The equivalence of uniqueness with differentiability is Rockafellar (1970, Thm. 25.1). ■

Theorem 15.4.1 (Benveniste–Scheinkman). Let $X \subseteq \mathbb{R}^n$ be convex, $V: X \rightarrow \mathbb{R}$ concave, and $x_0 \in \text{Int } X$. Suppose there is a neighbourhood $D \subseteq X$ of x_0 and a concave function $W: D \rightarrow \mathbb{R}$, differentiable at x_0 , with

$$W(x_0) = V(x_0) \quad \text{and} \quad W(x) \leq V(x) \quad \text{for all } x \in D. \quad (15.7)$$

Then V is differentiable at x_0 and $\nabla V(x_0) = \nabla W(x_0)$.

Proof. Let p satisfy (15.6) for V at x_0 , which [Lemma 15.4.1](#) provides. For $x \in D$,

$$W(x) \leq V(x) \leq V(x_0) + \langle p, x - x_0 \rangle = W(x_0) + \langle p, x - x_0 \rangle,$$

so p is a supergradient of W at x_0 . But W is differentiable at x_0 , so by [Lemma 15.4.1](#) its supergradient there is unique and equals $\nabla W(x_0)$; hence $p = \nabla W(x_0)$. Since p was an arbitrary supergradient of V , the supergradient of V at x_0 is unique, and [Lemma 15.4.1](#) makes V differentiable there with $\nabla V(x_0) = \nabla W(x_0)$. ■

This is Benveniste and Scheinkman (1979). The function W is constructed from the optimal policy: fixing the control at its optimal value $u^* = h(x_0)$ and letting the state vary gives $W(x) := r(x, u^*) +$

$\beta V(g(x, u^*))$, which is feasible hence dominated by V , and which agrees with V at x_0 because u^* is optimal there. Differentiability of V is thereby inherited from differentiability of the *primitives* r and g , and the argument is a dynamic instance of the envelope logic of [Theorem 12.3.2](#): the derivative of an optimised value equals the partial derivative holding the optimiser fixed.

Theorem 15.4.2 (Euler equation and envelope formula). Take $n = k = 1$, let the constraint set be $\Gamma(x) = \mathbb{R}$ or an interval whose interior contains the optimal control, and suppose r and g are continuously differentiable, V is concave, and the optimal policy h is interior. Then at every $x \in \text{Int } X$ the optimal control $u^* = h(x)$ satisfies the *first-order condition, dynamic*

$$\frac{\partial r}{\partial u}(x, u^*) + \beta V'(g(x, u^*)) \frac{\partial g}{\partial u}(x, u^*) = 0, \quad (15.8)$$

and V is differentiable at x with the *envelope formula*

$$V'(x) = \frac{\partial r}{\partial x}(x, u^*) + \beta V'(g(x, u^*)) \frac{\partial g}{\partial x}(x, u^*). \quad (15.9)$$

Proof. Equation (15.8) is [Theorem 7.2.1](#) applied to the differentiable function $u \mapsto r(x, u) + \beta V(g(x, u))$ at its interior maximiser, V being differentiable by [Theorem 15.4.1](#) applied at $g(x, u^*)$.

For (15.9), put $W(z) := r(z, u^*) + \beta V(g(z, u^*))$ on a neighbourhood of x , with $u^* = h(x)$ held fixed. Then $W(x) = V(x)$ because u^* is optimal at x , and $W(z) \leq V(z)$ for z near x because u^* is merely feasible at z . So [Theorem 15.4.1](#) gives $V'(x) = W'(x) = r_x(x, u^*) + \beta V'(g(x, u^*))g_x(x, u^*)$. ■

Remark 15.4.1 (The derivation by differentiating the policy). The derivation on (Fukuda 2026b, slide 18) differentiates $V(x) = r(x, h(x)) + \beta V(g(x, h(x)))$ directly, obtaining

$$V'(x) = \underbrace{\left[r_u + \beta V'(g) g_u \right]}_{=0 \text{ by (15.8)}} h'(x) + r_x + \beta V'(g) g_x,$$

so that the term in h' drops out and (15.9) remains. The computation is exactly right in content and presupposes what is at issue: that V and h are differentiable. [Theorem 15.4.1](#) establishes the conclusion without differentiating h at all, and therefore without assuming the policy is smooth — which it need not be, since [Theorem 15.3.1](#) delivers only upper hemicontinuity.

Corollary 15.4.1 (Euler equation). In the setting of [Theorem 15.4.2](#), suppose $r(x, u) = \tilde{r}(u)$ does not depend on x and $g(x, u) = x - u$. Then

$$\tilde{r}'(u_t) = \beta \tilde{r}'(u_{t+1}) \quad \text{for every } t. \quad (15.10)$$

Proof. Here $r_u = \tilde{r}'$, $g_u = -1$, $r_x = 0$ and $g_x = 1$, so (15.8) reads $\tilde{r}'(u) = \beta V'(x')$ and (15.9) reads $V'(x) = \beta V'(x')$. Applying the first at date t and at date $t+1$,

$$\tilde{r}'(u_t) = \beta V'(x_{t+1}) \stackrel{\text{envelope}}{=} \beta \cdot \beta V'(x_{t+2}) \stackrel{\text{FOC}}{=} \beta \tilde{r}'(u_{t+1}),$$

which is (15.10). ■

The chain in the proof — first-order condition, envelope, first-order condition — is the standard route from a Bellman equation to an Euler equation, and it eliminates the unknown V' entirely: the Euler equation involves only primitives (Fukuda 2026b, slide 32). Its economic reading is a no-arbitrage condition along the optimal path. Consuming Δ less today costs $\tilde{r}'(u_t)\Delta$ in current utility and returns $\beta \tilde{r}'(u_{t+1})\Delta$ in discounted utility tomorrow; at an optimum the two are equal, and the equality is precisely the first-order condition of the perturbed problem in Δ at $\Delta = 0$.

15.5 The Cake-Eating Problem

Problem. An agent endowed with a cake of size $w_0 > 0$ chooses consumption $(c_t)_{t \geq 0}$ to maximise $\sum_{t \geq 0} \beta^t \log c_t$ subject to $w_{t+1} = w_t - c_t$ and $c_t \geq 0$, $w_t \geq 0$.

There are two ways to assign the state and the control, and they are not computationally equivalent (Fukuda 2026b, slides 21–22). Taking the control to be consumption gives $V(w) = \max_{c \in [0, w]} \{\log c + \beta V(w - c)\}$, in which $w - c$ generally falls between grid points and must be interpolated. Taking the control to be *tomorrow's cake* gives

$$V(w) = \max_{w' \in [0, w]} \{\log(w - w') + \beta V(w')\}, \quad (15.11)$$

in which the transition is the identity on the grid. The second formulation is the one implemented.

Proposition 15.5.1 (Closed form). The cake-eating problem has value function and policy

$$V(w) = \frac{\log(1 - \beta)}{1 - \beta} + \frac{\beta \log \beta}{(1 - \beta)^2} + \frac{\log w}{1 - \beta}, \quad h(w) = (1 - \beta)w, \quad (15.12)$$

so that $c_t = (1 - \beta)\beta^t w_0$ and $w_{t+1} = \beta w_t$.

Proof. From the Euler equation. Corollary 15.4.1 with $\tilde{r} = \log$ gives $1/c_t = \beta/c_{t+1}$, that is $c_{t+1} = \beta c_t$, hence $c_\tau = \beta^{\tau-t} c_t$. Summing the budget identity $w_t = \sum_{\tau \geq t} c_\tau$ — which holds because the cake is exhausted in the limit, $\lim_t w_t = 0$ being forced by log utility and $\beta < 1$ — gives

$$w_t = \sum_{\tau=t}^{\infty} \beta^{\tau-t} c_t = \frac{c_t}{1 - \beta}, \quad \text{so} \quad c_t = (1 - \beta)w_t,$$

which is the stated policy, and then $w_{t+1} = w_t - c_t = \beta w_t$ and $c_t = (1 - \beta)\beta^t w_0$.

Value. Substituting the optimal path into the objective,

$$\begin{aligned} V(w_0) &= \sum_{t=0}^{\infty} \beta^t \log((1 - \beta)\beta^t w_0) = \log(1 - \beta) \sum_{t \geq 0} \beta^t + \log \beta \sum_{t \geq 0} t \beta^t + \log w_0 \sum_{t \geq 0} \beta^t \\ &= \frac{\log(1 - \beta)}{1 - \beta} + \frac{\beta \log \beta}{(1 - \beta)^2} + \frac{\log w_0}{1 - \beta}, \end{aligned}$$

using $\sum_{t \geq 0} \beta^t = 1/(1 - \beta)$ and

$$\sum_{t \geq 0} t \beta^t = \beta \sum_{t \geq 0} t \beta^{t-1} = \beta \frac{d}{d\beta} \sum_{t \geq 0} \beta^t = \beta \frac{d}{d\beta} \frac{1}{1 - \beta} = \frac{\beta}{(1 - \beta)^2},$$

the term-by-term differentiation being legitimate on $|\beta| < 1$, where the power series converges uniformly on compact subsets.

Verification. Write $V(w) = A + B \log w$ with $B = 1/(1 - \beta)$ and A as in (15.12). The objective in (15.11) is strictly concave in w' on $(0, w)$ and tends to $-\infty$ at both endpoints, so its unique maximiser is interior and solves $1/(w - w') = \beta B/w'$, that is $w' = \beta B w / (1 + \beta B) = \beta w$ since $\beta B = \beta/(1 - \beta)$ and $1 + \beta B = 1/(1 - \beta)$. Substituting back,

$$\log((1 - \beta)w) + \beta(A + B \log(\beta w)) = \underbrace{\log(1 - \beta) + \beta A + \beta B \log \beta}_{=A} + \underbrace{(1 + \beta B) \log w}_{=B}$$

the first bracket equalling A because $A(1 - \beta) = \log(1 - \beta) + \beta \log \beta / (1 - \beta)$ by (15.12). Hence $TV = V$. Optimality of the associated plan follows from (15.4): along it $w_m = \beta^m w_0$, so $\beta^m V(w_m) = \beta^m (A + B \log w_0) + m B \beta^m \log \beta \rightarrow 0$, and the transversality condition of Remark 15.2.1 holds even though Assumption 15.1.1(iii) fails. ■

The eaten fraction $1 - \beta$ is independent of the cake size — a consequence of the homotheticity of log utility, under which the problem scales — and independent of time, which is stationarity. The value function inherits the logarithm from the return function, which is why the guess $A + B \log w$ works and why it would fail for a return function outside the CRRA class.

Remark 15.5.1 (What the numerical solution adds). Solving (15.11) by value function iteration on a grid of 2000 points from 0.001 to $w_0 = 1$ with $\beta = 0.85$, starting from $W_0 \equiv 0$, reproduces (15.12) to grid accuracy (Fukuda 2026b, slides 23–30); this is `DP_cake_eating.m`, and `DP_cake_eating_CRRA.m` repeats it for CRRA utility while `DP_cake_eating_plot_iterations.m` displays the successive iterates W_m converging to V . The lower endpoint is not zero because $\log 0 = -\infty$; the truncation is

a discretisation of the state space, not an approximation of the economics, and the computed policy inherits an error of the order of the grid spacing near the boundary. The exercise is worth doing on a problem with a known closed form precisely because the discrepancy between (15.12) and the computed value is then pure numerical error and can be attributed.

15.6 The Ramsey Growth Model

Problem. A planner chooses consumption to maximise $\sum_{t \geq 0} \beta^t u(c_t)$ subject to

$$k_{t+1} = f(k_t) + (1 - \delta)k_t - c_t, \quad f(k) = Ak^\alpha, \quad \alpha \in (0, 1), \quad A > 0, \quad \delta \in (0, 1], \quad (15.13)$$

with $k_0 > 0$ given, $c_t \geq 0$ and $k_t \geq 0$; u is strictly increasing, strictly concave and continuously differentiable. In recursive form,

$$V(k) = \max_{k' \in [0, f(k) + (1 - \delta)k]} \{u(f(k) + (1 - \delta)k - k') + \beta V(k')\}. \quad (15.14)$$

Proposition 15.6.1 (Euler equation and steady state). Along an interior optimal path of (15.14),

$$u'(c_t) = \beta u'(c_{t+1})(f'(k_{t+1}) + 1 - \delta), \quad (15.15)$$

and the unique interior steady state (k^*, c^*) with $c_t \equiv c^*$, $k_t \equiv k^*$ is

$$k^* = \left(\frac{\alpha A}{\beta^{-1} - (1 - \delta)} \right)^{\frac{1}{1 - \alpha}}, \quad c^* = A(k^*)^\alpha - \delta k^*. \quad (15.16)$$

Proof. Apply Theorem 15.4.2 with state k , control k' , $r(k, k') = u(f(k) + (1 - \delta)k - k')$ and $g(k, k') = k'$. Then $r_{k'} = -u'(c)$ and $g_{k'} = 1$, so (15.8) gives

$$u'(c_t) = \beta V'(k_{t+1}); \quad (15.17)$$

and $r_k = u'(c)(f'(k) + 1 - \delta)$ with $g_k = 0$, so (15.9) gives

$$V'(k_t) = u'(c_t)(f'(k_t) + 1 - \delta). \quad (15.18)$$

Substituting (15.18) at $t + 1$ into (15.17) yields (15.15). Concavity of V , required by Theorem 15.4.2, follows from concavity of u and of the constraint set: Proposition 13.5.1 applies to each iterate $T^m W_0$ started from a concave W_0 , and Theorem 15.3.1 makes the uniform limit V concave, uniform limits preserving the inequality in Definition 9.3.1.

At a steady state, $c_t = c_{t+1}$ cancels u' from (15.15), leaving $1 = \beta(f'(k^*) + 1 - \delta)$, that is $\alpha A(k^*)^{\alpha-1} = \beta^{-1} - (1 - \delta)$, which rearranges to the first half of (15.16); the exponent $1/(1 - \alpha)$ is positive and the right-hand side is positive because $\beta^{-1} > 1 \geq 1 - \delta$, so k^* exists and is unique. Setting $k_{t+1} = k_t = k^*$ in (15.13) gives $c^* = f(k^*) - \delta k^*$. ■

Equation (15.15) is the intertemporal condition of the one-sector growth model: the marginal utility cost of investing a unit today equals its discounted marginal utility return, the return being the marginal product of capital plus the undepreciated remainder. The steady-state condition $\beta(f'(k^*) + 1 - \delta) = 1$ is the modified golden rule: the net marginal product $f'(k^*) - \delta$ equals the rate of time preference $\beta^{-1} - 1$. Capital is therefore below the golden-rule level that maximises steady-state consumption, at which $f'(k) = \delta$, and the wedge is impatience. Comparative statics follow directly: k^* is increasing in β , in A and in α , and decreasing in δ .

Unlike Section 15.5 the model has no closed form for general u and $\delta < 1$; the exception is $\delta = 1$ with $u = \log$, where the policy is $k' = \alpha\beta(Ak^\alpha)$. This is why the numerical treatment of (Fukuda 2026b, slides 37–41) restricts the grid to $[k_{\min}, k_{\max}]$ containing k^* : on that compact set the return function is bounded, Assumption 15.1.1 holds, and Theorem 15.3.1 applies as stated. The program `deterministic_Ramsey_growth.m` computes k^* from (15.16) first and then lays 1000 nodes over $[0.25k^*, 1.75k^*]$, so the compact set is chosen by the theory rather than guessed; infeasible pairs (k, k') receive a large penalty in place of an undefined utility, which is the discrete analogue of the constraint $c_t \geq 0$.

15.7 The Lucas Asset-Pricing Model

Environment. A representative agent in an endowment economy consumes a single non-storable good produced by a single tree. The endowment $(y_t)_{t \geq 0}$ is Markov, $y_{t+1} = G(y_t, \varepsilon_{t+1})$ with (ε_t) i.i.d. with distribution φ ; in the numerical implementation $\log y_{t+1} = \alpha \log y_t + \sigma \varepsilon_{t+1}$ with ε_t standard normal. The agent ranks streams by $\mathbb{E}_0 [\sum_{t \geq 0} \beta^t u(c_t)]$ with u of the CRRA form. A share $\pi_t \in [0, 1]$ of the tree, chosen at $t - 1$, is the agent's only store of value, and an ex-dividend claim bought at price p_t entitles the holder to y_{t+1} and to the resale price p_{t+1} .

The agent's problem. With the share as state alongside the endowment,

$$v(\pi, y) = \max_{\pi'} \left\{ u((y + p(y))\pi - p(y)\pi') + \beta \int v(\pi', G(y, z)) \varphi(dz) \right\}, \quad (15.19)$$

the budget constraint $c + p(y)\pi' \leq (y + p(y))\pi$ binding because u is strictly increasing. The agent takes the pricing function $p(\cdot)$ as given; the guess that the price depends on the history only through the current endowment is verified by the construction below, and is legitimate because the endowment is Markov.

Theorem 15.7.1 (Consumption-based asset pricing). In a competitive equilibrium of the Lucas economy the pricing function satisfies

$$p(y) = \beta \int \frac{u'(G(y, z))}{u'(y)} (G(y, z) + p(G(y, z))) \varphi(dz), \quad (15.20)$$

equivalently, in sequence form,

$$p_t = \mathbb{E}_t \left[\underbrace{\beta \frac{u'(y_{t+1})}{u'(y_t)}}_{\text{stochastic discount factor}} (y_{t+1} + p_{t+1}) \right]. \quad (15.21)$$

Proof. The first-order condition of (15.19) in π' is

$$u'(c) p(y) = \beta \int v_1(\pi', G(y, z)) \varphi(dz), \quad v_1 := \frac{\partial v}{\partial \pi}, \quad (15.22)$$

the cost in current marginal utility of an extra share equalling its discounted expected marginal value. Differentiating (15.19) in the state π with π' held at its optimum — which is legitimate by Theorem 15.4.1, v being concave in π as the value of a concave problem on a convex constraint set — gives the envelope formula

$$v_1(\pi, y) = u'(c)(y + p(y)), \quad (15.23)$$

an extra share being worth the marginal utility of the dividend plus the resale value it delivers.

Two market-clearing conditions pin down the equilibrium. The good is non-storable, so $c_t = y_t$; and with a representative agent there is no trade, so $\pi_t = 1$ for every t . Imposing $c = y$ and $\pi = \pi' = 1$ and substituting (15.23) into (15.22),

$$u'(y)p(y) = \beta \int u'(G(y, z)) (G(y, z) + p(G(y, z))) \varphi(dz),$$

which is (15.20) after division by $u'(y) > 0$. Rewriting the integral as a conditional expectation gives (15.21). ■

Equation (15.21) is the central identity of asset pricing: the price of any claim is the conditional expectation of its payoff weighted by the *stochastic discount factor* $m_{t+1} = \beta u'(y_{t+1})/u'(y_t)$. Prices are high for claims that pay off when consumption is low, because marginal utility is then high; that is the entire economics, and the equity premium is the statement that the covariance between m_{t+1} and equity payoffs is empirically far too small to match observed returns for plausible γ .

Theorem 15.7.2 (Existence of the pricing function). Suppose the map $y \mapsto \beta \int u'(G(y, z))G(y, z) \varphi(dz)$ defines a bounded continuous function h on $\mathbb{R}_+$. Then the operator

$$(\mathcal{T}f)(y) := h(y) + \beta \int f(G(y, z))\varphi(dz) \quad (15.24)$$

is a contraction of modulus β on $\mathcal{B}(\mathbb{R}_+, \mathbb{R})$, so it has a unique fixed point f^* , reached from any starting point, and the equilibrium price is recovered as

$$p^*(y) = \frac{f^*(y)}{u'(y)}. \quad (15.25)$$

Proof. Setting $f(y) := u'(y)p(y)$ turns (15.20) into $f = \mathcal{T}f$, because the integrand $u'(G(y, z))(G(y, z) + p(G(y, z)))$ splits as $u'(G(y, z))G(y, z)$, whose integral is $h(y)/\beta$, plus $f(G(y, z))$. This is Lucas' change of unknown, and its point is that the resulting operator is *affine* in f , whereas (15.20) in the unknown p is not.

For the contraction property, take $f_1, f_2 \in \mathcal{B}(\mathbb{R}_+, \mathbb{R})$. For each y ,

$$\begin{aligned} |(\mathcal{T}f_1)(y) - (\mathcal{T}f_2)(y)| &= \beta \left| \int [f_1(G(y, z)) - f_2(G(y, z))] \varphi(dz) \right| \\ &\leq \beta \int |f_1 - f_2|(G(y, z)) \varphi(dz) \leq \beta \|f_1 - f_2\|_\infty, \end{aligned}$$

the first inequality by the triangle inequality for integrals and the second because φ is a probability measure. Taking the supremum over y gives $\|\mathcal{T}f_1 - \mathcal{T}f_2\|_\infty \leq \beta \|f_1 - f_2\|_\infty$. Theorem 6.3.2 and Theorem 6.4.1 give the rest, and (15.25) inverts the substitution. ■

Example 15.7.1 (Logarithmic utility). Let $u = \log$, so $u'(c) = 1/c$. Then

$$h(y) = \beta \int \frac{1}{G(y, z)} G(y, z) \varphi(dz) = \beta \int \varphi(dz) = \beta,$$

a constant independent of y and of the law of motion G . The functional equation becomes $f = \beta + \beta f$ for a constant f , whose solution is $f^* = \beta/(1 - \beta)$, and uniqueness in Theorem 15.7.2 rules out any other bounded solution. Hence

$$p^*(y) = \frac{f^*(y)}{u'(y)} = \frac{\beta}{1 - \beta} y. \quad (15.26)$$

The price-dividend ratio is the constant $\beta/(1 - \beta)$: with logarithmic utility the income and substitution effects of a change in expected future dividends cancel exactly, so no feature of the endowment process — persistence α , volatility σ , even the shape of φ — affects the ratio. This is the cleanest statement of why the log case is a knife-edge, and why matching observed variation in price-dividend ratios requires $\gamma \neq 1$.

Remark 15.7.1 (Numerical implementation). Iterating (15.24) requires evaluating f at $G(y, z) = y^\alpha r$, $r = e^{\sigma \varepsilon}$ log-normal, which is not a grid point; `Lucas_Tree_DP.m` and `Lucas_Tree_DP_Octave.m` interpolate with a spline (Fukuda 2026b, slide 61). Two approximations therefore compound with the contraction error (15.5): the integral is replaced by an average over 500 draws, contributing a Monte Carlo error of order $500^{-1/2}$ by Remark 14.7.1, and f off the grid is replaced by a spline. Neither is controlled by (15.5), which bounds only the distance between the exact iterates and the exact fixed point. The grid is centred using the stationary moments of the endowment: from $\log y_{t+1} = \alpha \log y_t + \sigma \varepsilon_{t+1}$ with $|\alpha| < 1$, $\mathbb{E}[\log y_t] = 0$ and $\text{Var}(\log y_t) = \sigma^2/(1 - \alpha^2)$ in the stationary distribution, and the grid spans four standard deviations either side. Checking the computed p^* against (15.26) in the log case is the available validation, and it is the reason that case is run first.

15.8 Chapter Summary

A stationary discounted sequential problem reduces to a functional equation in the state alone. Bellman's principle, made precise in Theorem 15.2.1, has two halves: the value function solves the Bellman

equation, and — given boundedness, which supplies the transversality condition — any bounded solution of the Bellman equation is the value function and any plan attaining the maximum period by period is optimal. The Bellman operator satisfies Blackwell’s monotonicity and discounting conditions, hence is a contraction of modulus β on the Banach space of bounded continuous functions; [Theorem 6.4.1](#) then delivers existence, uniqueness, convergence of value function iteration from any starting point, and the two error bounds that make the algorithm usable. Berge’s theorem supplies continuity of the value function and upper hemicontinuity of the policy correspondence at each iterate.

Characterisation requires differentiability of the value function, which the Benveniste–Scheinkman theorem obtains without assuming it: a concave function sandwiched above a differentiable concave function that touches it at an interior point is differentiable there, with the same derivative. Applied with the control fixed at its optimum, this yields the envelope formula, and combining the envelope formula with the first-order condition eliminates the value function and produces the Euler equation in primitives alone.

The three applications instantiate the same machinery. In the cake-eating problem the Euler equation $c_{t+1} = \beta c_t$ integrates to the policy $h(w) = (1 - \beta)w$ and the value function $A + \log w/(1 - \beta)$, verified directly because unbounded log utility falls outside the standing hypotheses. In the Ramsey model the Euler equation $u'(c_t) = \beta u'(c_{t+1})(f'(k_{t+1}) + 1 - \delta)$ has the modified golden rule $\beta(f'(k^*) + 1 - \delta) = 1$ as its steady state, placing capital below the golden-rule level by exactly the rate of time preference. In the Lucas economy the first-order and envelope conditions, combined with the market-clearing conditions $c = y$ and $\pi = 1$, give the consumption-based pricing equation and the stochastic discount factor $\beta u'(y_{t+1})/u'(y_t)$; substituting $f = u'p$ linearises the functional equation into a contraction, and under log utility the price-dividend ratio is the constant $\beta/(1 - \beta)$ regardless of the endowment process.

Exercise 15.8.1. Complete the proof of [Theorem 15.2.1\(i\)](#) by justifying the interchange of suprema in detail: show that for a bounded objective, $\sup_{u, (u_t)_{t \geq 1}} \{\Phi(u, (u_t))\} = \sup_u \{\sup_{(u_t)_{t \geq 1}} \Phi(u, (u_t))\}$, and exhibit an unbounded example in which the two sides differ. Relate the example to [Remark 15.2.1](#).

Exercise 15.8.2. Solve the Ramsey model in closed form for $u = \log$ and $\delta = 1$ by guessing $V(k) = A + B \log k$. Show $B = \alpha/(1 - \alpha\beta)$ and that the policy is $k' = \alpha\beta A k^\alpha$, so the saving rate is the constant $\alpha\beta$. Verify that the steady state of this policy agrees with [\(15.16\)](#) at $\delta = 1$, and explain why the constant saving rate is a knife-edge property of the pairing of log utility with full depreciation.

Exercise 15.8.3. In the Lucas economy with $u(c) = c^{1-\gamma}/(1-\gamma)$ and $\log y_{t+1} = \alpha \log y_t + \sigma \varepsilon_{t+1}$, use [Proposition 10.7.1](#) to compute $h(y)$ in closed form, and show that the price-dividend ratio depends on y unless $\gamma = 1$ or $\alpha = 0$. Determine the sign of $\partial(p^*(y)/y)/\partial y$ as a function of γ and interpret it in terms of the relative strength of income and substitution effects.

Chapter 16

Continuous-Time Dynamic Optimisation

16.1 From Sequences to Paths

The chapter preceding treated the choice of a sequence; this one treats the choice of a function of time. The change is not merely notational. A sequence is a point in a space of countably many coordinates and can be perturbed one coordinate at a time; a path must be perturbed by another path, and the first-order condition is therefore an equation that must hold for *every* admissible perturbation. Converting such a family of scalar conditions into a pointwise differential equation is the office of the following lemma, and every result in this chapter rests on it.

Lemma 16.1.1 (Fundamental lemma of the calculus of variations). Let $t_0 < t_1$ and let $g \in \mathcal{C}([t_0, t_1], \mathbb{R})$ satisfy

$$\int_{t_0}^{t_1} g(t)h(t) dt = 0 \quad \text{for every } h \in \mathcal{C}([t_0, t_1], \mathbb{R}) \text{ with } h(t_0) = h(t_1) = 0. \quad (16.1)$$

Then $g \equiv 0$ on $[t_0, t_1]$.

Proof. Suppose $g(s) \neq 0$ for some $s \in (t_0, t_1)$; without loss of generality $g(s) > 0$, otherwise replace g by $-g$. By continuity there are α, β with $t_0 < \alpha < s < \beta < t_1$ and $g(t) > \frac{1}{2}g(s)$ for every $t \in [\alpha, \beta]$. Define

$$h(t) := \begin{cases} (t - \alpha)(\beta - t), & t \in [\alpha, \beta], \\ 0, & \text{otherwise.} \end{cases}$$

This h is continuous on $[t_0, t_1]$ — it vanishes at α and β , where the two branches meet — non-negative, and satisfies $h(t_0) = h(t_1) = 0$. Then

$$\int_{t_0}^{t_1} g(t)h(t) dt = \int_{\alpha}^{\beta} g(t)h(t) dt > \frac{g(s)}{2} \int_{\alpha}^{\beta} (t - \alpha)(\beta - t) dt = \frac{g(s)}{2} \cdot \frac{(\beta - \alpha)^3}{6} > 0,$$

using [Proposition 10.2.3](#) for the strict monotonicity of the integral of a continuous function that is strictly positive on $[\alpha, \beta]$. This contradicts (16.1). Hence g vanishes on the open interval, and by continuity at the endpoints as well. ■

This is Proposition 7.30 of Fukuda (2026d, p. 288). Read in the language of [Chapter 11](#), it says that the only continuous function orthogonal, in the L^2 inner product $\langle g, h \rangle = \int gh$, to every continuous function vanishing at the endpoints is the zero function: the set of such h is dense in $L^2[t_0, t_1]$, so its orthogonal complement is trivial by [Proposition 11.2.1](#). The elementary proof above avoids the density argument and is the one used here because it also delivers the conclusion for the piecewise-defined perturbations that appear in constrained problems.

16.2 The Calculus of Variations

Definition 16.2.1 (Variational problem). Given $f: [t_0, t_1] \times \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$ continuously differentiable and endpoints $x_0, x_1 \in \mathbb{R}$, the *calculus of variations* is

$$\max_{x(\cdot)} \int_{t_0}^{t_1} f(t, x(t), \dot{x}(t)) dt \quad \text{subject to} \quad x(t_0) = x_0, x(t_1) = x_1, \quad (16.2)$$

the maximum being over continuously differentiable paths $x: [t_0, t_1] \rightarrow \mathbb{R}$.

Theorem 16.2.1 (Euler–Lagrange equation). Let x^* solve (16.2) and suppose $t \mapsto f_{\dot{x}}(t, x^*(t), \dot{x}^*(t))$ is continuously differentiable. Then

$$f_x(t, x^*(t), \dot{x}^*(t)) - \frac{d}{dt} [f_{\dot{x}}(t, x^*(t), \dot{x}^*(t))] = 0 \quad \text{for every } t \in [t_0, t_1]. \quad (16.3)$$

Proof. Fix a continuously differentiable h with $h(t_0) = h(t_1) = 0$, so that $x^* + ah$ is admissible for every $a \in \mathbb{R}$, and set

$$J(a) := \int_{t_0}^{t_1} f(t, x^*(t) + ah(t), \dot{x}^*(t) + a\dot{h}(t)) dt.$$

Optimality of x^* gives $J(0) \geq J(a)$ for all a , so $a = 0$ is an interior maximiser of J and Theorem 7.2.1 gives $J'(0) = 0$, provided J is differentiable. It is, by Lemma 10.5.1: the integrand is continuously differentiable in a and its a -derivative is continuous in (t, a) , hence bounded on the compact set $[t_0, t_1] \times [-1, 1]$. Differentiating under the integral sign and using Theorem 8.3.1,

$$J'(a) = \int_{t_0}^{t_1} \left\{ f_x(t, x, \dot{x})h(t) + f_{\dot{x}}(t, x, \dot{x})\dot{h}(t) \right\} dt, \quad x = x^* + ah.$$

Integrating the second term by parts (Theorem 10.3.2),

$$\int_{t_0}^{t_1} f_{\dot{x}}\dot{h} dt = f_{\dot{x}}h|_{t_0}^{t_1} - \int_{t_0}^{t_1} \frac{df_{\dot{x}}}{dt} h dt = - \int_{t_0}^{t_1} \frac{df_{\dot{x}}}{dt} h dt,$$

the boundary term vanishing because $h(t_0) = h(t_1) = 0$. Hence

$$0 = J'(0) = \int_{t_0}^{t_1} \left[f_x(t, x^*, \dot{x}^*) - \frac{d}{dt} f_{\dot{x}}(t, x^*, \dot{x}^*) \right] h(t) dt$$

for every admissible h , and Lemma 16.1.1 gives (16.3). ■

The derivative $\frac{d}{dt} f_{\dot{x}}$ in (16.3) is a total derivative along the path, not a partial derivative: expanded by the chain rule it is $f_{\dot{x}t} + f_{\dot{x}x}\dot{x}^* + f_{\dot{x}\dot{x}}\dot{x}^*$, so the Euler–Lagrange equation is a second-order ordinary differential equation in x^* , and the two endpoint conditions of (16.2) are exactly what a second-order equation requires. That count is the structural reason a variational problem with one free endpoint needs a substitute boundary condition, and the substitute is the transversality condition of Section 16.3.

Remark 16.2.1 (Necessary, not sufficient). Theorem 16.2.1 is a first-order necessary condition and nothing more: it is Theorem 7.2.1 applied along a one-parameter family, so it identifies stationary paths, among which are minima and saddles. Sufficiency requires concavity, exactly as in Theorem 7.2.3. If $(x, \dot{x}) \mapsto f(t, x, \dot{x})$ is concave for each t , then a path satisfying (16.3) is a global maximiser, because for any admissible x ,

$$\int_{t_0}^{t_1} [f(t, x, \dot{x}) - f(t, x^*, \dot{x}^*)] dt \leq \int_{t_0}^{t_1} [f_x \cdot (x - x^*) + f_{\dot{x}} \cdot (\dot{x} - \dot{x}^*)] dt = 0,$$

the inequality by Theorem 9.3.1 and the final equality by the same integration by parts as in the proof, using (16.3) and $x(t_i) = x^*(t_i)$. The concavity hypothesis is satisfied in the economic problems below because U is concave and the constraint is linear in the control.

16.3 Optimal Control

Optimal control generalises [Definition 16.2.1](#) by separating the control from the rate of change of the state, which permits constraints on the control and transitions that are not of the form $\dot{x} = u$.

Definition 16.3.1 (Optimal control problem). Given f and g continuously differentiable,

$$\max_{u(\cdot)} \int_{t_0}^{t_1} f(t, x(t), u(t)) dt \quad \text{subject to} \quad \dot{x}(t) = g(t, x(t), u(t)), \quad x(t_0) = x_0. \quad (16.4)$$

The function x is the *state*, u the *control*, and the *Hamiltonian* is

$$\mathcal{H}(t, x, u, \lambda) := f(t, x, u) + \lambda(t) g(t, x, u), \quad (16.5)$$

with λ the *co-state variable*.

Theorem 16.3.1 (Maximum principle, interior case). Let (u^*, x^*) solve (16.4) with $t_1 < \infty$ and $x(t_1)$ free, and suppose u^* is interior to the set of admissible controls. Then there is a continuously differentiable $\lambda: [t_0, t_1] \rightarrow \mathbb{R}$ such that, for every t ,

$$\frac{\partial \mathcal{H}}{\partial u} = 0 \iff f_u + \lambda g_u = 0, \quad (16.6)$$

$$-\frac{\partial \mathcal{H}}{\partial x} = \dot{\lambda} \iff \dot{\lambda}(t) = -(f_x + \lambda g_x), \quad (16.7)$$

$$\frac{\partial \mathcal{H}}{\partial \lambda} = \dot{x} \iff \dot{x} = g, \quad (16.8)$$

together with the initial condition $x(t_0) = x_0$ and the *transversality condition* $\lambda(t_1) = 0$.

Proof. Perturb the control, $u = u^* + ah$ for a fixed continuously differentiable h , and let $y(t, a)$ be the state path it induces through (16.8), so $y(t, 0) = x^*(t)$ and $y(t_0, a) = x_0$ for every a . Let λ be any continuously differentiable function, to be chosen. Since the constraint holds identically along every admissible path, the objective is unchanged by adding $\lambda(g - \dot{x}) \equiv 0$ to the integrand, and integration by parts ([Theorem 10.3.2](#)) gives

$$\begin{aligned} J(a) &:= \int_{t_0}^{t_1} \left\{ f(t, y, u^* + ah) + \lambda g(t, y, u^* + ah) - \lambda \dot{y} \right\} dt \\ &= \int_{t_0}^{t_1} \left\{ f + \lambda g + \dot{\lambda} y \right\} dt - \lambda(t_1) y(t_1, a) + \lambda(t_0) \underbrace{y(t_0, a)}_{=x_0}. \end{aligned}$$

Differentiating at $a = 0$, legitimate by [Lemma 10.5.1](#), and writing $y_a := \partial y / \partial a$ evaluated at $a = 0$,

$$J'(0) = \int_{t_0}^{t_1} \left\{ (f_u + \lambda g_u) h + (f_x + \lambda g_x + \dot{\lambda}) y_a \right\} dt - \lambda(t_1) y_a(t_1), \quad (16.9)$$

the term in $\lambda(t_0)$ contributing nothing because $y(t_0, a) = x_0$ for all a .

The function λ is still free. Choose it to solve the linear differential equation (16.7) with terminal condition $\lambda(t_1) = 0$; a solution exists and is unique on $[t_0, t_1]$ because the equation is linear in λ with continuous coefficients, and the integrating-factor representation $\lambda(t) = \int_t^{t_1} f_x(s) \exp(\int_t^s g_x(\tau) d\tau) ds$ exhibits it. This choice annihilates the second and third terms of (16.9), and $J'(0) = 0$ by optimality, leaving

$$\int_{t_0}^{t_1} (f_u + \lambda g_u) h(t) dt = 0 \quad \text{for every admissible } h.$$

[Lemma 16.1.1](#) then gives (16.6) pointwise. Equation (16.8) is the constraint, restated as a partial derivative of $\mathcal{H}$. ■

Remark 16.3.1 (What the three conditions say). The co-state $\lambda(t)$ is the shadow price of the state at time t , the continuous-time counterpart of the multiplier of [Theorem 12.3.1](#) and of the derivative $V'(x)$ of [Theorem 15.4.2](#). The optimality condition (16.6) equates the marginal current return of the control to its marginal cost in terms of the state it consumes, valued at λ . The co-state equation (16.7) is an asset-pricing condition: rearranged as $-\dot{\lambda} = f_x + \lambda g_x$, it says the capital loss on a unit of the state equals its current dividend plus its contribution to future accumulation. The transversality condition $\lambda(t_1) = 0$ says a unit of the state has no value at the terminal date when the terminal state is free: anything left over is wasted, so nothing should be left over.

When $t_1 = \infty$ the terminal condition must be replaced. The standard form is $\lim_{t \rightarrow \infty} \lambda(t)x(t) = 0$ in present-value terms, or $\lim_{t \rightarrow \infty} e^{-\rho t} \mu(t)x(t) = 0$ in current-value terms, and [Section 16.5](#) shows what goes wrong without it: the first-order conditions leave a free constant that only the transversality condition pins down.

Proposition 16.3.1 (Current-value Hamiltonian). Consider the discounted problem

$$\max_{u(\cdot)} \int_{t_0}^{t_1} e^{-\rho t} f(t, x(t), u(t)) dt \quad \text{subject to} \quad \dot{x} = g(t, x, u), \quad x(t_0) = x_0, \quad (16.10)$$

and define the *current-value Hamiltonian* $H := e^{\rho t} \mathcal{H} = f(t, x, u) + \mu(t)g(t, x, u)$ with $\mu(t) := e^{\rho t} \lambda(t)$. Then (16.6)–(16.8) become

$$\frac{\partial H}{\partial u} = 0, \quad \dot{\mu}(t) = \rho \mu(t) - \frac{\partial H}{\partial x}, \quad \frac{\partial H}{\partial \mu} = \dot{x}. \quad (16.11)$$

Proof. The optimality condition is unaffected by multiplying $\mathcal{H}$ by the positive factor $e^{\rho t}$, which does not depend on u . For the co-state equation, (16.7) reads $-\dot{\lambda} = \partial \mathcal{H} / \partial x$, and multiplying by $e^{\rho t}$ gives $-e^{\rho t} \dot{\lambda} = \partial H / \partial x$. Since $\mu = e^{\rho t} \lambda$, the product rule ([Proposition 7.3.2](#)) gives

$$\dot{\mu} = \rho e^{\rho t} \lambda + e^{\rho t} \dot{\lambda} = \rho \mu - \frac{\partial H}{\partial x}.$$

The state equation is unchanged since $\partial H / \partial \mu = g$. ■

The current-value formulation removes the explicit $e^{-\rho t}$ from every equation and leaves an autonomous system whenever f and g do not depend on t directly. That is the practical reason it is preferred in economics: the resulting pair of differential equations in (x, μ) has a phase diagram, and the steady state and saddle path of [Section 16.6](#) are features of that diagram.

Proposition 16.3.2 (Calculus of variations as a special case). The problem (16.2) is (16.4) with $u := \dot{x}$ and $g(t, x, u) = u$, and the maximum principle then reproduces the Euler–Lagrange equation (16.3).

Proof. With $g = u$ the Hamiltonian is $\mathcal{H} = f(t, x, u) + \lambda u$, so (16.6) reads $f_u + \lambda = 0$ and (16.7) reads $\dot{\lambda} = -f_x$. Differentiating the first along the path and using $u = \dot{x}$ from (16.8),

$$\dot{\lambda} = -\frac{d}{dt} f_u(t, x, u) = -\frac{d}{dt} f_{\dot{x}}(t, x, \dot{x}),$$

and eliminating $\dot{\lambda}$ between this and $\dot{\lambda} = -f_x$ gives $f_x = \frac{d}{dt} f_{\dot{x}}$. ■

16.4 The Hamilton–Jacobi–Bellman Equation

The third route to the same conditions is dynamic programming, obtained as the continuous-time limit of [Chapter 15](#).

Heuristic derivation. Let the period length be $\delta > 0$ and the per-period discount factor $1/(1 + \delta\rho)$. The discrete Bellman equation for the value $W(t, x)$ of the problem started at date t in state x is

$$W(t, x) = \max_u \left\{ \left(\frac{1}{1 + \delta\rho} \right)^{\delta t} r(x, u) \delta + W(t + \delta, x + g(x, u)\delta) \right\},$$

the state moving by $g(x, u)\delta$ over an interval of length δ . Assuming W differentiable and expanding to first order by [Theorem 8.5.2](#),

$$W(t + \delta, x + g\delta) = W(t, x) + W_t(t, x)\delta + W_x(t, x)g(x, u)\delta + o(\delta).$$

Substituting, cancelling $W(t, x)$, dividing by δ and letting $\delta \downarrow 0$ — using $\lim_{\delta \rightarrow 0}(1 + \delta\rho)^{-\delta t/\delta} = \lim_{\delta \rightarrow 0}(1 + \delta\rho)^{-\rho t/(\delta\rho)} = e^{-\rho t}$ — gives

$$0 = W_t(t, x) + \max_u \{e^{-\rho t}r(x, u) + W_x(t, x)g(x, u)\}.$$

Stationarity makes $W(t, x) = e^{-\rho t}V(x)$ where $V(x) := \max \int_0^\infty e^{-\rho t}r(x_t, u_t) dt$ from $x_0 = x$; then $W_t = -\rho e^{-\rho t}V(x)$ and $W_x = e^{-\rho t}V'(x)$, and dividing through by $e^{-\rho t}$ leaves the following.

Definition 16.4.1 (HJB equation). The *Hamilton–Jacobi–Bellman equation* of the stationary infinite-horizon problem $V(x) = \max \int_0^\infty e^{-\rho t}r(x_t, u_t) dt$ subject to $\dot{x} = g(x, u)$, $x_0 = x$, is

$$\rho V(x) = \max_u \{r(x, u) + V'(x)g(x, u)\}. \quad (16.12)$$

Equation (16.12) is the asset-pricing form of the Bellman equation and was met in (10.6): the required return ρV on the “asset” x equals the current dividend r plus the expected capital gain $V'(x)\dot{x}$. Comparing (16.12) with (16.5) identifies the maximand as the current-value Hamiltonian with $\mu = V'(x)$: the co-state is the derivative of the value function, exactly as λ was the shadow price in [Remark 16.3.1](#) and as V' was in (15.18). The three methods of this chapter — variations, the maximum principle, dynamic programming — are therefore three presentations of one set of conditions, and which is most convenient depends on the problem rather than on the mathematics.

Remark 16.4.1 (The status of the derivation). The passage to the limit above assumes what it should prove: that W is differentiable, that the limit and the maximum interchange, and that the discrete problems converge to the continuous one. None of these is automatic. The value function of a control problem is in general only continuous, and (16.12) then holds in the weaker *viscosity* sense rather than pointwise. Where a classical solution of (16.12) does exist and the associated policy is admissible, a verification argument establishes optimality directly, without any limit: this is the content of [Proposition 16.5.2](#) below, and it is why guess-and-verify is the honest way to use (16.12).

16.5 The Continuous-Time Cake-Eating Problem

Problem. Maximise $\int_0^\infty e^{-rt} \log c(t) dt$ subject to $\dot{k}(t) = -c(t)$, $k(0) = k_0 > 0$, $c(t) > 0$, together with the transversality condition $\lim_{t \rightarrow \infty} k(t) = 0$.

Proposition 16.5.1 (Solution by three routes). The problem has the unique solution

$$k^*(t) = k_0 e^{-rt}, \quad c^*(t) = rk_0 e^{-rt}, \quad V(k) = \frac{\log r - 1 + \log k}{r}, \quad (16.13)$$

and the optimal feedback rule is $c = rk$.

Proof. Variational route. Substituting $c = -\dot{k}$ turns the problem into (16.2) with $f(t, k, \dot{k}) = e^{-rt} \log(-\dot{k})$. Here $f_k = 0$ and $f_{\dot{k}} = -e^{-rt}/(-\dot{k}) \cdot (-1) = e^{-rt}/\dot{k}$, so (16.3) reduces to

$$\frac{d}{dt} \left(\frac{e^{-rt}}{\dot{k}^*(t)} \right) = 0,$$

that is $e^{-rt}/\dot{k}^*(t)$ is constant in t , so $\dot{k}^*(t) = ae^{-rt}$ for some $a \in \mathbb{R}$. Integrating from 0 to t by [Theorem 10.3.1](#),

$$k^*(t) = k_0 + \int_0^t ae^{-r\tau} d\tau = k_0 + \frac{a}{r}(1 - e^{-rt}).$$

Letting $t \rightarrow \infty$ and imposing $\lim_t k^*(t) = 0$ gives $k_0 + a/r = 0$, that is $a = -rk_0$, whence $k^*(t) = k_0 e^{-rt}$ and $c^*(t) = -\dot{k}^*(t) = rk_0 e^{-rt}$.

Hamiltonian route. The current-value Hamiltonian is $H = \log c - \mu c$. Condition (16.11) gives $1/c = \mu$; and $\partial H/\partial k = 0$, so $\dot{\mu} = r\mu$. This separable equation integrates to $\mu(t) = \mu(0)e^{rt}$, hence $c(t) = 1/\mu(t) = ae^{-rt}$ with $a = 1/\mu(0)$. The state equation and the transversality condition then reproduce the previous paragraph with the sign of a reversed, giving $a = rk_0$ and the same paths. With the present-value Hamiltonian $\mathcal{H} = e^{-rt} \log c - \lambda c$ the co-state equation is $-\dot{\lambda} = \partial \mathcal{H}/\partial k = 0$, so λ is constant and $e^{-rt}/c(t) = \lambda$ delivers $c(t) = ae^{-rt}$ directly.

Dynamic programming route. See Proposition 16.5.2. ■

Proposition 16.5.2 (Verification via HJB). The function $V(k) = (\log r - 1 + \log k)/r$ solves (16.12) for the cake problem, the maximiser is $c = rk$, and the resulting path is optimal.

Proof. Here $r(k, c) = \log c$ and $g(k, c) = -c$, so (16.12) reads $rV(k) = \max_{c>0} \{\log c - V'(k)c\}$. Guess $V(k) = A + B \log k$, so $V'(k) = B/k > 0$ for $B > 0$; the objective is then strictly concave in c with $\lim_{c \downarrow 0}$ derivative $+\infty$, so by Theorem 7.2.3 the unique maximiser solves $1/c = B/k$, that is $c = k/B$. Substituting,

$$r(A + B \log k) = \log \frac{k}{B} - \frac{B}{k} \cdot \frac{k}{B} = \log k - \log B - 1.$$

Matching the coefficient of $\log k$ gives $rB = 1$, so $B = 1/r$; matching constants gives $rA = -\log B - 1 = \log r - 1$, so $A = (\log r - 1)/r$. The feedback rule is $c = k/B = rk$, so the state obeys $\dot{k} = -rk$, giving $k^*(t) = k_0 e^{-rt}$ and $c^*(t) = rk_0 e^{-rt}$, which satisfy the transversality condition.

For optimality, let (c, k) be any admissible path. Since V is concave, Theorem 9.3.1 and the HJB equation give, for every t ,

$$\frac{d}{dt} [e^{-rt} V(k(t))] = e^{-rt} [V'(k) \dot{k} - rV(k)] = e^{-rt} [-V'(k)c - rV(k)] \leq -e^{-rt} \log c(t),$$

the inequality because $rV(k) \geq \log c - V'(k)c$ for every feasible c , by the maximum in (16.12). Integrating from 0 to T ,

$$\int_0^T e^{-rt} \log c(t) dt \leq V(k_0) - e^{-rT} V(k(T)),$$

and letting $T \rightarrow \infty$ along any path satisfying the transversality condition kills the last term, giving $\int_0^\infty e^{-rt} \log c dt \leq V(k_0)$, with equality along the candidate path because the inequality above is then an equality at every t . ■

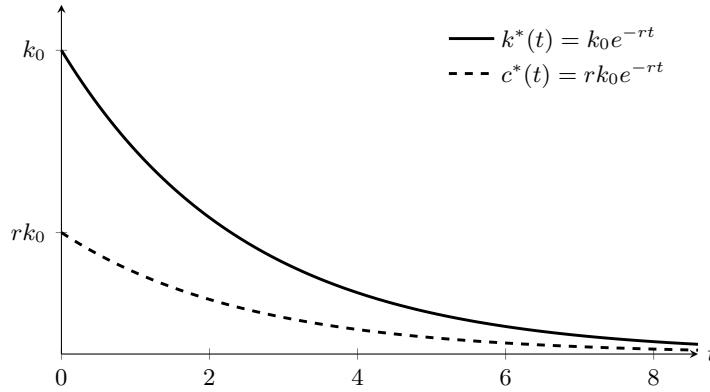


Figure 16.1: The optimal paths (16.13). Both decay at the common rate r , so the ratio $c^*(t)/k^*(t)$ is constant and equal to r : the feedback rule $c = rk$ is what makes the two curves scaled copies of each other. The cake is exhausted only in the limit, which is the content of the transversality condition $\lim_{t \rightarrow \infty} k(t) = 0$.

Remark 16.5.1 (The discrete-time limit). Compare (16.13), drawn in Figure 16.1, with (15.12). In discrete time with period length Δ and discount factor $\beta = e^{-r\Delta}$, the policy is $c = (1 - \beta)w$ per

period, so the consumption *rate* is

$$\frac{(1 - e^{-r\Delta})w}{\Delta} \xrightarrow{\Delta \downarrow 0} rw,$$

by [Theorem 7.4.1](#) applied to $\Delta \mapsto e^{-r\Delta}$, which is exactly the feedback rule $c = rk$. Correspondingly the cake decays at rate $\beta^{t/\Delta} = e^{-rt}$, matching $k^*(t) = k_0 e^{-rt}$. The rate of time preference r in continuous time and the discount factor β in discrete time are related by $\beta = 1/(1 + r\Delta) + o(\Delta)$, which is the substitution made in the derivation of [\(16.12\)](#). The agreement is not automatic — it requires the transversality conditions to correspond, which they do here because both take the form “the state is asymptotically exhausted”.

16.6 The Ramsey Model in Continuous Time

Problem. A planner maximises $\int_0^\infty e^{-\rho t} u(c(t)) dt$ subject to

$$\dot{k}(t) = f(k(t)) - \delta k(t) - c(t), \quad k(0) = k_0 > 0, \quad c(t) \geq 0, \quad k(t) \geq 0, \quad (16.14)$$

with f strictly increasing, strictly concave and satisfying the Inada conditions, u strictly increasing, strictly concave and twice differentiable, $\rho > 0$ and $\delta \in [0, 1]$.

Theorem 16.6.1 (Keynes–Ramsey rule and the modified golden rule). Along an interior optimal path of [\(16.14\)](#),

$$\frac{\dot{c}(t)}{c(t)} = \frac{f'(k(t)) - \delta - \rho}{\gamma(c(t))}, \quad \gamma(c) := -\frac{c u''(c)}{u'(c)}, \quad (16.15)$$

and the unique interior steady state (k^*, c^*) satisfies

$$f'(k^*) = \rho + \delta, \quad c^* = f(k^*) - \delta k^*. \quad (16.16)$$

Proof. The current-value Hamiltonian is $H = u(c) + \mu(f(k) - \delta k - c)$. By [Proposition 16.3.1](#),

$$\frac{\partial H}{\partial c} = u'(c) - \mu = 0, \quad \dot{\mu} = \rho\mu - \frac{\partial H}{\partial k} = \rho\mu - \mu(f'(k) - \delta) = \mu(\rho + \delta - f'(k)). \quad (16.17)$$

Differentiating $u'(c) = \mu$ in t by [Proposition 7.3.4](#) gives $u''(c)\dot{c} = \dot{\mu} = u'(c)(\rho + \delta - f'(k))$, so

$$\frac{c u''(c)}{u'(c)} \cdot \frac{\dot{c}}{c} = \rho + \delta - f'(k), \quad \text{that is} \quad -\gamma(c) \frac{\dot{c}}{c} = \rho + \delta - f'(k),$$

which is [\(16.15\)](#). At a steady state $\dot{c} = 0$, and since $\gamma(c) > 0$ by strict concavity of u , the numerator must vanish: $f'(k^*) = \rho + \delta$. Strict concavity of f makes f' strictly decreasing, and the Inada conditions make it range over $(0, \infty)$, so k^* exists and is unique. Setting $\dot{k} = 0$ in [\(16.14\)](#) gives $c^* = f(k^*) - \delta k^*$. ■

Remark 16.6.1 (Reading the Keynes–Ramsey rule). Equation [\(16.15\)](#) decomposes consumption growth into a rate of return and a willingness to substitute. Consumption rises when the net marginal product of capital $f'(k) - \delta$ exceeds the rate of time preference ρ , and the responsiveness is the elasticity of intertemporal substitution $1/\gamma$. For CRRA utility γ is constant and [\(16.15\)](#) is the continuous-time counterpart of [\(15.15\)](#); at $\rho = \beta^{-1} - 1$ the two steady-state conditions $f'(k^*) = \rho + \delta$ and $\beta(f'(k^*) + 1 - \delta) = 1$ coincide, as they must.

The modified golden rule places k^* strictly below the golden-rule stock k^g defined by $f'(k^g) = \delta$, which maximises steady-state consumption $f(k) - \delta k$. The gap is impatience: a planner with $\rho = 0$ would accumulate to k^g , and the discounted problem sacrifices steady-state consumption for consumption today. That the discounted planner is not dynamically inefficient — unlike a decentralised overlapping-generations economy, which can over-accumulate past k^g — follows from $f'(k^*) = \rho + \delta > \delta$.

Remark 16.6.2 (Saddle-path stability). The pair (16.14) and (16.15) is an autonomous planar system in (k, c) . Linearising about (k^*, c^*) by Theorem 8.5.2, the Jacobian is

$$J = \begin{pmatrix} f'(k^*) - \delta & -1 \\ -\frac{c^* f''(k^*)}{\gamma(c^*)} & 0 \end{pmatrix} = \begin{pmatrix} \rho & -1 \\ -\frac{c^* f''(k^*)}{\gamma(c^*)} & 0 \end{pmatrix},$$

whose determinant is $c^* f''(k^*)/\gamma(c^*) < 0$ because $f'' < 0$. By Definition 2.5.1 the eigenvalues of a 2×2 matrix are the roots of $p_J(\lambda) = \lambda^2 - (\text{tr } J)\lambda + \det J$, so $\lambda_1 \lambda_2 = \det J < 0$ and the discriminant $(\text{tr } J)^2 - 4 \det J > 0$: the eigenvalues are real and of opposite signs. The steady state is therefore a saddle, with exactly one one-dimensional stable manifold through it, drawn in Figure 16.2. Since k_0 is given and $c(0)$ is free, the initial consumption level is determined uniquely as the one placing $(k_0, c(0))$ on that stable manifold. Any other choice violates either feasibility, if consumption is too high and capital hits zero in finite time, or the transversality condition $\lim_{t \rightarrow \infty} e^{-\rho t} \mu(t) k(t) = 0$, if consumption is too low and capital grows without bound. The transversality condition therefore plays in continuous time exactly the selection role that it played in Remark 15.2.1.

The count of eigenvalues against the count of jump variables is the continuous-time form of the Blanchard–Kahn condition of Remark 2.6.1, with “outside the unit circle” replaced by “positive real part”: here one predetermined variable k and one jump variable c face one negative and one positive eigenvalue, so the equilibrium path exists and is unique.

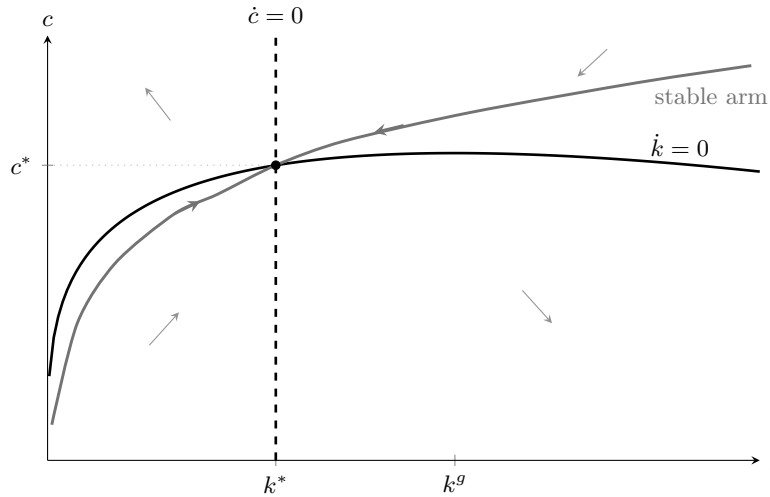


Figure 16.2: The two loci of Theorem 16.6.1 and the stable manifold of Remark 16.6.2. Consumption rises to the left of k^* and falls to its right; capital rises below the hump and falls above it. The grey arrows give the direction of motion in each of the four regions. The modified golden rule places k^* strictly below the golden-rule stock k^g at which $f'(k) = \delta$, the wedge being the rate of time preference ρ .

16.7 Chapter Summary

Continuous-time optimisation replaces the choice of a sequence by the choice of a path, and the first-order condition accordingly holds against every admissible perturbation. The fundamental lemma converts such a family of integral conditions into a pointwise differential equation, and it is the single analytic tool behind all three methods presented.

The calculus of variations perturbs the path directly and yields the Euler–Lagrange equation $f_x = \frac{d}{dt} f_{\dot{x}}$, a second-order differential equation matched to two endpoint conditions. It is necessary, not sufficient; concavity of f in $(x, \dot{x})$ upgrades it to sufficient. Optimal control separates the control from the state and perturbs the control, producing the Hamiltonian $\mathcal{H} = f + \lambda g$ and three conditions — optimality in the control, a differential equation for the co-state, and the state equation — together with $\lambda(t_1) = 0$ when the terminal state is free. The co-state is the shadow price of the state, and the co-state equation is a no-arbitrage condition on that price. The current-value Hamiltonian removes the

discount factor from the equations and leaves an autonomous system whose phase diagram carries the dynamics. The calculus of variations is the case $g = u$. Dynamic programming gives the Hamilton–Jacobi–Bellman equation $\rho V(x) = \max_u \{r(x, u) + V'(x)g(x, u)\}$, in which the co-state is identified as $V'(x)$; the heuristic passage from discrete time assumes differentiability of the value function, and the rigorous route is verification against a candidate.

The continuous-time cake-eating problem yields $c = rk$, $k(t) = k_0 e^{-rt}$ by all three methods, and matches the discrete-time policy $c = (1 - \beta)w$ in the limit of short periods. In each derivation the first-order conditions leave one free constant and only the transversality condition fixes it. The continuous-time Ramsey model yields the Keynes–Ramsey rule $\dot{c}/c = (f'(k) - \delta - \rho)/\gamma$ and the modified golden rule $f'(k^*) = \rho + \delta$, which places capital strictly below the golden-rule level by the amount of impatience; the steady state is a saddle, and initial consumption is the unique level placing the economy on the stable manifold.

Exercise 16.7.1. Prove [Lemma 16.1.1](#) under the weaker hypothesis that [\(16.1\)](#) holds only for infinitely differentiable h supported in (t_0, t_1) , by replacing the quadratic bump with $h(t) = \exp(-1/((t - \alpha)(\beta - t)))$ on $[\alpha, \beta]$ and 0 elsewhere, and verifying that this h is smooth. Explain why the strengthening matters for problems in which the admissible perturbations are constrained.

Exercise 16.7.2. Solve $\max \int_0^T e^{-\rho t} u(c) dt$ subject to $\dot{a} = ra - c$, $a(0) = a_0$ and $a(T) = 0$, with $u(c) = c^{1-\gamma}/(1-\gamma)$. Show that $\dot{c}/c = (r - \rho)/\gamma$, solve for $c(0)$ from the terminal condition, and verify that the solution converges as $T \rightarrow \infty$ to the infinite-horizon solution under the transversality condition $\lim_{t \rightarrow \infty} e^{-\rho t} u'(c(t))a(t) = 0$. Identify precisely where the terminal condition $a(T) = 0$ replaces $\lambda(T) = 0$ of [Theorem 16.3.1](#) and why both cannot be imposed.

Exercise 16.7.3. Consider the Hotelling problem $\max \int_0^\infty e^{-\rho t} p(t)q(t) dt$ subject to $\dot{S} = -q$, $S(0) = S_0$, where $p(t)$ is a given price path and $q \geq 0$ is extraction. Show that the maximum principle gives $\dot{\mu} = \rho\mu$, so the shadow price of the resource grows at the rate of interest, and deduce Hotelling’s rule that the price of the extracted resource net of extraction cost must grow at rate ρ along any interior path. State what fails when extraction is constrained to a capacity $q \leq \bar{q}$, and relate the answer to [Remark 12.1.1](#).

Appendix

Appendix A

The Implicit Function Theorem

[Theorem 8.6.1](#) was stated in [Chapter 8](#) and its proof deferred. It is given here in full, because the theorem carries the comparative statics of every model in this volume and because the argument is the most substantial application of [Theorem 6.4.1](#) outside dynamic programming.

A.1 Two Lemmas

Lemma A.1.1 (Neumann invertibility). Let $M \in M_{m \times m}(\mathbb{R})$ satisfy $\|I - M\| < 1$ in the operator norm. Then M is invertible, $M^{-1} = \sum_{j=0}^{\infty} (I - M)^j$, and $\|M^{-1}\| \leq (1 - \|I - M\|)^{-1}$.

Proof. Write $N := I - M$ and $q := \|N\| < 1$. Submultiplicativity of the operator norm gives $\|N^j\| \leq q^j$, so the partial sums $S_k := \sum_{j=0}^k N^j$ satisfy $\|S_k - S_l\| \leq \sum_{j=l+1}^k q^j \leq q^{l+1}/(1 - q)$ for $k > l$ and are therefore Cauchy. The space $M_{m \times m}(\mathbb{R})$, identified with $\mathbb{R}^{m^2}$, is complete by [Theorem 6.3.1](#), so $S_k \rightarrow S$ for some matrix S with $\|S\| \leq \sum_{j=0}^{\infty} q^j = (1 - q)^{-1}$. Since $S_k(I - N) = (I - N)S_k = I - N^{k+1}$ and $\|N^{k+1}\| \leq q^{k+1} \rightarrow 0$, passing to the limit gives $SM = MS = I$. ■

Lemma A.1.2 (Mean value inequality). Let $U \subseteq \mathbb{R}^m$ be open and convex and let $\Phi: U \rightarrow \mathbb{R}^m$ be continuously differentiable with $\|D\Phi(z)\| \leq \kappa$ for every $z \in U$. Then $\|\Phi(z_1) - \Phi(z_2)\| \leq \kappa \|z_1 - z_2\|$ for all $z_1, z_2 \in U$.

Proof. Convexity places the segment $s \mapsto z_2 + s(z_1 - z_2)$, $s \in [0, 1]$, inside U . The map $s \mapsto \Phi(z_2 + s(z_1 - z_2))$ is continuously differentiable with derivative $D\Phi(z_2 + s(z_1 - z_2))(z_1 - z_2)$ by [Theorem 8.3.1](#), so applying [Theorem 10.3.1](#) to each coordinate,

$$\Phi(z_1) - \Phi(z_2) = \int_0^1 D\Phi(z_2 + s(z_1 - z_2))(z_1 - z_2) ds.$$

Taking norms and using the triangle inequality for integrals gives the bound. The scalar mean value theorem ([Theorem 7.2.5](#)) cannot be applied coordinate by coordinate here, since each coordinate would produce its own intermediate point and no single point serves for all of them; the integral form avoids the difficulty, and that is why it is stated separately. ■

A.2 Proof of the Theorem

Proof of [Theorem 8.6.1](#). Write $A := \nabla_y f(\bar{x}, \bar{y})$, invertible by hypothesis, and define

$$\Phi(x, y) := y - A^{-1}f(x, y), \quad (x, y) \in \Lambda. \quad (\text{A.1})$$

Because A^{-1} is injective, $f(x, y) = 0$ if and only if $\Phi(x, y) = y$: the solutions of the system are the fixed points of $\Phi(x, \cdot)$.

Step 1: $\Phi(x, \cdot)$ is a contraction. Differentiating (A.1) in y gives $\nabla_y \Phi(x, y) = I - A^{-1}\nabla_y f(x, y)$, which vanishes at $(\bar{x}, \bar{y})$. Since f is continuously differentiable, $\nabla_y f$ is continuous, so there are $r > 0$ and $\delta > 0$

with $\overline{B_\delta(\bar{x})} \times \overline{B_r(\bar{y})} \subseteq \Lambda$ and

$$\|I - A^{-1}\nabla_y f(x, y)\| \leq \frac{1}{2} \quad \text{whenever } \|x - \bar{x}\| \leq \delta, \|y - \bar{y}\| \leq r. \quad (\text{A.2})$$

Lemma A.1.2, applied on the convex set $B_r(\bar{y})$ and extended to its closure by continuity, gives

$$\|\Phi(x, y_1) - \Phi(x, y_2)\| \leq \frac{1}{2} \|y_1 - y_2\|, \quad y_1, y_2 \in \overline{B_r(\bar{y})}, \|x - \bar{x}\| \leq \delta. \quad (\text{A.3})$$

Step 2: the ball is invariant. Since $f(\bar{x}, \bar{y}) = 0$ we have $\Phi(\bar{x}, \bar{y}) = \bar{y}$, and $x \mapsto \Phi(x, \bar{y})$ is continuous, so after shrinking δ we may assume $\|\Phi(x, \bar{y}) - \bar{y}\| \leq r/2$ for $\|x - \bar{x}\| \leq \delta$. For such x and any $y \in \overline{B_r(\bar{y})}$,

$$\|\Phi(x, y) - \bar{y}\| \leq \|\Phi(x, y) - \Phi(x, \bar{y})\| + \|\Phi(x, \bar{y}) - \bar{y}\| \leq \frac{1}{2} \|y - \bar{y}\| + \frac{r}{2} \leq r,$$

so $\Phi(x, \cdot)$ maps $\overline{B_r(\bar{y})}$ into itself.

Step 3: existence and uniqueness of e . The set $\overline{B_r(\bar{y})}$ is closed in the complete space $\mathbb{R}^m$, hence complete by [Proposition 6.3.2](#). By (A.3) and Step 2, $\Phi(x, \cdot)$ is a contraction of modulus $\frac{1}{2}$ of that set into itself, so [Theorem 6.4.1](#) supplies a unique fixed point, called $e(x)$. Setting $\Gamma := B_\delta(\bar{x})$ gives $e: \Gamma \rightarrow \overline{B_r(\bar{y})}$ with $f(x, e(x)) = 0$, which is (i); uniqueness of the fixed point at $x = \bar{x}$ forces $e(\bar{x}) = \bar{y}$, which is (ii). Uniqueness also yields the final claim of the theorem: inside $\Gamma \times B_r(\bar{y})$ the solution set of $f = 0$ is exactly the graph of e .

Step 4: (iii). Estimate (A.2) says $\|I - A^{-1}\nabla_y f(x, y)\| \leq \frac{1}{2} < 1$, so [Lemma A.1.1](#) makes $A^{-1}\nabla_y f(x, y)$ invertible and hence $\nabla_y f(x, y)$ invertible throughout the box, in particular at $(x, e(x))$.

Step 5: e is Lipschitz. For $x_1, x_2 \in \Gamma$, using $e(x_i) = \Phi(x_i, e(x_i))$ and then (A.3),

$$\|e(x_1) - e(x_2)\| \leq \underbrace{\|\Phi(x_1, e(x_1)) - \Phi(x_1, e(x_2))\|}_{\leq \frac{1}{2} \|e(x_1) - e(x_2)\|} + \|\Phi(x_1, e(x_2)) - \Phi(x_2, e(x_2))\|,$$

and the second term equals $\|A^{-1}[f(x_2, e(x_2)) - f(x_1, e(x_2))]\| \leq \|A^{-1}\| M \|x_1 - x_2\|$, where M is the supremum of $\|\nabla_x f\|$ over the closed box — finite by [Corollary 5.4.1](#) — and [Lemma A.1.2](#) is applied in the x variable. Rearranging,

$$\|e(x_1) - e(x_2)\| \leq L \|x_1 - x_2\|, \quad L := 2 \|A^{-1}\| M; \quad (\text{A.4})$$

in particular e is continuous.

Step 6: differentiability and (iv). Fix $x \in \Gamma$ and write $y := e(x)$, $B := \nabla_y f(x, y)$ — invertible by Step 4 — and $C := \nabla_x f(x, y)$. For small h put $k := e(x+h) - e(x)$, so $\|k\| \leq L \|h\|$ by (A.4). Differentiability of f at (x, y) gives

$$0 = f(x+h, y+k) - f(x, y) = Ch + Bk + o(\|h\| + \|k\|),$$

and $\|h\| + \|k\| \leq (1+L)\|h\|$ converts the remainder into $o(\|h\|)$. Multiplying by B^{-1} ,

$$k = -B^{-1}Ch + o(\|h\|),$$

which is precisely the statement that e is differentiable at x with $De(x) = -B^{-1}C$, that is (8.9). Continuity of De follows because $x \mapsto \nabla_y f(x, e(x))$ and $x \mapsto \nabla_x f(x, e(x))$ are continuous, by continuity of Df and of e , and because matrix inversion is continuous on the invertible matrices — itself a consequence of [Lemma A.1.1](#). Hence e is continuously differentiable. ■

Remark A.2.1 (Deriving the inverse function theorem). [Corollary 8.6.1](#) follows by applying [Theorem 8.6.1](#) to $F(z, x) := g(x) - z$ on $\mathbb{R}^n \times A$ at the point $(g(\bar{x}), \bar{x})$: here the derivative in the second block is $\nabla_x F = Dg(\bar{x})$, nonsingular by hypothesis, so there is a continuously differentiable e with $g(e(z)) = z$ near $g(\bar{x})$, and (8.9) gives $De(z) = -[Dg(e(z))]^{-1}(-I) = [Dg(e(z))]^{-1}$.

Appendix B

Matrix Differentiation

The identities below are used repeatedly in Chapters 8, 11 and 14. All vectors are columns, gradients of scalar functions are columns, and Df denotes the Jacobian, so that $Df(x) = \nabla f(x)^\top$ for scalar f . This convention is the one fixed in the front matter, and it is the only source of transposition errors in what follows.

B.1 Linear and Quadratic Forms

Proposition B.1.1. Let $a \in \mathbb{R}^n$, $A \in M_{n \times n}(\mathbb{R})$, $B \in M_{m \times n}(\mathbb{R})$, $x \in \mathbb{R}^n$ and $y \in \mathbb{R}^n$.

- (i) $\nabla(\langle a, x \rangle) = a$ and $D(Bx) = B$.
- (ii) $\nabla(x^\top Ax) = (A + A^\top)x$; in particular $\nabla(x^\top Ax) = 2Ax$ when A is symmetric, and $\nabla \|x\|^2 = 2x$.
- (iii) $D^2(x^\top Ax) = A + A^\top$, a constant matrix.
- (iv) If $\varphi(x) := f(Bx)$ with f differentiable, then $\nabla \varphi(x) = B^\top \nabla f(Bx)$.
- (v) $\nabla_x(x^\top Ay) = Ay$ and $\nabla_y(x^\top Ay) = A^\top x$.

Proof. (i) $\langle a, x \rangle = \sum_i a_i x_i$ has $\partial/\partial x_j = a_j$; the i -th coordinate of Bx is $\sum_j B_{ij} x_j$, whose partial derivative in x_j is B_{ij} .

(ii) Writing $q(x) = \sum_{i,j} A_{ij} x_i x_j$,

$$\frac{\partial q}{\partial x_k} = \sum_j A_{kj} x_j + \sum_i A_{ik} x_i = (Ax)_k + (A^\top x)_k,$$

the two sums arising from the occurrences of x_k in the first and in the second factor. Taking $A = I$ gives $\nabla \|x\|^2 = 2x$.

(iii) Differentiate (ii) once more: $x \mapsto (A + A^\top)x$ is linear with Jacobian $A + A^\top$ by (i).

(iv) [Theorem 8.3.1](#) gives $D\varphi(x) = Df(Bx)B$, and transposing, $\nabla \varphi(x) = B^\top \nabla f(Bx)$. The transpose is where the convention matters: Df is a row and ∇f a column.

(v) $x^\top Ay = \sum_{i,j} x_i A_{ij} y_j$ is linear in x for fixed y with coefficient vector Ay , and linear in y for fixed x with coefficient vector $A^\top x$; apply (i). ■

Example B.1.1 (Where each identity is used). Identity (ii) with symmetric A produced (8.3) and the second-order conditions of [Theorem 8.5.3](#). Identity (iv) with $f = \|\cdot\|^2$ gives the least-squares normal equations: minimising $\|y - X\beta\|^2$ over β has gradient $-2X^\top(y - X\beta)$, whose vanishing is $X^\top X\beta = X^\top y$, the calculus route to [Proposition 8.8.1](#) and the algebraic counterpart of the projection argument of [Proposition 11.5.1](#). Identity (iii) shows that the Hessian of a quadratic form is constant, so a quadratic objective is concave globally or not at all — which is why [Theorem 8.5.4](#) needs to be checked only once for such objectives rather than at every point.

B.2 Inverse, Trace and Determinant

Proposition B.2.1. Let $X \in M_{n \times n}(\mathbb{R})$ be invertible and $A \in M_{n \times n}(\mathbb{R})$.

(i) If $t \mapsto X(t)$ is differentiable with $X(t)$ invertible, then

$$\frac{d}{dt}X(t)^{-1} = -X(t)^{-1} \left(\frac{d}{dt}X(t) \right) X(t)^{-1}. \quad (\text{B.1})$$

(ii) $\partial \text{tr}(AX) / \partial X = A^\top$, entrywise.

(iii) $\partial \log \det X / \partial X = (X^{-1})^\top$.

Proof. (i) Differentiate $X(t)X(t)^{-1} = I$ by the product rule (Proposition 7.3.2, applied entrywise), obtaining $\dot{X}X^{-1} + X \frac{d}{dt}(X^{-1}) = 0$, and multiply on the left by X^{-1} .

(ii) $\text{tr}(AX) = \sum_{i,j} A_{ij}X_{ji}$, whose partial derivative in X_{ji} is A_{ij} ; assembling these into the matrix indexed by (j, i) gives $A^\top$.

(iii) Jacobi's formula $d(\det X) = \det X \cdot \text{tr}(X^{-1} dX)$, a consequence of the cofactor expansion of Definition 2.4.1, gives $d \log \det X = \text{tr}(X^{-1} dX)$; comparing with (ii) at $A = X^{-1}$ gives the stated gradient. A derivation of Jacobi's formula is in Axler (2024, Ch. 10). ■

Identity (B.1) is the matrix analogue of $\frac{d}{dt}(1/x) = -\dot{x}/x^2$, and it is what the Riccati recursion (14.26) and the comparative statics formula (8.9) reduce to in the scalar case. Identity (iii) is the first-order condition of Gaussian maximum likelihood with respect to a covariance matrix, and it is the reason the log-likelihood is concave in the precision matrix rather than in the covariance matrix.

Appendix C

The Multivariate Normal Distribution

[Theorem 14.6.1](#) and [Proposition 14.5.1](#) are the scalar and bivariate cases of one statement, recorded here in general form because econometric and macroeconomic applications use it with several signals and several states at once.

C.1 Definition and Affine Images

Definition C.1.1 (Multivariate normal). A random vector $X = (X_1, \dots, X_n)^\top$ is *multivariate normal distribution* with mean $\mu \in \mathbb{R}^n$ and covariance $\Sigma \succ 0$, written $X \sim \mathcal{N}(\mu, \Sigma)$, if it has density

$$f(x) = \frac{1}{(2\pi)^{n/2}(\det \Sigma)^{1/2}} \exp\left(-\frac{1}{2}(x - \mu)^\top \Sigma^{-1}(x - \mu)\right), \quad x \in \mathbb{R}^n. \quad (\text{C.1})$$

That (C.1) integrates to one is [Proposition 10.7.2](#) combined with the substitution $z = \Sigma^{-1/2}(x - \mu)$, whose Jacobian determinant is $(\det \Sigma)^{-1/2}$ by [Theorem 10.4.2](#); the symmetric square root $\Sigma^{1/2} = Q\Lambda^{1/2}Q^\top$ exists by [Theorem 2.5.1](#) and has strictly positive diagonal $\Lambda^{1/2}$ by [Proposition 2.5.1](#).

Proposition C.1.1 (Affine images and the moment generating function). Let $X \sim \mathcal{N}(\mu, \Sigma)$, $b \in \mathbb{R}^m$ and $B \in M_{m \times n}(\mathbb{R})$ of rank m . Then

$$b + BX \sim \mathcal{N}(b + B\mu, B\Sigma B^\top), \quad \mathbb{E}[e^{\langle t, X \rangle}] = \exp\left(\langle t, \mu \rangle + \frac{1}{2}t^\top \Sigma t\right) \quad \text{for } t \in \mathbb{R}^n. \quad (\text{C.2})$$

In particular every linear combination $\langle a, X \rangle$ is univariate normal with mean $\langle a, \mu \rangle$ and variance $a^\top \Sigma a$.

Proof. For the moment generating function, complete the square in the exponent of (C.1): with $\nu := \mu + \Sigma t$,

$$\langle t, x \rangle - \frac{1}{2}(x - \mu)^\top \Sigma^{-1}(x - \mu) = \langle t, \mu \rangle + \frac{1}{2}t^\top \Sigma t - \frac{1}{2}(x - \nu)^\top \Sigma^{-1}(x - \nu),$$

as multiplying out both sides and using the symmetry of Σ^{-1} verifies. Integrating the remaining Gaussian kernel gives 1 and leaves the stated formula. For the affine image, the moment generating function of $b + BX$ at $s \in \mathbb{R}^m$ is

$$\mathbb{E}[e^{\langle s, b + BX \rangle}] = e^{\langle s, b \rangle} \mathbb{E}[e^{\langle B^\top s, X \rangle}] = \exp\left(\langle s, b + B\mu \rangle + \frac{1}{2}s^\top B\Sigma B^\top s\right),$$

which is the moment generating function of $\mathcal{N}(b + B\mu, B\Sigma B^\top)$; a distribution on $\mathbb{R}^m$ whose moment generating function is finite on a neighbourhood of the origin is determined by it (Ash and Doléans-Dade 1999, Ch. 5). The rank condition keeps $B\Sigma B^\top$ positive definite. ■

Corollary C.1.1 (Uncorrelated implies independent). Let $(X_1^\top, X_2^\top)^\top \sim \mathcal{N}(\mu, \Sigma)$, with Σ partitioned into blocks $\Sigma_{11}, \Sigma_{12}, \Sigma_{21}, \Sigma_{22}$ conformably. Then X_1 and X_2 are independent if and only if $\Sigma_{12} = 0$.

Proof. Independence implies zero covariance for any square-integrable pair, by [Proposition 14.2.1\(iv\)](#). Conversely, if $\Sigma_{12} = 0$ then Σ and Σ^{-1} are block diagonal and $\det \Sigma = \det \Sigma_{11} \det \Sigma_{22}$, so (C.1) factors into the product of the two marginal densities. ■

The corollary is false without joint normality. Let Z be standard normal and let S be independent of Z with $\mathbb{P}(S = 1) = \mathbb{P}(S = -1) = \frac{1}{2}$. Then SZ is standard normal and $\text{Cov}(Z, SZ) = \mathbb{E}[S]\mathbb{E}[Z^2] = 0$, yet $|Z| = |SZ|$, so the two are far from independent. The pair (Z, SZ) is not jointly normal, since $Z + SZ$ equals 0 with probability $\frac{1}{2}$ and no non-degenerate normal variable has an atom.

C.2 The Conditional Law

Theorem C.2.1 (Conditional distribution). Let $(X_1^\top, X_2^\top)^\top$ be multivariate normal with mean $(\mu_1^\top, \mu_2^\top)^\top$ and covariance blocks $\Sigma_{11}, \Sigma_{12}, \Sigma_{21}, \Sigma_{22}$, with Σ_{22} invertible. Then

$$X_1 \mid X_2 = x_2 \sim \mathcal{N}\left(\mu_1 + \Sigma_{12}\Sigma_{22}^{-1}(x_2 - \mu_2), \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21}\right). \quad (\text{C.3})$$

The conditional mean is affine in x_2 and the conditional covariance does not depend on x_2 .

Proof. Set $Z := X_1 - \Sigma_{12}\Sigma_{22}^{-1}X_2$. By [Proposition C.1.1](#) the pair $(Z^\top, X_2^\top)^\top$ is an affine image of $(X_1^\top, X_2^\top)^\top$ and is therefore jointly normal, and

$$\text{Cov}(Z, X_2) = \Sigma_{12} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{22} = 0.$$

[Corollary C.1.1](#) makes Z independent of X_2 . Hence, conditionally on $X_2 = x_2$, the vector $X_1 = Z + \Sigma_{12}\Sigma_{22}^{-1}x_2$ is Z shifted by a constant, so its conditional law is normal with mean

$$\mathbb{E}[Z] + \Sigma_{12}\Sigma_{22}^{-1}x_2 = \mu_1 - \Sigma_{12}\Sigma_{22}^{-1}\mu_2 + \Sigma_{12}\Sigma_{22}^{-1}x_2$$

and covariance

$$\text{Var}(Z) = \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21} + \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{22}\Sigma_{22}^{-1}\Sigma_{21} = \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21},$$

which together give (C.3). ■

Remark C.2.1 (Recovering the results of [Chapter 14](#)). Take the blocks to be scalars with $X_1 = Y$ and $X_2 = X$. Then $\Sigma_{12}\Sigma_{22}^{-1} = \text{Cov}(X, Y)/\text{Var}(X)$ and (C.3) is (14.21), so [Proposition 14.5.1](#) is the bivariate case.

Take instead $X_1 = X \sim \mathcal{N}(\mu_x, \sigma_x^2)$ and $X_2 = Y = X + \varepsilon$ with $\varepsilon \sim \mathcal{N}(0, \sigma_\varepsilon^2)$ independent of X . The pair is jointly normal by [Proposition C.1.1](#), with $\Sigma_{11} = \Sigma_{12} = \sigma_x^2$ and $\Sigma_{22} = \sigma_x^2 + \sigma_\varepsilon^2$, so (C.3) gives conditional mean $\mu_x + \frac{\sigma_x^2}{\sigma_x^2 + \sigma_\varepsilon^2}(y - \mu_x)$ and conditional variance $\sigma_x^2 - \frac{\sigma_x^4}{\sigma_x^2 + \sigma_\varepsilon^2} = \frac{\sigma_x^2 \sigma_\varepsilon^2}{\sigma_x^2 + \sigma_\varepsilon^2}$, which is (14.22). The completing-the-square derivation given there is therefore an instance of the orthogonalisation argument used above, and the statement that precisions add is the identity $(\Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21})^{-1} = \sigma_x^{-2} + \sigma_\varepsilon^{-2}$ in this parametrisation.

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